

mielke-geometroynamics: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometroynamics_F00001	7	■ 1.000	A	A	potential 1-form A via electron interference.
mielke-geometroynamics_F00002	7	■ 0.980	$\operatorname{SL}(5, \mathbb{R})$	$\mathrm{SL}(5, \mathbb{R})$	model with a primordial $\mathrm{SL}(5, \mathbb{R})$ gauge group is presented. After a tiny
mielke-geometroynamics_F00003	15	■ 0.898	$\mathbb{R}$	$\mathbb{R}$	13 Topological $\mathrm{SL}(5, \mathbb{R})$ Gauge-Invariant Action
mielke-geometroynamics_F00004	17	■ 1.000	β	β	and Fermi’s theory of weak interactions, being the cause for the β -decay of certain
mielke-geometroynamics_F00005	17	■ 0.965	$\mathrm{W}^{\pm}$	$W^{\pm}$	empirical evidence by the detection of the mediating $W^{\pm}$ and Z^0 bosons as the
mielke-geometroynamics_F00006	17	■ 0.965	Z^0	Z^0	empirical evidence by the detection of the mediating $W^{\pm}$ and Z^0 bosons as the
mielke-geometroynamics_F00007	19	■ 1.000	1×10^{-12}	1×10^{-12}	mass is measured nowadays with a precision better than 1×10^{-12} ; this accuracy
mielke-geometroynamics_F00008	20	■ 1.000	∂_j	∂_j	to the electromagnetic fields if the partial derivatives ∂_j in the Dirac equation are
mielke-geometroynamics_F00009	21	■ 1.000	$D_j := \partial_j + i A_j$	$D_j := \partial_j + i A_j$	replaced by covariant derivatives $D_j := \partial_j + i A_j$. In the latter concept, a “minimal”
mielke-geometroynamics_F00010	21	■ 1.000	A_j	A_j	substitution of the electromagnetic vector potential A_j is taken into consideration.
mielke-geometroynamics_F00011	21	■ 1.000	$\psi \rightarrow e^{i\theta} \psi$	$\psi \rightarrow e^{i\theta} \psi$	remains invariant under the <i>local</i> transformation $\psi \rightarrow e^{i\theta} \psi$ of the phase of the
mielke-geometroynamics_F00012	21	■ 1.000	ψ	ψ	wave function ψ and the simultaneous transformation $A_j \rightarrow A_j - \partial_j \theta$ of the four-
mielke-geometroynamics_F00013	21	■ 1.000	$A_j \rightarrow A_j - \partial_j \theta$	$A_j \rightarrow A_j - \partial_j \theta$	wave function ψ and the simultaneous transformation $A_j \rightarrow A_j - \partial_j \theta$ of the four-
mielke-geometroynamics_F00014	22	■ 0.526	$n > 4$	$n > 4$	($n > 4$) spaces and was, to a certain extent, anticipated by the pioneering works of
mielke-geometroynamics_F00015	23	■ 0.343	$\{ \}^{+}$	$+$	an SU(3) nonet that consists of the massive spin-2 ⁺ mesons f, f′, A ₂ , K*(1420).
mielke-geometroynamics_F00016	23	■ 0.343	$A_2, \bar{K}^*(1420)$	A ₂ , K*(1420)	an SU(3) nonet that consists of the massive spin-2 ⁺ mesons f, f′, A ₂ , K*(1420).
mielke-geometroynamics_F00017	23	■ 0.987	$\operatorname{SL}(2, \mathbb{C})$	$\mathrm{SL}(2, \mathbb{C})$	$\mathrm{SL}(2, \mathbb{C})$, respectively. If we apply these findings to <i>strong gravity</i> , then it is obvious
mielke-geometroynamics_F00018	23	■ 0.697	$\operatorname{SU}(3)$	$\mathrm{SU}(3)$	to tie together the internal symmetries, such as the SU(3) one, with the spacetime
mielke-geometroynamics_F00019	23	■ 0.997	$\operatorname{SL}(2, \mathbb{C}) \subset \operatorname{SL}(2, \mathbb{C}) \otimes \operatorname{SU}(3) \subset \operatorname{SL}(6, \mathbb{C})$	$\mathrm{SL}(2, \mathbb{C}) \subset \mathrm{SL}(2, \mathbb{C}) \otimes \mathrm{SU}(3) \subset \mathrm{SL}(6, \mathbb{C})$	chain $\mathrm{SL}(2, \mathbb{C}) \subset \mathrm{SL}(2, \mathbb{C}) \otimes \mathrm{SU}(3) \subset \mathrm{SL}(6, \mathbb{C})$. If these symmetries are regarded
mielke-geometroynamics_F00020	24	■ 0.834	$\operatorname{SL}(6, \mathbb{C})$	$\mathrm{SL}(6, \mathbb{C})$	as local ones, we get an $\mathrm{SL}(6, \mathbb{C})$ gauge theory of strong interaction (SALAM 1973)
mielke-geometroynamics_F00021	24	■ 1.000	$\ell := \ell^* / \sqrt{\kappa}$	$\ell := \ell^* / \sqrt{\kappa}$	a further length $\ell := \ell^* / \sqrt{\kappa}$ is introduced into particle physics that corresponds to
mielke-geometroynamics_F00022	24	■ 0.995	$G_S = G_N / \kappa$	$G_S = G_N / \kappa$	the scaled “Newtonian” coupling constant $G_S = G_N / \kappa$ of strong gravity. SALAM &
mielke-geometroynamics_F00023	25	■ 1.000	$\sqrt{s} = 540 \text{ GeV}$	$\sqrt{s} = 540 \text{ GeV}$	associated missing energy at $\sqrt{s} = 540 \text{ GeV}$. Phys Lett B 122:103–116

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mielke-geometrodynamics_F00024	30	■ 1.000	$\mathrm{x}:\mathrm{M}\rightarrow\mathrm{R}^{\mathrm{n}}$	$\mathbf{x}:\mathbf{M}\rightarrow\mathbb{R}^n$	achieve this, we consider injective (reversible) mappings $\mathbf{x}:\mathbf{M}\rightarrow\mathbb{R}^n$, whose range
mielke-geometrodynamics_F00025	30	■ 1.000	E^{n}	$\mathbf{E}^n$	extends to an open subset of the usual Euclidean space $\mathbf{E}^n$. Such a map together with
mielke-geometrodynamics_F00026	30	■ 0.993	U_{i}	$\mathbf{U}_i$	its domain $\mathbf{U}_i$ is called an n-dimensional chart. By the projection π_i onto the single
mielke-geometrodynamics_F00027	30	■ 0.993	π_{i}	π_i	its domain $\mathbf{U}_i$ is called an n-dimensional chart. By the projection π_i onto the single
mielke-geometrodynamics_F00028	30	■ 1.000	R^{n}	$\mathbb{R}^n$	axes of the coordinate system of $\mathbb{R}^n$, the <i>local coordinates</i> $x_i=\pi_i\circ\mathbf{x}$, $i=1,\ldots,n$
mielke-geometrodynamics_F00029	30	■ 1.000	$\mathrm{x}_{\mathrm{i}}=\pi_{\mathrm{i}}\circ\mathrm{x}$, $i=1,\ldots,\mathrm{n}$	$\mathbf{x}_i=\pi_i\circ\mathbf{x}, i=1,\ldots,n$	axes of the coordinate system of $\mathbb{R}^n$, the <i>local coordinates</i> $x_i=\pi_i\circ\mathbf{x}$, $i=1,\ldots,n$
mielke-geometrodynamics_F00030	30	■ 1.000	S^{n}	$\mathbf{S}^n$	sphere $\mathbf{S}^n$). However, it is possible to construct a collection of charts of identical
mielke-geometrodynamics_F00031	31	■ 0.996	$\mathrm{U}_{\mathrm{i}}\subset\mathrm{M}$	$\mathbf{U}_i\subset\mathbf{M}$	dimensions so that the set-theoretic union of their domains $\mathbf{U}_i\subset\mathbf{M}$ cover $\mathbf{M}$ com-
mielke-geometrodynamics_F00032	31	■ 0.999	$\mathscr{D}(\mathrm{M})$	$\mathscr{D}(\mathbf{M})$	form an infinite-dimensional group with respect to composition, i.e., the group $\mathscr{D}(\mathbf{M})$
mielke-geometrodynamics_F00033	31	■ 1.000	$(\mathrm{U}_{\mathrm{i}},\mathrm{x}_{\mathrm{i}})$	$(\mathbf{U}_i,\mathbf{x}_i)$	A collection $(\mathbf{U}_i,\mathbf{x}_i)$ of compatible charts is called a $\mathbf{C}^\infty$ -atlas of $\mathbf{M}$. (The common
mielke-geometrodynamics_F00034	31	■ 1.000	C^{∞}	$\mathbf{C}^\infty$	A collection $(\mathbf{U}_i,\mathbf{x}_i)$ of compatible charts is called a $\mathbf{C}^\infty$ -atlas of $\mathbf{M}$. (The common
mielke-geometrodynamics_F00035	31	■ 0.997	C^{r}	$\mathbf{C}^r$	of class $\mathbf{C}^r$). This atlas again can be extended unequivocally to a <i>complete</i> one, which
mielke-geometrodynamics_F00036	32	■ 1.000	$\mathrm{E}^{\mathrm{n}(\mathrm{n}+1)/2}$	$\mathbf{E}^{n(n+1)/2}$	locally isometric embeddings of analytic Riemannian manifolds into $\mathbf{E}^{n(n+1)/2}$ are
mielke-geometrodynamics_F00037	32	■ 0.936	$N=n(3\ n+11)\ / \ 2$	$N=n(3n+11)/2$	dimension $N=n(3n+11)/2$ was proven by NASH (1956); for a noncompact one,
mielke-geometrodynamics_F00038	32	■ 1.000	$N=n(n+1)(3\ n+11)\ / \ 2$	$N=n(n+1)(3n+11)/2$	the extravagant dimension $N=n(n+1)(3n+11)/2$ results; cf. CHEN (2000) for a
mielke-geometrodynamics_F00039	32	■ 1.000	E^4	$\mathbb{E}^4$	surface of a rotating Kerr–Newman black hole into $\mathbb{E}^4$.
mielke-geometrodynamics_F00040	32	■ 0.997	$\mathrm{T}_{\mathrm{m}}(\mathrm{M})$	$\mathbf{T}_{\mathbf{m}}(\mathbf{M})$	<i>space</i> $\mathbf{T}_{\mathbf{m}}(\mathbf{M})$ attached to a point $\mathbf{m}\in\mathbf{M}$. Contrary to the given illustration (see
mielke-geometrodynamics_F00041	32	■ 0.997	$\mathrm{m}\in\mathrm{M}$	$\mathbf{m}\in\mathbf{M}$	<i>space</i> $\mathbf{T}_{\mathbf{m}}(\mathbf{M})$ attached to a point $\mathbf{m}\in\mathbf{M}$. Contrary to the given illustration (see
mielke-geometrodynamics_F00042	32	■ 0.573	$\mathrm{s}(\mathrm{t})$	$\mathbf{s}(\mathbf{t})$	to achieve this, a curve $\mathbf{s}(\mathbf{t})$ on the manifold is chosen that passes through a point
mielke-geometrodynamics_F00043	33	■ 0.985	$\mathrm{e}(\mathrm{m})$	$\mathbf{e}(\mathbf{m})$	tiable functions f to the space $\mathbb{R}$ of real numbers, defines the <i>tangent vector</i> $\mathbf{e}(\mathbf{m})$ or
mielke-geometrodynamics_F00044	33	■ 0.999	$\mathrm{e}_{\alpha}(\mathrm{m})$, $\alpha=1,\ldots,\mathrm{n}$	$\mathbf{e}_{\alpha}(\mathbf{m}), \alpha=1,\ldots,n$	independent vectors $\mathbf{e}_{\alpha}(\mathbf{m}), \alpha=1,\ldots,n$. It follows from (2.1.2) that the partial
mielke-geometrodynamics_F00045	33	■ 1.000	$\partial_{\mathrm{i}}:=\partial/\partial\mathrm{x}^{\mathrm{i}}$	$\partial_i:=\partial/\partial\mathbf{x}^i$	derivates $\partial_i:=\partial/\partial\mathbf{x}^i$ constitute a local basis for $\mathbf{T}_{\mathbf{m}}(\mathbf{M})$ with respect to a <i>holonomic</i>
mielke-geometrodynamics_F00046	33	■ 1.000	$[\mathrm{e}_{\mathrm{i}}(\mathrm{m}),\mathrm{e}_{\mathrm{j}}(\mathrm{m})]=[\partial_{\mathrm{i}},\partial_{\mathrm{j}}]=0$	$[\mathbf{e}_i(\mathbf{m}),\mathbf{e}_j(\mathbf{m})]=[\partial_i,\partial_j]=0$	coordinate system. By definition, the relation $[\mathbf{e}_i(\mathbf{m}),\mathbf{e}_j(\mathbf{m})]=[\partial_i,\partial_j]=0$ is satis-
mielke-geometrodynamics_F00047	33	■ 1.000	$\mathrm{e}_{\alpha}^{\mathrm{i}}(\mathrm{m})$	$\mathbf{e}_{\alpha}^i(m)$	The occurring nonsingular matrices $\mathbf{e}_{\alpha}^i(m)$ depend on the point in question and
mielke-geometrodynamics_F00048	33	■ 0.628	$\mathrm{T}(\mathrm{M})$	$\mathbf{T}(\mathbf{M})$	In addition to the tangent bundle $\mathbf{T}(\mathbf{M})$, a so-called <i>dual</i> bundle $\mathbf{T}^*(\mathbf{M})=\bigcup_{\mathbf{m}\in\mathbf{M}}\mathbf{T}_{\mathbf{m}}^*(\mathbf{M})$
mielke-geometrodynamics_F00049	33	■ 0.628	$\mathrm{T}^{\ast}(\mathrm{M})=\bigcup_{\mathbf{m}\in\mathbf{M}}\mathrm{T}_{\mathbf{m}}^{\ast}(\mathrm{M})$	$\mathbf{T}^*(\mathbf{M})=\bigcup_{\mathbf{m}\in\mathbf{M}}\mathbf{T}_{\mathbf{m}}^*(\mathbf{M})$	In addition to the tangent bundle $\mathbf{T}(\mathbf{M})$, a so-called <i>dual</i> bundle $\mathbf{T}^*(\mathbf{M})=\bigcup_{\mathbf{m}\in\mathbf{M}}\mathbf{T}_{\mathbf{m}}^*(\mathbf{M})$
mielke-geometrodynamics_F00050	33	■ 1.000	$\mathrm{T}_{\mathrm{m}}^{\ast}(\mathrm{M})$	$\mathbf{T}_{\mathbf{m}}^*(\mathbf{M})$	consisting of all possible cotangent spaces $\mathbf{T}_{\mathbf{m}}^*(\mathbf{M})$ on the manifold can be constructed.

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mielke-geometrodynamics_F00051	33	■ 1.000	$\vartheta^{\alpha}, \alpha=1, \ldots, n$	$\vartheta^{\alpha}, \alpha = 1, \ldots, n$	Let $\vartheta^{\alpha}, \alpha = 1, \ldots, n$, be a basis of $T_{\mathfrak{m}}^*(M)$. Then the “duality” ³ requires that ϑ^{α}
mielke-geometrodynamics_F00052	33	■ 1.000	ϑ^{α}	ϑ^{α}	Let $\vartheta^{\alpha}, \alpha = 1, \ldots, n$, be a basis of $T_{\mathfrak{m}}^*(M)$. Then the “duality” ³ requires that ϑ^{α}
mielke-geometrodynamics_F00053	33	■ 0.999	$\mathrm{d}x^i$	$\mathrm{d}x^i$	by $\mathrm{d}x^i$. In terms of the rigid basis of E^n , the 1-forms
mielke-geometrodynamics_F00054	34	■ 0.998	$\mathrm{d} \vartheta^{\alpha}$	$\mathrm{d}\vartheta^{\alpha}$	derivative $\mathrm{d}\vartheta^{\alpha}$, in contrast to that of $\mathrm{d}x^j$, is generally different from zero. Following
mielke-geometrodynamics_F00055	34	■ 0.998	$\mathrm{d}x^j$	$\mathrm{d}x^j$	derivative $\mathrm{d}\vartheta^{\alpha}$, in contrast to that of $\mathrm{d}x^j$, is generally different from zero. Following
mielke-geometrodynamics_F00056	34	■ 0.599	$\mathrm{E}_{\cdot j}^{\alpha}$	$E_{\cdot j}^{\alpha}$	(2.2.1), $E_{\cdot j}^{\alpha}$ can be understood as the dual tetrad field “reciprocal” to e_{β}^i (cf. (2.1.3)).
mielke-geometrodynamics_F00057	34	■ 0.599	e_{β}^i	e_{β}^i	(2.2.1), $E_{\cdot j}^{\alpha}$ can be understood as the dual tetrad field “reciprocal” to e_{β}^i (cf. (2.1.3)).
mielke-geometrodynamics_F00058	34	■ 0.979	$\otimes_{\mathrm{s}, \mathrm{a}}$	$\otimes_{\mathrm{s}, \mathrm{a}}$	Completely symmetric or antisymmetric products are denoted by $\otimes_{\mathrm{s}, \mathrm{a}}$ or, even more
mielke-geometrodynamics_F00059	34	■ 1.000	$\vee$	$\vee$	commonly, by the symbols $\vee$ and $\wedge$, respectively. Note that a tensor is a geometric
mielke-geometrodynamics_F00060	34	■ 1.000	$\wedge$	$\wedge$	commonly, by the symbols $\vee$ and $\wedge$, respectively. Note that a tensor is a geometric
mielke-geometrodynamics_F00061	34	■ 0.962	$\mathrm{T}_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(\mathfrak{m})$	$T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(\mathfrak{m})$	The quantities $T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(\mathfrak{m})$ are called its p covariant and q contravariant components,
mielke-geometrodynamics_F00062	34	■ 0.919	$\mathrm{T}_{\mathfrak{p}}^{\mathfrak{q}}(\mathfrak{M})$	$T_{\mathfrak{p}}^{\mathfrak{q}}(M)$	and $T_{\mathfrak{p}}^{\mathfrak{q}}(M)$ may be regarded as a bundle associated with $L(M)$ having the manifold
mielke-geometrodynamics_F00063	34	■ 0.919	$\mathrm{L}(\mathfrak{M})$	$L(M)$	and $T_{\mathfrak{p}}^{\mathfrak{q}}(M)$ may be regarded as a bundle associated with $L(M)$ having the manifold
mielke-geometrodynamics_F00064	34	■ 0.708	$\mathrm{D}^{(\mathfrak{p}, \mathfrak{q})}$	$D^{(\mathfrak{p}, \mathfrak{q})}$	M as a base, the tensor representation $D^{(\mathfrak{p}, \mathfrak{q})}$ of $\mathrm{GL}(n, \mathbb{R})$, i.e., $\rho^{(\mathfrak{p}, \mathfrak{q})}(\mathrm{GL}(n, \mathbb{R}))$ as a
mielke-geometrodynamics_F00065	34	■ 0.708	$\rho^{(\mathfrak{p}, \mathfrak{q})}(\mathrm{GL}(\mathfrak{n}, \mathbb{R}))$	$\rho^{(\mathfrak{p}, \mathfrak{q})}(\mathrm{GL}(\mathfrak{n}, \mathbb{R}))$	M as a base, the tensor representation $D^{(\mathfrak{p}, \mathfrak{q})}$ of $\mathrm{GL}(n, \mathbb{R})$, i.e., $\rho^{(\mathfrak{p}, \mathfrak{q})}(\mathrm{GL}(n, \mathbb{R}))$ as a
mielke-geometrodynamics_F00066	34	■ 0.907	$\otimes^{\mathfrak{p}} \mathbb{R}^n \otimes^{\mathfrak{q}} \mathbb{R}^n$	$\otimes^{\mathfrak{p}} \mathbb{R}^n \otimes^{\mathfrak{q}} \mathbb{R}^n$	structure group and the (Cartesian) product $\otimes^{\mathfrak{p}} \mathbb{R}^n \otimes^{\mathfrak{q}} \mathbb{R}^n$ as a typical fiber.
mielke-geometrodynamics_F00067	34	■ 1.000	$\mathrm{g}_{\mu \nu}$	$\mathfrak{g}_{\mu \nu}$	the scale of the manifold. Due to the postulated symmetry of the components $\mathfrak{g}_{\mu \nu}$,
mielke-geometrodynamics_F00068	35	■ 0.651	$(\mathfrak{p}, 0)$	$(\mathfrak{p}, 0)$	(p, 0). Usually, $\wedge := \otimes_{\mathrm{a}}$ is an abbreviation for the antisymmetrized tensor product,
mielke-geometrodynamics_F00069	35	■ 0.651	$\wedge := \otimes_{\mathrm{a}}$	$\wedge := \otimes_{\mathrm{a}}$	(p, 0). Usually, $\wedge := \otimes_{\mathrm{a}}$ is an abbreviation for the antisymmetrized tensor product,
mielke-geometrodynamics_F00070	35	■ 1.000	(1939, 1957)	(1939, 1957)	(1939, 1957) of the representations of the transformation group in flat spacetime, i.e.,
mielke-geometrodynamics_F00071	36	■ 1.000	$\mathscr{H}$	$\mathcal{H}$	rays in a Hilbert space $\mathcal{H}$, which is to be transformed according to an irreducible
mielke-geometrodynamics_F00072	36	■ 0.997	φ	φ	assumed that φ at a point $\mathfrak{m}$ is equal to the value $\varphi(\mathfrak{m})$ in a complex vector space
mielke-geometrodynamics_F00073	36	■ 0.997	$\varphi(\mathfrak{m})$	$\varphi(\mathfrak{m})$	assumed that φ at a point $\mathfrak{m}$ is equal to the value $\varphi(\mathfrak{m})$ in a complex vector space
mielke-geometrodynamics_F00074	36	■ 1.000	$\mathrm{V}_{\mathfrak{m}}$	$V_{\mathfrak{m}}$	$V_{\mathfrak{m}}$. Additionally, it can be postulated that a vector space of this kind is attached
mielke-geometrodynamics_F00075	36	■ 0.941	π	π	Definition A fiber bundle (F, M, π) consists of two C^{∞} -differentiable manifolds F
mielke-geometrodynamics_F00076	36	■ 0.362	$\mathrm{F}_{\mathfrak{m}} := \pi^{-1}(\mathfrak{m})$	$F_{\mathfrak{m}} := -\pi^{-1}(\mathfrak{m})$	and π the <i>projection</i> . The closed submanifold $F_{\mathfrak{m}} := \pi^{-1}(\mathfrak{m})$ will be called a <i>typical</i>
mielke-geometrodynamics_F00077	37	■ 0.980	$\pi^{-1}(\mathfrak{U}) \approx \mathfrak{U} \times \mathrm{F}_{\mathfrak{m}}$	$\pi^{-1}(U) \approx U \times F_{\mathfrak{m}}$	point $\mathfrak{m} \in M$ has a neighborhood U such that $\pi^{-1}(U) \approx U \times F_{\mathfrak{m}}$ is isomorphic to the

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mielke-geometrodynamics_F00078	37	0.686	$\left(\mathrm{U} \times \mathrm{F}_{\mathrm{m}}, \mathrm{U}, \pi_{\mathrm{u}}\right)$	$(\mathrm{U} \times \mathrm{F}_{\mathrm{m}}, \mathrm{U}, \pi_{\mathrm{u}})$	product bundle $(\mathrm{U} \times \mathrm{F}_{\mathrm{m}}, \mathrm{U}, \pi_{\mathrm{u}})$. The mapping
mielke-geometrodynamics_F00079	37	0.637	$\mathrm{P}(\mathrm{M}, \mathrm{G}, \pi, \delta)$	$\mathrm{P}(\mathrm{M}, \mathrm{G}, \pi, \delta)$	group, then the collection $\mathrm{P}(\mathrm{M}, \mathrm{G}, \pi, \delta)$ is called a <i>principal fiber bundle</i> if
mielke-geometrodynamics_F00080	37	0.790	$(\mathrm{P}, \mathrm{M}, \pi)$	$(\mathrm{P}, \mathrm{M}, \pi)$	(i) $(\mathrm{P}, \mathrm{M}, \pi)$ is a fiber bundle with typical fiber G ;
mielke-geometrodynamics_F00081	37	0.866	$\mathrm{P}, \mathrm{G}, \delta$	$\mathrm{P}, \mathrm{G}, \delta$	(ii) $(\mathrm{P}, \mathrm{G}, \delta)$ is a G -manifold (whereby G is acting on P from the right);
mielke-geometrodynamics_F00082	37	0.662	$\mathrm{U} \subset$	$\mathrm{U} \subset$	(iii) P is <i>locally</i> trivial, which means that every point of M has a neighborhood $\mathrm{U} \subset$
mielke-geometrodynamics_F00083	37	0.941	$\iota: \mathrm{U} \times \mathrm{G} \rightarrow \pi^{-1}(\mathrm{U})$	$\iota: \mathrm{U} \times \mathrm{G} \rightarrow \pi^{-1}(\mathrm{U})$	M together with an isomorphism $\iota: \mathrm{U} \times \mathrm{G} \rightarrow \pi^{-1}(\mathrm{U})$ for which the following
mielke-geometrodynamics_F00084	37	0.857	G	G	A manifold is called a <i>G-manifold</i> $(\mathrm{P}, \mathrm{G}, \delta)$ if G acts on it as a free transformation
mielke-geometrodynamics_F00085	37	0.970	$\mathrm{p}_2 = \mathrm{g} \mathrm{p}_1$	$\mathrm{p}_2 = \mathrm{g} \mathrm{p}_1$	there always existed a g that would provide a relation $\mathrm{p}_2 = \mathrm{g} \mathrm{p}_1$ for given p_1
mielke-geometrodynamics_F00086	37	0.970	p_1	p_1	there always existed a g that would provide a relation $\mathrm{p}_2 = \mathrm{g} \mathrm{p}_1$ for given p_1
mielke-geometrodynamics_F00087	37	0.666	p_2	p_2	and p_2 , then G considered as a manifold would be isomorphic to the base M .)
mielke-geometrodynamics_F00088	38	0.985	π, δ	π, δ	this purpose, the fiber bundles $\mathrm{V}(\mathrm{M}, \mathrm{F}; \mathrm{G}, \mathrm{P})$ associated with $\mathrm{P}(\mathrm{M}, \mathrm{G}, \pi, \delta)$ are
mielke-geometrodynamics_F00089	38	0.999	$\mathrm{P} \times \mathrm{F}$	$\mathrm{P} \times \mathrm{F}$	i.e., on the product space $\mathrm{P} \times \mathrm{F}$, an action from the right-hand side is defined by
mielke-geometrodynamics_F00090	38	0.678	$\mathrm{V} = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$	$\mathrm{V} = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$	by $\mathrm{V} = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$. The isomorphism $\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$ concerning the original domain
mielke-geometrodynamics_F00091	38	0.678	$\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$	$\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$	by $\mathrm{V} = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$. The isomorphism $\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$ concerning the original domain
mielke-geometrodynamics_F00092	38	0.307	$\pi_{\mathrm{v}}^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$	$\pi_{\mathrm{v}}^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$	induce the isomorphism $\pi_{\mathrm{v}}^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$ in the associated bundle. Provided that
mielke-geometrodynamics_F00093	38	0.907	$\pi_{\mathrm{v}}^{-1}(\mathrm{U})$	$\pi_{\mathrm{v}}^{-1}(\mathrm{U})$	$\pi_{\mathrm{v}}^{-1}(\mathrm{U})$ is an open submanifold of V , then V can be equipped with a differentiable
mielke-geometrodynamics_F00094	38	0.998	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	occurs, but its linear representation $\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$, i.e., the Lie homomorphism
mielke-geometrodynamics_F00095	38	0.852	$\mathrm{N} \times \mathrm{N}$	$\mathrm{N} \times \mathrm{N}$	of G into the general linear group of complex $\mathrm{N} \times \mathrm{N}$ matrices. If their representation
mielke-geometrodynamics_F00096	38	0.780	$\mathbb{C}^{\mathrm{N}}$	$\mathbb{C}^{\mathrm{N}}$	space, the N -dimensional vector space $\mathbb{C}^{\mathrm{N}}$ over the field of complex numbers, is
mielke-geometrodynamics_F00097	38	1.000	$\mathrm{T}_{\mathbb{C}}^*(\mathrm{M})$	$\mathrm{T}_{\mathbb{C}}^*(\mathrm{M})$	with the complexified ⁴ cotangent bundle $\mathrm{T}_{\mathbb{C}}^*(\mathrm{M})$, further vector bundles will come
mielke-geometrodynamics_F00098	38	1.000	ρ	ρ	ρ , i.e., the cross section
mielke-geometrodynamics_F00099	38	0.986	$\mathrm{N} \geq 1$	$\mathrm{N} \geq 1$	For $\mathrm{N} \geq 1$, these fields form an infinite-dimensional vector space over $\mathbb{C}^{\mathrm{N}}$. In
mielke-geometrodynamics_F00100	38	1.000	$\mathrm{b} = (\mathrm{b}_{\mathrm{A}}): \mathrm{U} \subset \mathrm{M} \rightarrow$	$\mathrm{b} = (\mathrm{b}_{\mathrm{A}}): \mathrm{U} \subset \mathrm{M} \rightarrow$	(2.4.3), the bundle coordinates relative to the local basis fields $\mathrm{b} = (\mathrm{b}_{\mathrm{A}}): \mathrm{U} \subset \mathrm{M} \rightarrow$
mielke-geometrodynamics_F00101	38	1.000	V^{ρ}	V^{ρ}	V^{ρ} are denoted by
mielke-geometrodynamics_F00102	39	1.000	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	Let the Lie homomorphism $\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$ be explicitly given by the nonsin-
mielke-geometrodynamics_F00103	39	0.964	$\rho_{\mathrm{A}}^{\mathrm{B}}(\mathrm{g}(\xi))$	$\rho_{\mathrm{A}}^{\mathrm{B}}(\mathrm{g}(\xi))$	gular complex $\mathrm{N} \times \mathrm{N}$ matrices $\rho_{\mathrm{A}}^{\mathrm{B}}(\mathrm{g}(\xi))$, where ξ^j ($j = 1, \dots, \dim \mathrm{G}$) prescribes a
mielke-geometrodynamics_F00104	39	0.964	ξ^j ($j = 1, \dots, \dim \mathrm{G}$)	ξ^j ($j = 1, \dots, \dim \mathrm{G}$)	gular complex $\mathrm{N} \times \mathrm{N}$ matrices $\rho_{\mathrm{A}}^{\mathrm{B}}(\mathrm{g}(\xi))$, where ξ^j ($j = 1, \dots, \dim \mathrm{G}$) prescribes a

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mielke-geometrodynamics_F00105	39	■ 1.000	$\rho=\mathrm{id}$	$\rho = \mathrm{id}$	identical representation, i.e., for $\rho = \mathrm{id}$, the infinitesimal generators I_j form a basis
mielke-geometrodynamics_F00106	39	■ 1.000	I_{j}	I_j	identical representation, i.e., for $\rho = \mathrm{id}$, the infinitesimal generators I_j form a basis
mielke-geometrodynamics_F00107	39	■ 0.938	$\mathfrak{g}$	$\mathfrak{g}$	of the Lie algebra $\mathfrak{g}$ of G . As a representation of I_j , the infinitesimal operators inherit
mielke-geometrodynamics_F00108	39	■ 1.000	$\mathrm{c}_{\mathrm{ij}}^{\mathrm{k}}$	c_{ij}^k	These algebraic relations are determined by the structure constants c_{ij}^k of the Lie
mielke-geometrodynamics_F00109	39	■ 0.960	V^{Ad}	V^{Ad}	vector bundle V^{Ad} respecting the adjoint representation of G as a structure group.
mielke-geometrodynamics_F00110	39	■ 0.983	$\mathrm{s}(\mathrm{t}) \subset \mathrm{M}$	$s(t) \subset M$	in a differentiable manner along a given curve $s(t) \subset M$ in the base manifold. In order
mielke-geometrodynamics_F00111	39	■ 1.000	$\mathrm{T}_{\mathrm{p}}^*(\mathrm{P})$	$T_p^*(P)$	cotangent bundle $T_p^*(P)$ at the point $p \in P$ (see KN I, Chap. 1). Then decompose it
mielke-geometrodynamics_F00112	39	■ 1.000	$\mathrm{p} \in \mathrm{P}$	$p \in P$	cotangent bundle $T_p^*(P)$ at the point $p \in P$ (see KN I, Chap. 1). Then decompose it
mielke-geometrodynamics_F00113	40	■ 1.000	$\mathrm{T}_{\mathrm{p}}^*(\mathrm{P})=\mathrm{H}_{\mathrm{p}} \oplus \mathrm{V}_{\mathrm{p}}$	$T_p^*(P) = H_p \oplus V_p$	into the direct sum of horizontal and the vertical subspaces, i.e., $T_p^*(P) = H_p \oplus V_p$.
mielke-geometrodynamics_F00114	40	■ 1.000	$\mathrm{F}_{\mathrm{m}}(\mathrm{M})$	$F_m(M)$	i.e., in the cotangent space of the bundle $F_m(M)$ of typical fibers on M .
mielke-geometrodynamics_F00115	40	■ 0.999	$\omega(\mathrm{e}) \in \mathscr{C}$	$\omega(e) \in \mathcal{C}$	This will be achieved by the assertion of a 1-form $\omega(e) \in \mathcal{C}$ with values in the
mielke-geometrodynamics_F00116	40	■ 1.000	$\mathrm{e}_{\mathrm{v}}=\mathrm{pdg} \in \mathrm{V}_{\mathrm{p}}$	$e_v = \mathrm{pdg} \in V_p$	(ii) For vertical elements $e_v = \mathrm{pdg} \in V_p$, it will be mapped to the left-invariant
mielke-geometrodynamics_F00117	40	■ 1.000	$\bigwedge^{\mathrm{p}} \mathrm{T}_{\mathbb{C}}^*(\mathrm{M}) \otimes V^{\rho}$	$\bigwedge^p T_{\mathbb{C}}^*(M) \otimes V^\rho$	connection in an associated vector bundle $\bigwedge^p T_{\mathbb{C}}^*(M) \otimes V^\rho$ is defined via the <i>covari-</i>
mielke-geometrodynamics_F00118	40	■ 1.000	$\mathrm{T}_{\mathbb{C}}^*(\mathrm{M}) \otimes V^{\rho}$	$T_{\mathbb{C}}^*(M) \otimes V^\rho$	onto those of the product bundle $T_{\mathbb{C}}^*(M) \otimes V^\rho$. This operator will be defined in such
mielke-geometrodynamics_F00119	41	■ 1.000	V^{ρ}	V^ρ	expressed as a linear transformation of the selfsame basis of V^ρ , i.e.,
mielke-geometrodynamics_F00120	41	■ 0.856	b	b	The image of Db will be referred to as the <i>(G-)covariant exterior derivative</i> of b .
mielke-geometrodynamics_F00121	41	■ 0.641	$\mathrm{b}_{\mathrm{A}}, \mathrm{A}=1, \ldots, \mathrm{N}$	$b_A, A = 1, \dots, N$	restriction to a neighborhood U of M . Let $b_A, A = 1, \dots, N$ be a local basis for the
mielke-geometrodynamics_F00122	41	■ 0.996	$\mathrm{V}_{\mathbb{U}}^{\rho}$	V_U^ρ	cross section V_U^ρ restricted to U . With respect to this local system of reference, the
mielke-geometrodynamics_F00123	41	■ 0.998	b_{A}	b_A	resulting effect of D on b_A can be expanded as follows:
mielke-geometrodynamics_F00124	41	■ 0.998	$\left[\omega_{\mathrm{A}}^{\mathrm{B}}\right]$	$\left[\omega_A^B\right]$	matrix $[\omega_A^B]$ acquires values in $T_{\mathbb{C}}^*(M)$ and will consequently be called a <i>connection</i>
mielke-geometrodynamics_F00125	41	■ 1.000	$\Gamma_{\alpha}^{\mathrm{j}}(\mathrm{m})$	$\Gamma_{\alpha}^j(m)$	The coefficients $\Gamma_{\alpha}^j(m)$ will turn out to be generalized gauge potentials as they
mielke-geometrodynamics_F00126	42	■ 0.997	$\mathrm{dd} \equiv 0$	$dd \equiv 0$	For differential forms in general, the identity $dd \equiv 0$ holds, which may be regarded as
mielke-geometrodynamics_F00127	42	■ 1.000	$\partial \partial \equiv 0$	$\partial \partial \equiv 0$	a counterpart to the homological relation $\partial \partial \equiv 0$. Figuratively speaking, this means
mielke-geometrodynamics_F00128	42	■ 0.905	Ω	Ω	that the curvature 2-form Ω satisfies the <i>second structure equation</i> of É. CARTAN:
mielke-geometrodynamics_F00129	42	■ 1.000	$\mathrm{G}=\mathrm{U}(1)$	$G = U(1)$	which can be formulated in a fiber bundle with abelian structure group $G = U(1)$,
mielke-geometrodynamics_F00130	43	■ 1.000	$\circlearrowleft=\partial \mathrm{U} \subset \mathrm{M}$	$\circlearrowleft = \partial U \subset M$	along an <i>infinitesimal</i> closed curve $\circlearrowleft = \partial U \subset M$. The very concept of a connection
mielke-geometrodynamics_F00131	43	■ 1.000	$\partial \mathrm{U}$	∂U	closed curve ∂U . Figure 2.2 shows this for a parallel displacement of a pair of tangent

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mielke-geometrodynamics_F00132	43	■ 1.000	$\mathrm{e}(\mathrm{m}) \in \mathrm{T}_{\mathrm{m}}(\mathrm{M})$	$\mathrm{e}(\mathrm{m}) \in \mathrm{T}_{\mathrm{m}}(\mathrm{M})$	$\mathrm{T}_{\mathrm{m}}(\mathrm{M})$ onto itself. These are generated by displacements of $\mathrm{e}(\mathrm{m}) \in \mathrm{T}_{\mathrm{m}}(\mathrm{M})$ along
mielke-geometrodynamics_F00133	44	■ 0.996	$\mathrm{G}(\mathrm{p})$	$\mathrm{G}(\mathrm{p})$	(i) $\mathrm{G}(\mathrm{p})$ is equivariant, i.e.,
mielke-geometrodynamics_F00134	44	■ 0.897	$\mathrm{F}_{\mathrm{m}} = \pi^{-1}(\mathrm{m})$	$\mathrm{F}_{\mathrm{m}} = \pi^{-1}(\mathrm{m})$	(ii) $\mathrm{G}(\mathrm{p})$ preserves each fiber $\mathrm{F}_{\mathrm{m}} = \pi^{-1}(\mathrm{m})$, i.e., acts trivially on the base space,
mielke-geometrodynamics_F00135	44	■ 1.000	$\mathrm{mathscr{G}}_{\mathrm{P}}$	$\mathcal{G}_{\mathrm{P}}$	position, they constitute an infinite-dimensional group, the group $\mathcal{G}_{\mathrm{P}}$ of <i>local</i>
mielke-geometrodynamics_F00136	45	■ 0.513	$\exp : \mathrm{mathfrak{g}} \rightarrow \mathrm{G}$	$\exp : \mathfrak{g} \rightarrow \mathrm{G}$	The exponential mapping $\exp : \mathfrak{g} \rightarrow \mathrm{G}$ from the Lie algebra $\mathfrak{g}$ into the structure
mielke-geometrodynamics_F00137	45	■ 0.614	$\mathrm{P} \times_{\mathrm{Ad}} \mathrm{G}$	$P \times_{\mathrm{Ad}} \mathrm{G}$	group G of P induces a corresponding mapping within $P \times_{\mathrm{Ad}} \mathrm{G}$. As a result,
mielke-geometrodynamics_F00138	45	■ 1.000	$\theta^k(\mathrm{m}) \in C^{\infty}(\mathrm{M})$	$\theta^k(\mathrm{m}) \in C^{\infty}(\mathrm{M})$	Here $\theta^k(\mathrm{m}) \in C^{\infty}(\mathrm{M})$ denote real functions on the base space. Gauge trans-
mielke-geometrodynamics_F00139	45	■ 0.984	$\mathrm{V}^{\rho}(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P})$	$\mathrm{V}^{\rho}(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P})$	formations within an associate vector bundle $\mathrm{V}^{\rho}(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P})$, i.e.,
mielke-geometrodynamics_F00140	45	■ 1.000	$\mathrm{mathscr{G}}_{\mathrm{V}}$	$\mathcal{G}_{\mathrm{V}}$	elements of $\mathcal{G}_{\mathrm{V}}$, are represented by the same expression to the extent that L^{k}
mielke-geometrodynamics_F00141	45	■ 1.000	L^{k}	L^{k}	elements of $\mathcal{G}_{\mathrm{V}}$, are represented by the same expression to the extent that L^{k}
mielke-geometrodynamics_F00142	45	■ 0.482	$\rho : \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	$\rho : \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	$\rho : \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$.
mielke-geometrodynamics_F00143	45	■ 1.000	$\bar{\varphi}$	$\bar{\varphi}$	It has to be emphasized that the bundle coordinates $\bar{\varphi}$ of the vector bundle constructed
mielke-geometrodynamics_F00144	45	■ 1.000	$\bar{\rho}$	$\bar{\rho}$	on the conjugate (“Dirac adjoint”) representation $\bar{\rho}$ transforms according to
mielke-geometrodynamics_F00145	46	■ 0.798	$\mathrm{V}(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P})$	$\mathrm{V}(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P})$	Let $\mathrm{V}(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P})$ be a complex vector bundle that is associated with
mielke-geometrodynamics_F00146	46	■ 0.438	γ_{k}	γ_{k}	2k-forms γ_{k} on P whose inverse images after the projection on the base space M read
mielke-geometrodynamics_F00147	46	■ 1.000	$\mathrm{k} \leq n/2$	$\mathrm{k} \leq n/2$	The number of curvature 2-forms $\mathcal{Q}$ in the exterior product is $\mathrm{k} \leq n/2$. From the
mielke-geometrodynamics_F00148	46	■ 1.000	ω	ω	of the latter is not determined by the particular choice of the connection ω , but depends
mielke-geometrodynamics_F00149	46	■ 0.959	k^{th}	k^{th}	precisely (see: KN II, Theorem 3.1), it can be stated that the abstractly defined k^{th}
mielke-geometrodynamics_F00150	46	■ 0.570	$\mathrm{c}_{\mathrm{k}}(\mathrm{V})$	$\mathrm{c}_{\mathrm{k}}(\mathrm{V})$	Chern class $\mathrm{c}_{\mathrm{k}}(\mathrm{V})$ of a complex vector bundle V is represented by the closed 2k-form
mielke-geometrodynamics_F00151	46	■ 0.546	$\mathrm{V}^{\mathbb{R}}$	$\mathrm{V}^{\mathbb{R}}$	In order to obtain the characteristic classes of a real vector bundle $\mathrm{V}^{\mathbb{R}}$ over M with
mielke-geometrodynamics_F00152	46	■ 0.983	$\mathbb{R}^{\mathrm{N}}$	$\mathbb{R}^{\mathrm{N}}$	the typical fiber $\mathbb{R}^{\mathrm{N}}$, it will be enlarged to a complex vector bundle V with the typical
mielke-geometrodynamics_F00153	46	■ 0.753	$\mathbb{C}^{\mathrm{N}}$	$\mathbb{C}^{\mathrm{N}}$	fiber $\mathbb{C}^{\mathrm{N}}$. The latter arises from the complexification of each fiber of $\mathrm{V}^{\mathbb{R}}$. Then the
mielke-geometrodynamics_F00154	46	■ 0.753	$\mathrm{V}^{\mathbb{R}}$	$\mathrm{V}^{\mathbb{R}}$	fiber $\mathbb{C}^{\mathrm{N}}$. The latter arises from the complexification of each fiber of $\mathrm{V}^{\mathbb{R}}$. Then the
mielke-geometrodynamics_F00155	47	■ 1.000	$\mathrm{s}=0$	$\mathrm{s} = 0$	the Euclidean signature $\mathrm{s} = 0$, we find locally, cf. DANIEL & VIALLET (1980),
mielke-geometrodynamics_F00156	47	■ 1.000	$\mathrm{P}(\mathrm{S}^4, \mathrm{SU}(\mathrm{f}), \pi, \delta)$	$\mathrm{P}(\mathrm{S}^4, \mathrm{SU}(\mathrm{f}), \pi, \delta)$	It is typical for the principal fiber bundles $\mathrm{P}(\mathrm{S}^4, \mathrm{SU}(\mathrm{f}), \pi, \delta)$, which are to be found in
mielke-geometrodynamics_F00157	47	■ 1.000	$\det \mathrm{G}=1$	$\det \mathrm{G} = 1$	for which $\det \mathrm{G} = 1$ holds. Inasmuch as the trace of the Lie-algebra-valued curvature
mielke-geometrodynamics_F00158	47	■ 0.565	$\mathrm{Tr}(\Omega \wedge \Omega)$	$\mathrm{Tr}(\Omega \wedge \Omega)$	2-form $\mathcal{Q}$ vanishes in such instances, the gauge-invariant 4-form $\mathrm{Tr}(\mathcal{Q} \wedge \mathcal{Q})$ suffices

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mielke-geometrodynamics_F00159	47	0.892	$\stackrel{*}{\pi}(\gamma_2)$	$\pi^*(\gamma_2)$	The integration of the second Chern class $\pi^*(\gamma_2)$ over the base space yields a
mielke-geometrodynamics_F00160	47	0.992	$\mathrm{G}=\mathrm{SU}(\mathrm{f})$	$G = \mathrm{SU}(f)$	index”; see JACKIW (1977). Concerning bundles with structure group $G = \mathrm{SU}(f)$,
mielke-geometrodynamics_F00161	47	0.827	$\mathrm{SU}(2)$	$\mathrm{SU}(2)$	et al. (1975) is correct, since in their paper, the isospin group $\mathrm{SU}(2)$ has been enlarged
mielke-geometrodynamics_F00162	47	0.749	$\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$	$\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$	to the real structure group $\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$, due to a special isomorphism
mielke-geometrodynamics_F00163	47	0.995	γ_2	γ_2	of the closed form γ_2 onto M is even an exact one. This is a property that it shares
mielke-geometrodynamics_F00164	48	1.000	$\mathrm{F}_{\alpha\beta}^j$	$F_{\alpha\beta}^j$	that the field strengths $F_{\alpha\beta}^j$ (components of the curvature form) have to vanish faster
mielke-geometrodynamics_F00165	48	1.000	$ \mathrm{x} ^{-2}$	$ \mathbf{x} ^{-2}$	than $ \mathbf{x} ^{-2}$ at infinity, since otherwise, the integral (2.8.13) would not exist. In order to
mielke-geometrodynamics_F00166	49	1.000	$ \mathrm{x} \rightarrow \infty$	$ x \rightarrow \infty$	should hold for $ x \rightarrow \infty$. Here $\tilde{\omega}$ denotes a Maurer–Cartan connection, i.e., a left-
mielke-geometrodynamics_F00167	49	1.000	$\stackrel{\infty}{\omega}$	$\tilde{\omega}$	should hold for $ x \rightarrow \infty$. Here $\tilde{\omega}$ denotes a Maurer–Cartan connection, i.e., a left-
mielke-geometrodynamics_F00168	49	0.978	$\partial\mathrm{M}$	∂M	If the boundary ∂M is chosen in such a way that only a pure gauge connection $\tilde{\omega}$
mielke-geometrodynamics_F00169	49	0.999	$\mathrm{G}=\mathrm{SU}(2)$	$G = \mathrm{SU}(2)$	in full compliance with (2.8.15); cf. JACKIW (1980). If $G = \mathrm{SU}(2)$ serves as a struc-
mielke-geometrodynamics_F00170	49	1.000	S^3	S^3	S^3 topologically, then it can be shown that $c_2(M)$ determines the “mapping degree”
mielke-geometrodynamics_F00171	49	1.000	$\mathrm{c}_2(\mathrm{M})$	$c_2(M)$	S^3 topologically, then it can be shown that $c_2(M)$ determines the “mapping degree”
mielke-geometrodynamics_F00172	49	0.954	$\mathrm{S}^3 \approx \mathrm{SU}(2)$	$S^3 \approx \mathrm{SU}(2)$	or the <i>winding number</i> ⁶ of the mapping of S^3 in $S^3 \approx \mathrm{SU}(2)$. These considerations
mielke-geometrodynamics_F00173	49	0.995	n^{th}	n^{th}	⁶ Speaking more generally, the n^{th} -homotopy group of the n -dimensional sphere S^n is given by the
mielke-geometrodynamics_F00174	49	1.000	$\mathbb{Z}$	$\mathbb{Z}$	group $\mathbb{Z}$ of integers, i.e., by $\pi_n(S^n) = \mathbb{Z}$. As such, it determines the winding number of the mapping
mielke-geometrodynamics_F00175	49	1.000	$\pi_n(S^n)=\mathbb{Z}$	$\pi_n(S^n) = \mathbb{Z}$	group $\mathbb{Z}$ of integers, i.e., by $\pi_n(S^n) = \mathbb{Z}$. As such, it determines the winding number of the mapping
mielke-geometrodynamics_F00176	49	1.000	$\mathrm{S}^n \rightarrow \mathrm{S}^n$	$S^n \rightarrow S^n$	$S^n \rightarrow S^n$.
mielke-geometrodynamics_F00177	52	1.000	$\delta\varphi$	$\delta\varphi$	The operator $\delta\varphi$ of <i>small variations</i> $\varphi + \delta\varphi$ of the fields φ is satisfying the Leibniz
mielke-geometrodynamics_F00178	52	1.000	$\varphi+\delta\varphi$	$\varphi + \delta\varphi$	The operator $\delta\varphi$ of <i>small variations</i> $\varphi + \delta\varphi$ of the fields φ is satisfying the Leibniz
mielke-geometrodynamics_F00179	53	0.937	$(n-1)$	$(n - 1)$	be the $(n - 1)$ -form of the <i>field momenta</i> which are <i>canonically conjugated</i> to $d\varphi$. It
mielke-geometrodynamics_F00180	53	0.937	$d\varphi$	$d\varphi$	be the $(n - 1)$ -form of the <i>field momenta</i> which are <i>canonically conjugated</i> to $d\varphi$. It
mielke-geometrodynamics_F00181	53	1.000	δ	δ	after δ has been interchanged with the integration. The last term is an exact form.
mielke-geometrodynamics_F00182	53	1.000	s	s	where s is the signature of the pseudo-Riemannian manifold.
mielke-geometrodynamics_F00183	53	1.000	$d\,d\equiv 0$	$dd \equiv 0$	where the last term vanishes identically, due to $dd \equiv 0$. Following the physically
mielke-geometrodynamics_F00184	53	0.913	x^i	$\mathbf{x}^i$	quently does not depend upon the coordinates x^i explicitly, we get the well-known
mielke-geometrodynamics_F00185	54	1.000	$\varphi \rightarrow \psi := \left(\varphi, \Theta^{(1)} := d\varphi\right)$	$\varphi \rightarrow \psi := \left(\varphi, \Theta^{(1)} := d\varphi\right)$	via the substitution $\varphi \rightarrow \psi := (\varphi, \Theta^{(1)} := d\varphi)$ into a first order system of the

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mielke-geometrodynamics_F00186	54	■ 1.000	$\mathscr{D}(M)$	$\mathscr{D}(M)$	respect to the group $\mathscr{D}(M)$ of general coordinate-transformations of the spacetime
mielke-geometrodynamics_F00187	54	■ 1.000	$\mathscr{G}$	$\mathscr{G}$	labels, but as well with respect to the “internal covariance group” $\mathscr{G}$ of gauge trans-
mielke-geometrodynamics_F00188	54	■ 0.989	L_{mat}	L_{mat}	The invariance of L_{mat} with respect to $\mathscr{D}(M)$ is already guaranteed by the use
mielke-geometrodynamics_F00189	55	■ 1.000	$\bar{\varphi} \varphi$	$\bar{\varphi} \varphi$	depend upon $\bar{\varphi} \varphi$, where φ denotes a gauge-covariant operator
mielke-geometrodynamics_F00190	55	■ 1.000	$\mathscr{O}$	$\mathscr{O}$	depend upon $\bar{\varphi} \varphi$, where $\mathscr{O}$ denotes a gauge-covariant operator
mielke-geometrodynamics_F00191	55	■ 0.996	$\omega^{\prime}$	$\omega^{\prime}$	by means of a 1-form $\omega^{\prime}$ in such a way that the term GdG^{-1} resulting from gauge
mielke-geometrodynamics_F00192	55	■ 0.996	GdG^{-1}	GdG^{-1}	by means of a 1-form $\omega^{\prime}$ in such a way that the term GdG^{-1} resulting from gauge
mielke-geometrodynamics_F00193	55	■ 1.000	$\text{G} \in \mathscr{G}_V$	$\text{G} \in \mathscr{G}_V$	concerning the “actively” applied gauge transformation $\text{G} \in \mathscr{G}_V$. For the derivation
mielke-geometrodynamics_F00194	55	■ 1.000	$\text{G} \omega$	$\text{G} \omega$	matrix $\text{G} \omega$ of the connection relative to a basis $\text{G} \mathbf{b}$, which is locally “rotated” by
mielke-geometrodynamics_F00195	55	■ 1.000	$\text{G} \mathbf{b}$	$\text{G} \mathbf{b}$	matrix $\text{G} \omega$ of the connection relative to a basis $\text{G} \mathbf{b}$, which is locally “rotated” by
mielke-geometrodynamics_F00196	55	■ 1.000	G^{-1}	G^{-1}	If G is substituted by G^{-1} exactly the relation (3.2.3) comes into existence, which
mielke-geometrodynamics_F00197	56	■ 0.749	$\sigma_{\alpha}^* \omega$	$\sigma_{\alpha}^* \omega$	image or pull-back $\sigma_{\alpha}^* \omega$ due to the defining properties of a connection (see: KN I,
mielke-geometrodynamics_F00198	56	■ 0.818	$G^{-1} \omega$	$G^{-1} \omega$	The transformed connection $G^{-1} \omega$ deviates from the original one solely by a gauge-
mielke-geometrodynamics_F00199	56	■ 1.000	$\bar{\omega}$	$\bar{\omega}$	For a so-called “copy” $\bar{\omega}$ of a connection ω , this expression is to be taken as different
mielke-geometrodynamics_F00200	56	■ 1.000	$\bar{\Omega} = \text{G}^{-1} \Omega \text{G}$	$\bar{\Omega} = \text{G}^{-1} \Omega \text{G}$	from zero, although - according to the definition - its curvature $\bar{\Omega} = \text{G}^{-1} \Omega \text{G}$ is
mielke-geometrodynamics_F00201	56	■ 1.000	$d^* \omega \simeq 0$	$d^* \omega \simeq 0$	transversal or Coulomb condition $d^* \omega \simeq 0$ does not fix uniquely (large) gauge trans-
mielke-geometrodynamics_F00202	56	■ 1.000	$A = A_i dx^i$	$A = A_i dx^i$	tions of Maxwell, the electromagnetic four-potential $A = A_i dx^i$, a one-form, is the
mielke-geometrodynamics_F00203	56	■ 1.000	F	F	fundamental variable, whereas Faraday’s field strength F is a derived concept and
mielke-geometrodynamics_F00204	57	■ 0.992	$dd\Phi \equiv 0$	$dd\Phi \equiv 0$	On the other hand, $dd\Phi \equiv 0$ for any exterior p-form Φ , due to the Poincaré lemma.
mielke-geometrodynamics_F00205	57	■ 0.992	Φ	Φ	On the other hand, $dd\Phi \equiv 0$ for any exterior p-form Φ , due to the Poincaré lemma.
mielke-geometrodynamics_F00206	57	■ 1.000	$dF \equiv 0$	$dF \equiv 0$	Thus if $dF \equiv 0$, there exists locally a one-form A such that $F = dA$. However, the
mielke-geometrodynamics_F00207	57	■ 1.000	$F = dA$	$F = dA$	Thus if $dF \equiv 0$, there exists locally a one-form A such that $F = dA$. However, the
mielke-geometrodynamics_F00208	57	■ 0.981	$U(1)$	$U(1)$	local $U(1)$ gauge transformation
mielke-geometrodynamics_F00209	57	■ 1.000	$D \rightarrow \mathscr{D} := D + ieA$	$D \rightarrow \mathscr{D} := D + ieA$	Although, A is needed for the standard minimal coupling $D \rightarrow \mathscr{D} := D + ieA$ to
mielke-geometrodynamics_F00210	57	■ 1.000	L	L	For a Lagrangian formulation, let us depart from the four-form L which consist of
mielke-geometrodynamics_F00211	57	■ 0.980	Ψ	Ψ	the gauge field part and L_{mat} governing the matter field Ψ and its minimal coupling
mielke-geometrodynamics_F00212	57	■ 1.000	$S = \int L$	$S = \int L$	usually be assumed to be of first order in the fields. Stationarity of the action $S = \int L$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_F00213	57	■ 1.000	H	H	The excitation H is the field momentum conjugated to A , i.e.
mielke-geometrodynamics_F00214	58	■ 1.000	$d\,d\,H \equiv 0$	$ddH \equiv 0$	$ddH \equiv 0$, the field equations imposes ‘on shell’ that the electric current j is con-
mielke-geometrodynamics_F00215	58	■ 1.000	j	j	$ddH \equiv 0$, the field equations imposes ‘on shell’ that the electric current j is con-
mielke-geometrodynamics_F00216	58	■ 1.000	$C:=A \wedge F=A \wedge d\,A$	$C := A \wedge F = A \wedge dA$	where $C := A \wedge F = A \wedge dA$ is the abelian Chern–Simons term, known to violate
mielke-geometrodynamics_F00217	58	■ 1.000	P	P	parity P .
mielke-geometrodynamics_F00218	58	■ 0.885	$\stackrel{\theta}{L}=-\left(\cos \theta F \wedge { }^* F+\sin \theta F \wedge F\right) / 2$	$\stackrel{\theta}{L}=-\left(\cos \theta F \wedge { }^* F+\sin \theta F \wedge F\right) / 2$	$\stackrel{\theta}{L}=-\left(\cos \theta F \wedge { }^* F+\sin \theta F \wedge F\right) / 2$, where θ is the ‘vacuum’ angle of duality rotations or
mielke-geometrodynamics_F00219	58	■ 0.885	θ	θ	$\stackrel{\theta}{L}=-\left(\cos \theta F \wedge { }^* F+\sin \theta F \wedge F\right) / 2$, where θ is the ‘vacuum’ angle of duality rotations or
mielke-geometrodynamics_F00220	59	■ 0.950	$*$	$*$	The Hodge dual $*$ depends on the metric g and on the orientation of the manifold, as
mielke-geometrodynamics_F00221	59	■ 0.950	g	g	The Hodge dual $*$ depends on the metric g and on the orientation of the manifold, as
mielke-geometrodynamics_F00222	59	■ 1.000	η	η	where η is the standard volume four-form. The Pontryagin four-form can be written
mielke-geometrodynamics_F00223	59	■ 1.000	$\partial \mathfrak{D} / \partial t$	$\partial \mathfrak{D} / \partial t$	³ The displacement current $\partial \mathfrak{D} / \partial t$, where $\mathfrak{D} = \varepsilon \mathbf{E}$, was anticipated 1839 by James Mac Cullagh, cf.
mielke-geometrodynamics_F00224	59	■ 1.000	$\mathfrak{D}=\varepsilon \mathbf{E}$	$\mathfrak{D} = \varepsilon \mathbf{E}$	³ The displacement current $\partial \mathfrak{D} / \partial t$, where $\mathfrak{D} = \varepsilon \mathbf{E}$, was anticipated 1839 by James Mac Cullagh, cf.
mielke-geometrodynamics_F00225	60	■ 1.000	$\vartheta_{[\alpha} \wedge \Sigma_{\beta]}=0$	$\vartheta_{[\alpha} \wedge \Sigma_{\beta]} = 0$	and turns out to be symmetric, i.e., $\vartheta_{[\alpha} \wedge \Sigma_{\beta]} = 0$. For non-vanishing charge current
mielke-geometrodynamics_F00226	60	■ 1.000	$B=\frac{1}{2} B_{i j} d x^i \wedge d x^j$	$B=\frac{1}{2} B_{i j} d x^i \wedge d x^j$	two-form $B=\frac{1}{2} B_{i j} d x^i \wedge d x^j$ are varied <i>independently</i> , slightly reminiscent of the
mielke-geometrodynamics_F00227	60	■ 1.000	B	B	Independent variations with respect to A and B lead to $dB \cong 0$ and to the <i>constraint</i> of
mielke-geometrodynamics_F00228	60	■ 1.000	$d\,B \cong 0$	$dB \cong 0$	Independent variations with respect to A and B lead to $dB \cong 0$ and to the <i>constraint</i> of
mielke-geometrodynamics_F00229	60	■ 1.000	$F:=d\,A \cong 0$	$F := dA \cong 0$	vanishing field strength $F := dA \cong 0$. This implies, however, that such a primordial
mielke-geometrodynamics_F00230	60	■ 1.000	$d\,C$	dC	(3.3.22) only by a boundary term dC derived from a <i>Chern–Simons</i> (CS) three-form
mielke-geometrodynamics_F00231	60	■ 1.000	$H=d\,B$	$H = dB$	excitation $H = dB$, the Kalb–Ramond <i>axion</i> three-form.
mielke-geometrodynamics_F00232	61	■ 1.000	$\delta d\,C / \delta A=d\,F \equiv 0$	$\delta dC / \delta A = dF \equiv 0$	can be recovered via the variation of the Pontryagin term, e.g., $\delta dC / \delta A = dF \equiv 0$
mielke-geometrodynamics_F00233	61	■ 0.998	$\psi=\psi_i d x^i$	$\psi=\psi_i d x^i$	where $\psi=\psi_i d x^i$ is a one-form or a covector; cf. LUCCHESI et al. (1993). In a
mielke-geometrodynamics_F00234	61	■ 1.000	$s\,\psi=D\,\phi$	$s\psi = D\phi$	BRST quantization, due to the zero mode $s\psi = D\phi$ with $s\phi = 0$, the “topological”
mielke-geometrodynamics_F00235	61	■ 1.000	$s\,\phi=0$	$s\phi = 0$	BRST quantization, due to the zero mode $s\psi = D\phi$ with $s\phi = 0$, the “topological”
mielke-geometrodynamics_F00236	61	■ 1.000	$B\,F$	BF	to topological field theory is the BF formalism which, in the case of gravity, was
mielke-geometrodynamics_F00237	61	■ 1.000	$F={ }^* B$	$F = { }^* B$	the definition of the field strength (3.3.24), but as a bonus, from the relation $F = { }^* B$
mielke-geometrodynamics_F00238	61	■ 0.998	$j:=\delta L_{\text {matter }} / \delta A$	$j := \delta L_{\text{matter}} / \delta A$	where $j := \delta L_{\text{matter}} / \delta A$ is the matter current.
mielke-geometrodynamics_F00239	61	■ 1.000	$d\,j \cong 0$	$dj \cong 0$	which “on shell,” is conserved classically, i.e., $dj \cong 0$. In general, this includes

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mielke-geometrodynamics_F00240	61	■ 0.999	d A	dA	<i>Pauli-type terms</i> generated by the variation of the Lagrangian with respect to dA ; cf.
mielke-geometrodynamics_F00241	62	■ 1.000	V(B)	$V(B)$	The quadratic term in (3.3.28) can be generalized to a four-form potential $V(B)$.
mielke-geometrodynamics_F00242	62	■ 1.000	d B+j \cong \emptyset	$dB + j \cong 0$	Then the field equation $dB + j \cong 0$ together with the relation $F \cong \partial V/\partial B$ for
mielke-geometrodynamics_F00243	62	■ 1.000	F \cong \partial V / \partial B	$F \cong \partial V/\partial B$	Then the field equation $dB + j \cong 0$ together with the relation $F \cong \partial V/\partial B$ for
mielke-geometrodynamics_F00244	62	■ 1.000	-B \wedge F+B \wedge B \wedge B / 3	$-B \wedge F + B \wedge B \wedge B/3$	systems with a cubic term $-B \wedge F + B \wedge B \wedge B/3$ are directly related to Chern–Simons
mielke-geometrodynamics_F00245	62	■ 0.999	C:=A \wedge F-A \wedge A \wedge A / 3	$C := A \wedge F - A \wedge A \wedge A/3$	theories departing from the noninvariant three-form $C := A \wedge F - A \wedge A \wedge A/3$; cf.
mielke-geometrodynamics_F00246	62	■ 1.000	F=-B \wedge B	$F = -B \wedge B$	$F = -B \wedge B$ together with the Bianchi identity $DB = DF \equiv 0$ correspond, in 3D
mielke-geometrodynamics_F00247	62	■ 1.000	D B=D F \equiv \emptyset	$DB = DF \equiv 0$	$F = -B \wedge B$ together with the Bianchi identity $DB = DF \equiv 0$ correspond, in 3D
mielke-geometrodynamics_F00248	62	■ 0.999	\Gamma_{\alpha}^j(\mathrm{j})(\mathrm{m})	$\Gamma_{\alpha}^j(\mathbf{m})$	Speaking a priori, the connection ω and the corresponding gauge potentials $\Gamma_{\alpha}^j(\mathbf{m})$
mielke-geometrodynamics_F00249	62	■ 1.000	d \omega	$d\omega$	only of its first derivative $d\omega$. The only <i>gauge-covariant</i> object of this kind that meets
mielke-geometrodynamics_F00250	62	■ 0.668	\mathscr{F}	$\mathcal{F}$	4-forms into 0-forms, i.e., scalar functions. In quantized gauge theories, $\mathcal{F}$ and $^*\mathcal{F}$
mielke-geometrodynamics_F00251	62	■ 0.668	{ }^{\ast} \mathscr{F}	$^*\mathcal{F}$	4-forms into 0-forms, i.e., scalar functions. In quantized gauge theories, $\mathcal{F}$ and $^*\mathcal{F}$
mielke-geometrodynamics_F00252	63	■ 0.731	\Pi^{\Omega}	Π^{Ω}	The 2-form Π^{Ω} of the <i>field momenta</i> being <i>canonically conjugated</i> to Ω are formally
mielke-geometrodynamics_F00253	63	■ 0.894	n-1	$n-1$	$(n-1)$ -form τ of the matter current as a source. The latter is canonically conjugate
mielke-geometrodynamics_F00254	63	■ 0.894	\tau	τ	$(n-1)$ -form τ of the matter current as a source. The latter is canonically conjugate
mielke-geometrodynamics_F00255	64	■ 1.000	\delta L / \delta \omega	$\delta L/\delta\omega$	the variation $\delta L/\delta\omega$ of the total system. They are complemented by the Bianchi
mielke-geometrodynamics_F00256	64	■ 1.000	(1978,1980)	(1978, 1980)	it has been shown by HORNDESKI (1978, 1980) that a field theory of Yang–Mills
mielke-geometrodynamics_F00257	65	■ 1.000	\mathrm{H}^3	H^3	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00258	65	■ 1.000	\partial \mathrm{H}^3	$\partial\mathrm{H}^3$	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00259	65	■ 1.000	\iota	ι	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00260	65	■ 0.879	\mathrm{SU}(\mathrm{N})	$\mathrm{SU}(\mathrm{N})$	In more generalized versions, these theories make use of $\mathrm{SU}(\mathrm{N})$ as a structure group,
mielke-geometrodynamics_F00261	66	■ 0.971	L\left(\mathscr{F}, { }^{\ast} \mathscr{F}\right)	$L(\mathcal{F}, ^*\mathcal{F})$	$L(\mathcal{F}, ^*\mathcal{F})$ is of course a mere empirical one and can be answered only by means of
mielke-geometrodynamics_F00262	66	■ 0.435	\mathrm{I}^j=\mathrm{i} \sigma^j	$\mathbf{I}^j = \mathrm{i}\sigma^j$	with the infinitesimal generators $\mathbf{I}^j = \mathrm{i}\sigma^j$, thus obtaining the group $\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$
mielke-geometrodynamics_F00263	66	■ 0.435	\mathrm{SU}(2) \times \mathrm{SU}(2) \approx	$\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$	with the infinitesimal generators $\mathbf{I}^j = \mathrm{i}\sigma^j$, thus obtaining the group $\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$
mielke-geometrodynamics_F00264	66	■ 0.765	\mathrm{SO}(4)	$\mathrm{SO}(4)$	$\mathrm{SO}(4)$ by proceeding from the array of matrices
mielke-geometrodynamics_F00265	67	■ 0.998	\mathrm{h}=\mathrm{h}(\mathrm{x})	$\mathbf{h} = \mathbf{h}(\mathbf{x})$	i.e., $\mathbf{h} = \mathbf{h}(\mathbf{x})$ in the spherically symmetric case. A similar nonlinear equation occurs
mielke-geometrodynamics_F00266	67	■ 0.499	\mathrm{M}^4	M^4	cf. JACKIW et al. (1977). In a flat space M^4 with Euclidean signature $s = 0$, this is a

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mielke-geometrodynamics_F00267	67	■ 0.999	$\mathrm{k}=1$	$k = 1$	in the case of $k = 1$ (’T HOOFT 1977). The configuration as given above depends
mielke-geometrodynamics_F00268	67	■ 0.512	$5\mathrm{k}+4$	$5k + 4$	on $5k + 4$ parameters. From studies of related objects in algebraic topology it is
mielke-geometrodynamics_F00269	67	■ 1.000	$\mathbb{R}^4 \cup \{\infty\} \approx \mathrm{S}^4$	$\mathbb{R}^4 \cup \{\infty\} \approx S^4$	of the Yang–Mills equation in a Euclidean space $\mathbb{R}^4 \cup \{\infty\} \approx S^4$, which is confor-
mielke-geometrodynamics_F00270	68	■ 1.000	$ \text{-}\infty\rangle$	$ \text{--}\infty\rangle$	transition amplitude between the asymptotic states $ \text{--}\infty\rangle$ and $\langle\infty $. This ampli-
mielke-geometrodynamics_F00271	68	■ 1.000	$\langle\infty $	$\langle\infty $	transition amplitude between the asymptotic states $ \text{--}\infty\rangle$ and $\langle\infty $. This ampli-
mielke-geometrodynamics_F00272	68	■ 1.000	M_{ω}	$\mathcal{M}_{\omega}$	configuration space $\mathcal{M}_{\omega}$ of all inequivalent gauge fields, i.e., of all possible connec-
mielke-geometrodynamics_F00273	68	■ 0.991	$\mathrm{M}_{\omega}=\mathrm{C}/\mathcal{G}_p$	$\mathcal{M}_{\omega}=\mathcal{C}/\mathcal{G}_p$	$\mathcal{M}_{\omega}=\mathcal{C}/\mathcal{G}_p$ has to be taken as the configuration space proper. Under not too stringent
mielke-geometrodynamics_F00274	68	■ 0.999	$\mu[\omega]$	$\mu[\omega]$	can be shown that the “measure” $\mu[\omega]$ for the functional integration, as given by the
mielke-geometrodynamics_F00275	69	■ 0.976	S	S	regime of gauge fields, the so-called <i>S-duality</i> .
mielke-geometrodynamics_F00276	69	■ 1.000	$\operatorname{Spin}^c(4)$	$\operatorname{Spin}^c(4)$	2006) coupled to spinors ψ regarded as a lift from the frame bundle to $\operatorname{Spin}^c(4)$.
mielke-geometrodynamics_F00277	69	■ 0.988	$\psi=\psi_L+\psi_R=P_{-}\psi+P_{+}\psi$	$\psi=\psi_L+\psi_R=P_{-}\psi+P_{+}\psi$	cf. JOST et al. (1996), where here the decomposition $\psi=\psi_L+\psi_R=P_{-}\psi+P_{+}\psi$
mielke-geometrodynamics_F00278	70	■ 1.000	$\bar{\psi}$	$\bar{\psi}$	The variation with respect to $\bar{\psi}$ and A leads to the <i>convective</i> -type spinor equation
mielke-geometrodynamics_F00279	70	■ 1.000	R	R	(3.6.1). For a nonnegative scalar curvature R in the Weizenböck formula, all solutions
mielke-geometrodynamics_F00280	70	■ 1.000	$\psi=0$	$\psi = 0$	satisfy $\psi = 0$ and then are $U(1)$ instantons.
mielke-geometrodynamics_F00281	70	■ 0.996	$F^{\pm}$	$F^{\pm}$	$F^{\pm}$ resembles (MIELKE 1984) the modified (double) duality ansatz
mielke-geometrodynamics_F00282	71	■ 1.000	A_{α}^j	A_{α}^j	gauge potentials A_{α}^j in the Lagrangian n-form. However, there is only one massless
mielke-geometrodynamics_F00283	71	■ 1.000	φ^j	φ^j	fields φ^j with f components but also the vector fields $\Theta^{(1)j}$ will be coupled to the
mielke-geometrodynamics_F00284	71	■ 1.000	$\Theta^{(1)j}$	$\Theta^{(1)j}$	fields φ^j with f components but also the vector fields $\Theta^{(1)j}$ will be coupled to the
mielke-geometrodynamics_F00285	71	■ 0.902	$\phi^{(0)}$	$\phi^{(0)}$	the 0-form $\phi^{(0)}$ and the 1-form $\Theta^{(1)}$ with values in the adjoint representation of G.
mielke-geometrodynamics_F00286	71	■ 0.902	$\Theta^{(1)}$	$\Theta^{(1)}$	the 0-form $\phi^{(0)}$ and the 1-form $\Theta^{(1)}$ with values in the adjoint representation of G.
mielke-geometrodynamics_F00287	71	■ 1.000	$\mathrm{A}_{\alpha}^j(\mathbf{m})$	$A_{\alpha}^j(\mathbf{m})$	gies to the Higgs model. The coupling of the gauge potentials $A_{\alpha}^j(\mathbf{m})$ to the Higgs
mielke-geometrodynamics_F00288	71	■ 1.000	$\delta L_{\mathrm{H}}/\delta\bar{\phi}^{(0)}$	$\delta L_{\mathrm{H}}/\delta\bar{\phi}^{(0)}$	the G-equivalence principle. From the variations and $\delta L_{\mathrm{H}}/\delta\bar{\phi}^{(0)}$ and $\delta L_{\mathrm{H}}/\delta\bar{\Theta}^{(1)}$, the
mielke-geometrodynamics_F00289	71	■ 1.000	$\delta L_{\mathrm{H}}/\delta\bar{\Theta}^{(1)}$	$\delta L_{\mathrm{H}}/\delta\bar{\Theta}^{(1)}$	the G-equivalence principle. From the variations and $\delta L_{\mathrm{H}}/\delta\bar{\phi}^{(0)}$ and $\delta L_{\mathrm{H}}/\delta\bar{\Theta}^{(1)}$, the
mielke-geometrodynamics_F00290	72	■ 1.000	μ	μ	It should be noted that the parameter μ in (3.7.5) complies formally with an imaginary
mielke-geometrodynamics_F00291	72	■ 1.000	$\varphi=0$	$\varphi = 0$	mass of the Klein–Gordon-type field. However, the trivial field configuration $\varphi = 0$
mielke-geometrodynamics_F00292	72	■ 1.000	$ \varphi $	$ \varphi $	This is a consequence of the <i>nonlinear term</i> in (3.7.1), which is quartic in $ \varphi $.
mielke-geometrodynamics_F00293	72	■ 1.000	${}^{\lambda}\varphi$	${}^{\lambda}\varphi$	quantum-mechanical state ${}^{\lambda}\varphi$, for which the vacuum expectation value

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mielke-geometrodynamics_F00294	72	■1.000	$\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$	$\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$	$\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$ along with the gauge fields $A_{\alpha}^{\mathfrak{j}}$ acquire a real, i.e., physical, mass																												
mielke-geometrodynamics_F00295	72	■1.000	$\mathrm{m}_{\mathrm{H}}=\sqrt{2}\,\mu$	$\mathfrak{m}_{\mathrm{H}}=\sqrt{2}\mu$	that is determined by the curvature of the potential at the bottom, i.e., by $\mathfrak{m}_{\mathrm{H}}=\sqrt{2}\mu$,																												
mielke-geometrodynamics_F00296	72	■0.882	G/H	G/H	field φ such that its range is an orbit G/H of G in V^{ρ} with respect to a subgroup H,																												
mielke-geometrodynamics_F00297	72	■1.000	$\mathrm{v}_{\mathrm{0}}\in V^{\rho}$	$\mathfrak{v}_0\in V^{\rho}$	Let $\mathfrak{v}_0\in V^{\rho}$ be a fixed vector for which the cross section $\overset{\circ}{\varphi}_{\min}:=\sigma(\mathfrak{v}_0)\in C^{\infty}(V^{\rho})$																												
mielke-geometrodynamics_F00298	72	■1.000	$\stackrel{\circ}{\varphi}_{\min}:=\sigma(\mathrm{v}_0)\in C^{\infty}\left(V^{\rho}\right)$	$\overset{\circ}{\varphi}_{\min}:=\sigma(\mathfrak{v}_0)\in C^{\infty}\left(V^{\rho}\right)$	Let $\mathfrak{v}_0\in V^{\rho}$ be a fixed vector for which the cross section $\overset{\circ}{\varphi}_{\min}:=\sigma(\mathfrak{v}_0)\in C^{\infty}(V^{\rho})$																												
mielke-geometrodynamics_F00299	72	■0.683	$\mathrm{a}\,\mathrm{g}\in\mathrm{G}$	$\mathfrak{a}\mathfrak{g}\in\mathrm{G}$	to find a $\mathfrak{g}\in\mathrm{G}$ that connects any two vectors $\mathfrak{v}$ and $\mathfrak{v}_0$ from V^{ρ} via the relation																												
mielke-geometrodynamics_F00300	72	■0.683	v_{0}	$\mathfrak{v}_0$	to find a $\mathfrak{g}\in\mathrm{G}$ that connects any two vectors $\mathfrak{v}$ and $\mathfrak{v}_0$ from V^{ρ} via the relation																												
mielke-geometrodynamics_F00301	72	■1.000	$\mathrm{v}=\rho(\mathrm{g})\cdot\mathrm{v}_{\mathrm{0}}$	$\mathfrak{v}=\rho(\mathfrak{g})\cdot\mathfrak{v}_0$	$\mathfrak{v}=\rho(\mathfrak{g})\cdot\mathfrak{v}_0$. An invariant subgroup of G, the so-called isotropy group																												
mielke-geometrodynamics_F00302	73	■0.996	$\stackrel{\circ}{\varphi}_{\min}=\sigma(\mathrm{v}_0)$	$\overset{\circ}{\varphi}_{\min}=\sigma(\mathfrak{v}_0)$	can be <i>uniquely</i> adjoined to the fixed $\overset{\circ}{\varphi}_{\min}=\sigma(\mathfrak{v}_0)$. Then																												
mielke-geometrodynamics_F00303	73	■1.000	$\mathrm{P}(\mathrm{G},\mathrm{M},\pi,\delta)$	$\mathrm{P}(\mathrm{G},\mathrm{M},\pi,\delta)$	is a subbundle of $\mathrm{P}(\mathrm{G},\mathrm{M},\pi,\delta)$ over the same base space; its structure group is H																												
mielke-geometrodynamics_F00304	73	■0.702	$\mathfrak{h}$	$\mathfrak{h}$	Proposition A connection ω in P is reducible to an $\mathfrak{h}$ -valued connection ω_{H} in Q if																												
mielke-geometrodynamics_F00305	73	■0.702	ω_{H}	ω_{H}	Proposition A connection ω in P is reducible to an $\mathfrak{h}$ -valued connection ω_{H} in Q if																												
mielke-geometrodynamics_F00306	73	■0.702	Q	Q	Proposition A connection ω in P is reducible to an $\mathfrak{h}$ -valued connection ω_{H} in Q if																												
mielke-geometrodynamics_F00307	73	■1.000	ω_{0}	ω_0	theories of gravity that will be developed later on. Let ω_0 be a timelike component																												
mielke-geometrodynamics_F00308	74	■1.000	ϕ	ϕ	of the connection can be identified with the Higgs field ϕ without self-interaction:																												
mielke-geometrodynamics_F00309	74	■1.000	$\lambda\rightarrow 0$	$\lambda\rightarrow 0$	limit $\lambda\rightarrow 0$ may be obtained by identifying ω_0 in the interaction-free Yang–Mills																												
mielke-geometrodynamics_F00310	74	■1.000	$\stackrel{*}{\sigma}\vec{\omega}$	$\overset{*}{\sigma}\vec{\omega}$	provides the static Yang–Mills potential $\overset{*}{\sigma}\vec{\omega}$ of the monopole. After this identifica-																												
mielke-geometrodynamics_F00311	74	■0.503	(1981,1983)	(1981,1983)	(1977), TCHRAKIAN (1981, 1983), PRASAD & ROSSI (1981a, b). The topological																												
mielke-geometrodynamics_F00312	75	■1.000	σ	σ	In this context, it is worthwhile to mention that in the nonlinear σ -model, a variant																												
mielke-geometrodynamics_F00313	75	■0.657	$\mathbb{R}^2\times\mathrm{S}^2$	$\mathbb{R}^2\times\mathrm{S}^2$	<table><tr><th>Gauge Field Terminology</th><th>Fiber Bundle Terminology</th></tr><tr><td>spacetime</td><td>base space</td></tr><tr><td>space of phase factors</td><td>bundle space</td></tr><tr><td>symmetry or gauge group</td><td>structure group</td></tr><tr><td>gauge transformations</td><td>inner automorphisms of the bundle</td></tr><tr><td>gauge principle</td><td>G-equivalence principle</td></tr><tr><td>classical fields</td><td>cross section of a vector bundle</td></tr><tr><td>gauge potential</td><td>connection 1-form</td></tr><tr><td>gauge field strength</td><td>curvature 2-form</td></tr><tr><td>electromagnetism</td><td>dynamical theory with a connection in a U(1)-bundle</td></tr><tr><td>electromagnetism with monopoles</td><td>connection in a nontrivial U(1)-bundle over $\mathbb{R}^2\times\mathrm{S}^2$</td></tr><tr><td>Dirac’s monopole quantization</td><td>classification of U(1)-bundles according to the first Chern class</td></tr><tr><td>Yang–Mills theory</td><td>dynamical theory with a connection in a SU(2)-bundle</td></tr><tr><td>instanton number</td><td>classification of SU(2)-bundles over S^4 according to the second Chern class</td></tr></table>	Gauge Field Terminology	Fiber Bundle Terminology	spacetime	base space	space of phase factors	bundle space	symmetry or gauge group	structure group	gauge transformations	inner automorphisms of the bundle	gauge principle	G-equivalence principle	classical fields	cross section of a vector bundle	gauge potential	connection 1-form	gauge field strength	curvature 2-form	electromagnetism	dynamical theory with a connection in a U(1)-bundle	electromagnetism with monopoles	connection in a nontrivial U(1)-bundle over $\mathbb{R}^2\times\mathrm{S}^2$	Dirac’s monopole quantization	classification of U(1)-bundles according to the first Chern class	Yang–Mills theory	dynamical theory with a connection in a SU(2)-bundle	instanton number	classification of SU(2)-bundles over S^4 according to the second Chern class
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electromagnetism with monopoles	connection in a nontrivial U(1)-bundle over $\mathbb{R}^2\times\mathrm{S}^2$																																
Dirac’s monopole quantization	classification of U(1)-bundles according to the first Chern class																																
Yang–Mills theory	dynamical theory with a connection in a SU(2)-bundle																																
instanton number	classification of SU(2)-bundles over S^4 according to the second Chern class																																

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					<table><tr><th>Gauge Field Terminology</th><th>Fiber Bundle Terminology</th></tr><tr><td>spacetime space of phase factors symmetry or gauge group gauge transformations gauge principle classical fields gauge potential gauge field strength</td><td>base space bundle space structure group inner automorphisms of the bundle G-equivalence principle cross section of a vector bundle connection 1-form curvature 2-form</td></tr><tr><td>electromagnetism electromagnetism with monopoles Dirac's monopole quantization</td><td>dynamical theory with a connection in a U(1)-bundle connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$ classification of U(1)-bundles according to the first Chern class</td></tr><tr><td>Yang–Mills theory instanton number</td><td>dynamical theory with a connection in a SU(2)-bundle classification of SU(2)-bundles over S^4 according to the second Chern class</td></tr></table>	Gauge Field Terminology	Fiber Bundle Terminology	spacetime space of phase factors symmetry or gauge group gauge transformations gauge principle classical fields gauge potential gauge field strength	base space bundle space structure group inner automorphisms of the bundle G-equivalence principle cross section of a vector bundle connection 1-form curvature 2-form	electromagnetism electromagnetism with monopoles Dirac's monopole quantization	dynamical theory with a connection in a U(1)-bundle connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$ classification of U(1)-bundles according to the first Chern class	Yang–Mills theory instanton number	dynamical theory with a connection in a SU(2)-bundle classification of SU(2)-bundles over S^4 according to the second Chern class	
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mielke-geometrodynamics_F00314	75	0.657	S^4	S^4										
mielke-geometrodynamics_F00315	78	1.000	$\mathrm{N}=2$	$N = 2$	N= 2 supersymmetric Yang-Mills theory. Nuclear Phys B 426(1):19–52									
mielke-geometrodynamics_F00316	79	1.000	ℓ^*	ℓ^*	diminutive coupling constant given by the Planck length ℓ^* . It may even provide the									
mielke-geometrodynamics_F00317	80	0.998	(1980,1981)	(1980, 1981)	those theories of gravity, was worked out in more detail by HEHL (1980, 1981) using									
mielke-geometrodynamics_F00318	81	0.951	$\mathrm{A}(\mathrm{n}, \mathbb{R})$	$A(n, \mathbb{R})$	and to start off deductively from the affine group $A(n, \mathbb{R})$ as structure group. As a									
mielke-geometrodynamics_F00319	81	1.000	A^{n}	A^n	Euclidean space E^n , regarded as an affine space A^n .									
mielke-geometrodynamics_F00320	81	0.999	$\mathrm{a}=\left(\mathrm{a}^{\alpha}\right) \in \mathbb{R}^{\mathrm{n}}$	$\mathbf{a} = (\mathbf{a}^\alpha) \in \mathbb{R}^n$	Let $\mathbf{a} = (\mathbf{a}^\alpha) \in \mathbb{R}^n$ be a row vector. According to its definition, the action of an									
mielke-geometrodynamics_F00321	81	0.992	$\mathrm{x} \in \mathrm{A}^{\mathrm{n}}$	$\mathbf{x} \in A^n$	element of $A(n, \mathbb{R})$ on a vector $\mathbf{x} \in A^n$ can expressed as the composition of a general									
mielke-geometrodynamics_F00322	81	1.000	$\tilde{x}:=\mathrm{binom}{x}{1} \in \mathbb{R}^{\mathrm{n}+1}$	$\tilde{x} := \begin{pmatrix} x \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}$	In the following, it will be convenient to introduce a vector $\tilde{x} := \begin{pmatrix} x \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}$ with									
mielke-geometrodynamics_F00323	81	1.000	$\mathrm{n}+1$	$n + 1$	$n + 1$ components and to replace the group action (4.1.1) by the abstract transfor-									
mielke-geometrodynamics_F00324	81	0.998	$\tilde{\mathrm{x}}$	$\tilde{\mathbf{x}}$	mations of $\tilde{\mathbf{x}}$ via the matrices									
mielke-geometrodynamics_F00325	81	0.890	$\mathrm{GL}(\mathrm{n}+1, \mathbb{R})$	$GL(n + 1, \mathbb{R})$	This faithful representation of $A(n,\mathbb{R})$ by a subgroup of $GL(n + 1, \mathbb{R})$ shows clearly									
mielke-geometrodynamics_F00326	81	0.623	$\mathrm{a} \in \mathbb{R}^{\mathrm{n}}$	$\mathbf{a} \in \mathbb{R}^n$	of the translations $\mathbf{a} \in \mathbb{R}^n$ and the linear transformations $\mathbf{g} \in GL(n, \mathbb{R})$. The rule for									
mielke-geometrodynamics_F00327	81	0.623	$\mathrm{g} \in \mathrm{GL}(\mathrm{n}, \mathbb{R})$	$\mathbf{g} \in GL(n, \mathbb{R})$	of the translations $\mathbf{a} \in \mathbb{R}^n$ and the linear transformations $\mathbf{g} \in GL(n, \mathbb{R})$. The rule for									
mielke-geometrodynamics_F00328	81	1.000	$0(1, \mathrm{n}-1) \subset \mathrm{GL}(\mathrm{n}, \mathbb{R})$	$0(1, n - 1) \subset GL(n, \mathbb{R})$	$0(1, n - 1) \subset GL(n, \mathbb{R})$. The invariance of the (flat) metric ground form of the									
mielke-geometrodynamics_F00329	81	1.000	$\mathfrak{a}(\mathrm{n}, \mathbb{R})$	$\mathfrak{a}(n, \mathbb{R})$	$\mathfrak{a}(n, \mathbb{R})$ of the affine group, the analogous decomposition									
mielke-geometrodynamics_F00330	82	0.999	$L_{\alpha \beta} \in \mathfrak{gl}(\mathrm{n}, \mathbb{R})$	$L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$	In these formulas, $L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$ and $P_\alpha \in \mathbb{R}^n$ denote the infinitesimal generators of									
mielke-geometrodynamics_F00331	82	0.999	$P_{\alpha} \in \mathbb{R}^{\mathrm{n}}$	$P_\alpha \in \mathbb{R}^n$	In these formulas, $L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$ and $P_\alpha \in \mathbb{R}^n$ denote the infinitesimal generators of									
mielke-geometrodynamics_F00332	82	1.000	$\mathrm{GL}(\mathrm{n}, \mathbb{R})$	$GL(n, \mathbb{R})$	for the tangent space. The general linear group $GL(n,\mathbb{R})$ acts on this basis as follows:									
mielke-geometrodynamics_F00333	82	0.833	$\mathrm{L}_{\mathrm{m}}(\mathrm{M})$	$L_m(M)$	It thus generates a space $L_m(M)$ of <i>linear frames of reference</i> at the point m. Let									
mielke-geometrodynamics_F00334	82	0.993	$\mathrm{U} \subset \mathrm{M}$	$U \subset M$	remember that in a neighborhood $U \subset M$ of a point $m \in M$ with coordinates x^i ,									

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mielke-geometrodynamics_F00335	82	■ 0.994	$\mathrm{e}_{\mathrm{a}}(\mathrm{m})$	$\mathbf{e}_a(\mathbf{m})$	a basis vector $\mathbf{e}_a(\mathbf{m})$ of $T_{\mathbf{m}}(\mathbf{M})$ may always be written in the way given above.
mielke-geometrodynamics_F00336	82	■ 0.852	$\mathbb{R}^{\mathrm{n}^2}$	$\mathbb{R}^{n^2}$	an open neighborhood of $\mathbb{R}^{n^2}$. Since $\mathbf{e}_{\alpha}^j(\mathbf{m})$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00337	82	■ 0.852	$\mathrm{e}_{\alpha}^j(\mathrm{m})$	$\mathbf{e}_{\alpha}^j(\mathbf{m})$	an open neighborhood of $\mathbb{R}^{n^2}$. Since $\mathbf{e}_{\alpha}^j(\mathbf{m})$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00338	82	■ 0.852	$n \times n$	$n \times n$	an open neighborhood of $\mathbb{R}^{n^2}$. Since $\mathbf{e}_{\alpha}^j(\mathbf{m})$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00339	82	■ 0.987	$\left(\mathrm{x}^i, \mathrm{e}_{\alpha}^{-j}\right)$	$(\mathbf{x}^i, \mathbf{e}_{\alpha}^{-j})$	the collection $(x^i, \mathbf{e}_{\alpha}^j)$ may be regarded as a choice of coordinates of the Cartesian
mielke-geometrodynamics_F00340	82	■ 0.840	$\mathrm{U} \times \mathrm{GL}(\mathrm{n}, \mathbb{R})$	$\mathbf{U} \times \mathrm{GL}(\mathbf{n}, \mathbb{R})$	product $\mathbf{U} \times \mathrm{GL}(\mathbf{n}, \mathbb{R})$. These coordinates arise from the original domain $\pi^{-1}(\mathbf{U})$, for
mielke-geometrodynamics_F00341	82	■ 0.840	$\pi^{-1}(\mathrm{U})$	$\pi^{-1}(\mathbf{U})$	product $\mathbf{U} \times \mathrm{GL}(\mathbf{n}, \mathbb{R})$. These coordinates arise from the original domain $\pi^{-1}(\mathbf{U})$, for
mielke-geometrodynamics_F00342	83	■ 0.858	$\operatorname{GL}(\mathrm{n})$	$\mathrm{GL}(\mathbf{n})$	which the projection π of the principal fiber bundle with the structure group $\mathrm{GL}(\mathbf{n})$,
mielke-geometrodynamics_F00343	83	■ 0.899	ϑ	ϑ	of linear frames. This is the case if an $\mathbb{R}^n$ -valued 1-form ϑ exists that not only is
mielke-geometrodynamics_F00344	83	■ 0.825	$\mathrm{n}=\operatorname{dim} \mathrm{M}$	$\mathbf{n} = \dim \mathbf{M}$	this close contact and due to the fact that ϑ has rank $\mathbf{n} = \dim \mathbf{M}$, it is natural to call
mielke-geometrodynamics_F00345	83	■ 0.394	$\mathbb{R}^n$	$\mathbb{R}^n$	associated with $L(\mathbf{M})$. Its standard fiber is $\mathbb{R}^n$, and thus $T(\mathbf{M})$ has the same dimension
mielke-geometrodynamics_F00346	83	■ 0.569	$\mathrm{A}(\mathrm{M}) / \mathbb{R}^n$	$A(\mathbf{M})/\mathbb{R}^n$	The possibility of identifying $L(\mathbf{M})$ with the quotient bundle $A(\mathbf{M})/\mathbb{R}^n$ results from
mielke-geometrodynamics_F00347	84	■ 0.999	$\mathrm{e}_{\alpha}(\mathrm{m})$	$\mathbf{e}_{\alpha}(\mathbf{m})$	the basis vectors $\mathbf{e}_{\alpha}(\mathbf{m})$ in the tangent space $T_{\mathbf{m}}(\mathbf{M})$ at the point $\mathbf{m} \in \mathbf{M}$ appears to
mielke-geometrodynamics_F00348	84	■ 0.593	$\mathrm{T}=\mathbb{R}^n$	$\mathbf{T} = \mathbb{R}^n$	group $\mathbf{T} = \mathbb{R}^n$ of translations can be regarded as the quotient space $A(\mathbf{n}, \mathbb{R})/\mathrm{GL}(\mathbf{n}, \mathbb{R})$,
mielke-geometrodynamics_F00349	84	■ 0.593	$\mathrm{A}(\mathrm{n}, \mathbb{R}) / \mathrm{GL}(\mathrm{n}, \mathbb{R})$	$A(\mathbf{n}, \mathbb{R})/\mathrm{GL}(\mathbf{n}, \mathbb{R})$	group $\mathbf{T} = \mathbb{R}^n$ of translations can be regarded as the quotient space $A(\mathbf{n}, \mathbb{R})/\mathrm{GL}(\mathbf{n}, \mathbb{R})$,
mielke-geometrodynamics_F00350	84	■ 1.000	$\mathrm{T}=\mathrm{P} / \mathrm{L}$	$\mathbf{T} = \mathbf{P} / \mathbf{L}$	realized by translations acting on the coset space $\mathbf{T} = \mathbf{P} / \mathbf{L}$, which is isomorphic to the
mielke-geometrodynamics_F00351	84	■ 0.550	$\widetilde{\omega}$	$\widetilde{\omega}$	connection. Let $\widetilde{\omega}$ be the 1-form of such a <i>generalized affine connection</i> , and let
mielke-geometrodynamics_F00352	84	■ 0.999	$\chi: \mathrm{L}(\mathrm{M}) \rightarrow 0_{\mathrm{m}} \times \mathrm{L}(\mathrm{M}) \subset \mathrm{A}(\mathrm{M})$	$\chi: L(\mathbf{M}) \rightarrow 0_{\mathbf{m}} \times L(\mathbf{M}) \subset A(\mathbf{M})$	$\chi: L(\mathbf{M}) \rightarrow 0_{\mathbf{m}} \times L(\mathbf{M}) \subset A(\mathbf{M})$ be the natural injection of linear frames into the
mielke-geometrodynamics_F00353	85	■ 0.448	$\stackrel{*}{\chi} \widetilde{\omega}$	$\overset{*}{\chi} \widetilde{\omega}$	image $\overset{*}{\chi} \widetilde{\omega}$ of the connection $\widetilde{\omega}$ is an $\mathfrak{a}(\mathbf{n}, \mathbb{R})$ -valued 1-form in $L(\mathbf{M})$ that decomposes
mielke-geometrodynamics_F00354	85	■ 0.507	ω^{L}	$\omega^{\mathbf{L}}$	tation (4.1.2) of $A(\mathbf{n}, \mathbb{R})$. In the above formula, $\omega^{\mathbf{L}}$ denotes a $\mathfrak{gl}(\mathbf{n}, \mathbb{R})$ -valued 1-form
mielke-geometrodynamics_F00355	85	■ 0.507	$\mathfrak{gl}(\mathrm{n}, \mathbb{R})$	$\mathfrak{gl}(\mathbf{n}, \mathbb{R})$	tation (4.1.2) of $A(\mathbf{n}, \mathbb{R})$. In the above formula, $\omega^{\mathbf{L}}$ denotes a $\mathfrak{gl}(\mathbf{n}, \mathbb{R})$ -valued 1-form
mielke-geometrodynamics_F00356	85	■ 0.912	ω^{T}	$\omega^{\mathbf{T}}$	in $\mathbf{M}$, while $\omega^{\mathbf{T}}$ is an $\mathbb{R}^n$ -valued 1-form, i.e., a tensorial form of the representation
mielke-geometrodynamics_F00357	85	■ 0.990	$\mathrm{H} \subset \mathrm{G}$	$\mathbf{H} \subset \mathbf{G}$	structure with respect to a subgroup $\mathbf{H} \subset \mathbf{G}$.
mielke-geometrodynamics_F00358	85	■ 0.944	$\widetilde{\Omega}$	$\widetilde{\Omega}$	The curvature $\widetilde{\Omega}$ in $A(\mathbf{M})$ can be derived from $\widetilde{\omega}$ by an analogous application of
mielke-geometrodynamics_F00359	85	■ 1.000	$\left[\omega^{\mathrm{T}}, \omega^{\mathrm{T}}\right]$	$[\omega^{\mathbf{T}}, \omega^{\mathbf{T}}]$	The commutator $[\omega^{\mathbf{T}}, \omega^{\mathbf{T}}]$ occurring in the calculation
mielke-geometrodynamics_F00360	85	■ 0.989	$\omega^{\mathrm{T}}, \omega^{\mathrm{L}}$	$\omega^{\mathbf{T}}, \omega^{\mathbf{L}}$	which is induced by $\omega^{\mathbf{L}}$. It is of importance to realize that the commutator $[\omega^{\mathbf{T}}, \omega^{\mathbf{L}}]$

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mielke-geometrodynamics_F00361	86	■ 1.000	$\mathscr{A}(n, \mathbb{R})$	$\mathscr{A}(n, \mathbb{R})$	of local translations are subgroups of $\mathscr{A}(n, \mathbb{R})$. Taking the cross section in the asso-
mielke-geometrodynamics_F00362	86	■ 0.996	C^{∞}	C^{∞}	ciated bundle is abbreviated by C^{∞} , and Ad denotes the adjoint representation with
mielke-geometrodynamics_F00363	86	■ 0.998	$GL(n, \mathbb{R})$	$GL(n, \mathbb{R})$	respect to $GL(n, \mathbb{R})$. Due to its construction, the group of local translations $\mathcal{T}(n, \mathbb{R})$
mielke-geometrodynamics_F00364	86	■ 0.998	$\mathscr{T}(n, \mathbb{R})$	$\mathcal{T}(n, \mathbb{R})$	respect to $GL(n, \mathbb{R})$. Due to its construction, the group of local translations $\mathcal{T}(n, \mathbb{R})$
mielke-geometrodynamics_F00365	86	■ 1.000	$\operatorname{Diff}(n, \mathbb{R})$	$\operatorname{Diff}(n, \mathbb{R})$	is <i>locally</i> isomorphic to the group of active diffeomorphisms $\operatorname{Diff}(n, \mathbb{R})$ of the man-
mielke-geometrodynamics_F00366	86	■ 0.905	$(n, \mathbb{R})$	$(n, \mathbb{R})$	$\operatorname{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine
mielke-geometrodynamics_F00367	86	■ 0.905	$\left(n + n^2\right)$	$(n + n^2)$	$\operatorname{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine
mielke-geometrodynamics_F00368	86	■ 0.905	$A(n, \mathbb{R})_{\mathrm{H}}$	$A(n, \mathbb{R})_{\mathrm{H}}$	$\operatorname{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine
mielke-geometrodynamics_F00369	86	■ 0.999	$P_i = \partial_i := \partial / \partial x^i$	$P_i = \partial_i := \partial / \partial x^i$	transformations as a subgroup, generated by the vector fields $P_i = \partial_i := \partial / \partial x^i$ and
mielke-geometrodynamics_F00370	86	■ 1.000	$L^i{}_j = x^i \partial_j$	$L^i{}_j = x^i \partial_j$	$L^i{}_j = x^i \partial_j$. Note that differentiable coordinate transformations, which leave exterior
mielke-geometrodynamics_F00371	86	■ 1.000	χ	χ	the natural injection χ , this formula decomposes into
mielke-geometrodynamics_F00372	87	■ 1.000	$\operatorname{V}^{\mathrm{id}}$	$\operatorname{V}^{\mathrm{id}}$	in $\operatorname{V}^{\mathrm{id}}$ consists of an orbit $\mathbb{R}^n$ of $A(n, \mathbb{R})$. The isotropy group of the zero section
mielke-geometrodynamics_F00373	87	■ 0.928	$\stackrel{\circ}{\varphi} = \sigma(0_{\mathrm{m}})$	$\stackrel{\circ}{\varphi} = \sigma(0_{\mathrm{m}})$	$\stackrel{\circ}{\varphi} = \sigma(0_{\mathrm{m}})$ is $\mathrm{H} = \mathrm{GL}(n, \mathbb{R})$, i.e., the general linear group. Then the subbundle
mielke-geometrodynamics_F00374	87	■ 0.928	$\mathrm{H} = \mathrm{GL}(n, \mathbb{R})$	$\mathrm{H} = \mathrm{GL}(n, \mathbb{R})$	$\stackrel{\circ}{\varphi} = \sigma(0_{\mathrm{m}})$ is $\mathrm{H} = \mathrm{GL}(n, \mathbb{R})$, i.e., the general linear group. Then the subbundle
mielke-geometrodynamics_F00375	87	■ 0.863	$\widetilde{\vartheta}$	$\widetilde{\vartheta}$	bution $\widetilde{\vartheta}$ transforms under $\mathscr{A}_{\mathrm{p}}$ like
mielke-geometrodynamics_F00376	87	■ 0.863	$\mathscr{A}_{\mathrm{p}}$	$\mathscr{A}_{\mathrm{p}}$	bution $\widetilde{\vartheta}$ transforms under $\mathscr{A}_{\mathrm{p}}$ like
mielke-geometrodynamics_F00377	88	■ 0.968	$\mathrm{D} = (\stackrel{*}{\chi} \widetilde{\mathrm{D}})$	$\mathrm{D} = (\stackrel{*}{\chi} \widetilde{\mathrm{D}})$	to $\mathrm{D} = (\stackrel{*}{\chi} \widetilde{\mathrm{D}})$, it follows that the torsion 2-form transforms like
mielke-geometrodynamics_F00378	88	■ 0.999	$\mathscr{G}_{\mathrm{p}}$	$\mathscr{G}_{\mathrm{p}}$	i.e., as a vector with respect to the subgroup $\mathscr{G}_{\mathrm{p}}$ of linear gauge transformations. As
mielke-geometrodynamics_F00379	89	■ 0.993	$\widetilde{\widetilde{\mathrm{D}}} = \mathrm{D}$	$\widetilde{\widetilde{\mathrm{D}}} = \mathrm{D}$	$\widetilde{\widetilde{\mathrm{D}}} = \mathrm{D}$ it remains valid) the affine connection to the linear $\mathfrak{gl}(n, \mathbb{R})$ -valued connection
mielke-geometrodynamics_F00380	90	■ 1.000	$\mathscr{C} \subset \mathbf{M}$	$\mathscr{C} \subset \mathbf{M}$	to a smooth curve $\mathscr{C} \subset \mathbf{M}$ that is parametrized by $s(t)$. In order to satisfy
mielke-geometrodynamics_F00381	90	■ 1.000	$\mathrm{F}(\mathbf{m}, \vartheta) : \mathbf{M} \times \mathrm{T}^*(\mathbf{M}) \rightarrow \mathbb{R}^+$	$\mathrm{F}(\mathbf{m}, \vartheta) : \mathbf{M} \times \mathrm{T}^*(\mathbf{M}) \rightarrow \mathbb{R}^+$	where $\mathrm{F}(\mathbf{m}, \vartheta) : \mathbf{M} \times \mathrm{T}^*(\mathbf{M}) \rightarrow \mathbb{R}^+$ is a smooth, <i>positive</i> , and <i>homogeneous</i> funda-
mielke-geometrodynamics_F00382	91	■ 1.000	$\mathscr{C}$	$\mathscr{C}$	ance of (4.3.4) versus admissible reparametrizations of the curve $\mathscr{C}$. Generally, these
mielke-geometrodynamics_F00383	91	■ 1.000	$(2, 0)$	$(2, 0)$	Subsequently, a covariant symmetric tensor field of degree (2, 0) is imprinted onto
mielke-geometrodynamics_F00384	91	■ 1.000	$\mathrm{g}_{ij} \in \mathrm{C}^{\infty}(\mathbf{M})$	$\mathrm{g}_{ij} \in \mathrm{C}^{\infty}(\mathbf{M})$	Indeed, the functions $\mathrm{g}_{ij} \in \mathrm{C}^{\infty}(\mathbf{M})$ describe “concerning the chosen, arbitrary coor-
mielke-geometrodynamics_F00385	92	■ 0.689	e_{α}^i	e_{α}^i	1-forms ϑ admitted. Since the e_{α}^i have been considered as local bundle coordinates
mielke-geometrodynamics_F00386	92	■ 1.000	$\mathfrak{D}(s, n - s)$	$\mathfrak{D}(s, n - s)$	$\mathfrak{D}(s, n - s)$ -valued connection $\omega^{\mathfrak{g}}$, the constraint

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mielke-geometrodynamics_F00387	92	■ 1.000	ω^{g}	ω^{g}	$\mathfrak{O}(\mathrm{s}, \mathrm{n} - \mathrm{s})$ -valued connection $\omega^{\mathfrak{g}}$, the constraint
mielke-geometrodynamics_F00388	92	■ 0.532	$\phi^{\{\circ\}}$	$\phi^{(\circ)}$	is a necessary and sufficient condition. For tensor-valued 0-forms $\phi^{(\circ)}$, the exterior
mielke-geometrodynamics_F00389	92	■ 0.651	$\Gamma_{\mathrm{i}}^{\alpha\beta}$	$\Gamma_{\mathrm{i}}^{\alpha\beta}$	where the $\Gamma_{\mathrm{i}}^{\alpha\beta}$ denote the (local) <i>Ricci rotation coefficients</i> . Due to the fact that
mielke-geometrodynamics_F00390	92	■ 1.000	$L_{\alpha\beta}$	$L_{\alpha\beta}$	the infinitesimal generators $L_{\alpha\beta}$ become skew after the restriction to the orthogonal
mielke-geometrodynamics_F00391	92	■ 1.000	$\omega^{\alpha\beta} = -\omega^{\beta\alpha}$	$\omega^{\alpha\beta} = -\omega^{\beta\alpha}$	subgroup, the constraint (4.3.14) analogously yields $\omega^{\alpha\beta} = -\omega^{\beta\alpha}$, i.e., the skew-
mielke-geometrodynamics_F00392	92	■ 0.988	$\mathrm{m}_{\mathrm{o}} \in \mathrm{M}$	$\mathbf{m}_{\mathrm{o}} \in \mathrm{M}$	neighborhood of a point $\mathbf{m}_{\mathrm{o}} \in \mathrm{M}$ depends entirely on Minkowski’s metric tensor o_{ij}
mielke-geometrodynamics_F00393	92	■ 0.988	$o_{\mathrm{i} \mathrm{j}}$	o_{ij}	neighborhood of a point $\mathbf{m}_{\mathrm{o}} \in \mathrm{M}$ depends entirely on Minkowski’s metric tensor o_{ij}
mielke-geometrodynamics_F00394	93	■ 1.000	$\mathrm{H} = \mathrm{O}(\mathrm{s}, \mathrm{n} - \mathrm{s})$	$\mathrm{H} = \mathrm{O}(\mathrm{s}, \mathrm{n} - \mathrm{s})$	gauge theory has only the orthogonal “isotropy group” $\mathrm{H} = \mathrm{O}(\mathrm{s}, \mathrm{n} - \mathrm{s})$ as resid-
mielke-geometrodynamics_F00395	93	■ 1.000	$\mathfrak{o}(\mathrm{s}, \mathrm{n} - \mathrm{s})$	$\mathfrak{o}(\mathrm{s}, \mathrm{n} - \mathrm{s})$	the reduction of the linear connection ω^{L} to an $\mathfrak{o}(\mathrm{s}, \mathrm{n} - \mathrm{s})$ -valued connection $\omega^{\mathfrak{g}}$
mielke-geometrodynamics_F00396	93	■ 0.999	$\mathrm{L}^{\mathrm{g}}(\mathrm{M})$	$\mathrm{L}^{\mathrm{g}}(\mathrm{M})$	in $\mathrm{L}^{\mathfrak{g}}(\mathrm{M})$. This consistency does not, as is implied by HANSON & REGGE (1979),
mielke-geometrodynamics_F00397	93	■ 1.000	$\mathrm{D}\vartheta = 0$	$\mathrm{D}\vartheta = 0$	demand the vanishing of the torsion, i.e., $\mathrm{D}\vartheta = 0$. In a Lagrangian formulation of
mielke-geometrodynamics_F00398	93	■ 0.995	Θ	Θ	Higgs field ϑ , i.e., the canonical <i>orthogonal</i> 1-form, ⁴ and the resulting torsion Θ
mielke-geometrodynamics_F00399	93	■ 0.913	ϑ^{g}	ϑ^{g}	⁴ As for the nomenclature, the 1-form ϑ , which is restricted by (4.3.10), will not be denoted by $\vartheta^{\mathfrak{g}}$
mielke-geometrodynamics_F00400	93	■ 1.000	Ω^{g}	Ω^{g}	and thus differs from $\omega^{\mathfrak{g}}$ and $\Omega^{\mathfrak{g}}$, since we prefer to stick to (4.3.10) or (4.3.9), respectively, as
mielke-geometrodynamics_F00401	94	■ 0.997	$\omega^{\{\}}$	$\omega^{\{\}}$	Here $\omega^{\{\}}$ denotes the 1-form of the torsion-free Christoffel-type connection for which
mielke-geometrodynamics_F00402	94	■ 0.890	K	K	between the 2-form Θ and the 1-form K of the so-called <i>contortion</i> , having values
mielke-geometrodynamics_F00403	94	■ 1.000	U_{n}	U_{n}	denoted by U_{n} , in contrast to a purely Riemannian manifold, which is symbolized by V_{n} .
mielke-geometrodynamics_F00404	94	■ 1.000	V_{n}	V_{n}	denoted by U_{n} , in contrast to a purely Riemannian manifold, which is symbolized by V_{n} .
mielke-geometrodynamics_F00405	95	■ 0.456	$R_{\mathrm{i} \mathrm{j} \mathrm{c}}^{\{\dots \mathrm{d}\}}$	$R_{ijc}^{\{\dots d\}}$	In this formula, $R_{ijc}^{\{\dots d\}}$ are the mixed components of the Riemannian curvature tensor,
mielke-geometrodynamics_F00406	95	■ 0.999	$R_{\mathrm{i} \mathrm{j} \mathrm{k} \mathrm{l}}^{\{\}}$	$R_{ijkl}^{\{\}}$	holonomic components $R_{ijkl}^{\{\}}$ have the following symmetries with respect to an inter-
mielke-geometrodynamics_F00407	95	■ 0.666	$S \ S \ B$	SSB	cation of a Higgs–Kibble-type mechanism of <i>SSB</i> that resulted in the construction of
mielke-geometrodynamics_F00408	95	■ 1.000	$\mathrm{O}(1, \mathrm{n} - 1)$	$\mathrm{O}(1, \mathrm{n} - 1)$	$\mathrm{O}(1, \mathrm{n} - 1)$ as the “residual” symmetry of the theory via the general linear group
mielke-geometrodynamics_F00409	95	■ 0.778	$\operatorname{GL}(\mathrm{n}, \mathbb{R})$	$\mathrm{GL}(\mathrm{n}, \mathbb{R})$	$\mathrm{GL}(\mathrm{n}, \mathbb{R})$.
mielke-geometrodynamics_F00410	96	■ 1.000	$\mathrm{R}_{\alpha\beta}^{\cdots \mathrm{cd}}(\mathrm{m})$	$\mathrm{R}_{\alpha\beta}^{\cdots \mathrm{cd}}(\mathrm{m})$	spacetime-dependent “structure constants” $\mathrm{R}_{\alpha\beta}^{\cdots \mathrm{cd}}(\mathrm{m})$ occur in the generic case. (The
mielke-geometrodynamics_F00411	96	■ 1.000	$\vartheta, \Theta, \Omega^{\mathrm{g}}$	$\vartheta, \Theta, \Omega^{\mathrm{g}}$	However, contrasting with $\vartheta, \Theta, \Omega^{\mathfrak{g}}$ (see (4.2.20), (4.2.14), and (4.2.13)), the
mielke-geometrodynamics_F00412	97	■ 1.000	L_{g}	L_{g}	only via of the forms Θ and $\Omega^{\mathfrak{g}}$. The variation of L_{g} , regarded as a total differential,
mielke-geometrodynamics_F00413	97	■ 1.000	t	t	defines the <i>translational</i> and (<i>Lorentz</i> -)rotational canonical gauge field momenta Π^{T}

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mielke-geometrodynamics_F00414	97	■ 1.000	Π^{T}	Π^{T}	defines the <i>translational</i> and (<i>Lorentz</i> -)rotational canonical gauge field momenta Π^{T}
mielke-geometrodynamics_F00415	97	■ 0.999	Π^{L}	Π^{L}	and Π^{L} . Due to the special role of ϑ , there occurs additionally the (n-1)-form E ,
mielke-geometrodynamics_F00416	97	■ 0.999	E	E	and Π^{L} . Due to the special role of ϑ , there occurs additionally the (n-1)-form E ,
mielke-geometrodynamics_F00417	97	■ 0.998	$\Pi_{\square}^{\mathrm{T}}$	$\Pi_{\square}^{\mathrm{T}}$	where the “contortional” 1-form $\Pi_{\square}^{\mathrm{T}}$, having values in the Lie algebra of the
mielke-geometrodynamics_F00418	97	■ 0.931	$\Pi^{\mathrm{T}} := \left[\Pi_{\square}^{\mathrm{T}}, \vartheta \right]$	$\Pi^{\mathrm{T}} := \left[\Pi_{\square}^{\mathrm{T}}, \vartheta \right]$	$\Pi^{\mathrm{T}} := [\Pi_{\square}^{\mathrm{T}}, \vartheta]$. This is in complete analogy to the relation (4.3.24) for the contor-
mielke-geometrodynamics_F00419	97	■ 1.000	$\delta \vartheta$	$\delta \vartheta$	The application of Hamilton’s principle to the independent variations of S for $\delta \vartheta$
mielke-geometrodynamics_F00420	97	■ 1.000	$\delta \omega^{\mathrm{g}}$	$\delta \omega^{\mathrm{g}}$	and $\delta \omega^{\mathrm{g}}$ leads to the <i>Euler–Lagrange</i> equations
mielke-geometrodynamics_F00421	98	■ 0.844	Σ	Σ	fied with the 3-form Σ , the energy–momentum current, and the 3-form τ of the spin
mielke-geometrodynamics_F00422	98	■ 0.999	$\lambda^{\varphi} = \mathbf{o}$	$\lambda^{\varphi} = \mathbf{o}$	If the Euler–Lagrange equations $\lambda^{\varphi} = \mathbf{o}$ of the matter fields are satisfied, it will
mielke-geometrodynamics_F00423	98	■ 1.000	τ_{s}	τ_{s}	turn out that Σ and τ_{s} are the field momenta of matter that are <i>canonically conjugate</i>
mielke-geometrodynamics_F00424	98	■ 1.000	Υ	Υ	hypermomentum current Υ of the matter fields in such a generalization (HEHL &
mielke-geometrodynamics_F00425	100	■ 1.000	$\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$	$\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$	symmetric connection 1-form $\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$ is defined by $\delta L = \delta \Gamma^{\alpha\beta} \wedge (\partial L / \partial \Gamma^{\alpha\beta})$.
mielke-geometrodynamics_F00426	100	■ 1.000	$\delta L = \delta \Gamma^{\alpha\beta} \wedge (\partial L / \partial \Gamma^{\alpha\beta})$	$\delta L = \delta \Gamma^{\alpha\beta} \wedge (\partial L / \partial \Gamma^{\alpha\beta})$	symmetric connection 1-form $\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$ is defined by $\delta L = \delta \Gamma^{\alpha\beta} \wedge (\partial L / \partial \Gamma^{\alpha\beta})$.
mielke-geometrodynamics_F00427	100	■ 0.629	$S = \int_M L_{\mathrm{mat}}$	$S = \int_M L_{\mathrm{mat}}$	The action $S = \int_M L_{\mathrm{mat}}$ for the matter Lagrangian is, by construction, invari-
mielke-geometrodynamics_F00428	100	■ 1.000	$\operatorname{Diff}(M)$	$\operatorname{Diff}(M)$	ant under the group $\operatorname{Diff}(M)$ of coordinate transformations and local frame rota-
mielke-geometrodynamics_F00429	100	■ 0.924	$\mathcal{T} \subset \operatorname{Diff}(M)$	$\mathcal{T} \subset \operatorname{Diff}(M)$	one-parameter group of <i>local</i> translations $\mathcal{T} \subset \operatorname{Diff}(M)$, we employ the $SO(1, 3)$ -
mielke-geometrodynamics_F00430	100	■ 0.924	$SO(1, 3)$	$SO(1, 3)$	one-parameter group of <i>local</i> translations $\mathcal{T} \subset \operatorname{Diff}(M)$, we employ the $SO(1, 3)$ -
mielke-geometrodynamics_F00431	100	■ 0.590	$\mathbb{L}_{\xi} := \xi \rfloor D + D \xi \rfloor$	$\mathbb{L}_{\xi} := \xi \rfloor D + D \xi \rfloor$	covariant Lie derivative $\mathbb{L}_{\xi} := \xi \rfloor D + D \xi \rfloor$ on M with respect to an arbitrary vector
mielke-geometrodynamics_F00432	100	■ 0.590	M	M	covariant Lie derivative $\mathbb{L}_{\xi} := \xi \rfloor D + D \xi \rfloor$ on M with respect to an arbitrary vector
mielke-geometrodynamics_F00433	100	■ 1.000	ξ	ξ	field ξ . Since $Dg_{\alpha\beta} = 0$, we obtain
mielke-geometrodynamics_F00434	100	■ 1.000	$Dg_{\alpha\beta} = 0$	$Dg_{\alpha\beta} = 0$	field ξ . Since $Dg_{\alpha\beta} = 0$, we obtain
mielke-geometrodynamics_F00435	101	■ 0.779	$\mathbb{L}_{\xi} L = D \xi \rfloor L$	$\mathbb{L}_{\xi} L = D \xi \rfloor L$	$\mathbb{L}_{\xi} L = D \xi \rfloor L$. Comparing the boundary term, we can read off the identity
mielke-geometrodynamics_F00436	101	■ 1.000	$\xi \rightarrow e_{\alpha}$	$\xi \rightarrow e_{\alpha}$	If we replace $\xi \rightarrow e_{\alpha}$ by the basis frame, then (4.5.4) yields directly the explicit
mielke-geometrodynamics_F00437	101	■ 0.995	$F_{\alpha}^{\mathrm{Max}} = (e_{\alpha} \rfloor F) \wedge j$	$F_{\alpha}^{\mathrm{Max}} = (e_{\alpha} \rfloor F) \wedge j$	calculus $F_{\alpha}^{\mathrm{Max}} = (e_{\alpha} \rfloor F) \wedge j$.
mielke-geometrodynamics_F00438	101	■ 1.000	$\simeq$	$\simeq$	no field equations are invoked. <i>Weak</i> identities, which are denoted by $\simeq$, hold only
mielke-geometrodynamics_F00439	101	■ 0.999	$\delta L / \delta \Psi = 0$	$\delta L / \delta \Psi = 0$	if the matter field equation $\delta L / \delta \Psi = 0$ is satisfied.
mielke-geometrodynamics_F00440	101	■ 0.972	$\delta D \Psi = D \delta \Psi + \delta \Gamma_{\alpha}^{\beta} \wedge I^{\alpha}_{\beta} \Psi$	$\delta D \Psi = D \delta \Psi + \delta \Gamma_{\alpha}^{\beta} \wedge I^{\alpha}_{\beta} \Psi$	Since $\delta D \Psi = D \delta \Psi + \delta \Gamma_{\alpha}^{\beta} \wedge I^{\alpha}_{\beta} \Psi$, this is equivalent to

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mielke-geometrodynamics_F00456	103	■ 1.000	$\xi=\partial / \partial \varphi$	$\xi = \partial / \partial \varphi$	solution, if $\xi = \partial / \partial t$ or $\xi = \partial / \partial \varphi$, respectively, is chosen. Here no “factor of two”
mielke-geometrodynamics_F00457	104	■ 1.000	$\pm (\Lambda / 3) r^3$	$\pm (\Lambda / 3) r^3$	complex each contribute <i>one-half</i> of the mass, so that the divergent terms $\pm (\Lambda / 3) r^3$
mielke-geometrodynamics_F00458	104	■ 1.000	Λ	Λ	proportional to the cosmological constant Λ exactly cancel each other.
mielke-geometrodynamics_F00459	104	■ 0.539	$\Gamma^{(T) \alpha}$	$\Gamma^{(T) \alpha}$	includes the true translational potential $\Gamma^{(T) \alpha}$ and the $GL(n, \mathbb{R})$ -gauge connection
mielke-geometrodynamics_F00460	104	■ 1.000	$\Gamma_{\alpha}^{(L) \beta}$	$\Gamma_{\alpha}^{(L) \beta}$	$\Gamma_{\alpha}^{(L) \beta}$, as can be deduced from the Möbius-type five-dimensional representation of
mielke-geometrodynamics_F00461	104	■ 1.000	$A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$	$A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$	the affine gauge group $A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$.
mielke-geometrodynamics_F00462	104	■ 0.559	$\widetilde{\xi}$	$\widetilde{\xi}$	view: the condition for a parallel transport of an affine vector $\widetilde{\xi}$ around a small closed
mielke-geometrodynamics_F00463	104	■ 0.993	$\widetilde{\Gamma}$	$\widetilde{\Gamma}$	loop by means of $\widetilde{\Gamma}$ reads
mielke-geometrodynamics_F00464	104	■ 0.272	$\mathbb{L}_y := y \rfloor D + Dy$	$\mathbb{L}_y := y \rfloor D + Dy$	where $\mathbb{L}_y := y \rfloor D + Dy \rfloor$ denotes the gauge-covariant Lie derivative and $R^{(T) \alpha}$ the
mielke-geometrodynamics_F00465	104	■ 0.272	$R^{(T) \alpha}$	$R^{(T) \alpha}$	where $\mathbb{L}_y := y \rfloor D + Dy \rfloor$ denotes the gauge-covariant Lie derivative and $R^{(T) \alpha}$ the
mielke-geometrodynamics_F00466	104	■ 1.000	y	y	loop parametrized by y yields
mielke-geometrodynamics_F00467	105	■ 1.000	$A = A_i^J \lambda_J dx^i$	$A = A_i^J \lambda_J dx^i$	where $A = A_i^J \lambda_J dx^i$ is a Yang–Mills-type connection and P is the principal value
mielke-geometrodynamics_F00468	105	■ 0.996	γ	γ	of the integral. If the loop γ lies in a field-free region, i.e., one with $U(1)$ field
mielke-geometrodynamics_F00469	105	■ 1.000	$F := dA = 0$	$F := dA = 0$	strength $F := dA = 0$ or Yang–Mills curvature $F = dA + A \wedge A = 0$, but encloses
mielke-geometrodynamics_F00470	105	■ 1.000	$F = dA + A \wedge A = 0$	$F = dA + A \wedge A = 0$	strength $F := dA = 0$ or Yang–Mills curvature $F = dA + A \wedge A = 0$, but encloses
mielke-geometrodynamics_F00471	105	■ 1.000	$F \neq 0$	$F \neq 0$	a “confined” region with nontrivial “magnetic” flux $F \neq 0$, the potential A can still
mielke-geometrodynamics_F00472	105	■ 1.000	$A(4, \mathbb{R})$	$A(4, \mathbb{R})$	arising from a gauge theory of the affine structure group $A(4, \mathbb{R})$.
mielke-geometrodynamics_F00473	105	■ 0.924	$GL(4, \mathbb{R})$	$GL(4, \mathbb{R})$	For a so-called manifold Ψ carrying no $GL(4, \mathbb{R})$ -excitations, i.e., no spin, no
mielke-geometrodynamics_F00474	105	■ 1.000	$\Gamma^{(T) \alpha} \stackrel{*}{=} 0$	$\Gamma^{(T) \alpha} \stackrel{*}{=} 0$	of the cone there is $\Gamma^{(T) \alpha} \stackrel{*}{=} 0$ locally.
mielke-geometrodynamics_F00475	105	■ 0.651	(1993, 1994)	(1993, 1994)	time manifold; cf. ANANDAN (1993, 1994). Because the coframe is nondegenerate
mielke-geometrodynamics_F00476	105	■ 1.000	P_{α}	P_{α}	Since the physical dimension of P_{α} is $2\pi \hbar / \ell$, the gravitational analogue of Dirac’s
mielke-geometrodynamics_F00477	105	■ 1.000	$2 \pi \hbar / \ell$	$2\pi \hbar / \ell$	Since the physical dimension of P_{α} is $2\pi \hbar / \ell$, the gravitational analogue of Dirac’s
mielke-geometrodynamics_F00478	105	■ 1.000	$\mathcal{M}_n = n M_{\text{Planck}}$	$\mathcal{M}_n = n M_{\text{Planck}}$	i.e., the mass would turn out to be a multiple $\mathcal{M}_n = n M_{\text{Planck}}$ of the (huge) Planck
mielke-geometrodynamics_F00479	109	■ 0.485	$\Gamma^{\alpha \beta} = \Gamma^{\{j \alpha \beta} - K^{\alpha \beta}$	$\Gamma^{\alpha \beta} = \Gamma^{\{j \alpha \beta} - K^{\alpha \beta}$	The Riemann–Cartan (RC) connection $\Gamma^{\alpha \beta} = \Gamma^{\{j \alpha \beta} - K^{\alpha \beta}$, a one-form, can be
mielke-geometrodynamics_F00480	109	■ 0.469	$\Gamma^{\{\alpha \beta\}}$	$\Gamma^{\{\alpha \beta\}}$	split into the unique Levi-Civita connection $\Gamma^{\{j \alpha \beta}$ of Riemannian geometry and
mielke-geometrodynamics_F00481	109	■ 1.000	$K_{\alpha \beta} = -K_{\beta \alpha}$	$K_{\alpha \beta} = -K_{\beta \alpha}$	a <i>contortion</i> one-form $K_{\alpha \beta} = -K_{\beta \alpha}$, a tensor-valued one-form $K = i K^{\alpha \beta} \sigma_{\alpha \beta} / 4$,
mielke-geometrodynamics_F00482	109	■ 1.000	$K = i K^{\alpha \beta} \sigma_{\alpha \beta} / 4$	$K = i K^{\alpha \beta} \sigma_{\alpha \beta} / 4$	a <i>contortion</i> one-form $K_{\alpha \beta} = -K_{\beta \alpha}$, a tensor-valued one-form $K = i K^{\alpha \beta} \sigma_{\alpha \beta} / 4$,

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mielke-geometrodynamics_F00483	109	■ 1.000	$T^{\alpha}=K^{\alpha}{}_{\beta}\wedge\vartheta^{\beta}$	$T^{\alpha}=K^{\alpha}{}_{\beta}\wedge\vartheta^{\beta}$	implicitly related to torsion via $T^{\alpha}=K^{\alpha}{}_{\beta}\wedge\vartheta^{\beta}$. Recently, it has been stressed by
mielke-geometrodynamics_F00484	109	■ 1.000	$\Gamma=\Gamma^{\{\}}-K$	$\Gamma=\Gamma^{\{\}}-K$	WEINBERG (2005) that this RC connection $\Gamma=\Gamma^{\{\}}-K$ is just a <i>deformation</i> (or
mielke-geometrodynamics_F00485	109	■ 1.000	$\Gamma^{\{\}}$	$\Gamma^{\{\}}$	field redefinition) of the Christoffel connection $\Gamma^{\{\}}$ by the contortion, at least from
mielke-geometrodynamics_F00486	110	■ 1.000	$D>2$	$D>2$	renormalizable in $D>2$ without Chern–Simons (CS) terms (ALVES et al. 1999),
mielke-geometrodynamics_F00487	110	■ 1.000	j_5	j_5	equation relates torsion to spin, i.e., to the axial current j_5 in the case of Dirac fields,
mielke-geometrodynamics_F00488	110	■ 0.972	$\bar{\psi}:=\psi^{\dagger}\gamma_0$	$\bar{\psi}:=\psi^{\dagger}\gamma_0$	$\bar{\psi}:=\psi^{\dagger}\gamma_0$ denotes the Dirac adjoint and $D\psi:=d\psi+\Gamma\wedge\psi$ is the exterior covari-
mielke-geometrodynamics_F00489	110	■ 0.972	$D\psi:=d\psi+\Gamma\wedge\psi$	$D\psi:=d\psi+\Gamma\wedge\psi$	$\bar{\psi}:=\psi^{\dagger}\gamma_0$ denotes the Dirac adjoint and $D\psi:=d\psi+\Gamma\wedge\psi$ is the exterior covari-
mielke-geometrodynamics_F00490	110	■ 1.000	$\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$	$\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$	where $\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$ is the Clifford-algebra-valued coframe satisfying $D\gamma=[\gamma,K]$
mielke-geometrodynamics_F00491	110	■ 1.000	$D\gamma=[\gamma,K]$	$D\gamma=[\gamma,K]$	where $\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$ is the Clifford-algebra-valued coframe satisfying $D\gamma=[\gamma,K]$
mielke-geometrodynamics_F00492	110	■ 1.000	$=\gamma_{\alpha}T^{\alpha}$	$=\gamma_{\alpha}T^{\alpha}$	$=\gamma_{\alpha}T^{\alpha}$, and $T^{\alpha}:=D\vartheta^{\alpha}$ is the torsion two-form.
mielke-geometrodynamics_F00493	110	■ 1.000	$T^{\alpha}:=D\vartheta^{\alpha}$	$T^{\alpha}:=D\vartheta^{\alpha}$	$=\gamma_{\alpha}T^{\alpha}$, and $T^{\alpha}:=D\vartheta^{\alpha}$ is the torsion two-form.
mielke-geometrodynamics_F00494	110	■ 1.000	$L_D=\bar{L}_D=L_D^{\dagger}$	$L_D=\bar{L}_D=L_D^{\dagger}$	Since $L_D=\bar{L}_D=L_D^{\dagger}$ even in an unholonomic frame, the minimal coupling pro-
mielke-geometrodynamics_F00495	111	■ 1.000	$\tau_{\alpha\beta\gamma}=\tau_{[\alpha\beta\gamma]}$	$\tau_{\alpha\beta\gamma}=\tau_{[\alpha\beta\gamma]}$	with <i>totally antisymmetric</i> components $\tau_{\alpha\beta\gamma}=\tau_{[\alpha\beta\gamma]}$. Eventually, torsion merely
mielke-geometrodynamics_F00496	111	■ 0.975	$\mu_{\alpha}=\vartheta_{\alpha}\wedge*j_5/4$	$\mu_{\alpha}=\vartheta_{\alpha}\wedge*j_5/4$	couples to the <i>spin–energy potential</i> $\mu_{\alpha}=\vartheta_{\alpha}\wedge*j_5/4$, i.e., to a two-form propor-
mielke-geometrodynamics_F00497	111	■ 0.792	$\mathrm{GR}_{\parallel}$	$\mathrm{GR}_{\parallel}$	Transitions from EC theory or GR to $\mathrm{GR}_{\parallel}$ can be <i>generated</i> via the Chern–Simons-
mielke-geometrodynamics_F00498	111	■ 0.701	$C_{\mathrm{TT}^*}:=\vartheta^{\alpha}\wedge*T_{\alpha}$	$C_{\mathrm{TT}^*}:=\vartheta^{\alpha}\wedge*T_{\alpha}$	type term $C_{\mathrm{TT}^*}:=\vartheta^{\alpha}\wedge*T_{\alpha}$, involving the Hodge dual of the torsion. On the material
mielke-geometrodynamics_F00499	111	■ 1.000	$L_D\rightarrow L_D+dU$	$L_D\rightarrow L_D+dU$	side, a related change of the Dirac Lagrangian can be induced via $L_D\rightarrow L_D+dU$,
mielke-geometrodynamics_F00500	111	■ 1.000	$U=\vartheta^{\alpha}\wedge\mu_{\alpha}$	$U=\vartheta^{\alpha}\wedge\mu_{\alpha}$	where $U=\vartheta^{\alpha}\wedge\mu_{\alpha}$ features as a superpotential. Then the corresponding boundary
mielke-geometrodynamics_F00501	111	■ 1.000	$\hat{\tau}_{\alpha\beta}$	$\hat{\tau}_{\alpha\beta}$	sions (R1) and (R2) of MIELKE et al. (1989), such that the relocalized spin $\hat{\tau}_{\alpha\beta}$
mielke-geometrodynamics_F00502	111	■ 1.000	$U=\vartheta^{\alpha}\wedge\vartheta_{\alpha}\wedge*j_5/4$	$U=\vartheta^{\alpha}\wedge\vartheta_{\alpha}\wedge*j_5/4$	vanishes. Observe that for Dirac spinors, $U=\vartheta^{\alpha}\wedge\vartheta_{\alpha}\wedge*j_5/4$ is trivial “on shell.”
mielke-geometrodynamics_F00503	111	■ 1.000	$\kappa=8\pi G_{\mathrm{N}}$	$\kappa=8\pi G_{\mathrm{N}}$	and the orthogonal tetrads, whereas $\kappa=8\pi G_{\mathrm{N}}$ is the gravitational constant in natural
mielke-geometrodynamics_F00504	112	■ 1.000	Σ_{α}	Σ_{α}	coupled to the canonical energy–momentum current Σ_{α} of matter, is obtained by
mielke-geometrodynamics_F00505	112	■ 1.000	L_{EC}	L_{EC}	varying the matter coupled Lagrangian L_{EC} with respect to the coframe ϑ^{α} .
mielke-geometrodynamics_F00506	112	■ 0.973	$\widehat{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+e_{\alpha}\rfloor T^{\beta}$	$\widehat{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+e_{\alpha}\rfloor T^{\beta}$	with respect to the transposed connection $\widehat{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+e_{\alpha}\rfloor T^{\beta}$. Observe that the
mielke-geometrodynamics_F00507	113	■ 1.000	$\sigma:=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}=\frac{i}{2}\gamma\wedge\gamma$	$\sigma:=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}=\frac{i}{2}\gamma\wedge\gamma$	involving the unit-curvature two-form $\sigma:=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}=\frac{i}{2}\gamma\wedge\gamma$, the equiva-
mielke-geometrodynamics_F00508	113	■ 1.000	$\Theta=0$	$\Theta=0$	for vanishing torsion, i.e., for $\Theta=0$ in Einstein’s standard GR.
mielke-geometrodynamics_F00509	113	■ 1.000	$\Gamma^{\alpha\beta}$	$\Gamma^{\alpha\beta}$	By varying (5.3.1) with respect to the linear connection $\Gamma^{\alpha\beta}$, we obtain the second

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mielke-geometrodynamics_F00510	113	0.839	$\tilde{\mathrm{T}}_{\mu\nu}$	$\tilde{\mathrm{T}}_{\mu\nu}$	$\tilde{\mathrm{T}}_{\mu\nu}$ contains additional terms depending quadratically on the spin angular momen-
mielke-geometrodynamics_F00511	113	1.000	$\tau_{\mathrm{s}}=\mathrm{o}$	$\tau_{\mathrm{s}} = \mathbf{0}$	tum of matter; cf. HEHL et al. (1976). In the instance that $\tau_{\mathrm{s}} = \mathbf{0}$, the EC equation
mielke-geometrodynamics_F00512	113	1.000	ℓ^{*4}	ℓ^{*4}	terms in (5.3.12) are proportional to ℓ^{*4} and could thus, with a modified Planck
mielke-geometrodynamics_F00513	114	1.000	$\ell^{*}=10^{-33} \mathrm{~cm}$	$\ell^{*} = 10^{-33} \text{ cm}$	length $\ell^{*} = 10^{-33} \text{ cm}$, play a role on the macroscopic scale only in cosmology; cf.,
mielke-geometrodynamics_F00514	114	1.000	10^{80}	10^{80}	be threatened by a final singularity, since the gravitational collapse of 10^{80} baryons
mielke-geometrodynamics_F00515	114	0.528	$R^{\{\backslash\lambda\alpha\backslash\beta\}}$	$R^{\{\lambda\alpha\beta\}}$	into the Riemannian curvature $R^{[\alpha\beta]}$ plus contortion pieces.
mielke-geometrodynamics_F00516	114	1.000	$\{ \}^{\{i\}} \mathrm{T}_{\alpha}$	$^{(i)}T_{\alpha}$	allelism as a consequence. Here $^{(i)}T_{\alpha}$ are the three irreducible pieces of the torsion.
mielke-geometrodynamics_F00517	114	1.000	$\mathrm{\mathscr{A}}$	$\mathscr{A}$	When only this axial torsion $\mathscr{A}$ enters algebraically, the EC Lagrangian
mielke-geometrodynamics_F00518	114	1.000	L_{HE}	L_{HE}	generalizes ¹ the metric Hilbert–Einstein Lagrangian L_{HE} to an RC spacetime with
mielke-geometrodynamics_F00519	114	0.647	$\mathrm{T}^{\{\alpha\}} \wedge \{ \}^{*} \left(\{ \}^{\{1\}} \mathrm{T}_{\alpha} - 2^{\{2\}} \mathrm{T}_{\alpha} - \frac{1}{2} \{ \}^{\{3\}} \mathrm{T}_{\alpha} \right)$	$T^{\alpha} \wedge * \left(^{(1)}T_{\alpha} - 2^{(2)}T_{\alpha} - \frac{1}{2} ^{(3)}T_{\alpha} \right)$	dure, since the particular combination $T^{\alpha} \wedge * \left(^{(1)}T_{\alpha} - 2^{(2)}T_{\alpha} - \frac{1}{2} ^{(3)}T_{\alpha} \right)$ of irreducible pieces is
mielke-geometrodynamics_F00520	114	0.558	$d \mathrm{C}_{\mathrm{TT}}^{\{ \}^{*}}$	dC_{TT}^{*}	the space of gravity theories, the term dC_{TT}^{*} interrelates GR with its teleparallelism equivalent
mielke-geometrodynamics_F00521	115	1.000	$G_{\alpha}^{\{\backslash\backslash\}}=G_{\alpha}\{ \}^{\{\beta\}} \eta_{\beta}$	$G_{\alpha}^{\{\}} = G_{\alpha}^{\beta} \eta_{\beta}$	can be decomposed into the Einstein three-form $G_{\alpha}^{\{\}} = G_{\alpha}^{\beta} \eta_{\beta}$ with respect to the
mielke-geometrodynamics_F00522	115	1.000	$\ell=\sqrt{\kappa}$	$\ell = \sqrt{\kappa}$	coupled via the “bare” fundamental length $\ell = \sqrt{\kappa}$. Then “on shell,” EC theory
mielke-geometrodynamics_F00523	115	1.000	$k \rightarrow \infty$	$k \rightarrow \infty$	bounded in the ultraviolet limit $k \rightarrow \infty$. Quite generally, the renormalization flow is
mielke-geometrodynamics_F00524	115	1.000	Γ_k	Γ_k	for the effective action Γ_k , i.e.,
mielke-geometrodynamics_F00525	116	0.999	$k \partial_k$	$k \partial_k$	where $k \partial_k$ is a scale derivative returning the Euler degree of homogeneous functionals.
mielke-geometrodynamics_F00526	116	1.000	k	k	where k is the renormalization group (RG) scale in momentum space and Λ the
mielke-geometrodynamics_F00527	116	1.000	ρ_{Λ}	ρ_{Λ}	cosmological constant related to dark energy (DE) of density ρ_{Λ} ; see also BJORKEN
mielke-geometrodynamics_F00528	116	1.000	d_{N}	d_{N}	where d_{N} is the anomalous dimension of the running Newton coupling g_{N} .
mielke-geometrodynamics_F00529	116	1.000	g_{N}	g_{N}	where d_{N} is the anomalous dimension of the running Newton coupling g_{N} .
mielke-geometrodynamics_F00530	116	1.000	g_{N}^{*}	$g_{\mathrm{N}*}$	stants (5.4.3) run into some nontrivial fixed points $g_{\mathrm{N}*}$ and λ_{*} , depending on the
mielke-geometrodynamics_F00531	116	1.000	λ_{*}	λ_{*}	stants (5.4.3) run into some nontrivial fixed points $g_{\mathrm{N}*}$ and λ_{*} , depending on the
mielke-geometrodynamics_F00532	116	1.000	$R^{\{n\}}$	R^n	high-order polynomials R^n of the Ricci scalar, similarly as in the classically bifur-
mielke-geometrodynamics_F00533	116	1.000	$f(R)$	$f(R)$	cating $f(R)$ models (SCHUNCK et al. 2005), but then the issue of physical ghosts or
mielke-geometrodynamics_F00534	117	1.000	κ	κ	also scales with the gravitational constant κ , but inversely compared to the Hilbert–
mielke-geometrodynamics_F00535	117	1.000	$1 / k^4$	$1/k^4$	bilinears is mandatory, including the $1/k^4$ dependence in the functional integral.
mielke-geometrodynamics_F00536	117	1.000	f_{*}^2	f_{*}^2	and the corresponding dimensionless renormalized running coupling f_{*}^2 becomes

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mielke-geometrodynamics_F00537	117	■ 1.000	$R_{\alpha\beta} \equiv 0$	$R_{\alpha\beta} \equiv 0$	One way to avoid such anomalies is to employ curvature constraints like $R_{\alpha\beta} \equiv 0$,
mielke-geometrodynamics_F00538	117	■ 1.000	$SL(5, \mathbb{R})$	$SL(5, \mathbb{R})$	from a “minimalist” $SL(5, \mathbb{R})$ gauge model that includes only a “bare” Pontryagin-
mielke-geometrodynamics_F00539	118	■ 1.000	$\Delta c / c \simeq 10^{-17}$	$\Delta c / c \simeq 10^{-17}$	Lorentz-invariant, nowadays measured at the unprecedented $\Delta c / c \simeq 10^{-17}$ level
mielke-geometrodynamics_F00540	118	■ 1.000	10^{-13}	10^{-13}	et al. 2012) at the 10^{-13} level. Moreover, the upper bound for a violation of the
mielke-geometrodynamics_F00541	118	■ 1.000	10^{-8}	10^{-8}	quarks, is about 10^{-8} ; cf. HUGHES (1993).
mielke-geometrodynamics_F00542	118	■ 0.979	$\bar{\nu}_e$	$\bar{\nu}_e$	Assuming that not only the dominant delayed antineutrinos $\bar{\nu}_e$ but at least one prompt
mielke-geometrodynamics_F00543	118	■ 0.813	ν_e	ν_e	neutrino ν_e was in the pulse, one could infer that the equivalence principle is valid
mielke-geometrodynamics_F00544	118	■ 1.000	10^{-6}	10^{-6}	(GUZZO et al. 2002) for electron–neutrinos ν_e up to 10^{-6} .
mielke-geometrodynamics_F00545	118	■ 0.996	CPT	CPT	Moreover, a test of the CPT theorem for the neutral kaon system indirectly
mielke-geometrodynamics_F00546	118	■ 1.000	K^0	K^0	yielded an upper bound for the relative difference of the <i>inertial masses</i> of the K^0
mielke-geometrodynamics_F00547	118	■ 1.000	$\bar{K}^0$	$\bar{K}^0$	meson and its antiparticle $\bar{K}^0$ of less than 10^{-18} , when employing (MURAYAMA
mielke-geometrodynamics_F00548	118	■ 1.000	10^{-18}	10^{-18}	meson and its antiparticle $\bar{K}^0$ of less than 10^{-18} , when employing (MURAYAMA
mielke-geometrodynamics_F00549	120	■ 0.761	$(5, \mathbb{R})$	$(5, \mathbb{R})$	Mielke EW (2012) Einstein-Weyl gravity from a topological gauge $SL(5, \mathbb{R})$ invariant action. Adv
mielke-geometrodynamics_F00550	122	■ 1.000	(1986, 1991)	(1986, 1991)	On the other hand, Ashtekar’s reformulation ASHTEKAR (1986, 1991) of general
mielke-geometrodynamics_F00551	122	■ 0.996	dC_{TT}	dC_{TT}	ary term dC_{TT} constructed from a <i>translational Chern–Simons term</i> (CS) multiplied
mielke-geometrodynamics_F00552	122	■ 0.987	$(\mathrm{GR}_{\parallel})$	$(\mathrm{GR}_{\parallel})$	as was already suggested by EINSTEIN (1928), a theory with <i>teleparallelism</i> ($\mathrm{GR}_{\parallel}$).
mielke-geometrodynamics_F00553	123	■ 1.000	$\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \text{ cm}$	$\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \text{ cm}$	scale $\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \text{ cm}$, a topologically rich spectrum of dislocations
mielke-geometrodynamics_F00554	123	■ 1.000	q	q	In the one-dimensional harmonic oscillator model (MIELKE 1998) with q as gen-
mielke-geometrodynamics_F00555	123	■ 0.994	$\mathcal{C} = q^2/2$	$\mathcal{C} = q^2/2$	term derived from the <i>Chern–Simons</i> (CS)-type term $\mathcal{C} = q^2/2$ as <i>generating func-</i>
mielke-geometrodynamics_F00556	123	■ 0.999	$\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$	$\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$	<i>tion</i> . After quantization, the corresponding operator $\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$
mielke-geometrodynamics_F00557	123	■ 0.999	$\Psi = N \exp(i\theta \int \mathcal{C})$	$\Psi = N \exp\left(i\theta \int \mathcal{C}\right)$	shown that $\Psi = N \exp(i\theta \int \mathcal{C})$ is, up to an overall factor, the unique solution of the
mielke-geometrodynamics_F00558	123	■ 1.000	$P := \mathbb{R}^4 \ltimes SO(1, 3)$	$P := \mathbb{R}^4 \ltimes SO(1, 3)$	Since the Poincaré group $P := \mathbb{R}^4 \ltimes SO(1, 3)$ is the semidirect product of trans-
mielke-geometrodynamics_F00559	124	■ 1.000	$\mathcal{A} := *(\vartheta_{\alpha} \wedge T^{\alpha}) =$	$\mathcal{A} := *(\vartheta_{\alpha} \wedge T^{\alpha}) =$	They are <i>translational</i> and Lorentz-rotational CS terms, where $\mathcal{A} := *(\vartheta_{\alpha} \wedge T^{\alpha}) =$
mielke-geometrodynamics_F00560	124	■ 1.000	$\mathcal{A}_i dx^i$	$\mathcal{A}_i dx^i$	$\mathcal{A}_i dx^i$ is the <i>axial torsion</i> one-form. In the first <i>translational</i> CS three-form, there
mielke-geometrodynamics_F00561	124	■ 1.000	ℓ	ℓ	necessarily occurs a fundamental length ℓ for dimensional reasons.
mielke-geometrodynamics_F00562	124	■ 0.907	$SO(1, 4)$	$SO(1, 4)$	the de Sitter group $SO(1, 4)$ or $SO(2, 3)$ as a subgroup, both CS three-forms are
mielke-geometrodynamics_F00563	124	■ 0.907	$SO(2, 3)$	$SO(2, 3)$	the de Sitter group $SO(1, 4)$ or $SO(2, 3)$ as a subgroup, both CS three-forms are

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mielke-geometrodynamics_F00564	124	■ 1.000	$g_{\{\alpha \beta\}}$	$g_{\alpha\beta}$	free Pontryagin form, in the NY term (6.2.4), a metric $g_{\alpha\beta}$ is needed to raise and
mielke-geometrodynamics_F00565	124	■ 1.000	$T_{\{\alpha\}}=g_{\{\alpha \beta\}} T^{\{\beta\}}$	$T_{\alpha} = g_{\alpha\beta} T^{\beta}$	lower the indices, for instance in $T_{\alpha} = g_{\alpha\beta} T^{\beta}$.
mielke-geometrodynamics_F00566	124	■ 1.000	$\vartheta_{\{\alpha\}}$	ϑ_{α}	ϑ_{α} , on the first Bianchi identity
mielke-geometrodynamics_F00567	125	■ 1.000	$R_{\{\alpha \beta\}}^{\{\ \ \ \}}$	$R_{\alpha\beta}^{\{\ \ \}}$	of general relativity (GR), where $R_{\alpha\beta}^{\{\ \ \}}$ denotes the Riemannian curvature with respect
mielke-geometrodynamics_F00568	125	■ 1.000	$\Gamma_{\{\alpha \beta\}}^{\{\ \ \}}$	$\Gamma_{\alpha\beta}^{\{\ \}}$	to the Levi-Civita connection $\Gamma_{\alpha\beta}^{\{\ \}}$.
mielke-geometrodynamics_F00569	125	■ 1.000	$K_{\{\alpha \beta\}}$	$K_{\alpha\beta}$	is dual to the contortion one-form $K_{\alpha\beta}$, which features in the decomposition $\Gamma_{\alpha\beta} =$
mielke-geometrodynamics_F00570	125	■ 1.000	$\Gamma_{\{\alpha \beta\}}=$	$\Gamma_{\alpha\beta} =$	is dual to the contortion one-form $K_{\alpha\beta}$, which features in the decomposition $\Gamma_{\alpha\beta} =$
mielke-geometrodynamics_F00571	125	■ 0.985	$\left.\left.\left.\left.\Gamma_{\{\beta \alpha\}}=\Gamma_{\{\alpha \beta\}}-K_{\{\alpha \beta\}}=\Gamma_{\{\alpha \beta\}}^{\{\ \ \}}+e_{\alpha\rfloor}T_{\beta}+(e_{\alpha\rfloor}e_{\beta\rfloor}T_{\gamma}\right)\wedge\vartheta^{\gamma}\right.\right.\right.$	$-\Gamma_{\beta\alpha} = \Gamma_{\alpha\beta}^{\{\ \ \}} - K_{\alpha\beta} = \Gamma_{\alpha\beta}^{\{\ \ \}} + e_{\alpha\rfloor} T_{\beta} + (e_{\alpha\rfloor} e_{\beta\rfloor} T_{\gamma}) \wedge \vartheta^{\gamma}$	$-\Gamma_{\beta\alpha} = \Gamma_{\alpha\beta}^{\{\ \ \}} - K_{\alpha\beta} = \Gamma_{\alpha\beta}^{\{\ \ \}} + e_{\alpha\rfloor} T_{\beta} + (e_{\alpha\rfloor} e_{\beta\rfloor} T_{\gamma}) \wedge \vartheta^{\gamma}$ of the RC connection.
mielke-geometrodynamics_F00572	125	■ 0.971	$C_{\mathrm{TT}^*}:=\vartheta^{\alpha}\wedge{}^*T_{\alpha}/2\ell^2$	$C_{\mathrm{TT}^*} := \vartheta^{\alpha} \wedge {}^*T_{\alpha}/2\ell^2$	where $C_{\mathrm{TT}^*} := \vartheta^{\alpha} \wedge {}^*T_{\alpha}/2\ell^2$ is a dual CS term, proper <i>teleparallelism</i> (GR) spec-
mielke-geometrodynamics_F00573	125	■ 1.000	$R^{\{\alpha \beta\}}=0$	$R^{\alpha\beta} = 0$	the proper constraint $R^{\alpha\beta} = 0$ has to be imposed by subtracting $R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$ from
mielke-geometrodynamics_F00574	125	■ 1.000	$R^{\{\alpha \beta\}}\wedge\lambda_{\{\alpha \beta\}}$	$R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$	the proper constraint $R^{\alpha\beta} = 0$ has to be imposed by subtracting $R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$ from
mielke-geometrodynamics_F00575	125	■ 1.000	$\lambda_{\{\alpha \beta\}}=-\lambda_{\{\beta \alpha\}}$	$\lambda_{\alpha\beta} = -\lambda_{\beta\alpha}$	(6.2.9), where the 2-form $\lambda_{\alpha\beta} = -\lambda_{\beta\alpha}$ is a Lagrange multiplier, see HEHL et al.
mielke-geometrodynamics_F00576	126	■ 1.000	$\vartheta^{\{\alpha\}}, \Gamma^{\{\alpha \beta\}}$	$\vartheta^{\alpha}, \Gamma^{\alpha\beta}$	By varying this Lagrangian independently with respect to ϑ^{α} , $\Gamma^{\alpha\beta}$, and the mul-
mielke-geometrodynamics_F00577	126	■ 1.000	$\lambda_{\{\alpha \beta\}}$	$\lambda_{\alpha\beta}$	tiplier $\lambda_{\alpha\beta}$, we obtain as field equations
mielke-geometrodynamics_F00578	126	■ 0.856	$L_{\{\ \ \}}$	$L_{ }$	first field Eq. (6.2.13) is the same as that of the unconstrained Lagrangian $L_{ }$. The
mielke-geometrodynamics_F00579	126	■ 1.000	$\lambda_{\{\alpha \beta\}}=\left(\gamma/\ell^2\right)\eta_{\{\alpha \beta\}}$	$\lambda_{\alpha\beta} = (\gamma/\ell^2) \eta_{\alpha\beta}$	<i>spinless</i> matter, a possible solution is $\lambda_{\alpha\beta} = (\gamma/\ell^2) \eta_{\alpha\beta}$, as will be shown below.
mielke-geometrodynamics_F00580	126	■ 0.982	$\boldsymbol{G R}_{\{\ \ \}}$	$\boldsymbol{GR}_{ }$	6.2.2 Yang–Mills-Type Formulation of Complex GR
mielke-geometrodynamics_F00581	127	■ 1.000	$\sigma=\pm\mathrm{i} \gamma$	$\sigma = \pm i \gamma$	of BAEKLER et al. (1982); if $\sigma = \pm i \gamma$ is chosen for one of the free parameters there.
mielke-geometrodynamics_F00582	127	■ 1.000	$\theta_{\mathrm{T}}=\pm\mathrm{i}$	$\theta_{\mathrm{T}} = \pm i$	version (6.2.9) of Einstein’s GR for the choice $\theta_{\mathrm{T}} = \pm i$ into a <i>chiral gauge theory</i>
mielke-geometrodynamics_F00583	127	■ 0.984	$\stackrel{\pm}{T}^{\alpha}:=\frac{1}{2}\left(T^{\alpha} \pm \mathrm{i} * T^{\alpha}\right)$	$\stackrel{(\pm)}{T}^{\alpha} := \frac{1}{2} (T^{\alpha} \pm i {}^*T^{\alpha})$	where $\stackrel{(\pm)}{T}^{\alpha} := \frac{1}{2} (T^{\alpha} \pm i {}^*T^{\alpha})$ denotes the self-dual or anti-self-dual torsion. The
mielke-geometrodynamics_F00584	128	■ 1.000	$\lambda_{\{\alpha \beta\}}^{(\tau)}$	$^{(\tau)}\lambda_{\alpha\beta}$	et al. 1989) an additional piece $^{(\tau)}\lambda_{\alpha\beta}$ in the ansatz for the Lagrangian multiplier,
mielke-geometrodynamics_F00585	128	■ 1.000	$D^{(\tau)} \lambda_{\{\alpha \beta\}}=\tau_{\{\alpha \beta\}}$	$D^{(\tau)} \lambda_{\alpha\beta} = \tau_{\alpha\beta}$	provided it satisfies $D^{(\tau)} \lambda_{\alpha\beta} = \tau_{\alpha\beta}$.
mielke-geometrodynamics_F00586	128	■ 0.584	$\stackrel{\pm}{\Pi}^{\alpha}$	$\stackrel{(\pm)}{\Pi}^{\alpha}$	The complex translational field momenta $\stackrel{(\pm)}{\Pi}^{\alpha}$ replace the role of the torsion if we
mielke-geometrodynamics_F00587	128	■ 0.731	$\delta L_{ } / \delta \vartheta^{\alpha}$	$\delta \stackrel{(\pm)}{L}_{ } / \delta \vartheta^{\alpha}$	allow us to rederive the field Eq. (6.2.24) via $\delta \stackrel{(\pm)}{L}_{ } / \delta \vartheta^{\alpha}$ from a variational principle.

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mielke-geometrodynamics_F00588	128	■ 1.000	$R_{\alpha\beta}=0$	$R_{\alpha\beta} = 0$	Cartan spacetime with $R_{\alpha\beta} = 0$ and in view of the first Noether identity (6.8.11),
mielke-geometrodynamics_F00589	128	■ 0.999	$\stackrel{\scriptscriptstyle (\pm)}{\Pi}_{\alpha}$	$\stackrel{\scriptscriptstyle (\pm)}{\Pi}_{\alpha}$	terms of the momenta $\stackrel{\scriptscriptstyle (\pm)}{\Pi}_{\alpha}$, it would be convenient to have a <i>relocalized</i> energy–momentum current
mielke-geometrodynamics_F00590	128	■ 0.121	$\stackrel{\scriptscriptstyle (\pm)}{\Sigma}_{\alpha}$	$\stackrel{\scriptscriptstyle (\pm)}{\Sigma}_{\alpha}$	$\stackrel{\scriptscriptstyle (\pm)}{\Sigma}_{\alpha}$ for which $\stackrel{\scriptscriptstyle (\pm)}{D}\stackrel{\scriptscriptstyle (\pm)}{\Sigma}_{\alpha} \cong 0$ holds.
mielke-geometrodynamics_F00591	128	■ 0.121	$\stackrel{\scriptscriptstyle (\pm)}{D}$	$\stackrel{\scriptscriptstyle (\pm)}{D}$	$\stackrel{\scriptscriptstyle (\pm)}{\Sigma}_{\alpha}$ for which $\stackrel{\scriptscriptstyle (\pm)}{D}\stackrel{\scriptscriptstyle (\pm)}{\Sigma}_{\alpha} \cong 0$ holds.
mielke-geometrodynamics_F00592	129	■ 0.821	$\stackrel{\scriptscriptstyle (\mp)}{A}_{\alpha}{}^{\beta}$	$\stackrel{\scriptscriptstyle (\mp)}{A}_{\alpha}{}^{\beta}$	valued 1-form $\stackrel{\scriptscriptstyle (\mp)}{A}_{\alpha}{}^{\beta}$ deforms our original Riemann–Cartan connection $\Gamma_{\alpha}{}^{\beta}$ similarly
mielke-geometrodynamics_F00593	129	■ 0.946	$\underline{\Gamma}^{\alpha\beta}_{\parallel} \rightarrow 0$	$\underline{\Gamma}^{\alpha\beta}_{\parallel} \rightarrow 0$	(1973), THIRRING (1978), HEHL et al. (1980). In the teleparallelism limit $\underline{\Gamma}^{\alpha\beta}_{\parallel} \rightarrow 0$,
mielke-geometrodynamics_F00594	129	■ 1.000	H_{α}	H_{α}	the “superpotential” H_{α} is the <i>Møller</i> or <i>Freud complex</i> (FREUD 1939)
mielke-geometrodynamics_F00595	129	■ 0.700	$\overparen{D}\eta_{\alpha} := D\eta_{\alpha} -$	$\widehat{D}\eta_{\alpha} := D\eta_{\alpha} -$	³ The <i>transposed</i> connection, which has the property (MIELKE et al. 1989) that $\widehat{D}\eta_{\alpha} := D\eta_{\alpha} -$
mielke-geometrodynamics_F00596	129	■ 0.521	$\left(e_{\alpha} \downarrow T^{\beta}\right) \wedge \eta_{\beta} \equiv 0$	$\left(e_{\alpha} \downarrow T^{\beta}\right) \wedge \eta_{\beta} \equiv 0$	$\left(e_{\alpha} \downarrow T^{\beta}\right) \wedge \eta_{\beta} \equiv 0$, may be regarded as a special real version of our Sen-type connection, for
mielke-geometrodynamics_F00597	129	■ 0.593	$\stackrel{\scriptscriptstyle (\pm)}{D}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \stackrel{\scriptscriptstyle (\pm)}{\Pi}{}^{\beta}$	$\stackrel{\scriptscriptstyle (\pm)}{D}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \stackrel{\scriptscriptstyle (\pm)}{\Pi}{}^{\beta}$	which $\stackrel{\scriptscriptstyle (\pm)}{D}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \stackrel{\scriptscriptstyle (\pm)}{\Pi}{}^{\beta}$ holds.
mielke-geometrodynamics_F00598	130	■ 1.000	E_{α}	E_{α}	et al. (2000) that E_{α} is a covariant conserved gravitational energy, i.e., that $DE_{\alpha} =$
mielke-geometrodynamics_F00599	130	■ 1.000	$D E_{\alpha} =$	$DE_{\alpha} =$	et al. (2000) that E_{α} is a covariant conserved gravitational energy, i.e., that $DE_{\alpha} =$
mielke-geometrodynamics_F00600	130	■ 1.000	$DDH_{\alpha} \equiv -R_{\alpha}{}^{\beta} \wedge H_{\beta} = 0$	$DDH_{\alpha} \equiv -R_{\alpha}{}^{\beta} \wedge H_{\beta} = 0$	$DDH_{\alpha} \equiv -R_{\alpha}{}^{\beta} \wedge H_{\beta} = 0$ due to the vanishing of the Ricci identity in a teleparallel
mielke-geometrodynamics_F00601	130	■ 0.984	$\mathbf{GR}_{\parallel}$	GR _∥	6.4 Hamiltonian Formulation of Complex GR_∥
mielke-geometrodynamics_F00602	130	■ 1.000	Σ_t	Σ_t	into a family of spacelike three-dimensional hypersurfaces Σ_t that are parametrized
mielke-geometrodynamics_F00603	130	■ 1.000	$x^0=t$	$x^0 = t$	by the coordinate time $x^0 = t$. Then there exists the future-directed <i>timelike</i> vector
mielke-geometrodynamics_F00604	130	■ 0.496	$u := -N^2 dt, \Longleftarrow$	$u := -N^2 dt, \Longleftrightarrow$	which is uniquely linked to the hypersurface orthogonal 1-form $u := -N^2 dt, \Longleftrightarrow$
mielke-geometrodynamics_F00605	130	■ 0.963	$du \wedge u = 0$	$du \wedge u = 0$	$du \wedge u = 0$ via $u = g(n, \cdot)$. The lapse function N and the shift vector N^A , which
mielke-geometrodynamics_F00606	130	■ 0.963	$u=g(n,$	$u = g(n,$	$du \wedge u = 0$ via $u = g(n, \cdot)$. The lapse function N and the shift vector N^A , which
mielke-geometrodynamics_F00607	130	■ 0.963	$)$. The lapse function N	$)$.The lapse function N	$du \wedge u = 0$ via $u = g(n, \cdot)$. The lapse function N and the shift vector N^A , which
mielke-geometrodynamics_F00608	130	■ 0.963	N^A	N^A	$du \wedge u = 0$ via $u = g(n, \cdot)$. The lapse function N and the shift vector N^A , which
mielke-geometrodynamics_F00609	130	■ 0.974	p	p	The <i>normal</i> part of a p -form Ψ is defined by
mielke-geometrodynamics_F00610	130	■ 0.997	$(3+1)$	$(3 + 1)$	operators. (For more details on this very concise (3+1) decomposition of exterior
mielke-geometrodynamics_F00611	131	■ 1.000	$L=L(\vartheta^{\alpha}, \Psi, D\Psi)$	$L = L(\vartheta^{\alpha}, \Psi, D\Psi)$	Let us consider a Lagrangian $L = L(\vartheta^{\alpha}, \Psi, D\Psi)$ which is of first differential
mielke-geometrodynamics_F00612	131	■ 0.973	$\left.\mathscr{K}=L_{\perp}:=n\right\rfloor L$	$\mathscr{K} = L_{\perp} := n\rfloor L$	where $\mathscr{K} = L_{\perp} := n\rfloor L$ denotes the normal part of the Lagrangian. Since the “time
mielke-geometrodynamics_F00613	131	■ 0.677	$\bullet$	$\bullet$	derivative” $\bullet$ is more precisely given by the Lie derivative $\ell_n := dn\rfloor + n\rfloor d$ along

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mielke-geometrodynamics_F00614	131	0.677	$\left.\ell_n:=d\right\rfloor+n\downarrow d$	$\ell_n := dn\rfloor + n\downarrow d$	derivative” • is more precisely given by the Lie derivative $\ell_n := dn\rfloor + n\rfloor d$ along
mielke-geometrodynamics_F00615	131	1.000	n	n	the timelike vector field n , we have
mielke-geometrodynamics_F00616	131	0.965	$\left.n^{\alpha}:=n\right\rfloor\vartheta^{\alpha}$	$n^{\alpha} := n\rfloor\vartheta^{\alpha}$	and transvect the resulting expression with the four-vector $n^{\alpha} := n\rfloor\vartheta^{\alpha}$ of <i>lapse and</i>
mielke-geometrodynamics_F00617	131	0.605	$\left.n^{\alpha}e_{\alpha}\right\rfloor\Psi=n\downarrow\Psi$	$n^{\alpha}e_{\alpha}\rfloor\Psi = n\downarrow\Psi$	<i>shift</i> , for which the useful relation $n^{\alpha}e_{\alpha}\rfloor\Psi = n\rfloor\Psi$ can be derived. For the projection
mielke-geometrodynamics_F00618	132	0.919	$\mathrm{L}_n=D\,n\downarrow+n\downarrow D$	$L_n = Dn\downarrow + n\downarrow D$	In introducing the gauge-covariant Lie derivative $L_n = Dn\rfloor + n\rfloor D$, separating
mielke-geometrodynamics_F00619	132	0.651	$n\rfloor\Psi=n\downarrow(dt\wedge n\rfloor\Psi)=n\rfloor\Psi$	$n\rfloor\Psi = n\downarrow(dt\wedge n\rfloor\Psi) = n\rfloor\Psi$	sion is related to the Hamiltonian (6.4.5). Indeed, using $n\rfloor\Psi = n\rfloor(dt\wedge n\rfloor\Psi) = n\rfloor\Psi$
mielke-geometrodynamics_F00620	132	0.479	$\left.\Gamma_{\perp}^{\alpha\beta}:=n\right\rfloor\Gamma^{\alpha\beta}$	$\Gamma_{\perp}^{\alpha\beta} := n\rfloor\Gamma^{\alpha\beta}$	where the normal vector $n^{\alpha} := n\rfloor\vartheta^{\alpha}$ comprises <i>lapse and shift</i> and $\Gamma_{\perp}^{\alpha\beta} := n\rfloor\Gamma^{\alpha\beta}$
mielke-geometrodynamics_F00621	132	1.000	$\mathscr{G}_{\alpha}$	$\mathscr{G}_{\alpha}$	is the normal part of the Lorentz connection and $\mathscr{G}_{\alpha}$ and $\mathscr{G}_{\alpha\beta}$ represent the tangential
mielke-geometrodynamics_F00622	132	1.000	$\mathscr{G}_{\alpha\beta}$	$\mathscr{G}_{\alpha\beta}$	is the normal part of the Lorentz connection and $\mathscr{G}_{\alpha}$ and $\mathscr{G}_{\alpha\beta}$ represent the tangential
mielke-geometrodynamics_F00623	133	0.709	$D_{\alpha}:=e_{\alpha}\rfloor D$	$D_{\alpha} := e_{\alpha}\rfloor D$	where $D_{\alpha} := e_{\alpha}\rfloor D$ are the anholonomic components of the exterior covariant
mielke-geometrodynamics_F00624	133	0.927	$D\lambda_{\alpha\beta}$	$D\lambda_{\alpha\beta}$	arise the extra tangential term $\underline{D}\lambda_{\alpha\beta}$ in the <i>Lorentz</i> generator $\mathscr{G}_{\alpha\beta}$, where
mielke-geometrodynamics_F00625	133	1.000	$\underline{\Psi}:=\Psi-dt\wedge n\rfloor\Psi$	$\underline{\Psi} := \Psi - dt\wedge n\rfloor\Psi$	$\underline{\Psi} := \Psi - dt\wedge n\rfloor\Psi$ in the notation of differential forms. However, as suggested
mielke-geometrodynamics_F00626	133	0.893	$\underline{\lambda}_{\alpha\beta}=(1/2\ell^2)*(\vartheta_{\alpha}\wedge\vartheta_{\beta})=-\left(1/N\ell^2\right)n_{[\alpha}*\vartheta_{\beta]}$	$\lambda_{\alpha\beta} = (1/2\ell^2) * (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = - (1/N\ell^2) n_{[\alpha} *\vartheta_{\beta]}$	by $\lambda_{\alpha\beta} = (1/2\ell^2) * (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = -(1/N\ell^2) n_{[\alpha} *\vartheta_{\beta]}$. This is another manifestation of
mielke-geometrodynamics_F00627	133	0.925	$d\stackrel{(\mp)}{C}_{\mathrm{TT}}$	$d\stackrel{(\mp)}{C}_{\mathrm{TT}}$	by the boundary term $d\stackrel{(\mp)}{C}_{\mathrm{TT}}$.
mielke-geometrodynamics_F00628	133	0.989	$(\alpha=\hat{0})$	$(\alpha = \hat{0})$	After this trivialization of the Lorentz constraints, the Hamiltonian $(\alpha = \hat{0})$ and
mielke-geometrodynamics_F00629	133	1.000	$(\alpha=B)$	$(\alpha = B)$	diffeomorphism constraints $(\alpha = B)$ of complexified $\mathrm{GR}_{\parallel}$, in terms of a complex
mielke-geometrodynamics_F00630	133	0.464	$\stackrel{(\pm)}{\mathscr{A}}_{\alpha}:=\stackrel{*}{\Pi}_{\alpha}$	$\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$	(1994). Here the tangential Ashtekar-type connection $\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$ is three-dual
mielke-geometrodynamics_F00631	133	0.130	$\stackrel{*}{-}_{\alpha}$	$\stackrel{*}{-}_{\alpha}$	ties $\stackrel{*}{\vartheta}_{\alpha}$, where we note that the spatial Hodge dual is involutive, i.e., $\stackrel{*}{*} = +1$.
mielke-geometrodynamics_F00632	133	0.130	$\stackrel{*}{*}=+1$	$\stackrel{*}{*} = +1$	ties $\stackrel{*}{\vartheta}_{\alpha}$, where we note that the spatial Hodge dual is involutive, i.e., $\stackrel{*}{*} = +1$.
mielke-geometrodynamics_F00633	133	1.000	$\underline{\vartheta}^{\hat{0}}=0$	$\underline{\vartheta}^{\hat{0}} = 0$	Moreover, by adopting tetrads in the temporal gauge $\underline{\vartheta}^{\hat{0}} = 0$ of SCHWINGER (1963)
mielke-geometrodynamics_F00634	133	0.858	$\stackrel{(\pm)}{\mathscr{A}}_{\hat{0}}=d^{(\pm)}\phi$	$\stackrel{(\pm)}{\mathscr{A}}_{\hat{0}} = d^{(\pm)}\phi$	and the gauge $\stackrel{(\pm)}{\mathscr{A}}_{\hat{0}} = d^{(\pm)}\phi$ of NESTER (1989), the Hamiltonian constraint
mielke-geometrodynamics_F00635	134	1.000	$\alpha=\hat{0}$	$\alpha = \hat{0}$	i.e., (6.4.19) for $\alpha = \hat{0}$, vanishes identically.
mielke-geometrodynamics_F00636	134	0.561	$\stackrel{*}{-}\vartheta_{\alpha}$	$\stackrel{*}{-}\vartheta_{\alpha}$	parts of the basic one-forms, i.e., the <i>triad densities</i> $\stackrel{*}{\vartheta}_{\alpha}$ and the tangential part of
mielke-geometrodynamics_F00637	134	0.677	$\stackrel{(\pm)}{\mathscr{A}}_{\alpha}:=\stackrel{*}{\Pi}_{\alpha}$	$\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$	the self-dual or anti-self-dual connection $\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$, the three-dual of the chiral
mielke-geometrodynamics_F00638	134	0.995	$\stackrel{\pm}{p}$	$\stackrel{\pm}{p}$	momenta $\stackrel{(\pm)}{\Pi}_{\alpha}$, become the generalized coordinates q and momenta $\stackrel{\pm}{p}$ of the bosonic

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mielke-geometrodynamics_F00639	134	■ 0.337	$\stackrel{*}{\stackrel{\pm}{\Pi}}_{\alpha}$	$\stackrel{*}{\stackrel{\pm}{\Pi}}_{\alpha}$	prevails (MIELKE 2002), where the complex field “momenta” $\stackrel{*}{\stackrel{\pm}{\Pi}}_{\alpha}$ become differ-
mielke-geometrodynamics_F00640	134	■ 0.316	$\underline{*}^{\vartheta^{\beta}}$	$\underline{*}^{\vartheta^{\beta}}$	ential operators, whereas the triad densities $\underline{*}\vartheta^{\beta}$ remain generalized coordinates q .
mielke-geometrodynamics_F00641	134	■ 1.000	$\alpha=B$	$\alpha = B$	For the remaining Gauss constraint (6.4.8) with $\alpha = B$, let us try the state vector
mielke-geometrodynamics_F00642	134	■ 0.731	$\underline{C}_{\mathrm{TT}}$	$\underline{C}_{\mathrm{TT}}$	Chern–Simons term $\underline{\underline{C}}_{\mathrm{TT}}$ in terms of the triads or triad densities and the tangential
mielke-geometrodynamics_F00643	134	■ 0.862	$\underline{*}^{\underline{\vartheta}^B}$	$\underline{*}^{\underline{\vartheta}^B}$	rewritten in terms of the triad densities $\underline{*}\vartheta^B$ with $B = 1, 2, 3$ and the tangential part
mielke-geometrodynamics_F00644	134	■ 0.862	$B=1,2,3$	$B = 1, 2, 3$	rewritten in terms of the triad densities $\underline{*}\vartheta^B$ with $B = 1, 2, 3$ and the tangential part
mielke-geometrodynamics_F00645	134	■ 0.637	$\stackrel{(\pm)}{\underline{T}}^B$	$\stackrel{(\pm)}{\underline{T}}^B$	momentum operator (6.4.22) returns the chiral torsion $\stackrel{(\pm)}{\underline{T}}^B$ as a prefactor:
mielke-geometrodynamics_F00646	134	■ 1.000	$\underline{\Gamma}^{\alpha\beta}_{\parallel} \stackrel{*}{=} 0$	$\underline{\Gamma}^{\alpha\beta}_{\parallel} \stackrel{*}{=} 0$	Since the corresponding connection $\underline{\Gamma}^{\alpha\beta}_{\parallel} \stackrel{*}{=} 0$ is of pure gauge type, the state vector
mielke-geometrodynamics_F00647	135	■ 0.999	$\Psi \wedge \stackrel{*}{\Phi} = \Phi \wedge \stackrel{*}{\Psi}$	$\Psi \wedge \stackrel{*}{\Phi} = \Phi \wedge \stackrel{*}{\Psi}$	where the rule $\Psi \wedge \stackrel{*}{\Phi} = \Phi \wedge \stackrel{*}{\Psi}$ for differential forms of the same degree has
mielke-geometrodynamics_F00648	135	■ 0.994	$SU(2)$	$SU(2)$	borrowed from $SU(2)$ Chern–Simons field theory (WITTEN 1989; GUADAGNINI
mielke-geometrodynamics_F00649	135	■ 1.000	$\mathcal{H}_A \Psi_A(A)=0$	$\mathcal{H}_A \Psi_A(A) = 0$	straint $\mathcal{H}_A \Psi_A(A) = 0$ of GR with cosmological constant in the nonperturbative
mielke-geometrodynamics_F00650	135	■ 1.000	$\ell \rightarrow 0$	$\ell \rightarrow 0$	ℓ , it is dimensionless and nonsingular even for $\ell \rightarrow 0$, due to $\dim \vartheta^B = [\ell]$.
mielke-geometrodynamics_F00651	135	■ 1.000	$\dim \vartheta^B = [\ell]$	$\dim \vartheta^B = [\ell]$	ℓ , it is dimensionless and nonsingular even for $\ell \rightarrow 0$, due to $\dim \vartheta^B = [\ell]$.
mielke-geometrodynamics_F00652	135	■ 0.937	$\underline{T}^{(\mp)B} = 0$	$\underline{T}^{(\mp)B} = 0$	configurations with $\underline{T}^{(\mp)B} = 0$, i.e., <i>self-dual or anti-self-dual</i> tangential torsion. Con-
mielke-geometrodynamics_F00653	135	■ 0.802	$\stackrel{(\pm)}{\underline{T}} = 0$	$\stackrel{(\pm)}{\underline{T}} = 0$	straint are dominated by <i>self-dual</i> torsion solutions satisfying $\stackrel{(\pm)}{\underline{T}} = 0$.
mielke-geometrodynamics_F00654	135	■ 1.000	$\mathbb{R}^4 \cup \infty = S^4$	$\mathbb{R}^4 \cup \infty = S^4$	$\mathbb{R}^4 \cup \infty = S^4$ with the spherically symmetric metric
mielke-geometrodynamics_F00655	136	■ 0.997	(r, ψ, θ, ϕ)	(r, ψ, θ, ϕ)	where (r, ψ, θ, ϕ) are the standard hyperspherical coordinates that parameterize the
mielke-geometrodynamics_F00656	136	■ 1.000	S^3	S^3	unit three-sphere S^3 . The function f carries physical dimension [length]. This space
mielke-geometrodynamics_F00657	136	■ 1.000	f	f	unit three-sphere S^3 . The function f carries physical dimension [length]. This space
mielke-geometrodynamics_F00658	136	■ 1.000	$df = \pm 2 h dr$	$df = \pm 2 h dr$	which is self-dual or anti-self-dual, provided that $df = \pm 2 h dr$. Here the self-dual
mielke-geometrodynamics_F00659	136	■ 0.998	$\eta_{\mu\nu}^A := \eta_0^A \delta_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$	$\eta_{\mu\nu}^A := \eta_0^A \delta_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$	generalized Levi-Civita symbol $\eta_{\mu\nu}^A := \eta_0^A \delta_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$ of ’T HOOFT (1991),
mielke-geometrodynamics_F00660	136	■ 1.000	$r \rightarrow \infty$	$r \rightarrow \infty$	radial infinity $r \rightarrow \infty$ yields
mielke-geometrodynamics_F00661	136	■ 0.954	$\stackrel{(\mp)}{\underline{T}}^B = 0$	$\stackrel{(\mp)}{\underline{T}}^B = 0$	self-dual, i.e., $\stackrel{(\mp)}{\underline{T}}^B = 0$, then integration over the chiral CS term $\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$ yields the
mielke-geometrodynamics_F00662	136	■ 0.954	$\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$	$\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$	self-dual, i.e., $\stackrel{(\mp)}{\underline{T}}^B = 0$, then integration over the chiral CS term $\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$ yields the
mielke-geometrodynamics_F00663	136	■ 1.000	$\Psi_{\parallel}(\vartheta) \propto e^{-C}$	$\Psi_{\parallel}(\vartheta) \propto e^{-C}$	Thus one can surmise that the state vector $\Psi_{\parallel}(\vartheta) \propto e^{-C}$ as a solution of the

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mielke-geometrodynamics_F00664	136	■ 1.000	$\underline{\vartheta^{\hat{0}}}=3\kappa d\theta_{\mathrm{L}}=\hbar d r=\pm d f / 2$	$\vartheta^{\hat{0}}=3\kappa d\theta_{\mathrm{L}}=\hbar d r=\pm d f / 2$	⁴ Interestingly, in the gauge $\vartheta^{\hat{0}}=3\kappa d\theta_{\mathrm{L}}=\hbar d r=\pm d f / 2$, such instantons are solutions to the
mielke-geometrodynamics_F00665	136	■ 0.991	$T^{\hat{0}}=0$	$T^{\hat{0}}=0$	topological Eq. (6.7.12), due to $T^{\hat{0}}=0$.
mielke-geometrodynamics_F00666	137	■ 1.000	$A=A_{\mu}^I\lambda_I$	$A=A_{\mu}^I\lambda_I$	From the <i>holonomy</i> of a Lie-algebra-valued connection one-form $A=A_{\mu}^I\lambda_I$
mielke-geometrodynamics_F00667	137	■ 0.967	$d x^{\mu} \in \mathscr{C}$	$d x^{\mu} \in \mathscr{C}$	$d x^{\mu} \in \mathscr{C}$, one can construct a <i>gauge-invariant</i> operator, or character, along the loop
mielke-geometrodynamics_F00668	137	■ 1.000	$\gamma^{\mu}(s)$	$\gamma^{\mu}(s)$	with coordinates $\gamma^{\mu}(s)$. Then the Wilson loop can be written as
mielke-geometrodynamics_F00669	137	■ 0.826	$\Psi(A)$	$\Psi(A)$	The state vector $\Psi(A)$ for the gauge potential can be transformed via the Wilson
mielke-geometrodynamics_F00670	137	■ 1.000	$\mathscr{C} / \mathscr{G}$	$\mathscr{C} / \mathscr{G}$	The integration is performed over a coset space $\mathscr{C} / \mathscr{G}$ in order to factor out gauge-
mielke-geometrodynamics_F00671	137	■ 1.000	$G(\gamma)$	$G(\gamma)$	of the Wilson loop is the <i>self-linking number</i> $G(\gamma)$ of Gauss. It is interesting to see
mielke-geometrodynamics_F00672	137	■ 1.000	$V(\gamma)$	$V(\gamma)$	of the Jones polynomial $V(\gamma)$. If we could formally transform these results of lattice
mielke-geometrodynamics_F00673	138	■ 1.000	$ \Psi\rangle$	$ \Psi\rangle$	state vector $ \Psi\rangle$ of the universe would be characterized by knot invariants, such as
mielke-geometrodynamics_F00674	138	■ 1.000	$\mathbb{R}^4$	$\mathbb{R}^4$	group $\mathbb{R}^4$ as structure group. Then, instead of Wilson-type loops, we encounter
mielke-geometrodynamics_F00675	138	■ 0.979	$\left.n^{\alpha}\right\rfloor=\left\lfloor n^{\alpha}\right\rfloor$	$n^{\alpha}=n\rfloor\vartheta^{\alpha}$	GUADAGNINI 1993) via one of the orthonormal legs $n^{\alpha}=n\rfloor\vartheta^{\alpha}$. In view of (6.4.25),
mielke-geometrodynamics_F00676	138	■ 1.000	$(1 / 16 \pi^2)$	$(1 / 16 \pi^2)$	It is well known from differential geometry that the Euler number $(1 / 16 \pi^2)$
mielke-geometrodynamics_F00677	138	■ 1.000	$\int R^{\alpha \beta} \wedge R_{\alpha \beta}^{(\star)}$	$\int R^{\alpha \beta} \wedge R_{\alpha \beta}^{(\star)}$	$\int R^{\alpha \beta} \wedge R_{\alpha \beta}^{(\star)}$ and the Pontryagin integral $(1 / 8 \pi^2) \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$ bear an intimate
mielke-geometrodynamics_F00678	138	■ 1.000	$(1 / 8 \pi^2) \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$	$(1 / 8 \pi^2) \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$	$\int R^{\alpha \beta} \wedge R_{\alpha \beta}^{(\star)}$ and the Pontryagin integral $(1 / 8 \pi^2) \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$ bear an intimate
mielke-geometrodynamics_F00679	138	■ 1.000	ϵ	ϵ	to the following argument: For a real constant ϵ with $0 \leq \epsilon \leq 1$, the RC connec-
mielke-geometrodynamics_F00680	138	■ 1.000	$0 \leq \epsilon \leq 1$	$0 \leq \epsilon \leq 1$	to the following argument: For a real constant ϵ with $0 \leq \epsilon \leq 1$, the RC connec-
mielke-geometrodynamics_F00681	138	■ 0.973	$\Gamma_{\alpha \beta}(\epsilon)=\Gamma_{\alpha \beta}^{\{\}}-\epsilon K_{\alpha \beta}$	$\Gamma_{\alpha \beta}(\epsilon)=\Gamma_{\alpha \beta}^{\{\}}-\epsilon K_{\alpha \beta}$	to $\Gamma_{\alpha \beta}(\epsilon)=\Gamma_{\alpha \beta}^{\{\}}-\epsilon K_{\alpha \beta}$ such that $\Gamma_{\alpha \beta}(0)=\Gamma_{\alpha \beta}^{\{\}}$. Provided a topological quantity
mielke-geometrodynamics_F00682	138	■ 0.973	$\Gamma_{\alpha \beta}(0)=\Gamma_{\alpha \beta}^{\{\}}$	$\Gamma_{\alpha \beta}(0)=\Gamma_{\alpha \beta}^{\{\}}$	to $\Gamma_{\alpha \beta}(\epsilon)=\Gamma_{\alpha \beta}^{\{\}}-\epsilon K_{\alpha \beta}$ such that $\Gamma_{\alpha \beta}(0)=\Gamma_{\alpha \beta}^{\{\}}$. Provided a topological quantity
mielke-geometrodynamics_F00683	138	■ 1.000	$\Gamma_{\alpha \beta}$	$\Gamma_{\alpha \beta}$	depends on the connection $\Gamma_{\alpha \beta}$, it must be invariant under this continuous defor-
mielke-geometrodynamics_F00684	138	■ 0.679	$T:=e_{\alpha} \rfloor T^{\alpha}=d \sigma$	$T:=e_{\alpha} \rfloor T^{\alpha}=d \sigma$	torsion $T:=e_{\alpha} \rfloor T^{\alpha}=d \sigma$ derived from a scalar field via a gradient (GREGORASH &
mielke-geometrodynamics_F00685	138	■ 1.000	N	N	manifold, which may enclose the singularity line of the dislocation N times, yields
mielke-geometrodynamics_F00686	139	■ 1.000	$\sigma=$	$\sigma=$	provided the scalar field is not single-valued, as, for instance, in the case of $\sigma=$
mielke-geometrodynamics_F00687	139	■ 0.997	$\arcsin(x)$	$\arcsin(x)$	$\arcsin(x)$.
mielke-geometrodynamics_F00688	139	■ 1.000	$\pi_3(S O(4))=$	$\pi_3(S O(4))=$	nontrivial winding number will occur as a result of the homotopy group $\pi_3(S O(4))=$
mielke-geometrodynamics_F00689	139	■ 1.000	$\mathbb{Z} \oplus \mathbb{Z}$	$\mathbb{Z} \oplus \mathbb{Z}$	$\mathbb{Z} \oplus \mathbb{Z}$. On the other hand, the 3D hypersurface admits the intriguing mapping $\ell \vartheta^A \rightarrow$
mielke-geometrodynamics_F00690	139	■ 1.000	$\ell \vartheta^A \rightarrow$	$\ell \vartheta^A \rightarrow$	$\mathbb{Z} \oplus \mathbb{Z}$. On the other hand, the 3D hypersurface admits the intriguing mapping $\ell \vartheta^A \rightarrow$

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mielke-geometrodynamics_F00691	139	■ 0.999	$\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$ $\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$	$\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$	$\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$ of the triads to the corresponding three-dual of the $\mathfrak{su}(2)$ -valued
mielke-geometrodynamics_F00692	139	■ 0.999	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	$\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$ of the triads to the corresponding three-dual of the $\mathfrak{su}(2)$ -valued
mielke-geometrodynamics_F00693	139	■ 0.718	$\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$	$\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$	connection (BAEKLER et al. 1992). Since this implies a mapping $\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$
mielke-geometrodynamics_F00694	139	■ 1.000	$\mathbb{R} \times W^3$	$\mathbb{R} \times W^3$	small scale of “quantum gravity,” the underlying spacetime manifold $\mathbb{R} \times W^3$ may
mielke-geometrodynamics_F00695	139	■ 1.000	$d j_5$	$d j_5$	In classical EC theory, the net axial current production $d j_5$ seems (CHANG &
mielke-geometrodynamics_F00696	139	■ 1.000	$d \mathscr{A} \wedge d \mathscr{A}$	$d \mathscr{A} \wedge d \mathscr{A}$	$d \mathscr{A} \wedge d \mathscr{A}$ involving the axial torsion. Since torsion instantons are characterized
mielke-geometrodynamics_F00697	140	■ 0.685	$\theta_{\mathrm{T}} = 2 / \gamma$	$\theta_{\mathrm{T}} = 2 / \gamma$	⁵ The translational angle $\theta_{\mathrm{T}} = 2 / \gamma$ is at times identified (FREIDEL et al. 2005) with the inverse
mielke-geometrodynamics_F00698	140	■ 0.985	$d C_{\mathrm{TT}}$	$d C_{\mathrm{TT}}$	lational Chern–Simons term $d C_{\mathrm{TT}}$ were considered earlier by MIELKE (1992).
mielke-geometrodynamics_F00699	141	■ 1.000	$\vartheta_{[\beta} \wedge H_{\alpha]}$	$\vartheta_{[\beta} \wedge H_{\alpha]}$	Note that the angular momentum part $\vartheta_{[\beta} \wedge H_{\alpha]}$ of the gauge fields induced by
mielke-geometrodynamics_F00700	141	■ 1.000	$D H_{\alpha\beta}$	$D H_{\alpha\beta}$	the NY term cancels identically against one part of $D H_{\alpha\beta}$. In the case of Dirac fields
mielke-geometrodynamics_F00701	141	■ 1.000	μ_{α}	μ_{α}	(MIELKE 2004), the <i>spin–energy potential</i> μ_{α} , a two-form, is related to the axial
mielke-geometrodynamics_F00702	141	■ 1.000	$j_5 = \bar{\psi} \gamma \gamma_5 \psi$	$j_5 = \bar{\psi} \gamma \gamma_5 \psi$	current three-form $j_5 = \bar{\psi} \gamma \gamma_5 \psi$ via
mielke-geometrodynamics_F00703	141	■ 1.000	$D \theta_{\mathrm{L}} \neq 0$	$D \theta_{\mathrm{L}} \neq 0$	that (6.7.6) for $D \theta_{\mathrm{L}} \neq 0$ provides torsion with a dynamical coupling to RC curva-
mielke-geometrodynamics_F00704	141	■ 1.000	$\mu_{\alpha} \wedge \vartheta^{\alpha} = 0$	$\mu_{\alpha} \wedge \vartheta^{\alpha} = 0$	i.e., $\mu_{\alpha} \wedge \vartheta^{\alpha} = 0$, in view of (6.7.8).
mielke-geometrodynamics_F00705	141	■ 0.726	$T = e_{\alpha} \rfloor T^{\alpha}$	$T = e_{\alpha} \rfloor T^{\alpha}$	where the vector torsion one-form $T = e_{\alpha} \rfloor T^{\alpha}$ enters as an intermediate source. Due
mielke-geometrodynamics_F00706	141	■ 0.815	$DD\theta = dd\theta \equiv 0$	$DD\theta = dd\theta \equiv 0$	to the Poincaré lemma $DD\theta = dd\theta \equiv 0$ for a (pseudo) scalar field, Eq. (6.7.10) has
mielke-geometrodynamics_F00707	141	■ 1.000	ϑ^{β}	ϑ^{β}	as a first integral. After a contraction with ϑ^{β} , the dual of the axial torsion one-form
mielke-geometrodynamics_F00708	141	■ 0.996	$\mathscr{A} := *(\vartheta^{\alpha} \wedge T_{\alpha})$	$\mathscr{A} := *(\vartheta^{\alpha} \wedge T_{\alpha})$	$\mathscr{A} := *(\vartheta^{\alpha} \wedge T_{\alpha})$, i.e., the translational CS term, turns out to be related to the volume
mielke-geometrodynamics_F00709	141	■ 0.999	θ_{L}	θ_{L}	a coupling to a kinetic term arising from the pseudoscalar field θ_{L} rescaling the
mielke-geometrodynamics_F00710	141	■ 1.000	$T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha\beta\gamma\mu} = 0$	$T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha\beta\gamma\mu} = 0$	$T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha\beta\gamma\mu} = 0$. Again we end up with a first-order equation for torsion, where,
mielke-geometrodynamics_F00711	142	■ 1.000	DH	DH	Part of the sources for the Yang–Mills-type divergence terms “ DH ” are the mate-
mielke-geometrodynamics_F00712	143	■ 1.000	L_g	L_g	aid of the coframe ϑ^{α} , in general gives us back the gauge Lagrangian L_g amended
mielke-geometrodynamics_F00713	143	■ 1.000	${}^{(i)}T^{\alpha}$	${}^{(i)}T^{\alpha}$	Lagrangian, which is at most <i>quadratic</i> in the irreducible pieces ${}^{(i)}T^{\alpha}$ and ${}^{(j)}R^{\alpha\beta}$
mielke-geometrodynamics_F00714	143	■ 1.000	${}^{(j)}R^{\alpha\beta}$	${}^{(j)}R^{\alpha\beta}$	Lagrangian, which is at most <i>quadratic</i> in the irreducible pieces ${}^{(i)}T^{\alpha}$ and ${}^{(j)}R^{\alpha\beta}$
mielke-geometrodynamics_F00715	143	■ 1.000	$\ell \simeq 10^{-32} \mathrm{cm}$	$\ell \simeq 10^{-32} \mathrm{cm}$	The gravitational coupling constant is absorbed in the Planck length $\ell \simeq 10^{-32} \mathrm{cm}$,
mielke-geometrodynamics_F00716	143	■ 1.000	$a_0, a_i (i=1, 2, 3), \kappa$	$a_0, a_i (i=1, 2, 3), \kappa$	whereas a_0, a_i ($i=1, 2, 3$), κ , and b_j ($j=1, \dots, 6$) are dimensionless coupling
mielke-geometrodynamics_F00717	143	■ 1.000	$b_j (j=1, \dots, 6)$	$b_j (j=1, \dots, 6)$	whereas a_0, a_i ($i=1, 2, 3$), κ , and b_j ($j=1, \dots, 6$) are dimensionless coupling

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mielke-geometrodynamics_F00718	144	■ 1.000	C: $Q \rightarrow Q^{\prime}=-Q$	$C:Q\rightarrow Q'=-Q$	a charge conjugation $C:Q\rightarrow Q'=-Q$, one obtains the same gravitational field.
mielke-geometrodynamics_F00719	144	■ 1.000	C	C	Maxwell equations are invariant under CPT, charge conjugation C has to be accom-
mielke-geometrodynamics_F00720	144	■ 0.963	T	T	panied by an additional time reversal T (particles moving backward in time $\stackrel{\wedge}{=}$ antipar-
mielke-geometrodynamics_F00721	144	■ 0.963	$\stackrel{\wedge}{=}$	$\stackrel{\wedge}{=}$	panied by an additional time reversal T (particles moving backward in time $\stackrel{\wedge}{=}$ antipar-
mielke-geometrodynamics_F00722	144	■ 1.000	$a_{\{0\}}=0, a_{\{i\}}=1$	$a_0=0, a_i=1$	by the choice $a_0=0, a_i=1$, and $b_j=1$. In order to obtain its complexified version,
mielke-geometrodynamics_F00723	144	■ 1.000	$b_{\{j\}}=1$	$b_j=1$	by the choice $a_0=0, a_i=1$, and $b_j=1$. In order to obtain its complexified version,
mielke-geometrodynamics_F00724	144	■ 1.000	θ_{T}	θ_{T}	parameters θ_{T} and θ_{L} :
mielke-geometrodynamics_F00725	145	■ 1.000	P: $\vartheta^{\prime A}=-\vartheta^A$	$P:\vartheta'^A=-\vartheta^A$	are invariant under space reflections $P:\vartheta'^A=-\vartheta^A$, where $A=1,2,3$, due to the
mielke-geometrodynamics_F00726	145	■ 1.000	A=1,2,3	$A=1,2,3$	are invariant under space reflections $P:\vartheta'^A=-\vartheta^A$, where $A=1,2,3$, due to the
mielke-geometrodynamics_F00727	145	■ 1.000	$\eta_{\alpha\beta\gamma\delta}$	$\eta_{\alpha\beta\gamma\delta}$	occurrence of the determinant of the metric in $\eta_{\alpha\beta\gamma\delta}$. In contradistinction to these
mielke-geometrodynamics_F00728	145	■ 1.000	$R_{\{\alpha\}}\wedge R_{\{\beta\}}\wedge R_{\{\alpha\}}$	$R_{\alpha}{}^{\beta}\wedge R_{\beta}{}^{\alpha}$	The term $R_{\alpha}{}^{\beta}\wedge R_{\beta}{}^{\alpha}$ is known to yield, after integration over spacetime, a topo-
mielke-geometrodynamics_F00729	145	■ 1.000	$\langle 0 R^{\alpha\beta} 0\rangle=0$	$\langle 0 R^{\alpha\beta} 0\rangle=0$	Even above the Planck scale, we can at most require $\langle 0 R^{\alpha\beta} 0\rangle=0$, but we could
mielke-geometrodynamics_F00730	151	■ 0.938	Ω_{C}	Ω_{C}	Let Ω_{C} be the 2-form of a generalized curvature tensor (NOMIZU 1972) that is
mielke-geometrodynamics_F00731	151	■ 0.929	$\operatorname{SL}(2,\mathbb{C})\approx\widetilde{\operatorname{SO}}(3,1)$	$\operatorname{SL}(2,\mathbb{C})\approx\widetilde{\operatorname{SO}}(3,1)$	¹ The bispinor representation of the covering group $\operatorname{SL}(2,\mathbb{C})\approx\widetilde{\operatorname{SO}}(3,1)$ is inapt for this purpose,
mielke-geometrodynamics_F00732	151	■ 1.000	$\sigma_{\alpha\beta}=\frac{1}{2}[\gamma_{\alpha},\gamma_{\beta}]$	$\sigma_{\alpha\beta}=\frac{1}{2}[\gamma_{\alpha},\gamma_{\beta}]$	since its infinitesimal generators $\sigma_{\alpha\beta}=\frac{1}{2}[\gamma_{\alpha},\gamma_{\beta}]$ are neither self-dual nor anti-self-dual. As a direct
mielke-geometrodynamics_F00733	151	■ 1.000	$\sigma_{\alpha\beta}^{(\star)}=\gamma^5\sigma_{\alpha\beta}$	$\sigma_{\alpha\beta}^{(\star)}=\gamma^5\sigma_{\alpha\beta}$	sum of the spinor and its conjugate representation, they rather satisfy the relation $\sigma_{\alpha\beta}^{(\star)}=\gamma^5\sigma_{\alpha\beta}$,
mielke-geometrodynamics_F00734	151	■ 0.829	$\gamma^5:=\frac{1}{4!}\varepsilon_{\mathrm{abcd}}\gamma^{\mathrm{a}}\gamma^{\mathrm{b}}\gamma^{\mathrm{c}}\gamma^{\mathrm{d}}$	$\gamma^5:=\frac{1}{4!}\varepsilon_{\mathrm{abcd}}\gamma^{\mathrm{a}}\gamma^{\mathrm{b}}\gamma^{\mathrm{c}}\gamma^{\mathrm{d}}$	whereby the matrix $\gamma^5:=\frac{1}{4!}\varepsilon_{\mathrm{abcd}}\gamma^{\mathrm{a}}\gamma^{\mathrm{b}}\gamma^{\mathrm{c}}\gamma^{\mathrm{d}}$ is, within the Clifford algebra, an axial Lorentz scalar.
mielke-geometrodynamics_F00735	152	■ 1.000	s=0	$s=0$	In a four-dimensional Riemannian manifold with the Euclidean signature $s=0$,
mielke-geometrodynamics_F00736	153	■ 1.000	$R_{\alpha\beta}$	$R_{\alpha\beta}$	where $R_{\alpha\beta}$ is the curvature two-form. In WEYL (1919), in the concluding chapter
mielke-geometrodynamics_F00737	153	■ 0.813	$\Gamma^{\alpha\beta}:=\Gamma_i^{\alpha\beta}dx^i$	$\Gamma^{\alpha\beta}:=\Gamma_i^{\alpha\beta}dx^i$	$\Gamma^{\alpha\beta}:=\Gamma_i^{\alpha\beta}dx^i$. In his purely affine approach, Yang did not consider the canonical
mielke-geometrodynamics_F00738	153	■ 0.977	$\vartheta^{\alpha}=E_{\{j\}}^{\alpha}dx^{\{j\}}$	$\vartheta^{\alpha}=E_j^{\alpha}dx^j$	with respect to the coframe $\vartheta^{\alpha}=E_j^{\alpha}dx^j$ dual to the local anholonomic frame e_{α} .
mielke-geometrodynamics_F00739	153	■ 0.977	e_{α}	e_{α}	with respect to the coframe $\vartheta^{\alpha}=E_j^{\alpha}dx^j$ dual to the local anholonomic frame e_{α} .
mielke-geometrodynamics_F00740	154	■ 1.000	G_{ij}	G_{ij}	which is dual to the usual Einstein tensor G_{ij} in Riemannian spacetime. In exterior
mielke-geometrodynamics_F00741	154	■ 0.494	$\operatorname{Ric}_{\alpha\beta}:=(-1)^s*\left(R_{(\alpha}{}^{\delta}\wedge\eta_{\delta \beta)}\right)$	$\operatorname{Ric}_{\alpha\beta}:=(-1)^s*\left(R_{(\alpha}{}^{\delta}\wedge\eta_{\delta \beta)}\right)$	$\operatorname{Ric}_{\alpha\beta}:=(-1)^s*(R_{(\alpha}{}^{\delta}\wedge\eta_{\delta \beta)})$.
mielke-geometrodynamics_F00742	154	■ 1.000	$\tau_{\alpha\beta}=\vartheta_{[\alpha}\wedge$	$\tau_{\alpha\beta}=\vartheta_{[\alpha}\wedge$	Macroscopically, the matter coupling ² to the spin current three-form $\tau_{\alpha\beta}=\vartheta_{[\alpha}\wedge$
mielke-geometrodynamics_F00743	154	■ 0.998	$\mu_{\beta]}$	$\mu_{\beta]}$	$\mu_{\beta]}$ seems to disfavor such models, a criticism which has already raised by FAIRCHILD

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mielke-geometrodynamics_F00744	154	■ 0.589	$\Gamma^{\{\beta\alpha\beta\}} \propto \partial g$	$\Gamma^{\{\beta\alpha\beta\}} \propto \partial g$	In Riemannian spacetime, with its Levi–Civita connection $\Gamma^{\{\}}^{\alpha\beta} \propto \partial g$, these
mielke-geometrodynamics_F00745	154	■ 1.000	$\{ \}^{\{(\star)\}}$	$(\star)$	and the Lie dual $(\star)$ of the curvature $R_{\alpha\beta}$ in a space(time) of signature s .
mielke-geometrodynamics_F00746	154	■ 1.000	$L_{\{\text{SKY}\}}^{\{(\star)\}} = 2 L_{\{\text{SKY}\}}$	$L_{\text{SKY}}^{(\star)} = 2L_{\text{SKY}}$	i.e., they lead to $L_{\text{SKY}}^{(\star)} = 2L_{\text{SKY}}$. Both second-order duality conditions satisfy Yang’s
mielke-geometrodynamics_F00747	154	■ 1.000	$j_{\{5\}} := \bar{\psi}^{\{\}} \gamma_{\{5\}} \psi$	$j_{\bar{5}} := \bar{\psi}^* \gamma_{\bar{5}} \psi$	current $j_{\bar{5}} := \bar{\psi}^* \gamma_{\bar{5}} \psi$.
mielke-geometrodynamics_F00748	155	■ 1.000	$D \eta_{\{\alpha\beta\}} = 0$	$D\eta_{\alpha\beta} = 0$	its Lie dual, due to $D\eta_{\alpha\beta} = 0$ in a Riemannian spacetime. More on this “vacuum
mielke-geometrodynamics_F00749	155	■ 1.000	$\zeta = \pm 1$	$\zeta = \pm 1$	whereby only $\zeta = \pm 1$ (and zero) are admitted.
mielke-geometrodynamics_F00750	155	■ 1.000	$\zeta = +1$	$\zeta = +1$	$\zeta = +1$: <i>self-double-dual curvature</i>
mielke-geometrodynamics_F00751	155	■ 1.000	$\zeta = -1$	$\zeta = -1$	$\zeta = -1$: <i>anti-self-double-dual curvature</i>
mielke-geometrodynamics_F00752	155	■ 0.999	$\mathscr{A} := \{ \}^{\{ \}} \left(\vartheta_{\{\alpha\}} \wedge T^{\{\alpha\}} \right)$	$\mathscr{A} := \{ \}^{\{ \}} (\vartheta_{\alpha} \wedge T^{\alpha})$	Riemannian curvature $R_{\alpha\beta}^{\{\}}$ and the axial torsion one-form $\mathscr{A} := \{ \}^{\{ \}} (\vartheta_{\alpha} \wedge T^{\alpha})$ is moti-
mielke-geometrodynamics_F00753	155	■ 0.638	$\left\langle dj_{\{5\}} \right\rangle = 2i^* m \left\langle \bar{\psi} \gamma_{\{5\}} \psi \right\rangle - \left(R_{\alpha\beta}^{\{\}} \wedge R^{\{\}}{}^{\alpha\beta} + \frac{1}{2} d\mathscr{A} \wedge d\mathscr{A} \right) /$	$\langle dj_{\bar{5}} \rangle = 2i^* m \langle \bar{\psi} \gamma_{\bar{5}} \psi \rangle - \left(R_{\alpha\beta}^{\{\}} \wedge R^{\{\}}{}^{\alpha\beta} + \frac{1}{2} d\mathscr{A} \wedge d\mathscr{A} \right) /$	vated by the <i>axial anomaly</i> $\langle dj_{\bar{5}} \rangle = 2i^* m \langle \bar{\psi} \gamma_{\bar{5}} \psi \rangle - \left(R_{\alpha\beta}^{\{\}} \wedge R^{\{\}}{}^{\alpha\beta} + \frac{1}{2} d\mathscr{A} \wedge d\mathscr{A} \right) /$
mielke-geometrodynamics_F00754	155	■ 1.000	$48 \pi^2$	$48\pi^2$	$48\pi^2$ in the coupling to Dirac fields ψ .
mielke-geometrodynamics_F00755	156	■ 1.000	$d^{\{4\}} x$	$d^4 x$	arises, where the four-dimensional integration over $d^4 x$ is already implied in our
mielke-geometrodynamics_F00756	156	■ 0.996	$\{ \}^{\{ \}} R_{\{\alpha\beta\}} = \theta_{\{\text{L}\}}^{\{ \}} R_{\{\alpha\beta\}}^{\{(\star)\}} + \left(\theta_{\{\text{T}\}}^{\{ \}} \right) \eta_{\{\alpha\beta\}} \wedge \left(\theta_{\{\text{L}\}}^{\{ \}} \right) \eta_{\{\alpha\beta\}} / 6$	${}^* R_{\alpha\beta} = \theta_{\text{L}}^* R_{\alpha\beta}^{(\star)} + (\theta_{\text{T}}^* \Lambda / 6) \eta_{\alpha\beta}$	1994) via a modified duality ${}^* R_{\alpha\beta} = \theta_{\text{L}}^* R_{\alpha\beta}^{(\star)} + (\theta_{\text{T}}^* \Lambda / 6) \eta_{\alpha\beta}$, which, however, explic-
mielke-geometrodynamics_F00757	156	■ 0.998	$\sigma_{\{\alpha\}} := \Sigma_{\{\alpha\}} - D^{\{\}} \mu_{\{\alpha\}}$	$\sigma_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha}$	& MAGGIOLO 2005) symmetrized energy–momentum current $\sigma_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha}$.)
mielke-geometrodynamics_F00758	156	■ 0.682	$\left\lfloor e_{\{\alpha\}} \right\rfloor \left\lfloor e_{\{\beta\}} \right\rfloor R^{\alpha\beta}$	$R := e_{\alpha} \rfloor e_{\beta} \rfloor R^{\alpha\beta}$	trace with $R := e_{\alpha} \rfloor e_{\beta} \rfloor R^{\alpha\beta}$ as irreducible pieces. Then the double-dual curvature
mielke-geometrodynamics_F00759	157	■ 1.000	$o_{\{i\}j}^{\{\text{Mink}\}}$	o_{ij}^{Mink}	with o_{ij}^{Mink} denoting the (pseudo)orthogonal Minkowski metric. Earlier, EINSTEIN &
mielke-geometrodynamics_F00760	157	■ 0.529	$D^{\{\}} G_{\{\alpha\}}^{\{\}} \equiv 0$	$D^{\{\}} G_{\alpha}^{\{\}} \equiv 0$	Due to the contracted Bianchi identity $D^{\{\}} G_{\alpha}^{\{\}} \equiv 0$ in Riemannian spacetime, the
mielke-geometrodynamics_F00761	157	■ 0.999	$R = -4 \Lambda$	$R = -4\Lambda$	curvature scalar is constant, i.e., more precisely, $R = -4\Lambda$. Thus, (7.4.6) is equivalent
mielke-geometrodynamics_F00762	157	■ 1.000	$\Lambda \simeq 1 / \kappa$	$\Lambda \simeq 1/\kappa$	with cosmological constant $\Lambda \simeq 1/\kappa$. It is crucial to note that the latter carries phys-
mielke-geometrodynamics_F00763	157	■ 1.000	$R = 0$	$R = 0$	sical Einstein–Fokker background due to $R = 0$ in double-dual Thompson spaces.
mielke-geometrodynamics_F00764	157	■ 1.000	$h_{\{i\}j} := g_{\{i\}j} - g_{\{i\}j}^{\text{dS}}$	$h_{ij} := g_{ij} - g_{ij}^{\text{dS}}$	On the other hand, the linearized Einstein equations for $h_{ij} := g_{ij} - g_{ij}^{\text{dS}}$ in a de
mielke-geometrodynamics_F00765	157	■ 1.000	$g_{\{i\}j}^{\{\text{dS}\}}$	g_{ij}^{dS}	Sitter background g_{ij}^{dS} with a positive cosmological constant are
mielke-geometrodynamics_F00766	158	■ 1.000	m / H	m/H	de Sitter space if m/H is kept finite in the massless limit. For gravitons, the mass
mielke-geometrodynamics_F00767	158	■ 1.000	$m \leq 7 \times 10^{-32} \text{ eV}$	$m \leq 7 \times 10^{-32} \text{ eV}$	(GOLDHABER & NIETO 2010) of $m \leq 7 \times 10^{-32} \text{ eV}$, but larger than the Hubble
mielke-geometrodynamics_F00768	158	■ 1.000	$H \simeq 10^{-33} \text{ eV}$	$H \simeq 10^{-33} \text{ eV}$	scale of $H \simeq 10^{-33} \text{ eV}$.

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mielke-geometrodynamics_F00769	158	■ 0.994	$\Lambda_{\text{obs}} > 0$	$\Lambda_{\text{obs}} > 0$	is induced via the tiny cosmological constant $\Lambda_{\text{obs}} > 0$ of our expanding universe.
mielke-geometrodynamics_F00770	158	■ 0.994	$1 / g^2$	$1/g^2$	constant $1/g^2$ is ‘traded’ for a dimensional one. This ‘dimensional transmutation’ à
mielke-geometrodynamics_F00771	159	■ 0.662	$\operatorname{SU}(2)$	$SU(2)$	Originally, Yang and Mills formulated their model as an $SU(2)$ gauge theory with
mielke-geometrodynamics_F00772	159	■ 0.968	1974)	1974)	1974) is an $SO(1, 3)$ gauge theory, whose covering gauge group, after a Wick rotation
mielke-geometrodynamics_F00773	159	■ 1.000	$\overline{SO}(4) \cong$	$\overline{SO}(4) \cong$	to Euclidean space in the path integral quantization, is locally isomorphic to $\overline{SO}(4) \cong$
mielke-geometrodynamics_F00774	159	■ 1.000	$SU(2) \otimes SU(2)$	$SU(2) \otimes SU(2)$	$SU(2) \otimes SU(2)$. This doubling of the (internal) unitary structure group $SU(2)$ was
mielke-geometrodynamics_F00775	159	■ 0.719	$\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$	$\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$	occur at the tiny <i>dimensionless scale</i> of $\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$, as suggested by observa-
mielke-geometrodynamics_F00776	159	■ 0.999	ΔL	ΔL	In perturbative quantum gravity (ALVAREZ 1989), there arise counterterms ΔL of
mielke-geometrodynamics_F00777	159	■ 0.995	$\operatorname{Ric}_{\alpha\beta} := \left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$	$\operatorname{Ric}_{\alpha\beta} := \left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$	version of the zero-form $\operatorname{Ric}_{\alpha\beta} := \left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$.
mielke-geometrodynamics_F00778	160	■ 1.000	$H_{\alpha\beta}$	$H_{\alpha\beta}$	Here the field momenta H_{α} and $H_{\alpha\beta}$ are understood as arising from a generating
mielke-geometrodynamics_F00779	160	■ 1.000	L_{eff}	L_{eff}	n-form G as part of some effective gauge Lagrangian L_{eff} that includes the countert-
mielke-geometrodynamics_F00780	160	■ 0.637	$n-2$	$n-2$	In our dynamical approach, the $(n-2)$ -forms H_{α} and $H_{\alpha\beta}$ will be gauge field
mielke-geometrodynamics_F00781	160	■ 1.000	$P := \mathbb{R}^4 \otimes$	$P := \mathbb{R}^4 \otimes$	tively. Due to the <i>semidirect product</i> structure of the Poincaré group $P := \mathbb{R}^4 \ltimes$
mielke-geometrodynamics_F00782	160	■ 0.945	$H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$	$H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$	$SO(1, 3)$, the gauge field momenta contribute to the gauge potentials via $H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$
mielke-geometrodynamics_F00783	160	■ 1.000	$H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$	$H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$	and $H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$ in the FR (7.7.2, 7.7.3) just in an <i>intertwined</i> manner.
mielke-geometrodynamics_F00784	160	■ 0.991	$\left(DH_{\alpha}{}^{\beta} \right)$	$\left(DH_{\alpha}{}^{\beta} \right)$	similarly as for Yang–Mills fields, a term proportional to $\left(DH_{\alpha}{}^{\beta} \right)$. However, ‘on
mielke-geometrodynamics_F00785	160	■ 0.388	$\left(H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$	$\left(H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$	$\left(H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$ due to an algebraic identity. In the FR (7.7.2) of the coframe,
mielke-geometrodynamics_F00786	160	■ 0.388	H^{β}	H^{β}	$\left(H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$ due to an algebraic identity. In the FR (7.7.2) of the coframe,
mielke-geometrodynamics_F00787	160	■ 0.644	$\left(DH^{\alpha} \right) \cong \left(E^{\alpha} \right)$	$\left(DH^{\alpha} \right) \cong \left(E^{\alpha} \right)$	the same situation arises, with the modification that the ‘on shell’ term $\left(DH^{\alpha} \right) \cong \left(E^{\alpha} \right)$
mielke-geometrodynamics_F00788	160	■ 1.000	$\widetilde{G}$	$\widetilde{G}$	order generation functional $\widetilde{G}$. When coupling to matter, FRs have to be handled with
mielke-geometrodynamics_F00789	161	■ 1.000	$\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$	$\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$	momenta $\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$ can be expanded as $\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$. (For the
mielke-geometrodynamics_F00790	161	■ 1.000	$\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$	$\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$	momenta $\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$ can be expanded as $\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$. (For the
mielke-geometrodynamics_F00791	161	■ 1.000	$\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$	$\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$	time being, the field momentum $\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$ conjugate to the torsion $T^{\alpha} :=$
mielke-geometrodynamics_F00792	161	■ 1.000	$T^{\alpha} :=$	$T^{\alpha} :=$	time being, the field momentum $\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$ conjugate to the torsion $T^{\alpha} :=$
mielke-geometrodynamics_F00793	161	■ 1.000	$D \vartheta^{\alpha}$	$D \vartheta^{\alpha}$	$D \vartheta^{\alpha}$ will be set to zero.)
mielke-geometrodynamics_F00794	161	■ 1.000	$\widetilde{H}_{\alpha\beta\mu\nu}$	$\widetilde{H}_{\alpha\beta\mu\nu}$	For a <i>fixed point</i> of the transformation (7.7.9) , the Hessian $\widetilde{H}_{\alpha\beta\mu\nu}$ obviously has

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mielke-geometrodynamics_F00795	161	■ 1.000	$\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$	$\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$	to vanish. This condition, i.e., $\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$, can be readily solved.
mielke-geometrodynamics_F00796	161	■ 1.000	$\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$	$\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$	If parity-violating terms such as $\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$ arising from the Nieh–Yan
mielke-geometrodynamics_F00797	162	■ 1.000	$\widetilde{L}$	$\widetilde{L}$	Lagrangian $\widetilde{L}$.
mielke-geometrodynamics_F00798	162	■ 1.000	$\eta_{\alpha\beta}$	$\eta_{\alpha\beta}$	Accordingly, $\eta_{\alpha\beta}$ and $\widetilde{H}_{\alpha\beta}$ interchange their roles as generalized coordinates and
mielke-geometrodynamics_F00799	162	■ 1.000	$\widetilde{H}_{\alpha\beta}$	$\widetilde{H}_{\alpha\beta}$	Accordingly, $\eta_{\alpha\beta}$ and $\widetilde{H}_{\alpha\beta}$ interchange their roles as generalized coordinates and
mielke-geometrodynamics_F00800	162	■ 1.000	$\widetilde{L} = L_{\mathrm{EC}}$	$\widetilde{L} = L_{\mathrm{EC}}$	momenta, respectively. If we had started from $\widetilde{L} = L_{\mathrm{EC}}$, then we would be led back
mielke-geometrodynamics_F00801	162	■ 1.000	$L = L_{\mathrm{EC}}$	$L = L_{\mathrm{EC}}$	to $L = L_{\mathrm{EC}}$ for the choice $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = 0$. In the case $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = i$, this
mielke-geometrodynamics_F00802	162	■ 1.000	$\theta_{\mathrm{T}}^* = 1$	$\theta_{\mathrm{T}}^* = 1$	to $L = L_{\mathrm{EC}}$ for the choice $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = 0$. In the case $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = i$, this
mielke-geometrodynamics_F00803	162	■ 1.000	$\theta_{\mathrm{T}} = 0$	$\theta_{\mathrm{T}} = 0$	to $L = L_{\mathrm{EC}}$ for the choice $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = 0$. In the case $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = i$, this
mielke-geometrodynamics_F00804	162	■ 1.000	$\theta_{\mathrm{T}} = i$	$\theta_{\mathrm{T}} = i$	to $L = L_{\mathrm{EC}}$ for the choice $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = 0$. In the case $\theta_{\mathrm{T}}^* = 1$ and $\theta_{\mathrm{T}} = i$, this
mielke-geometrodynamics_F00805	162	■ 0.436	$\left. \Gamma_{\alpha}{}^{\beta} \rightarrow \Gamma_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} \mp (i\ell^2/2) e_{\alpha} \right]^{(\mp)}{}^{\beta}$	$\left. \Gamma_{\alpha}{}^{\beta} \rightarrow \Gamma_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} \mp (i\ell^2/2) e_{\alpha} \right]^{(\mp)}{}^{\beta}$	ition $\Gamma_{\alpha}{}^{\beta} \rightarrow \Gamma_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} \mp (i\ell^2/2) e_{\alpha} \rfloor H^{\beta}$, induced by the translational Chern–
mielke-geometrodynamics_F00806	162	■ 0.999	idC_{TT}	idC_{TT}	Simons term idC_{TT} . The resulting Sen-type connection still couples <i>minimally</i> to the
mielke-geometrodynamics_F00807	162	■ 0.999	$**\Phi = (-1)^{p(4-p)+s} \Phi$	$**\Phi = (-1)^{p(4-p)+s} \Phi$	holds. Applied to p -forms, it is almost involutive, i.e., $**\Phi = (-1)^{p(4-p)+s} \Phi$. For
mielke-geometrodynamics_F00808	162	■ 0.997	$s := \mathrm{sig} = 1$	$s := \mathrm{sig} = 1$	spacetimes in which $s := \mathrm{sig} = 1$ holds, it induces an <i>almost complex structure</i> , and
mielke-geometrodynamics_F00809	163	■ 1.000	$\{\eta, \eta_{\alpha}, \eta_{\alpha\beta}, \eta_{\alpha\beta\gamma}, \eta_{\alpha\beta\gamma\delta}\}$	$\{\eta, \eta_{\alpha}, \eta_{\alpha\beta}, \eta_{\alpha\beta\gamma}, \eta_{\alpha\beta\gamma\delta}\}$	makes it possible to generate the so-called η , or dual, basis $\{\eta, \eta_{\alpha}, \eta_{\alpha\beta}, \eta_{\alpha\beta\gamma}, \eta_{\alpha\beta\gamma\delta}\}$
mielke-geometrodynamics_F00810	163	■ 0.925	$\eta_{\alpha} := e_{\alpha} \rfloor \eta = {}^* \vartheta_{\alpha}, \eta_{\alpha\beta} :=$	$\eta_{\alpha} := e_{\alpha} \rfloor \eta = {}^* \vartheta_{\alpha}, \eta_{\alpha\beta} :=$	of exterior forms via consecutive interior products: $\eta_{\alpha} := e_{\alpha} \rfloor \eta = {}^* \vartheta_{\alpha}, \eta_{\alpha\beta} :=$
mielke-geometrodynamics_F00811	163	■ 0.920	$e_{\beta} \rfloor \eta_{\alpha} = \eta_{\alpha\beta\gamma} \vartheta^{\gamma} = e_{\beta} \rfloor e_{\alpha} \rfloor \eta = {}^* (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = \frac{1}{2} \eta_{\alpha\beta\gamma\delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}$	$e_{\beta} \rfloor \eta_{\alpha} = \eta_{\alpha\beta\gamma} \vartheta^{\gamma} = e_{\beta} \rfloor e_{\alpha} \rfloor \eta = {}^* (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = \frac{1}{2} \eta_{\alpha\beta\gamma\delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}$	$e_{\beta} \rfloor \eta_{\alpha} = \eta_{\alpha\beta\gamma} \vartheta^{\gamma} = e_{\beta} \rfloor e_{\alpha} \rfloor \eta = {}^* (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = \frac{1}{2} \eta_{\alpha\beta\gamma\delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}$, and $\eta_{\alpha\beta\gamma} := e_{\gamma}$
mielke-geometrodynamics_F00812	163	■ 0.920	$\eta_{\alpha\beta\gamma} := e_{\gamma}$	$\eta_{\alpha\beta\gamma} := e_{\gamma}$	$e_{\beta} \rfloor \eta_{\alpha} = \eta_{\alpha\beta\gamma} \vartheta^{\gamma} = e_{\beta} \rfloor e_{\alpha} \rfloor \eta = {}^* (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = \frac{1}{2} \eta_{\alpha\beta\gamma\delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}$, and $\eta_{\alpha\beta\gamma} := e_{\gamma}$
mielke-geometrodynamics_F00813	163	■ 0.862	$\rfloor \eta_{\alpha\beta} = {}^* (\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma})$	$\rfloor \eta_{\alpha\beta} = {}^* (\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma})$	$\rfloor \eta_{\alpha\beta} = {}^* (\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma})$. Anholonomic indices are lowered by $o_{\alpha\beta} = e^i{}_{\alpha} e^j{}_{\beta} g_{ij}$,
mielke-geometrodynamics_F00814	163	■ 0.862	$o_{\alpha\beta} = e^i{}_{\alpha} e^j{}_{\beta} g_{ij}$	$o_{\alpha\beta} = e^i{}_{\alpha} e^j{}_{\beta} g_{ij}$	$\rfloor \eta_{\alpha\beta} = {}^* (\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma})$. Anholonomic indices are lowered by $o_{\alpha\beta} = e^i{}_{\alpha} e^j{}_{\beta} g_{ij}$,
mielke-geometrodynamics_F00815	163	■ 1.000	$o_{\alpha\beta}$	$o_{\alpha\beta}$	where $o_{\alpha\beta}$ denotes the Minkowski metric.
mielke-geometrodynamics_F00816	163	■ 1.000	$\Phi = \frac{1}{p!} \Phi_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}$	$\Phi = \frac{1}{p!} \Phi_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}$	dimensional manifold, the Hodge dual of a p -form $\Phi = \frac{1}{p!} \Phi_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}$

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mielke-geometrodynamics_F00817	163	0.769	$\{ \}^{\{ * \}} \backslash \Phi = (-1)^{p(4-p) + \text{sig}} \Phi$	$**\Phi = (-1)^{p(4-p) + \text{sig}} \Phi$	so that it is almost involutive, i.e., $**\Phi = (-1)^{p(4-p) + \text{sig}} \Phi$.
mielke-geometrodynamics_F00818	163	0.998	$\left\{ \mathbf{1}_4, \gamma_\alpha, \sigma_{\alpha\beta}, \gamma_5, \gamma_5 \gamma_\alpha \right\}$	$\{ \mathbf{1}_4, \gamma_\alpha, \sigma_{\alpha\beta}, \gamma_5, \gamma_5 \gamma_\alpha \}$	In the familiar Pauli representation, the 16 matrices $\{ \mathbf{1}_4, \gamma_\alpha, \sigma_{\alpha\beta}, \gamma_5, \gamma_5 \gamma_\alpha \}$, where
mielke-geometrodynamics_F00819	163	0.994	$\sigma_{\alpha\beta} := \frac{i}{2} \left(\gamma_\alpha \gamma_\beta - \gamma_\beta \gamma_\alpha \right)$	$\sigma_{\alpha\beta} := \frac{i}{2} (\gamma_\alpha \gamma_\beta - \gamma_\beta \gamma_\alpha)$	$\sigma_{\alpha\beta} := \frac{i}{2} (\gamma_\alpha \gamma_\beta - \gamma_\beta \gamma_\alpha)$ are the Lorentz generators, and $\gamma_5 = -i \gamma_0 \gamma_1 \gamma_2 \gamma_3$ with
mielke-geometrodynamics_F00820	163	0.994	$\gamma_5 = -i \gamma_{\hat{0}} \gamma_{\hat{1}} \gamma_{\hat{2}} \gamma_{\hat{3}}$	$\gamma_5 = -i \gamma_{\hat{0}} \gamma_{\hat{1}} \gamma_{\hat{2}} \gamma_{\hat{3}}$	$\sigma_{\alpha\beta} := \frac{i}{2} (\gamma_\alpha \gamma_\beta - \gamma_\beta \gamma_\alpha)$ are the Lorentz generators, and $\gamma_5 = -i \gamma_0 \gamma_1 \gamma_2 \gamma_3$ with
mielke-geometrodynamics_F00821	163	0.736	$\gamma_5^2 = (-1)^{\text{sig}+1} \mathbf{1}_4$	$\gamma_5^2 = (-1)^{\text{sig}+1} \mathbf{1}_4$	$\gamma_5^2 = (-1)^{\text{sig}+1} \mathbf{1}_4$ constitute a basis of the <i>Clifford algebra</i> in four dimensions with
mielke-geometrodynamics_F00822	163	1.000	$\text{Tr}(\gamma_\alpha \gamma_\beta) = 4 o_{\alpha\beta}$	$\text{Tr}(\gamma_\alpha \gamma_\beta) = 4 o_{\alpha\beta}$	They are normalized by the traces $\text{Tr}(\gamma_\alpha \gamma_\beta) = 4 o_{\alpha\beta}$ and $\text{Tr}(\sigma_{\alpha\beta} \sigma^{\gamma\delta}) = 8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$,
mielke-geometrodynamics_F00823	163	1.000	$\text{Tr}(\sigma_{\alpha\beta} \sigma^{\gamma\delta}) = 8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$	$\text{Tr}(\sigma_{\alpha\beta} \sigma^{\gamma\delta}) = 8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$	They are normalized by the traces $\text{Tr}(\gamma_\alpha \gamma_\beta) = 4 o_{\alpha\beta}$ and $\text{Tr}(\sigma_{\alpha\beta} \sigma^{\gamma\delta}) = 8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$,
mielke-geometrodynamics_F00824	163	1.000	$[\alpha\beta] = \frac{1}{2} (\alpha\beta - \beta\alpha)$	$[\alpha\beta] = \frac{1}{2} (\alpha\beta - \beta\alpha)$	where $[\alpha\beta] = \frac{1}{2} (\alpha\beta - \beta\alpha)$ denotes the antisymmetrization of indices. For the unit
mielke-geometrodynamics_F00825	164	0.988	Lie	Lie	whereas for the <i>Lie</i> (or right) dual $\sigma_{\alpha\beta}^{(*)} := \frac{1}{2} \sigma^{\gamma\delta} \eta_{\alpha\beta\gamma\delta} = i \gamma_5 \sigma_{\alpha\beta}$, we obtain the asso-
mielke-geometrodynamics_F00826	164	0.988	$\sigma_{\alpha\beta}^{(*)} := \frac{1}{2} \sigma^{\gamma\delta} \eta_{\alpha\beta\gamma\delta} = i \gamma_5 \sigma_{\alpha\beta}$	$\sigma_{\alpha\beta}^{(*)} := \frac{1}{2} \sigma^{\gamma\delta} \eta_{\alpha\beta\gamma\delta} = i \gamma_5 \sigma_{\alpha\beta}$	whereas for the <i>Lie</i> (or right) dual $\sigma_{\alpha\beta}^{(*)} := \frac{1}{2} \sigma^{\gamma\delta} \eta_{\alpha\beta\gamma\delta} = i \gamma_5 \sigma_{\alpha\beta}$, we obtain the asso-
mielke-geometrodynamics_F00827	164	0.998	$\sigma^{(*)}$	$\sigma^{(*)}$	$\sigma^{(*)}$ solely by multiplication by γ_5 , which here is regarded as just an anticommuting
mielke-geometrodynamics_F00828	164	0.998	γ_5	γ_5	$\sigma^{(*)}$ solely by multiplication by γ_5 , which here is regarded as just an anticommuting
mielke-geometrodynamics_F00829	164	0.995	$12 i^* \gamma$	$12 i^* \gamma$	relates this to $12 i^* \gamma$, the Hodge dual ³ for the basis of Clifforms has been recovered.
mielke-geometrodynamics_F00830	164	1.000	$P_\pm = \frac{1}{2} (1 \pm \gamma_5)$	$P_\pm = \frac{1}{2} (1 \pm \gamma_5)$	representation involves explicitly the <i>chirality projector</i> $P_\pm = \frac{1}{2} (1 \pm \gamma_5)$, obeying
mielke-geometrodynamics_F00831	164	1.000	$P_\pm P_\pm = P_\pm$	$P_\pm P_\pm = P_\pm$	$P_\pm P_\pm = P_\pm$.
mielke-geometrodynamics_F00832	164	0.981	$\gamma \rightarrow \tilde{\gamma} = e^\varphi \gamma$	$\gamma \rightarrow \tilde{\gamma} = e^\varphi \gamma$	(ATIYAH et al. 1978) under the Weyl rescaling $\gamma \rightarrow \tilde{\gamma} = e^\varphi \gamma$, where φ can be viewed
mielke-geometrodynamics_F00833	164	1.000	$\{ \}^{\{ \# \}}$	$\#$	$\#$ on <i>arbitrary</i> two-forms allows us to <i>reconstruct</i> a spacetime metric h , which is,
mielke-geometrodynamics_F00834	164	1.000	h	h	$\#$ on <i>arbitrary</i> two-forms allows us to <i>reconstruct</i> a spacetime metric h , which is,
mielke-geometrodynamics_F00835	164	0.524	$\{ \}^{\{ * \}} = -1$	$** = -1$	³ For Minkowski signature (sign = 1), the Hodge dual satisfies $** = -1$, and therefore, i^* is used
mielke-geometrodynamics_F00836	164	0.524	$i^{\{ * \}}$	i^*	³ For Minkowski signature (sign = 1), the Hodge dual satisfies $** = -1$, and therefore, i^* is used
mielke-geometrodynamics_F00837	165	1.000	$\Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta}$	$\Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta}$	of gravity. In terms of the Clifford-algebra-valued <i>connection</i> $\Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta}$, the
mielke-geometrodynamics_F00838	165	1.000	$\overline{SO}_0(1,3) \cong SL(2, C)$	$\overline{SO}_0(1,3) \cong SL(2, C)$	$\overline{SO}_0(1,3) \cong SL(2, C)$ -covariant exterior derivative
mielke-geometrodynamics_F00839	165	0.975	$[\Psi, \Phi] := \Psi \wedge \Phi - (-1)^{pq}$	$[\Psi, \Phi] := \Psi \wedge \Phi - (-1)^{pq}$	of p-forms employs the algebra-valued <i>form commutator</i> $[\Psi, \Phi] := \Psi \wedge \Phi - (-1)^{pq}$
mielke-geometrodynamics_F00840	165	0.755	$\Phi \wedge \Psi$	$\Phi \wedge \Psi$	$\Phi \wedge \Psi$.
mielke-geometrodynamics_F00841	165	1.000	$K := \frac{i}{4} K^{\alpha\beta} \sigma_{\alpha\beta}$	$K := \frac{i}{4} K^{\alpha\beta} \sigma_{\alpha\beta}$	involves $K := \frac{i}{4} K^{\alpha\beta} \sigma_{\alpha\beta}$, the contortion one-form.

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mielke-geometrodynamics_F00842	166	■ 1.000	$\Theta^{(2)}$	$\Theta^{(2)} :=$	The torsion two-form can be irreducibly decomposed into the trace part $\Theta^{(2)} :=$
mielke-geometrodynamics_F00843	166	■ 1.000	$\frac{1}{3} \gamma \wedge T$	$\frac{1}{3} \gamma \wedge T$	$\frac{1}{3} \gamma \wedge T$, the axial torsion $\Theta^{(3)} := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$, and the tensor torsion $\Theta^{(1)} :=$
mielke-geometrodynamics_F00844	166	■ 1.000	$\Theta^{(3)} := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$	$\Theta^{(3)} := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$	$\frac{1}{3} \gamma \wedge T$, the axial torsion $\Theta^{(3)} := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$, and the tensor torsion $\Theta^{(1)} :=$
mielke-geometrodynamics_F00845	166	■ 1.000	$\Theta^{(1)}$	$\Theta^{(1)} :=$	$\frac{1}{3} \gamma \wedge T$, the axial torsion $\Theta^{(3)} := -\frac{1}{3} * (\gamma \wedge \mathcal{A})$, and the tensor torsion $\Theta^{(1)} :=$
mielke-geometrodynamics_F00846	166	■ 1.000	$\Theta - \Theta^{(2)} - \Theta^{(3)}$	$\Theta - \Theta^{(2)} - \Theta^{(3)}$	$\Theta - \Theta^{(2)} - \Theta^{(3)}$, where the one-forms of the trace and axial vector torsion, respec-
mielke-geometrodynamics_F00847	166	■ 0.984	$\widetilde{\mathcal{A}} = \mathcal{A}$	$\widetilde{\mathcal{A}} = \mathcal{A}$	Under a Weyl rescaling, the axial torsion remains invariant, i.e., $\widetilde{\mathcal{A}} = \mathcal{A}$. Both
mielke-geometrodynamics_F00848	167	■ 1.000	$D\Omega^{g(\star)} \equiv 0$	$D\Omega^{g(\star)} \equiv 0$	i.e., $D\Omega^{g(\star)} \equiv 0$, provided that $D\eta_{\alpha\beta\gamma\delta} = 0$ holds as in spaces with vanishing Weyl
mielke-geometrodynamics_F00849	167	■ 1.000	$D\eta_{\alpha\beta\gamma\delta} = 0$	$D\eta_{\alpha\beta\gamma\delta} = 0$	i.e., $D\Omega^{g(\star)} \equiv 0$, provided that $D\eta_{\alpha\beta\gamma\delta} = 0$ holds as in spaces with vanishing Weyl
mielke-geometrodynamics_F00850	167	■ 0.995	$\mathcal{A} := *(\vartheta_\alpha \wedge T^\alpha) = \mathcal{A}_i dx^i$	$\mathcal{A} := *(\vartheta_\alpha \wedge T^\alpha) = \mathcal{A}_i dx^i$	where $\mathcal{A} := *(\vartheta_\alpha \wedge T^\alpha) = \mathcal{A}_i dx^i$ is the <i>axial torsion</i> one-form. The translational
mielke-geometrodynamics_F00851	168	■ 0.471	$R^{[\cdot]} := (-1)^{\text{sig} + 1} * (R^{[\cdot]\beta} \wedge \eta_{\beta\alpha})$	$R^{[\cdot]} := (-1)^{\text{sig} + 1} * (R^{[\cdot]\beta} \wedge \eta_{\beta\alpha})$	vature scalar $R^{[\cdot]} := (-1)^{\text{sig} + 1} * (R^{[\cdot]\beta} \wedge \eta_{\beta\alpha})$ and the axial torsion piece $d\mathcal{A} \wedge d\mathcal{A}$
mielke-geometrodynamics_F00852	168	■ 0.995	L_{SKY}	L_{SKY}	Lagrangian L_{SKY} as well as a Ricci-squared term and a scalar-curvature-squared
mielke-geometrodynamics_F00853	168	■ 0.827	$\text{Ric}_{\alpha\beta} := (-1)^{\text{sig}} * (R_{(\alpha}{}^\delta \wedge \eta_{\delta \beta)})$	$\text{Ric}_{\alpha\beta} := (-1)^{\text{sig}} * (R_{(\alpha}{}^\delta \wedge \eta_{\delta \beta)})$	$\text{Ric}_{\alpha\beta} := (-1)^{\text{sig}} * (R_{(\alpha}{}^\delta \wedge \eta_{\delta \beta)})$, is also known as the Gauss–Bonnet term.
mielke-geometrodynamics_F00854	169	■ 1.000	$G_\alpha := R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} / 2$	$G_\alpha := R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} / 2$	$G_\alpha := R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} / 2$ and the curvature scalar R . The latter is the zero-form
mielke-geometrodynamics_F00855	169	■ 0.894	$G^{\{\cdot\}} = G_\alpha^{\{\cdot\}\beta} \eta_\beta \gamma^\alpha$	$G^{\{\cdot\}} = G_\alpha^{\{\cdot\}\beta} \eta_\beta \gamma^\alpha$	decomposes into the Einstein three-form $G^{[\cdot]} = G_\alpha^{[\cdot]\beta} \eta_\beta \gamma^\alpha$ with respect to the
mielke-geometrodynamics_F00856	170	■ 0.983	2+1	2 + 1	Feynman RP (1981) The qualitative behavior of Yang-Mills theory in 2+1 dimensions. Nuclear
mielke-geometrodynamics_F00857	171	■ 0.998	$\mathfrak{m} \rightarrow 0$	$\mathfrak{m} \rightarrow 0$	Kogan II, Mouslopoulos S, Papazoglou A (2001) The $\mathfrak{m} \rightarrow 0$ limit for massive graviton in dS ₄ and
mielke-geometrodynamics_F00858	171	■ 0.998	dS_4	dS_4	Kogan II, Mouslopoulos S, Papazoglou A (2001) The $\mathfrak{m} \rightarrow 0$ limit for massive graviton in dS ₄ and
mielke-geometrodynamics_F00859	171	■ 0.999	AdS_4	AdS_4	AdS ₄ . How to circumvent the van Dam-Veltman-Zakharov discontinuity. Phys Lett B 503(1):173–
mielke-geometrodynamics_F00860	171	■ 0.878	5, $\mathbb{R}$	5, $\mathbb{R}$	Mielke EW (2012) Einstein-Weyl gravity from a topological SL(5, $\mathbb{R}$) gauge invariant action. Adv
mielke-geometrodynamics_F00861	172	■ 1.000	$R + R^2 + Q^2$	$R + R^2 + Q^2$	Zhytnikov V (1994) Wavelike exact solutions of $R + R^2 + Q^2$ gravity. J Math Phys 35(11):6001–
mielke-geometrodynamics_F00862	173	■ 0.999	(2014, 2006)	(2014, 2006)	“with flying colors”; see WILL (2014, 2006) for recent reviews. However, Einstein’s
mielke-geometrodynamics_F00863	173	■ 0.999	$\lambda_{\text{DE}} = (\hbar c / \rho_{\text{DE}})^{1/4} \simeq 85 \mu \text{ m}$	$\lambda_{\text{DE}} = (\hbar c / \rho_{\text{DE}})^{1/4} \simeq 85 \mu \text{ m}$	cal dark energy (DE) scale of $\lambda_{\text{DE}} = (\hbar c / \rho_{\text{DE}})^{1/4} \simeq 85 \mu \text{ m}$. Thus this “window” of
mielke-geometrodynamics_F00864	174	■ 1.000	$[\hbar] = [p][q]$	$[\hbar] = [p][q]$	dimensional transmutation due to the physical dimension $[\hbar] = [p][q]$ of Planck’s
mielke-geometrodynamics_F00865	174	■ 1.000	$\hbar$	$\hbar$	(reduced) constant $\hbar$.
mielke-geometrodynamics_F00866	174	■ 1.000	$C := \frac{i}{4} C^{\alpha\beta} \sigma_{\alpha\beta}$	$C := \frac{i}{4} C^{\alpha\beta} \sigma_{\alpha\beta}$	ghosts, into operator transformations. Let $C := \frac{i}{4} C^{\alpha\beta} \sigma_{\alpha\beta}$ denote the zero-form of the
mielke-geometrodynamics_F00867	174	■ 0.998	$\Psi := \frac{i}{4} \Psi_j^{\alpha\beta} \sigma_{\alpha\beta} dx^j$	$\Psi := \frac{i}{4} \Psi_j^{\alpha\beta} \sigma_{\alpha\beta} dx^j$	$\Psi := \frac{i}{4} \Psi_j^{\alpha\beta} \sigma_{\alpha\beta} dx^j$ the topological ghost one-form, and $\Phi := \frac{i}{4} \Phi^{\alpha\beta} \sigma_{\alpha\beta}$ the correspond-

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mielke-geometrodynamics_F00868	174	■ 0.998	$\Phi:=\frac{i}{4}\Phi^{\alpha\beta}\sigma_{\alpha\beta}$	$\Phi := \frac{i}{4}\Phi^{\alpha\beta}\sigma_{\alpha\beta}$	$\Psi:=\frac{i}{4}\Psi_i^{\alpha\beta}\sigma_{\alpha\beta}dx^j$ the topological ghost one-form, and $\Phi:=\frac{i}{4}\Phi^{\alpha\beta}\sigma_{\alpha\beta}$ the correspond-
mielke-geometrodynamics_F00869	174	■ 1.000	$\sigma_{\alpha\beta}$	$\sigma_{\alpha\beta}$	of the generator $\sigma_{\alpha\beta}$ of the linear (or Lorentz) group.
mielke-geometrodynamics_F00870	175	■ 1.000	Γ	Γ	This is consistent with the interpretation of Γ and Ω^g as connection one-form and
mielke-geometrodynamics_F00871	175	■ 1.000	Ω^g	Ω^g	This is consistent with the interpretation of Γ and Ω^g as connection one-form and
mielke-geometrodynamics_F00872	175	■ 1.000	C, Ψ	C, Ψ	number of degrees of freedom carried by the ghosts C, Ψ , and Φ is equal to that of
mielke-geometrodynamics_F00873	175	■ 1.000	$s^2=0$	$s^2 = 0$	variables, i.e., $s^2 = 0$.
mielke-geometrodynamics_F00874	175	■ 0.948	$^{\{2\}}$	2	(1988) by introducing the <i>graded</i> ² connection and curvature forms
mielke-geometrodynamics_F00875	175	■ 1.000	$s\,s=0$	$ss = 0$	Moreover, a straightforward proof of the nilpotency $ss = 0$ of the BRST operator
mielke-geometrodynamics_F00876	175	■ 1.000	$(d\oplus s)(d\oplus s)\equiv 0$	$(d\oplus s)(d\oplus s)\equiv 0$	s follows simply from $(d\oplus s)(d\oplus s)\equiv 0$ as a result of the graded Bianchi identity
mielke-geometrodynamics_F00877	175	■ 1.000	$[s,d]:=s\,d+d\,s=0$	$[s,d]:=sd+ds=0$	(8.2.3), the anticommutation of the graded commutator $[s,d]:=sd+ds=0$, and
mielke-geometrodynamics_F00878	175	■ 0.615	$\bar{C}, \bar{\chi}$	$\bar{C}, \bar{\chi}$	ism, where the Lorentz-algebra-valued antighosts $\bar{C}, \bar{\chi}$, and $\bar{\Phi}$ obey the following
mielke-geometrodynamics_F00879	175	■ 0.615	$\bar{\Phi}$	$\bar{\Phi}$	ism, where the Lorentz-algebra-valued antighosts $\bar{C}, \bar{\chi}$, and $\bar{\Phi}$ obey the following
mielke-geometrodynamics_F00880	175	■ 1.000	$[\Psi,\Phi]:=\Psi\wedge\Phi-$	$[\Psi,\Phi]:=\Psi\wedge\Phi-$	ghost number g , such that the graded commutator is more generally defined by $[\Psi,\Phi]:=\Psi\wedge\Phi-$
mielke-geometrodynamics_F00881	175	■ 0.510	$(-1)^{\left(p_1+g_1\right)\left(p_2+g_2\right)}\Phi\wedge\Psi$	$(-1)^{(p_1+g_1)(p_2+g_2)}\Phi\wedge\Psi$	$(-1)^{(p_1+g_1)(p_2+g_2)}\Phi\wedge\Psi$, (CHANG & SOO 1992).
mielke-geometrodynamics_F00882	176	■ 1.000	$\bar{\eta}$	$\bar{\eta}$	By construction, s is nilpotent for the antighosts, since the Lagrange multiplier $\bar{\eta}$,
mielke-geometrodynamics_F00883	176	■ 1.000	$\mathbb{B}$	$\mathbb{B}$	the self-dual two-forms $\mathbb{B}$, and β are auxiliary fields introduced as trivial pairs.
mielke-geometrodynamics_F00884	176	■ 0.998	$\alpha=\alpha_i dx^i$	$\alpha=\alpha_i dx^i$	By introducing a BRST gauge field $\alpha=\alpha_i dx^i$ with ghost number -1 and a
mielke-geometrodynamics_F00885	176	■ 0.999	λ	λ	commuting ghost λ of α , one can promote (DE CARVALHO & BAULIEU 1992) the
mielke-geometrodynamics_F00886	176	■ 0.999	α	α	commuting ghost λ of α , one can promote (DE CARVALHO & BAULIEU 1992) the
mielke-geometrodynamics_F00887	176	■ 1.000	$C\rightarrow(\alpha+$	$C\rightarrow(\alpha+$	promotion, which likewise, can be generated via the field redefinitions $C\rightarrow(\alpha+$
mielke-geometrodynamics_F00888	176	■ 0.712	$\lambda)C, \Psi\rightarrow(\alpha+\lambda)\Psi$	$\lambda)C, \Psi\rightarrow(\alpha+\lambda)\Psi$	$\lambda)C, \Psi\rightarrow(\alpha+\lambda)\Psi$ as well as $\Phi\rightarrow(\alpha+\lambda)^2\Phi$ of the ghosts. Thus, local BRST
mielke-geometrodynamics_F00889	176	■ 0.712	$\Phi\rightarrow(\alpha+\lambda)^2\Phi$	$\Phi\rightarrow(\alpha+\lambda)^2\Phi$	$\lambda)C, \Psi\rightarrow(\alpha+\lambda)\Psi$ as well as $\Phi\rightarrow(\alpha+\lambda)^2\Phi$ of the ghosts. Thus, local BRST
mielke-geometrodynamics_F00890	176	■ 1.000	$\gamma=\vartheta^\alpha\gamma_\alpha$	$\gamma=\vartheta^\alpha\gamma_\alpha$	In the affine or (broken) Poincaré gauge theory, the coframe $\gamma=\vartheta^\alpha\gamma_\alpha$ (locally
mielke-geometrodynamics_F00891	176	■ 1.000	$\widetilde{\Theta}:=\Theta\oplus\psi$	$\widetilde{\Theta}:=\Theta\oplus\psi$	respectively. In the graded torsion two-form $\widetilde{\Theta}:=\Theta\oplus\psi$, there occurs the transla-
mielke-geometrodynamics_F00892	176	■ 0.999	$\psi:=\psi_j^\alpha\gamma_\alpha dx^j$	$\psi:=\psi_j^\alpha\gamma_\alpha dx^j$	tional ‘‘Clifford’’ $\psi:=\psi_j^\alpha\gamma_\alpha dx^j$ of the corresponding topological ghost.
mielke-geometrodynamics_F00893	177	■ 0.885	$s\rightarrow\widetilde{s}=s+\mathbb{L}_\zeta$	$s\rightarrow\widetilde{s}=s+\mathbb{L}_\zeta$	alizing the BRST transformations s via $s\rightarrow\widetilde{s}=s+\mathbb{L}_\zeta$ involving the covariant Lie
mielke-geometrodynamics_F00894	177	■ 0.566	$\left.\left.\mathrm{L}_{\zeta}:=\zeta\lrcorner D-D\zeta\right\right\}$	$\mathbb{L}_\zeta:=\zeta\lrcorner D-D\zeta]$	derivative $\mathbb{L}_\zeta:=\zeta\lrcorner D-D\zeta]$ along an anticommuting ghost vector field $\zeta=\zeta^i\partial_i$.

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mielke-geometrodynamics_F00895	177	■ 0.566	$\backslash\mathrm{zeta}=\backslash\mathrm{zeta}^{\mathrm{i}}\ \backslash\mathrm{partial}_{\mathrm{i}}$	$\zeta = \zeta^i \partial_i$	derivative $\mathbb{L}_\zeta := \zeta \rfloor D - D\zeta \rfloor$ along an anticommuting ghost vector field $\zeta = \zeta^i \partial_i$.
mielke-geometrodynamics_F00896	177	■ 0.339	$\backslash\mathrm{mathfrak{L}}_{\backslash\mathrm{zeta}}$	$\mathfrak{L}_\zeta$	of $\mathbb{L}_\zeta$, since ζ has ghost number one. Its BRST transformation is $s\zeta = \phi + \mathbb{L}_\zeta \zeta$, such
mielke-geometrodynamics_F00897	177	■ 0.339	$\backslash\mathrm{zeta}$	ζ	of $\mathbb{L}_\zeta$, since ζ has ghost number one. Its BRST transformation is $s\zeta = \phi + \mathbb{L}_\zeta \zeta$, such
mielke-geometrodynamics_F00898	177	■ 0.339	$\mathrm{s}\ \backslash\mathrm{zeta}=\backslash\mathrm{phi}+\backslash\mathrm{mathfrak{L}}_{\backslash\mathrm{zeta}}\ \backslash\mathrm{zeta}$	$s\zeta = \phi + \mathfrak{L}_\zeta \zeta$	of $\mathbb{L}_\zeta$, since ζ has ghost number one. Its BRST transformation is $s\zeta = \phi + \mathbb{L}_\zeta \zeta$, such
mielke-geometrodynamics_F00899	177	■ 1.000	$\mathrm{p}+1$	$p + 1$	Its action on a p-form, reduces to a finite series with up to $p + 1$ terms. In effect, the
mielke-geometrodynamics_F00900	177	■ 0.497	$\backslash\mathrm{exp}\ (\backslash\mathrm{zeta}\backslash\mathrm{rfloor})\ \backslash\mathrm{widetilde}\{\backslash\mathrm{Omega}\}^{\mathrm{g}}$	$\exp(\zeta \rfloor) \widetilde{\Omega}^g$	by $\exp(\zeta \rfloor) \widetilde{\Omega}^g$ and $\exp(\zeta \rfloor)(\widetilde{\Theta} \oplus \phi)$, respectively, where $\phi := \phi^\alpha \gamma_\alpha$ are the ghosts of
mielke-geometrodynamics_F00901	177	■ 0.497	$\backslash\mathrm{left}.\backslash\mathrm{exp}\ (\backslash\mathrm{zeta}\backslash\mathrm{rfloor}\backslash\mathrm{right})(\backslash\mathrm{widetilde}\{\backslash\mathrm{Theta}\}\ \backslash\mathrm{oplus}\ \backslash\mathrm{phi})$	$\exp(\zeta \rfloor) (\widetilde{\Theta} \oplus \phi)$	by $\exp(\zeta \rfloor) \widetilde{\Omega}^g$ and $\exp(\zeta \rfloor)(\widetilde{\Theta} \oplus \phi)$, respectively, where $\phi := \phi^\alpha \gamma_\alpha$ are the ghosts of
mielke-geometrodynamics_F00902	177	■ 0.497	$\backslash\mathrm{phi}:=\backslash\mathrm{phi}^{\mathrm{a}}\backslash\mathrm{alpha}\ \backslash\mathrm{gamma}_{\backslash\mathrm{alpha}}$	$\phi := \phi^\alpha \gamma_\alpha$	by $\exp(\zeta \rfloor) \widetilde{\Omega}^g$ and $\exp(\zeta \rfloor)(\widetilde{\Theta} \oplus \phi)$, respectively, where $\phi := \phi^\alpha \gamma_\alpha$ are the ghosts of
mielke-geometrodynamics_F00903	177	■ 1.000	d	d	(BAULIEU 1987), the exterior derivative d suffers from the similarity transformation
mielke-geometrodynamics_F00904	177	■ 1.000	$\backslash\mathrm{Gamma}^{\mathrm{\{(\backslash\mathrm{mathrm}\{T\})\}}}=(\backslash\mathrm{gamma}-\mathrm{D}\ \backslash\mathrm{xi})\ /\ \backslash\mathrm{ell}$	$\Gamma^{(\mathrm{T})} = (\gamma - D\xi)/\ell$	<i>translational</i> connection $\Gamma^{(\mathrm{T})} = (\gamma - D\xi)/\ell$ and corresponding curvature $\Omega^{(\mathrm{T})} =$
mielke-geometrodynamics_F00905	177	■ 1.000	$\backslash\mathrm{Omega}^{\mathrm{\{(\backslash\mathrm{mathrm}\{T\})\}}} =$	$\Omega^{(\mathrm{T})} =$	<i>translational</i> connection $\Gamma^{(\mathrm{T})} = (\gamma - D\xi)/\ell$ and corresponding curvature $\Omega^{(\mathrm{T})} =$
mielke-geometrodynamics_F00906	177	■ 1.000	$\mathrm{D}\ \backslash\mathrm{Gamma}^{\mathrm{\{(\backslash\mathrm{mathrm}\{T\})\}}}=(\backslash\mathrm{Theta}-\mathrm{D}\ \mathrm{D}\ \backslash\mathrm{xi})\ /\ \backslash\mathrm{ell}$	$D\Gamma^{(\mathrm{T})} = (\Theta - DD\xi)/\ell$	$D\Gamma^{(\mathrm{T})} = (\Theta - DD\xi)/\ell$. Then the translational connection $\Gamma^{(\mathrm{T})}$ should be graded
mielke-geometrodynamics_F00907	177	■ 1.000	$\backslash\mathrm{Gamma}^{\mathrm{\{(\backslash\mathrm{mathrm}\{T\})\}}}$	$\Gamma^{(\mathrm{T})}$	$D\Gamma^{(\mathrm{T})} = (\Theta - DD\xi)/\ell$. Then the translational connection $\Gamma^{(\mathrm{T})}$ should be graded
mielke-geometrodynamics_F00908	177	■ 1.000	$\backslash\mathrm{gamma}\ \backslash\mathrm{rightarrow}\ \backslash\mathrm{widetilde}\{\backslash\mathrm{gamma}\}:=\backslash\mathrm{ell}\backslash\mathrm{left}(\backslash\mathrm{Gamma}^{\mathrm{\{(\backslash\mathrm{mathrm}\{T\})\}}}\ \backslash\mathrm{oplus}\ \mathrm{c}\backslash\mathrm{right})$	$\gamma \rightarrow \widetilde{\gamma} := \ell \left(\Gamma^{(\mathrm{T})} \oplus c \right)$	as well, e.g., by the substitution $\gamma \rightarrow \widetilde{\gamma} := \ell(\Gamma^{(\mathrm{T})} \oplus c)$ in the first structure equation.
mielke-geometrodynamics_F00909	177	■ 1.000	$\backslash\mathrm{xi}:=\backslash\mathrm{xi}^{\mathrm{a}}\backslash\mathrm{alpha}\ \backslash\mathrm{gamma}_{\backslash\mathrm{alpha}}$	$\xi := \xi^\alpha \gamma_\alpha$	The “quartet” of scalars $\xi := \xi^\alpha \gamma_\alpha$ corresponds to the “generalized radius vectors”
mielke-geometrodynamics_F00910	177	■ 1.000	$\mathrm{A}(4,\ \backslash\mathrm{mathbb}\{\mathrm{R}\})\ /\ \mathrm{G}\ \mathrm{L}(4,\ \backslash\mathrm{mathbb}\{\mathrm{R}\})\ \backslash\mathrm{approx}\ \backslash\mathrm{mathbb}\{\mathrm{R}\}^{\mathrm{\{4\}}}$	$A(4, \mathbb{R})/GL(4, \mathbb{R}) \approx \mathbb{R}^4$	of Cartan and lives on the coset space $A(4, \mathbb{R})/GL(4, \mathbb{R}) \approx \mathbb{R}^4$ of the affine group.
mielke-geometrodynamics_F00911	178	■ 1.000	$\backslash\mathrm{Gamma}^{\mathrm{\{(\backslash\mathrm{mathrm}\{T\})\}}}\ \backslash\mathrm{stackrel{*}{=}}\ 0$	$\Gamma^{(\mathrm{T})} \stackrel{*}{=} 0$	naturally arises in the affine gauge theory after, locally, the gauge $\Gamma^{(\mathrm{T})} \stackrel{*}{=} 0$ is imposed
mielke-geometrodynamics_F00912	178	■ 0.995	$\mathrm{S}\ \mathrm{U}(\mathrm{N})$	$SU(N)$	and replace the internal $SU(N)$ group by the linear group $SL(4, \mathbb{R})$ of the tangent
mielke-geometrodynamics_F00913	178	■ 0.995	$\mathrm{S}\ \mathrm{L}(4,\ \backslash\mathrm{mathbb}\{\mathrm{R}\})$	$SL(4, \mathbb{R})$	and replace the internal $SU(N)$ group by the linear group $SL(4, \mathbb{R})$ of the tangent
mielke-geometrodynamics_F00914	178	■ 0.899	$\backslash\mathrm{mathrm}\{\mathrm{SO}\}(1,3)$	$\mathrm{SO}(1,3)$	space embracing the Lorentz group $\mathrm{SO}(1,3)$ (or $\mathrm{SO}(4)$ in Euclidean space with
mielke-geometrodynamics_F00915	178	■ 0.467	$=0$	$= 0$	signature $\mathrm{sig} = 0$) as subgroup. Then one starts from the gravitational Pontryagin
mielke-geometrodynamics_F00916	178	■ 0.999	$\mathrm{s}\backslash\backslash\mathrm{cdots}\backslash\backslash$	$s\{\cdots\}$	This purely topological action can be amended by any s -exact four-form $s\{\cdots\}$,
mielke-geometrodynamics_F00917	178	■ 1.000	$\mathrm{d}^{\mathrm{\{*\}}}\ \backslash\mathrm{Gamma}=0,\ \mathrm{D}^{\mathrm{\{*\}}}\ \backslash\mathrm{Psi}=0$	$d^*\Gamma = 0, D^*\Psi = 0$	(1988), we may choose the Lorentz-type conditions $d^*\Gamma = 0, D^*\Psi = 0$ on the
mielke-geometrodynamics_F00918	178	■ 1.000	$\backslash\mathrm{Omega}^{\mathrm{\{g(\ \backslash\mathrm{pm})\}}} = 0$	$\Omega^{g(\pm)} = 0$	self-duality condition $\Omega^{g(\pm)} = 0$ on the curvature as <i>gauge constraints</i> consistently
mielke-geometrodynamics_F00919	178	■ 1.000	$\mathrm{C}\ \mathrm{P}$	CP	³ For the CP -invariant Euler term, a similar result would hold, i.e., $sL_{\mathrm{Euler}} =$
mielke-geometrodynamics_F00920	178	■ 1.000	$\mathrm{s}\ \mathrm{L}_{\backslash\mathrm{text}\ \{\mathrm{Euler}\}} =$	$sL_{\mathrm{Euler}} =$	³ For the CP -invariant Euler term, a similar result would hold, i.e., $sL_{\mathrm{Euler}} =$
mielke-geometrodynamics_F00921	178	■ 0.998	$(-1)^{\mathrm{\{ \backslash\mathrm{text}\ \{\mathrm{sig}\}+1\}}\ \mathrm{d}\backslash\mathrm{left}(\backslash\mathrm{Psi}^{\mathrm{\{ \backslash\mathrm{alpha}\ \backslash\mathrm{beta}\}}}\ \backslash\mathrm{wedge}\ \mathrm{R}_{\backslash\mathrm{alpha}\ \backslash\mathrm{beta}}^{\mathrm{\{ \backslash\mathrm{star}\}}}\backslash\mathrm{right})$	$(-1)^{\mathrm{sig}+1} d \left(\Psi^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \right)$	$(-1)^{\mathrm{sig}+1} d \left(\Psi^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \right)$. However, the latter is only partially metric-free, since it involves

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mielke-geometrodynamics_F00922	179	■ 1.000	$\backslash\mathrm{bar}\{\backslash\mathrm{chi}\}$	$\bar{\chi}$	with the four-form character of a Lagrangian, the auxiliary fields $\bar{\chi}$ and β have
mielke-geometrodynamics_F00923	179	■ 0.993	$\backslash\mathrm{widetilde}\{\Psi\}_{\mathrm{F}}:=\backslash\mathrm{bar}\{\mathrm{C}\}_{\alpha\beta}\backslash\mathrm{left}\{\mathrm{d}^*\backslash\mathrm{Gamma}^{\alpha\beta}+\frac{1}{2}\backslash\mathrm{B}^{\alpha\beta}\}\backslash\mathrm{mathbb{B}}^{\alpha\beta}\backslash\mathrm{right}\}$	$\tilde{\Psi}_{\mathrm{F}} := \bar{C}_{\alpha\beta} \left(d^* \Gamma^{\alpha\beta} + \frac{1}{2} * \mathbb{B}^{\alpha\beta} \right)$	$\tilde{\Psi}_{\mathrm{F}} := \bar{C}_{\alpha\beta} (d^* \Gamma^{\alpha\beta} + \frac{1}{2} * \mathbb{B}^{\alpha\beta})$.
mielke-geometrodynamics_F00924	179	■ 0.835	$\backslash\mathrm{cdots}\backslash$	$\{\cdots\}$	four-form $\{\cdots\}$ is able to induce a dependence on the metric that in is not present in
mielke-geometrodynamics_F00925	179	■ 0.998	$\mathrm{d}^*\backslash\mathrm{Gamma}=\backslash\mathrm{mathbb{B}}$	$d^* \Gamma = * \mathbb{B}$	$d^* \Gamma = * \mathbb{B}$ on the linear connection. Moreover, for vanishing real <i>gauge parameter</i>
mielke-geometrodynamics_F00926	179	■ 1.000	$\backslash\mathrm{rho}=0$	$\rho = 0$	$\rho = 0$, the equation of motion for the auxiliary Lagrange-multiplier-type two-form
mielke-geometrodynamics_F00927	180	■ 1.000	$\backslash\mathrm{rho}=1$	$\rho = 1$	In the case of the choice $\rho = 1$ of the real gauge parameter, the two-form β is present
mielke-geometrodynamics_F00928	180	■ 1.000	$\mathrm{D}\backslash\mathrm{Omega}^g\equiv 0$	$D\Omega^g \equiv 0$	Due to the second Bianchi identity $D\Omega^g \equiv 0$, the dual field equation
mielke-geometrodynamics_F00929	180	■ 0.960	$\mathrm{T}_{\mu\nu}:=2^*\backslash\mathrm{left}\{\partial\mathrm{L}/\partial\mathrm{g}_{\mu\nu}\}\backslash\mathrm{right}\}=0$	$T_{\mu\nu} := 2^* (\partial L / \partial g_{\mu\nu}) = 0$	i.e., $T_{\mu\nu} := 2^* (\partial L / \partial g_{\mu\nu}) = 0$. In turn, this implies that the BRST quantization is
mielke-geometrodynamics_F00930	181	■ 0.999	$\backslash\mathrm{mathscr{A}}:=\backslash\mathrm{left}\{\backslash\mathrm{vartheta}_{\alpha}\backslash\mathrm{wedge}\backslash\mathrm{right}\}$	$\mathscr{A} := * (\vartheta_{\alpha} \wedge$	respect to the Riemannian curvature $R_{\alpha\beta}^{\parallel}$ and the <i>axial torsion</i> one-form $\mathscr{A} := * (\vartheta_{\alpha} \wedge$
mielke-geometrodynamics_F00931	181	■ 0.999	T^{α}	T^{α}	T^{α}) is rather well motivated by the <i>axial anomaly</i> in RC spacetime
mielke-geometrodynamics_F00932	181	■ 1.000	$\backslash\mathrm{gamma}:=\backslash\mathrm{vartheta}^{\alpha}\backslash\mathrm{gamma}_{\alpha}$	$\gamma := \vartheta^{\alpha} \gamma_{\alpha}$	In addition, the coframe $\gamma := \vartheta^{\alpha} \gamma_{\alpha}$ is constrained by the Hilbert–de Donder gauge
mielke-geometrodynamics_F00933	181	■ 1.000	$\mathrm{d}^*\backslash\mathrm{gamma}=0$	$d^* \gamma = 0$	condition $d^* \gamma = 0$ complying with the BRST algebra of translations.
mielke-geometrodynamics_F00934	181	■ 1.000	$\backslash\mathrm{widetilde}\{\mathrm{L}\}_{\mathrm{FP}}:=\mathrm{L}_{\mathrm{FP}}+\Delta\mathrm{L}_{\mathrm{FP}}$	$\tilde{L}_{\mathrm{FP}} := L_{\mathrm{FP}} + \Delta L_{\mathrm{FP}}$	Then for $\rho = 0$ there arises from $\tilde{L}_{\mathrm{FP}} := L_{\mathrm{FP}} + \Delta L_{\mathrm{FP}}$ the <i>modified double duality</i>
mielke-geometrodynamics_F00935	181	■ 1.000	i	i	on the curvature two-form Ω^g , which is real due to the extra imaginary unit i .
mielke-geometrodynamics_F00936	181	■ 0.692	$\backslash\mathrm{stackrel{\pm}{\Omega}}^g$	$\stackrel{\pm}{\Omega}^g$	only one sign for the dual curvature $\stackrel{\pm}{\Omega}^g$ leads to a self-consistent constraint. This
mielke-geometrodynamics_F00937	182	■ 0.811	$\backslash\mathrm{theta}=\pi/2$	$\theta = \pi/2$	of the curvature, which for $\theta = \pi/2$ specializes to the self-dual curvature $\stackrel{+}{\Omega}^g$, and for
mielke-geometrodynamics_F00938	182	■ 0.811	$\backslash\mathrm{stackrel{+}{\Omega}}^g$	$\stackrel{+}{\Omega}^g$	of the curvature, which for $\theta = \pi/2$ specializes to the self-dual curvature $\stackrel{+}{\Omega}^g$, and for
mielke-geometrodynamics_F00939	182	■ 1.000	$\backslash\mathrm{theta}=3\pi/2$	$\theta = 3\pi/2$	$\theta = 3\pi/2$ to the anti-self-dual curvature $\bar{\Omega}^g$. Since the “unit”-curvature two-form σ
mielke-geometrodynamics_F00940	182	■ 1.000	$\backslash\mathrm{bar}\{\backslash\mathrm{Omega}\}^g$	$\bar{\Omega}^g$	$\theta = 3\pi/2$ to the anti-self-dual curvature $\bar{\Omega}^g$. Since the “unit”-curvature two-form σ
mielke-geometrodynamics_F00941	182	■ 1.000	$\backslash\mathrm{theta}_{\mathrm{L}}=\backslash\mathrm{theta}_{\mathrm{T}}=0$	$\theta_{\mathrm{L}} = \theta_{\mathrm{T}} = 0$	a special case of the Ansatz (8.6.4) for the choice $\theta_{\mathrm{L}} = \theta_{\mathrm{T}} = 0$ and $\theta_{\mathrm{L}}^* = \mp(-1)^{\mathrm{sig}}$. Interestingly
mielke-geometrodynamics_F00942	182	■ 1.000	$\backslash\mathrm{theta}_{\mathrm{L}}^{\star}=\mp(-1)^{\mathrm{sig}}$	$\theta_{\mathrm{L}}^* = \mp(-1)^{\mathrm{sig}}$	a special case of the Ansatz (8.6.4) for the choice $\theta_{\mathrm{L}} = \theta_{\mathrm{T}} = 0$ and $\theta_{\mathrm{L}}^* = \mp(-1)^{\mathrm{sig}}$. Interestingly
mielke-geometrodynamics_F00943	183	■ 0.983	$\mathrm{L}_{\mathrm{qPG}}$	L_{qPG}	tion parts of the gravitational Lagrangian L_{qPG} . In the field momenta, each of the
mielke-geometrodynamics_F00944	183	■ 1.000	$a_{(M)}$	$a_{(M)}$	the Lagrangian with individual dimensionless weights $a_{(M)}$ and $b_{(N)}$, respectively.
mielke-geometrodynamics_F00945	183	■ 1.000	$b_{(N)}$	$b_{(N)}$	the Lagrangian with individual dimensionless weights $a_{(M)}$ and $b_{(N)}$, respectively.
mielke-geometrodynamics_F00946	183	■ 1.000	$\backslash\mathrm{theta}_{\mathrm{T}},\backslash\mathrm{theta}_{\mathrm{T}}^{\star}$	$\theta_{\mathrm{T}}, \theta_{\mathrm{T}}^*$	for the rotational field momenta (MIELKE 1992; ZHYTNIKOV 1994), where $\theta_{\mathrm{T}}, \theta_{\mathrm{T}}^*$,
mielke-geometrodynamics_F00947	183	■ 1.000	$\backslash\mathrm{theta}_{\mathrm{L}}^{\star}$	θ_{L}^*	θ_{L} , and θ_{L}^* are dimensionless constants. The constraint (8.5.3) corresponds to the

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mielke-geometrodynamics_F00948	183	■ 1.000	$a_{\{0\}}=0, \ b_{(N)}=1$	$a_0 = 0, b_{(N)} = 1$	SKY Lagrangian with $a_0 = 0, b_{(N)} = 1$, and the choice $\theta_L^* = \mp(-1)^{\text{sig}}$ as well as
mielke-geometrodynamics_F00949	183	■ 0.359	$d\ C_{\mathrm{TT}}^* := d(\vartheta^\alpha \wedge {}^*T_\alpha) / 2\ \ell^2$	$dC_{\mathrm{TT}}^* := d(\vartheta^\alpha \wedge {}^*T_\alpha) / 2\ell^2$	$dC_{\mathrm{TT}}^* := d(\vartheta^\alpha \wedge {}^*T_\alpha) / 2\ell^2$ inducing the teleparallelism equivalence, as well as the
mielke-geometrodynamics_F00950	183	■ 1.000	$\frac{1}{2} s\ \bar{\chi}^{\alpha\beta} \wedge (H_{\alpha\beta} + \partial L_\theta / \partial R^{\alpha\beta})$	$\frac{1}{2} s \bar{\chi}^{\alpha\beta} \wedge (H_{\alpha\beta} + \partial L_\theta / \partial R^{\alpha\beta})$	BRST gauge constraint $\frac{1}{2} s \bar{\chi}^{\alpha\beta} \wedge (H_{\alpha\beta} + \partial L_\theta / \partial R^{\alpha\beta})$ generalizing the corresponding
mielke-geometrodynamics_F00951	183	■ 0.680	$\widetilde{L}_{\mathrm{FP}}$	$\widetilde{L}_{\mathrm{FP}}$	terms in the Faddeev–Popov Lagrangian $\widetilde{L}_{\mathrm{FP}}$ modified by (8.5.1).
mielke-geometrodynamics_F00952	184	■ 1.000	D	D	The covariant derivative D of the “unit”-curvature pieces proportional to $\eta_{\alpha\beta}$ and
mielke-geometrodynamics_F00953	184	■ 1.000	$\vartheta_\alpha \wedge \vartheta_\beta$	$\vartheta_\alpha \wedge \vartheta_\beta$	$\vartheta_\alpha \wedge \vartheta_\beta$ in the DD Ansatz is responsible for the two additional torsion terms. For
mielke-geometrodynamics_F00954	184	■ 0.999	$\tau_{\alpha\beta}=0$	$\tau_{\alpha\beta} = 0$	spinless matter, i.e., $\tau_{\alpha\beta} = 0$, one can algebraically resolve (8.6.9) using Eq. (A.1.26)
mielke-geometrodynamics_F00955	184	■ 1.000	$H_\alpha(**)$	$H_\alpha(**)$	of HEHL et al. (1995) for the translational momenta $H_\alpha(**)$ constrained by double
mielke-geometrodynamics_F00956	185	■ 1.000	$H_{\alpha\beta}(**) \wedge \vartheta^\alpha \wedge \vartheta^\beta$	$H_{\alpha\beta}(**) \wedge \vartheta^\alpha \wedge \vartheta^\beta$	The contraction $H_{\alpha\beta}(**) \wedge \vartheta^\alpha \wedge \vartheta^\beta$ of the DD Ansatz (8.6.4) together with the linear
mielke-geometrodynamics_F00957	185	■ 1.000	$\theta_{\mathrm{L}} \neq 0$	$\theta_{\mathrm{L}} \neq 0$	expansion (8.6.3) implies, for $\theta_{\mathrm{L}} \neq 0$, that the pseudoscalar curvature vanishes for
mielke-geometrodynamics_F00958	185	■ 1.000	$R_{\alpha\beta} \wedge \vartheta^\alpha \wedge \vartheta^\beta = 0$	$R_{\alpha\beta} \wedge \vartheta^\alpha \wedge \vartheta^\beta = 0$	consistency, i.e., $R_{\alpha\beta} \wedge \vartheta^\alpha \wedge \vartheta^\beta = 0$. Then the scalar curvature
mielke-geometrodynamics_F00959	185	■ 1.000	$\theta_{\mathrm{L}} = 0$	$\theta_{\mathrm{L}} = 0$	⁵ There occurs an interesting modification in the case that $\theta_{\mathrm{L}} = 0$, since then the pseudoscalar cur-
mielke-geometrodynamics_F00960	185	■ 1.000	$\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_\alpha \wedge \vartheta_\beta / 4\ell^2$	$\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_\alpha \wedge \vartheta_\beta / 4\ell^2$	vature four-form $\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_\alpha \wedge \vartheta_\beta / 4\ell^2$ needs to be subtracted from (8.6.12), which would induce
mielke-geometrodynamics_F00961	186	■ 1.000	$T^\alpha \simeq 0$	$T^\alpha \simeq 0$	vanishing torsion $T^\alpha \simeq 0$ via Lagrange multipliers in the gravitational gauge field
mielke-geometrodynamics_F00962	186	■ 1.000	$T^\alpha = 0$	$T^\alpha = 0$	Then obviously, $T^\alpha = 0$ will emerge by varying the Lagrange multiplier, and the
mielke-geometrodynamics_F00963	186	■ 1.000	λ_α	λ_α	second field equation (8.6.7) amounts to an algebraic equation for the two-form λ_α .
mielke-geometrodynamics_F00964	186	■ 0.382	$\Gamma^{[j\alpha\beta}$	$\Gamma^{[j\alpha\beta}$	which, generically, is of third order in the Levi-Civita connection $\Gamma^{[]\alpha\beta}$ of Rie-
mielke-geometrodynamics_F00965	186	■ 0.803	$\mu_\alpha = \frac{1}{4} \vartheta_\alpha \wedge {}^*j_5$	$\mu_\alpha = \frac{1}{4} \vartheta_\alpha \wedge {}^*j_5$	In the case of Dirac spinors, the <i>spin-energy potential</i> $\mu_\alpha = \frac{1}{4} \vartheta_\alpha \wedge {}^*j_5$ is dual to
mielke-geometrodynamics_F00966	186	■ 1.000	$\mu_\alpha \wedge \delta T^\alpha$	$\mu_\alpha \wedge \delta T^\alpha$	can likewise be obtained from the variation $\mu_\alpha \wedge \delta T^\alpha$.
mielke-geometrodynamics_F00967	186	■ 1.000	$DR_{\alpha\beta} \equiv 0, DR_{\alpha\beta}^{(*)} \equiv$	$DR_{\alpha\beta} \equiv 0, DR_{\alpha\beta}^{(*)} \equiv$	coupling constants in the θ -type boundary term (8.6.5): since $DR_{\alpha\beta} \equiv 0, DR_{\alpha\beta}^{(*)} \equiv$
mielke-geometrodynamics_F00968	186	■ 0.420	$0, D\ \eta_{\alpha\beta}=0$	$0, D\eta_{\alpha\beta} = 0$	0, $D\eta_{\alpha\beta} = 0$, and $D(\vartheta_\alpha \wedge \vartheta_\beta) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$ in a Riemannian spacetime, the
mielke-geometrodynamics_F00969	186	■ 0.420	$D(\vartheta_\alpha \wedge \vartheta_\beta) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$	$D(\vartheta_\alpha \wedge \vartheta_\beta) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$	0, $D\eta_{\alpha\beta} = 0$, and $D(\vartheta_\alpha \wedge \vartheta_\beta) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$ in a Riemannian spacetime, the
mielke-geometrodynamics_F00970	186	■ 1.000	$R_{\alpha\beta}^{(*)}$	$R_{\alpha\beta}^{(*)}$	the Lie dual $R_{\alpha\beta}^{(*)}$ of the curvature does not contribute in (8.6.8), due to the Bach–
mielke-geometrodynamics_F00971	186	■ 0.428	$G^{\{ \}} := R^{\{1\}\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \gamma^\alpha / 2$	$G^{\{ \}} := R^{\{1\}\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \gamma^\alpha / 2$	Here $G^{\{ \}} := R^{\{1\}\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \gamma^\alpha / 2$ is the Einstein three-form dual to the usual Einstein
mielke-geometrodynamics_F00972	186	■ 0.997	$G_{ij} := \operatorname{Ric}_{ij}^{\{ \}} - \frac{1}{2} g_{ij}$	$G_{ij} := \operatorname{Ric}_{ij}^{\{ \}} - \frac{1}{2} g_{ij}$	tensor $G_{ij} := \operatorname{Ric}_{ij}^{\{ \}} - \frac{1}{2} g_{ij}$ written in terms of the Riemannian connection $\Gamma^{\{ \}}$. Matter
mielke-geometrodynamics_F00973	186	■ 0.993	$\kappa_{\text{eff}} = \ell^2 / \theta_{\mathrm{T}}^*$	$\kappa_{\text{eff}} = \ell^2 / \theta_{\mathrm{T}}^*$	whereas the effective gravitational coupling constant $\kappa_{\text{eff}} = \ell^2 / \theta_{\mathrm{T}}^*$ depends on a

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mielke-geometrodynamics_F00974	186	■ 1.000	$\backslash\mathrm{theta}_{\mathrm{T}}^{\ast}$	$\theta_{\mathrm{T}}^{\ast}$	“bare” length scale ℓ “renormalized” by the vacuum angle $\theta_{\mathrm{T}}^{\ast}$.
mielke-geometrodynamics_F00975	187	■ 1.000	$\backslash\mathrm{widehat{\Sigma}}_{\alpha}$	$\widehat{\Sigma}_{\alpha}$	<i>Einstein equation</i> (8.6.18) to the symmetrized energy–momentum current $\widehat{\Sigma}_{\alpha}$ of
mielke-geometrodynamics_F00976	187	■ 0.987	$\backslash\mathrm{kappa}_{\mathrm{\text{eff}}}=8\pi\mathrm{G}_{\mathrm{N}}/\mathrm{c}^4$	$\kappa_{\mathrm{eff}}=8\pi G_{\mathrm{N}}/c^4$	matter requires, at least locally, the macroscopic Newtonian value $\kappa_{\mathrm{eff}}=8\pi G_{\mathrm{N}}/c^4$
mielke-geometrodynamics_F00977	187	■ 0.894	$\mathrm{L}_{\mathrm{\text{qPG}}}$	L_{qPG}	most quadratic Lagrangian L_{qPG} with ten ⁶ parameters $a_{(M)}$ and $b_{(N)}$.
mielke-geometrodynamics_F00978	187	■ 1.000	$\backslash\mathrm{angle}\mathrm{varphi}\mathrm{rangle}\propto 1/\mathrm{ell}$	$\langle\varphi\rangle\propto 1/\ell$	tive to associate this with a (spontaneous) <i>symmetry breaking</i> $\langle\varphi\rangle\propto 1/\ell$ of the scale
mielke-geometrodynamics_F00979	188	■ 1.000	a	a	When these covectors are induced by an axion a and dilaton φ as respective potentials,
mielke-geometrodynamics_F00980	188	■ 1.000	$\backslash\mathrm{Phi}=\mathrm{a}+\mathrm{i}\mathrm{f}_{\mathrm{varphi}}\exp\left(-\mathrm{varphi}/\mathrm{f}_{\mathrm{varphi}}\right)$	$\Phi=a+if_{\varphi}\exp(-\varphi/f_{\varphi})$	even combine into a single complex scalar, the <i>axidilaton</i> $\Phi=a+if_{\varphi}\exp(-\varphi/f_{\varphi})$;
mielke-geometrodynamics_F00981	192	■ 0.954	$\backslash\mathrm{Lambda}_{\mathrm{\text{eff}}}>0$	$\Lambda_{\mathrm{eff}}>0$	(here generalized to an n-dimensional manifold) and second for $\Lambda_{\mathrm{eff}}>0$ to the Nariai
mielke-geometrodynamics_F00982	193	■ 0.989	$\mathrm{N}=\mathrm{N}(\mathrm{r}\pm\mathrm{t})$	$N=N(r\pm t)$	with $N=N(r\pm t)$ denoting an arbitrary harmonic function of r and t ; cf. Ni (1975).
mielke-geometrodynamics_F00983	193	■ 0.989	r	r	with $N=N(r\pm t)$ denoting an arbitrary harmonic function of r and t ; cf. Ni (1975).
mielke-geometrodynamics_F00984	193	■ 0.998	$\backslash\mathrm{Lambda}_{\mathrm{\text{eff}}}=0$	$\Lambda_{\mathrm{eff}}=0$	spacetime, however, the Schwarzschild solution (9.1.1, 2) with $\Lambda_{\mathrm{eff}}=0$ is of para-
mielke-geometrodynamics_F00985	194	■ 0.358	$\{ \}^{\ast}$	$\ast$	nonsingular one, we first pass to a different coordinate system. Let $r^{\ast}$ be the “tortoise”
mielke-geometrodynamics_F00986	194	■ 0.996	$\mathrm{u}^2-\mathrm{v}^2=1$	$u^2-v^2=1$	$u^2-v^2=1$, that is, for $r=r^{\ast}=0$, this geometry becomes singular in real space-
mielke-geometrodynamics_F00987	194	■ 0.996	$\mathrm{r}=\mathrm{r}^{\ast}=\mathrm{o}$	$r=r^{\ast}=o$	$u^2-v^2=1$, that is, for $r=r^{\ast}=0$, this geometry becomes singular in real space-
mielke-geometrodynamics_F00988	194	■ 1.000	$\mathrm{\tilde{v}}:=\mathrm{i}v$	$\tilde{v}:=iv$	be avoided by making use of a purely “imaginary” coordinate $\tilde{v}:=iv$. The sphere
mielke-geometrodynamics_F00989	194	■ 0.531	S^2	S^2	S^2 given by $e^v=0$ deteriorates to a rotational axis, while the imaginary time axis
mielke-geometrodynamics_F00990	194	■ 0.531	$\mathrm{e}^v=0$	$e^v=0$	S^2 given by $e^v=0$ deteriorates to a rotational axis, while the imaginary time axis
mielke-geometrodynamics_F00991	194	■ 1.000	$\tau:=$	$\tau:=$	$\tau:=$ it represents an angle variable about this axis in Euclidean spacetime. (Corre-
mielke-geometrodynamics_F00992	194	■ 1.000	K_3	K_3	K_3 -surface represents the only compact manifold that is allowed by a metric with
mielke-geometrodynamics_F00993	195	■ 0.629	ω^g	ω^g	connection ω^g into a Christoffel-type connection $\omega^{\mathfrak{I}}$ and the remaining contortion
mielke-geometrodynamics_F00994	195	■ 0.975	$\mathrm{R}\mathrm{C}$	RC	2-form in an RC spacetime. Its Riemannian part may be decomposed further into
mielke-geometrodynamics_F00995	195	■ 0.917	$\Omega_{\mathrm{C}}^{\mathfrak{I}}$	$\Omega_{\mathrm{C}}^{\mathfrak{I}}$	where $\Omega_{\mathrm{C}}^{\mathfrak{I}}$ denotes Weyl’s conformal curvature 2-form in a Riemannian spacetime.
mielke-geometrodynamics_F00996	195	■ 1.000	$\mathrm{f}_i=\mathrm{o}$	$f_i=o$	choice $f_i=o$ and $\zeta=-1$, as discussed by BAEKLER et al. (1982), the duality ansatz
mielke-geometrodynamics_F00997	196	■ 1.000	$\mathrm{K}\wedge\mathrm{K}$	$K\wedge K$	fact that $K\wedge K$ is already self-double-dual.
mielke-geometrodynamics_F00998	196	■ 1.000	$\mu=0$	$\mu=0$	cosmos, vanishes without Schwarzschild’s source term, i.e., for $\mu=0$. This is in
mielke-geometrodynamics_F00999	196	■ 1.000	$\mathrm{m}\mathrm{u}$	mu	parameter mu vanishes in the metric ground form. Here the radial functions occurring
mielke-geometrodynamics_F01000	197	■ 1.000	$\kappa<0$	$\kappa<0$	For a real solution, $\kappa<0$ is necessary. In this case, torsion can exist only in an anti-de

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mielke-geometrodynamics_F01001	197	■ 1.000	$\mathrm{t} \rightarrow \mathrm{i} \tau$	$t \rightarrow -i\tau$	Weyl’s trick, i.e., the formal substitution $t \rightarrow -i\tau$ of the timelike variable. It follows
mielke-geometrodynamics_F01002	197	■ 0.687	$\mathrm{f}=\mathrm{T}_{\hat{0}\hat{1}}^{\hat{0}\hat{1}}$	$f = T_{01}^{\cdot\hat{0}}$	from the matrix representation (9.1.15) that the anholonomic components $f = T_{01}^{\cdot\hat{0}}$
mielke-geometrodynamics_F01003	197	■ 0.790	$\mathrm{g}=\mathrm{T}_{\hat{1}\hat{2}}^{\hat{1}\hat{2}}\mathrm{T}_{\hat{1}\hat{3}}^{\hat{1}\hat{3}}=\mathrm{T}_{\hat{1}\hat{2}}^{\hat{1}\hat{2}}\mathrm{T}_{\hat{1}\hat{3}}^{\hat{1}\hat{3}}$	$g = T_{12}^{\cdot\hat{2}} = T_{13}^{\cdot\hat{3}}$	and $g = T_{12}^{\cdot\hat{2}} = T_{13}^{\cdot\hat{3}}$ are the only ones that are then allowed to be nonzero. A solution
mielke-geometrodynamics_F01004	198	■ 1.000	$\mathrm{n}=4$	$n = 4$	ourselves to a manifold of dimension $n = 4$. The previous classification of the source-
mielke-geometrodynamics_F01005	198	■ 0.998	R^6	R^6	space R^6 . For then
mielke-geometrodynamics_F01006	199	■ 0.995	γ_2^g	γ_2^g	the form γ_2^g over the basis manifold, which provides for a topological invariant. It is
mielke-geometrodynamics_F01007	199	■ 1.000	$\mathrm{H}_{\mathrm{P}}(\mathrm{M},\mathbb{R})$	$H_P(M, \mathbb{R})$	rank of the pth homology group $H_P(M, \mathbb{R})$, i.e., from the so-called Betti numbers
mielke-geometrodynamics_F01008	199	■ 1.000	$\mathrm{b}:=\operatorname{dim}\mathrm{H}_{\mathrm{p}}(\mathrm{M},\mathbb{R})$	$b := \dim H_p(M, \mathbb{R})$	$b := \dim H_p(M, \mathbb{R})$, which were introduced by Riemann. Similarly as in the case of
mielke-geometrodynamics_F01009	199	■ 1.000	$\chi(\mathrm{M})$	$\chi(M)$	the vanishing of $\chi(M)$. In particular, it is known that a compact even-dimensional
mielke-geometrodynamics_F01010	199	■ 1.000	$\mathrm{s}=1$	$s = 1$	manifold admits a pseudo-Riemannian metric of signature $s = 1$ (“Lorentz metric”)
mielke-geometrodynamics_F01011	199	■ 0.996	$\alpha:\omega^g-\omega^g$	$\alpha : \omega^g - \omega^g$	$\alpha : \omega^g - \omega^g$ of the embedding of the boundary ∂M in M . Although it is exactly
mielke-geometrodynamics_F01012	200	■ 1.000	B^n	B^n	an open ball B^n is to be cut out of M and N in order to identify the then respectively
mielke-geometrodynamics_F01013	200	■ 1.000	$\partial\mathrm{B}^n=\mathrm{S}^{n-1}$	$\partial B^n = S^{n-1}$	existing boundary spheres $\partial B^n = S^{n-1}$ with each other via a diffeomorphism that
mielke-geometrodynamics_F01014	200	■ 0.736	$\mathrm{W}^n:=\mathrm{S}^1\times\mathrm{S}^{n-1}$	$W^n := S^1 \times S^{n-1}$	$W^n := S^1 \times S^{n-1}$ (WHEELER 1955), the manifold $M\#(S^1 \times S^{n-1})$ comes into exis-
mielke-geometrodynamics_F01015	200	■ 0.736	$\left(\mathrm{S}^1\times\mathrm{S}^{n-1}\right)$	$\left(S^1 \times S^{n-1}\right)$	$W^n := S^1 \times S^{n-1}$ (WHEELER 1955), the manifold $M\#(S^1 \times S^{n-1})$ comes into exis-
mielke-geometrodynamics_F01016	200	■ 0.980	$\mathbb{C}P^2$	$\mathbb{CP}^2$	number” of a surface (cf. SEIFERT & THREFALL 1934), and $\mathbb{CP}^2$ denotes the complex
mielke-geometrodynamics_F01017	200	■ 1.000	E^2	E^2	projective two-space in E^2 . For spaces with uneven dimensions,
mielke-geometrodynamics_F01018	201	■ 1.000	$\mathrm{T}^4=\mathrm{S}^1\times\mathrm{S}^1\times\mathrm{S}^1\times\mathrm{S}^1$	$T^4 = S^1 \times S^1 \times S^1 \times S^1$	by the flat torus $T^4 = S^1 \times S^1 \times S^1 \times S^1$. Solutions with anti-self-double-dual cur-
mielke-geometrodynamics_F01019	202	■ 0.982	$\mathscr{S}_*(\mathrm{M}):=\mathscr{M}/\tilde{\mathscr{D}}$	$\mathcal{S}_*(M) := \mathcal{M} / \tilde{\mathcal{D}}$	space” $\mathcal{S}_*(M) := \mathcal{M} / \tilde{\mathcal{D}}$ of all Riemannian metrics modulo the diffeomorphisms
mielke-geometrodynamics_F01020	203	■ 1.000	S^2	S^2	Euler–Poincaré topological invariant (9.3.1). The two-sphere part S^2 of these $\mathbb{R} \times$
mielke-geometrodynamics_F01021	203	■ 1.000	$\mathbb{R}\times$	$\mathbb{R} \times$	Euler–Poincaré topological invariant (9.3.1). The two-sphere part S^2 of these $\mathbb{R} \times$
mielke-geometrodynamics_F01022	203	■ 1.000	$S^1\times S^2$	$S^1 \times S^2$	$S^1 \times S^2$ topological configurations may even be knotted (MIELKE 1977a), and within
mielke-geometrodynamics_F01023	203	■ 1.000	$\mathbf{R}^4$	$\mathbf{R}^4$	piecewise linear structures (MILNOR 2011) on a noncompact $\mathbf{R}^4$. However, mani-
mielke-geometrodynamics_F01024	203	■ 1.000	U_4	U_4	Baekler P (1980) The unique spherically symmetric solution of the U_4 -theory of gravity in the
mielke-geometrodynamics_F01025	205	■ 0.999	$\mathrm{R}+\mathrm{R}^2$	$R + R^2$	Neville DE (1980) Birkhoff theorems for $R + R^2$ gravity theories with torsion. Phys Rev D
mielke-geometrodynamics_F01026	208	■ 0.837	C_{TL}	C_{TL}	term C_{TL} , which simulates to some extent Einstein’s theory in 3D, allowing us to
mielke-geometrodynamics_F01027	208	■ 0.994	$\varepsilon\rightarrow 1/\varepsilon$	$\varepsilon \rightarrow 1/\varepsilon$	$\varepsilon \rightarrow 1/\varepsilon$.

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mielke-geometrodynamics_F01028	208	0.674	$\backslash\Gamma^{\beta}_{\gamma}=\backslash\Gamma_{\gamma}^{\beta} d x^{\gamma}$	$\Gamma^{\beta\gamma}=\Gamma^{\beta\gamma}_j dx^j$	connection $\Gamma^{\beta\gamma}=\Gamma^{\beta\gamma}_j dx^j$, i.e.,																																																								
mielke-geometrodynamics_F01029	209	1.000	$\backslash\Gamma_{\alpha}^{\star}$	$\Gamma_{\alpha}^{\star}$	Since the coframe ϑ_{α} has physical dimensions [length] and the connection $\Gamma_{\alpha}^{\star}$ is																																																								
mielke-geometrodynamics_F01030	209	1.000	$\backslash\vartheta_{\alpha} / \ell$	ϑ_{α}/ℓ	dimensionless, one can use ϑ_{α}/ℓ and $\Gamma_{\alpha}^{\star}$ as <i>dimensionless</i> gauge potentials of local																																																								
mielke-geometrodynamics_F01031	209	1.000	$C=\backslash\operatorname{Tr}\{A \wedge F\}$	$C=\operatorname{Tr}\{A\wedge F\}$	The corresponding Chern–Simons three-forms of gauge type $C=\operatorname{Tr}\{A\wedge F\}$ are the																																																								
mielke-geometrodynamics_F01032	209	1.000	T_{α}	T_{α}	are uniquely related to the torsion T_{α} , the curvature $R_{\alpha}^{\star}$, and a cosmological term η_{α} ,																																																								
mielke-geometrodynamics_F01033	209	1.000	$R_{\alpha}^{\star}$	$R_{\alpha}^{\star}$	are uniquely related to the torsion T_{α} , the curvature $R_{\alpha}^{\star}$, and a cosmological term η_{α} ,																																																								
mielke-geometrodynamics_F01034	209	1.000	$\backslash\eta_{\alpha}$	η_{α}	are uniquely related to the torsion T_{α} , the curvature $R_{\alpha}^{\star}$, and a cosmological term η_{α} ,																																																								
mielke-geometrodynamics_F01035	210	1.000	$n^{\{2\}}$	n^2	<table><tr><th></th><th>Objects</th><th>p-form</th><th>Components</th><th>n = 4</th><th>3</th><th>2</th></tr><tr><td>ϑ^{α}</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_{\alpha}^{\star}$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^{α}</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_{α}</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_{α}</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	p -form	Components	n = 4	3	2	ϑ^{α}	Vector	1	n^2	16	9	1	$\Gamma_{\alpha}^{\star}$	Vector	1	n^2	16	9	1	T^{α}	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_{α}	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_{α}	Vector	$n-1$	n^2	16	9	4
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mielke-geometrodynamics_F01036	210	1.000	$n^{\{2\}}(n-1) / 2$	$n^2(n-1)/2$	<table><tr><th></th><th>Objects</th><th>p-form</th><th>Components</th><th>n = 4</th><th>3</th><th>2</th></tr><tr><td>ϑ^{α}</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_{\alpha}^{\star}$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^{α}</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_{α}</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_{α}</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	p -form	Components	n = 4	3	2	ϑ^{α}	Vector	1	n^2	16	9	1	$\Gamma_{\alpha}^{\star}$	Vector	1	n^2	16	9	1	T^{α}	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_{α}	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_{α}	Vector	$n-1$	n^2	16	9	4
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mielke-geometrodynamics_F01037	210	1.000	$R^{\{\alpha \beta\}}$	$R^{\alpha\beta}$	<table><tr><th></th><th>Objects</th><th>p-form</th><th>Components</th><th>n = 4</th><th>3</th><th>2</th></tr><tr><td>ϑ^{α}</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_{\alpha}^{\star}$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^{α}</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_{α}</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_{α}</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	p -form	Components	n = 4	3	2	ϑ^{α}	Vector	1	n^2	16	9	1	$\Gamma_{\alpha}^{\star}$	Vector	1	n^2	16	9	1	T^{α}	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_{α}	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_{α}	Vector	$n-1$	n^2	16	9	4
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mielke-geometrodynamics_F01038	210	1.000	$n^{\{2\}}(n-1)^{\{2\}} / 4$	$n^2(n-1)^2/4$	<table><tr><th></th><th>Objects</th><th>p-form</th><th>Components</th><th>n = 4</th><th>3</th><th>2</th></tr><tr><td>ϑ^{α}</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_{\alpha}^{\star}$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^{α}</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_{α}</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_{α}</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	p -form	Components	n = 4	3	2	ϑ^{α}	Vector	1	n^2	16	9	1	$\Gamma_{\alpha}^{\star}$	Vector	1	n^2	16	9	1	T^{α}	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_{α}	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_{α}	Vector	$n-1$	n^2	16	9	4
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	Objects	p -form	Components	n = 4	3	2																																																							
ϑ^α	Vector	1	n^2	16	9	1																																																							
$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1																																																							
T^α	Vector	2	$n^2(n-1)/2$	24	9	2																																																							
$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1																																																							
Σ_α	Vector	$n-1$	n^2	16	9	4																																																							
$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2																																																							
η_α	Vector	$n-1$	n^2	16	9	4																																																							
mielke-geometrodynamics_F01039	210	■ 1.000	$\tau_{\alpha\beta}$	$\tau_{\alpha\beta}$																																																									
mielke-geometrodynamics_F01040	210	■ 1.000	n=3	$n = 3$	components in various dimensions: Observe that only for $n = 3$ do all fields have																																																								
mielke-geometrodynamics_F01041	210	■ 0.872	$\mathrm{SO}(1,2)$	$\mathrm{SO}(1,2)$	dimensional Lorentz group $\mathrm{SO}(1,2)$ as a $9 = 5 \otimes 3 \otimes 1$ multiplet after bivectors are																																																								
mielke-geometrodynamics_F01042	210	■ 0.872	$9=5 \otimes 3 \otimes 1$	$9 = 5 \otimes 3 \otimes 1$	dimensional Lorentz group $\mathrm{SO}(1,2)$ as a $9 = 5 \otimes 3 \otimes 1$ multiplet after bivectors are																																																								
mielke-geometrodynamics_F01043	211	■ 1.000	$H=-\partial L / \partial F$	$H = -\partial L / \partial F$	are the translational and rotational field momenta (or the excitation $H = -\partial L / \partial F$																																																								
mielke-geometrodynamics_F01044	211	■ 1.000	$H_{\alpha}=H_{[\alpha\beta]} \vartheta^{\beta}$	$H_\alpha = H_{[\alpha\beta]} \vartheta^\beta$	(In the particular case of an antisymmetric field $H_\alpha = H_{[\alpha\beta]} \vartheta^\beta$, this is dual to the																																																								
mielke-geometrodynamics_F01045	211	■ 1.000	$S_{\alpha}=-^* H_{\alpha}$	$S_\alpha = -^* H_\alpha$	translational field strength, i.e., $S_\alpha = -^* H_\alpha$.) The sources for the gravitational field																																																								
mielke-geometrodynamics_F01046	211	■ 0.957	$\mathbb{R}^3 \otimes S_0(1,2)$	$\mathbb{R}^3 \otimes SO(1,2)$	$\mathbb{R}^3 \ltimes SO(1,2)$, we arrive at the MIELKE & BAEKLER (1991) (MB) model, which is																																																								
mielke-geometrodynamics_F01047	211	■ 1.000	$\theta_{\mathrm{T}}, \theta_{\mathrm{L}}$	$\theta_{\mathrm{T}}, \theta_{\mathrm{L}}$	Allowing for arbitrary “vacuum angles” θ_{T} , θ_{L} , and $\theta_{\mathrm{TL}} = -\chi$, the most general																																																								
mielke-geometrodynamics_F01048	211	■ 1.000	$\theta_{\mathrm{TL}}=-\chi$	$\theta_{\mathrm{TL}} = -\chi$	Allowing for arbitrary “vacuum angles” θ_{T} , θ_{L} , and $\theta_{\mathrm{TL}} = -\chi$, the most general																																																								
mielke-geometrodynamics_F01049	212	■ 0.997	$C_{\mathrm{T}}, C_{\mathrm{L}}$	$C_{\mathrm{T}}, C_{\mathrm{L}}$	where C_{T} , C_{L} , and C_{TL} are respectively the translational, rotational, and “mixed” CS																																																								
mielke-geometrodynamics_F01050	212	■ 1.000	C_{T}	C_{T}	tional term C_{T} is covariant, it appears that the MB model is only semi-topological,																																																								
mielke-geometrodynamics_F01051	212	■ 1.000	$\Gamma^{\star\alpha}$	$\Gamma^{\star\alpha}$	Consequently, varying the Lagrangian (10.2.2) with respect to ϑ^α and $\Gamma^{\star\alpha}$ yields																																																								
mielke-geometrodynamics_F01052	212	■ 1.000	$\theta_{\mathrm{T}} / 2 \ell$	$\theta_{\mathrm{T}} / 2\ell$	$\theta_{\mathrm{T}} / 2\ell$, resembling a cosmological constant familiar from 4D gravity.																																																								
mielke-geometrodynamics_F01053	212	■ 1.000	$\varepsilon=\theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$	$\varepsilon = \theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$	where the contortional constants $\varepsilon = \theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$ and $\rho = -\theta_{\mathrm{T}}^2 / A$ are related to the																																																								
mielke-geometrodynamics_F01054	212	■ 1.000	$\rho=-\theta_{\mathrm{T}}^2 / A$	$\rho = -\theta_{\mathrm{T}}^2 / A$	where the contortional constants $\varepsilon = \theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$ and $\rho = -\theta_{\mathrm{T}}^2 / A$ are related to the																																																								
mielke-geometrodynamics_F01055	212	■ 1.000	$A:=-(-1)^s \theta_{\mathrm{TL}}^2+2 \theta_{\mathrm{T}} \theta_{\mathrm{L}} \neq 0$	$A := -(-1)^s \theta_{\mathrm{TL}}^2 + 2 \theta_{\mathrm{T}} \theta_{\mathrm{L}} \neq 0$	$A := -(-1)^s \theta_{\mathrm{TL}}^2 + 2 \theta_{\mathrm{T}} \theta_{\mathrm{L}} \neq 0$ is nonvanishing.																																																								
mielke-geometrodynamics_F01056	213	■ 0.994	$K_{\alpha}^{\star}=-^* T_{\alpha} / 2=-(-1)^s(\varepsilon / 2 \ell) \vartheta_{\alpha}$	$K_\alpha^\star = -^* T_\alpha / 2 = -(-1)^s (\varepsilon / 2\ell) \vartheta_\alpha$	for the RC curvature. From the relation $K_\alpha^\star = -^* T_\alpha / 2 = -(-1)^s (\varepsilon / 2\ell) \vartheta_\alpha$ for the																																																								
mielke-geometrodynamics_F01057	213	■ 1.000	$\varepsilon=0$	$\varepsilon = 0$	$\varepsilon = 0$, i.e., in a purely Riemannian spacetime. Alternatively, for $\rho = 0$, i.e., in the																																																								
mielke-geometrodynamics_F01058	213	■ 1.000	$\Lambda_{\text {eff }}$	Λ_{eff}	cosmological term Λ_{eff} has the anti-de Sitter (AdS) metric																																																								
mielke-geometrodynamics_F01059	214	■ 1.000	J	J	respectively. Observe that the shift is proportional to the angular momentum J of																																																								
mielke-geometrodynamics_F01060	214	■ 1.000	$J^2 \leq M^2$	$J^2 \leq M^2$	the solution, which allows for $J^2 \leq M^2$ to interpret this configuration as a <i>rotating</i>																																																								
mielke-geometrodynamics_F01061	214	■ 1.000	$M=-1$	$M = -1$	normalization $M = -1$ and $J = 0$, it reduces to the AdS metric.																																																								
mielke-geometrodynamics_F01062	214	■ 1.000	$J=0$	$J = 0$	normalization $M = -1$ and $J = 0$, it reduces to the AdS metric.																																																								

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mielke-geometrodynamics_F01063 [conf 0.080]	214	■ 0.080	$\backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}^{\{\}}\star}\backslashright.\}$ $\backslashrightarrow \backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}\star}=\backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}\star}\backslashleft\{\{\}^{\{\}}\backslash\star\right.\}-K_{\alpha}^{\{\alpha\}^{\{\}}\star}$	$\Gamma_{\alpha}^{\{\}^{\{\}}^{\{\}}\star} \rightarrow \Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\}^{\{\}}^{\{\}}\star} - K_{\alpha}^{\star}$	rest essentially on a “prolongation” $\Gamma_{\alpha}^{\{\}^{\{\}}^{\{\}}\star} \rightarrow \Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\}^{\{\}}^{\{\}}\star} - K_{\alpha}^{\star}$ of the Riemannian to
mielke-geometrodynamics_F01064	214	■ 1.000	$\backslashvarepsilon$	ε	effective cosmological constant Λ_{eff} is related to ε and ρ via (10.2.10), the black hole
mielke-geometrodynamics_F01065	214	■ 1.000	$\backslash\Gamma_{\alpha}^{\{\alpha\}^{\{\}}\star}=\frac{1}{2}\backslasheta_{\alpha\beta\gamma}\backslashGamma_{\alpha}^{\{\alpha\}^{\{\}}\star}\backslashGamma_{\beta}^{\{\beta\}^{\{\}}\star}\backslashGamma_{\gamma}^{\{\gamma\}^{\{\}}\star}$	$\Gamma_{\alpha}^{\star} = \frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}$	where ε is a continuous parameter, involves the Lie dual $\Gamma_{\alpha}^{\star} = \frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}$ of the
mielke-geometrodynamics_F01066	214	■ 1.000	$\backslash\widetilde{\text{T}}_{\alpha}=0$	$\widetilde{T}_{\alpha} = 0$	Riemannian spacetime with $\widetilde{T}_{\alpha} = 0$ and deformed <i>teleparallelism</i> in the local gauge
mielke-geometrodynamics_F01067	214	■ 0.999	$\backslash\widetilde{\text{Gamma}}_{\alpha}^{\{\alpha\}^{\{\}}\star}\backslashstackrel{*}{=}\ 0$	$\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$	$\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$, equivalent to the covariant constraint of vanishing modified RC curvature,
mielke-geometrodynamics_F01068	214	■ 1.000	$\backslash\widetilde{\text{R}}_{\alpha}^{\{\alpha\}^{\{\}}\star}=0$	$\widetilde{R}_{\alpha}^{\star} = 0$	i.e., $\widetilde{R}_{\alpha}^{\star} = \hat{0}$. In the latter case, coframe and connection are locally <i>Lie dual</i> to each
mielke-geometrodynamics_F01069	215	■ 1.000	$1\ /\ \backslashvarepsilon$	$1/\varepsilon$	large rotational momentum proportional to $1/\varepsilon$ and conversely, resembling, to some
mielke-geometrodynamics_F01070	215	■ 0.915	$\backslashrho=(-1)^{\{\text{sign}\}}\backslashvarepsilon^{\{2\}}\ /\ 4$	$\rho = (-1)^{\text{sign}}\varepsilon^2/4$	as originally envisioned by CARTAN (1924). Here $\rho = (-1)^{\text{sign}}\varepsilon^2/4$ depends quadrat-
mielke-geometrodynamics_F01071	215	■ 0.518	$\text{R}_{\alpha}^{\{\alpha\}^{\{\}}\star}=(-1)^{\{\text{sign}\}}\backslashvarepsilon\ \text{T}_{\alpha}^{\{\alpha\}^{\{\}}\star}\ /\ 4\ \backslashell$	$R_{\alpha}^{\star} = (-1)^{\text{sign}}\varepsilon T_{\alpha}/4\ell$	should not ignore that $R_{\alpha}^{\star} = (-1)^{\text{sign}}\varepsilon T_{\alpha}/4\ell$ induces, for $\varepsilon \neq 0$, a constant-curvature
mielke-geometrodynamics_F01072	215	■ 0.518	$\backslashvarepsilon\ \backslashneq\ 0$	$\varepsilon \neq 0$	should not ignore that $R_{\alpha}^{\star} = (-1)^{\text{sign}}\varepsilon T_{\alpha}/4\ell$ induces, for $\varepsilon \neq 0$, a constant-curvature
mielke-geometrodynamics_F01073	216	■ 1.000	$\backslashvarepsilon\ \rightarrow\ -\varepsilon$	$\varepsilon \rightarrow -\varepsilon$	symmetry” $\varepsilon \rightarrow -\varepsilon$ in (10.3.9) and (10.3.10), we see that the limit $\gamma = (1 - \delta\varepsilon) \rightarrow$
mielke-geometrodynamics_F01074	216	■ 1.000	$\backslash\gamma=(1-\delta\varepsilon)\ \rightarrow$	$\gamma = (1 - \delta\varepsilon) \rightarrow$	symmetry” $\varepsilon \rightarrow -\varepsilon$ in (10.3.9) and (10.3.10), we see that the limit $\gamma = (1 - \delta\varepsilon) \rightarrow$
mielke-geometrodynamics_F01075	216	■ 1.000	$\backslash\gamma\ \backslashneq\ 0$	$\gamma \neq 0$	(10.2.1). For $\gamma \neq 0$, the vacuum solution (10.3.9) yields
mielke-geometrodynamics_F01076	216	■ 1.000	$\backslashvarepsilon\ \rightarrow\ 0$	$\varepsilon \rightarrow 0$	in the limit $\varepsilon \rightarrow 0$ of <i>weak</i> axial torsion coupling, the translational field momenta
mielke-geometrodynamics_F01077	216	■ 1.000	$\text{H}_{\alpha}^{\{\alpha\}^{\{\}}\star}=\backslash\gamma\ \backslashvartheta_{\alpha}\ /\ \backslashell$	$H_{\alpha}^{\star} = \gamma\vartheta_{\alpha}/\ell$	i.e., $H_{\alpha}^{\star} = \gamma\vartheta_{\alpha}/\ell$, similar to the case of the EC action. Conversely, in the limit
mielke-geometrodynamics_F01078	216	■ 1.000	$\backslashvarepsilon\ \rightarrow\ \infty$	$\varepsilon \rightarrow \infty$	$\varepsilon \rightarrow \infty$ of <i>strong</i> axial torsion coupling, the translational momenta (10.3.9) become
mielke-geometrodynamics_F01079	216	■ 1.000	$\text{H}_{\alpha}^{\{\alpha\}^{\{\}}\star}=-\backslash\gamma\ \backslashvartheta_{\alpha}\ /\ \delta\ell^{\{2\}}$	$H_{\alpha} = -\gamma\vartheta_{\alpha}/\delta\ell^2$	$H_{\alpha} = -\gamma\vartheta_{\alpha}/\delta\ell^2$, and the rotational momenta turn out to be $H_{\alpha}^{\star} = 0$. This implies
mielke-geometrodynamics_F01080	216	■ 1.000	$\text{H}_{\alpha}^{\{\alpha\}^{\{\}}\star}=0$	$H_{\alpha}^{\star} = 0$	$H_{\alpha} = -\gamma\vartheta_{\alpha}/\delta\ell^2$, and the rotational momenta turn out to be $H_{\alpha}^{\star} = 0$. This implies
mielke-geometrodynamics_F01081	217	■ 1.000	$\backslash\{\text{H},\ \text{d}\ \text{t},\ \text{d}\ \backslash\mathscr{A},\ \text{d}\ h\}$	$\{H, dt, d\mathscr{A}, dh\}$	a basis $\{H, dt, d\mathscr{A}, dh\}$ of one-forms can be constructed that, for nonzero torsion
mielke-geometrodynamics_F01082	217	■ 1.000	$\text{H}=0$	$H = 0$	For $H = 0$, the three one-forms on the right-hand side become linearly dependent,
mielke-geometrodynamics_F01083	217	■ 0.998	$\text{H}_{\alpha}\ \backslashneq\ 0$	$H_{\alpha} \neq 0$	thereby sending us back to the exact solution (10.3.9). However, for $H_{\alpha} \neq 0$, an exact
mielke-geometrodynamics_F01084	217	■ 1.000	$\text{d}\ \backslash\mathscr{A}$	$d\mathscr{A}$	can be proposed in which one coordinate “leg,” for instance $d\mathscr{A}$, needs to be con-
mielke-geometrodynamics_F01085	218	■ 1.000	$\text{d}^{\{\star\}}\ \backslashvartheta_{\alpha}=0$	$d^{\star}\vartheta_{\alpha} = 0$	generalizes the Hilbert–de Donder or <i>transversality condition</i> $d^{\star}\vartheta_{\alpha} = 0$ for the
mielke-geometrodynamics_F01086	218	■ 1.000	$\backslasheta_{\alpha}^{\{\alpha\}^{\{\}}\star}=\{\}^{\{\star\}}\ \backslashvartheta_{\alpha}$	$\eta_{\alpha} = ^{\star}\vartheta_{\alpha}$	To this end, we will employ the relation $\eta_{\alpha} = ^{\star}\vartheta_{\alpha}$ for the η_{α} basis and iterate the
mielke-geometrodynamics_F01087	219	■ 1.000	$\text{m}=\backslashvarepsilon\ /\ \backslashell$	$m = \varepsilon/\ell$	where $m = \varepsilon/\ell$, results in vacuum and the gauge-covariant Hilbert–de Donder or
mielke-geometrodynamics_F01088	219	■ 0.782	$\backslashvartheta^{\{\alpha\}}=\text{E}_{\{i\}}^{\{\}\{\alpha\}}\ \text{d}\ x^{\{i\}}$	$\vartheta^{\alpha} = E_i^{\alpha}dx^i$	is as follows: In 3D, the coframe $\vartheta^{\alpha} = E_i^{\alpha}dx^i$ has $3 \times 3 = 9$ components, of which

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mielke-geometrodynamics_F01089	219	0.782	$3 \times 3 = 9$	$3 \times 3 = 9$	is as follows: In 3D, the coframe $\vartheta^\alpha = E_i^\alpha dx^i$ has $3 \times 3 = 9$ components, of which
mielke-geometrodynamics_F01090	219	0.924	$S O(1,2)$	$SO(1, 2)$	3 are pure gauge due to the $SO(1, 2)$ Lorentz transformation $\vartheta'^\alpha = A_\beta^\alpha(x)\vartheta^\beta$. The
mielke-geometrodynamics_F01091	219	0.924	$\vartheta^{\prime \alpha}=\Lambda_{\beta}^{\alpha}(x) \vartheta^{\beta}$	$\vartheta'^\alpha = \Lambda_\beta^\alpha(x)\vartheta^\beta$	3 are pure gauge due to the $SO(1, 2)$ Lorentz transformation $\vartheta'^\alpha = A_\beta^\alpha(x)\vartheta^\beta$. The
mielke-geometrodynamics_F01092	220	0.992	$2 / \theta_{\mathrm{L}}$	$2/\theta_{\mathrm{L}}$	contributes to the mass with an “anyon”-type g-factor of $2/\theta_{\mathrm{L}}$; cf. JACKIW & NAIR
mielke-geometrodynamics_F01093	220	1.000	$\theta_{\mathrm{T}} \theta_{\mathrm{L}} / 2 \chi=1$	$\theta_{\mathrm{T}}\theta_{\mathrm{L}}/2\chi = 1$	to be constrained to $\theta_{\mathrm{T}}\theta_{\mathrm{L}}/2\chi = 1$, or at least to be positive in order to avoid nega-
mielke-geometrodynamics_F01094	220	1.000	$\theta_{\mathrm{T}} \neq 0$	$\theta_{\mathrm{T}} \neq 0$	$\theta_{\mathrm{T}} \neq 0$, the vacuum angle of the translational Chern–Simons term C_{T} . In any case,
mielke-geometrodynamics_F01095	221	1.000	$\varepsilon \neq \tilde{\varepsilon}$	$\varepsilon \neq \tilde{\varepsilon}$	Then, for <i>different</i> continuous parameters $\varepsilon \neq \tilde{\varepsilon}$, we obtain
mielke-geometrodynamics_F01096	221	1.000	$\tilde{\theta}_{\mathrm{L}}$	$\tilde{\theta}_{\mathrm{L}}$	parent. Consequently, the constants θ_{L} and $\tilde{\theta}_{\mathrm{L}}$ give rise to the two central charges
mielke-geometrodynamics_F01097	222	1.000	$C^{kl}=C^{lk}$	$C^{kl} = C^{lk}$	is the one-form associated with the symmetric Cotton tensor $C^{kl} = C^{lk}$. In general,
mielke-geometrodynamics_F01098	222	0.940	$\Gamma^{\{\lambda \alpha \beta\}}$	$\Gamma^{\{\lambda\alpha\beta}$	this is a third-order equation in the Levi-Civita connection $\Gamma^{\{\} \alpha\beta}$, i.e., of fourth order
mielke-geometrodynamics_F01099	222	1.000	$T_{\alpha}=D \vartheta_{\alpha}=0$	$T_\alpha = D\vartheta_\alpha = 0$	Due to $T_\alpha = D\vartheta_\alpha = 0$ in Riemannian spacetime, implying $H_\alpha = 0$ in 3D, the mod-
mielke-geometrodynamics_F01100	222	1.000	$H_{\alpha}=0$	$H_\alpha = 0$	Due to $T_\alpha = D\vartheta_\alpha = 0$ in Riemannian spacetime, implying $H_\alpha = 0$ in 3D, the mod-
mielke-geometrodynamics_F01101	222	0.891	$D H_{\alpha}^{\star}=\delta \ell D H_{\alpha}+(\gamma / \ell) D \vartheta_{\alpha} \equiv 0$	$DH_\alpha^\star = \delta\ell DH_\alpha + (\gamma/\ell)D\vartheta_\alpha \equiv 0$	ified S -duality leads to $DH_\alpha^\star = \delta\ell DH_\alpha + (\gamma/\ell)D\vartheta_\alpha \equiv 0$, and the two-form $D^{[\} H_{[\alpha\beta]}$
mielke-geometrodynamics_F01102	222	0.891	$D^{[4]} H_{[\alpha \beta]}$	$D^{[4]}H_{[\alpha\beta]}$	ified S -duality leads to $DH_\alpha^\star = \delta\ell DH_\alpha + (\gamma/\ell)D\vartheta_\alpha \equiv 0$, and the two-form $D^{[\} H_{[\alpha\beta]}$
mielke-geometrodynamics_F01103	222	0.539	$G_{\alpha}=R_{\alpha}^{\{\star\}}$	$G_\alpha = R_\alpha^{\{\} \star}$	in Riemannian spacetime. Since $G_\alpha = R_\alpha^{\{\} \star}$ is the Einstein current two-form in 3D,
mielke-geometrodynamics_F01104	222	0.990	$\widetilde{\Sigma}_{\alpha}:=\Sigma_{\alpha}-D^{\{\} \mu} \mu_{\alpha}$	$\widetilde{\Sigma}_\alpha := \Sigma_\alpha - D^{\{\} \mu}\mu_\alpha$	<i>symmetric</i> Belinfante–Rosenfeld two-form $\widetilde{\Sigma}_\alpha := \Sigma_\alpha - D^{[\} \mu_\alpha$ as source.
mielke-geometrodynamics_F01105	223	1.000	$E=p c_{\mathrm{F}}$	$E = pc_{\mathrm{F}}$	spectrum is $E = pc_{\mathrm{F}}$, where c_{F} is the effective speed of fermions in the 2D carbon
mielke-geometrodynamics_F01106	223	1.000	c_{F}	c_{F}	spectrum is $E = pc_{\mathrm{F}}$, where c_{F} is the effective speed of fermions in the 2D carbon
mielke-geometrodynamics_F01107	223	0.825	$\mathrm{SO}(3)$	$\mathrm{SO}(3)$	space(time): in the Euclidean case, the covering group of the rotation group $\mathrm{SO}(3)$
mielke-geometrodynamics_F01108	224	0.902	$s=1$	$s = 1$	On the other hand, for Lorentzian signature $s = 1$, the covering group of $\mathrm{SO}(1,2)$ is
mielke-geometrodynamics_F01109	224	0.942	$S L(2, \mathbb{R})$	$SL(2, \mathbb{R})$	isomorphic to the real group $SL(2, \mathbb{R})$, or as well to the symplectic algebra $\mathfrak{sp}(2, \mathbb{R})$.
mielke-geometrodynamics_F01110	224	0.942	$\mathfrak{s p}(2, \mathbb{R})$	$\mathfrak{sp}(2, \mathbb{R})$	isomorphic to the real group $SL(2, \mathbb{R})$, or as well to the symplectic algebra $\mathfrak{sp}(2, \mathbb{R})$.
mielke-geometrodynamics_F01111	224	1.000	$\gamma_{\alpha} \gamma_{\beta}+\gamma_{\beta} \gamma_{\alpha}=2 g_{\alpha \beta}$	$\gamma_\alpha\gamma_\beta + \gamma_\beta\gamma_\alpha = 2g_{\alpha\beta}$	thus providing a realization of the Clifford algebra $\gamma_\alpha\gamma_\beta + \gamma_\beta\gamma_\alpha = 2g_{\alpha\beta}$ in 3D.
mielke-geometrodynamics_F01112	224	1.000	$D \psi:=d \psi-\frac{1}{2} \gamma_{\alpha} \wedge \Gamma^{\star \alpha} \psi$	$D\psi := d\psi - \frac{1}{2}\gamma_\alpha \wedge \Gamma^{\star\alpha}\psi$	where $D\psi := d\psi - \frac{1}{2}\gamma_\alpha \wedge \Gamma^{\star\alpha}\psi$ denotes the covariant spinor derivative in 3D.
mielke-geometrodynamics_F01113	224	1.000	L_{D}	L_{D}	is obtained by varying L_{D} with respect to $\overline{\Psi}$. Since
mielke-geometrodynamics_F01114	224	1.000	$\bar{\Psi}$	$\bar{\Psi}$	is obtained by varying L_{D} with respect to $\overline{\Psi}$. Since
mielke-geometrodynamics_F01115	224	1.000	$\hat{D}$	$\hat{D}$	The curvature associated with the deformed covariant derivative $\hat{D}$ is

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mielke-geometrodynamics_F01116	225	■ 0.999	F:=d A	$F := dA$	field strength reads $F := dA$, i.e., it remains an exact two-form. In 3D, the Maxwell
mielke-geometrodynamics_F01117	225	■ 0.539	d\left(A \wedge^* F\right) / 2	$d(A \wedge^* F) / 2$	However, this three-form is gauge-invariant only modulo an exact form $d(A \wedge^* F)/2$.
mielke-geometrodynamics_F01118	225	■ 0.528	\square:=d^*\{*\} d^*\{*\}-\{*\}^*\{*\} d^*\{*\} d	$\square := d^* d^* - * d^* d$	where $\square := d^* d^* - * d^* d$ is the d'Alembertian in 2 + 1 dimensions. Since (10.6.3) is a
mielke-geometrodynamics_F01119	225	■ 1.000	\lambda=\hbar / c m_{\text {photon }}	$\lambda = \hbar / c m_{\text{photon}}$	reasons, via the Compton wavelength $\lambda = \hbar / c m_{\text{photon}}$ for massive photons.
mielke-geometrodynamics_F01120	226	■ 0.987	x^{\alpha}	x^α	vention that x^α and y^α are spacelike orthogonal vectors that span the (x, y) -plane
mielke-geometrodynamics_F01121	226	■ 0.987	y^{\alpha}	y^α	vention that x^α and y^α are spacelike orthogonal vectors that span the (x, y) -plane
mielke-geometrodynamics_F01122	226	■ 0.987	(x, y)	(x, y)	vention that x^α and y^α are spacelike orthogonal vectors that span the (x, y) -plane
mielke-geometrodynamics_F01123	226	■ 1.000	n^{\alpha}	n^α	Moreover, the vector n^α is a timelike unit vector normal to the hypersurface with
mielke-geometrodynamics_F01124	226	■ 1.000	n^{\alpha} n_{\alpha}=s	$n^\alpha n_\alpha = s$	$n^\alpha n_\alpha = s$, the signature s of our 3D spacetime. Following SOLENG (1992), ANANDAN
mielke-geometrodynamics_F01125	226	■ 1.000	\tau_{\alpha}^{\star}	τ_α^*	(1994), and BAKKE et al. (2009), we assume that the two-forms Σ_α and τ_α^* of the
mielke-geometrodynamics_F01126	227	■ 1.000	X	X	of the defect; cf. Fig. 10.2. From the specification (10.7.1) of the one-forms X and
mielke-geometrodynamics_F01127	227	■ 1.000	Y	Y	Y , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$
mielke-geometrodynamics_F01128	227	■ 1.000	\Sigma_{\hat{0}} \neq 0	$\Sigma_{\hat{0}} \neq 0$	Y , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$
mielke-geometrodynamics_F01129	227	■ 1.000	\tau_{\hat{1}\hat{2}}=	$\tau_{\hat{1}\hat{2}} =$	Y , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$
mielke-geometrodynamics_F01130	227	■ 0.975	-\tau_{\hat{2}\hat{1}} \neq 0	$-\tau_{\hat{2}\hat{1}} \neq 0$	$-\tau_{\hat{2}\hat{1}} \neq 0$.
mielke-geometrodynamics_F01131	227	■ 1.000	x^{[\alpha} y^{\beta]} R_{\alpha \beta}=R_{\hat{1}\hat{2}}=-R_{\hat{2}\hat{1}} \neq 0	$x^{[\alpha} y^{\beta]} R_{\alpha \beta} = R_{\hat{1}\hat{2}} = -R_{\hat{2}\hat{1}} \neq 0$	with x^α and y^α reveal that $x^{[\alpha} y^{\beta]} R_{\alpha \beta} = R_{\hat{1}\hat{2}} = -R_{\hat{2}\hat{1}} \neq 0$ are the only nonvanishing
mielke-geometrodynamics_F01132	227	■ 0.511	\left.N^{\alpha}=n\right\rfloor \vartheta^{\alpha}	$N^\alpha = n \rfloor \vartheta^\alpha$	We recall that $N^\alpha = n \rfloor \vartheta^\alpha$ is the lapse and shift vector in the (2+1) decomposition à
mielke-geometrodynamics_F01133	228	■ 1.000	d(\mathscr{A} \wedge d \mathscr{A})	$d(\mathscr{A} \wedge d\mathscr{A})$	Pointryagin-type term $d(\mathscr{A} \wedge d\mathscr{A})$ from the axial torsion. Moreover, the Nieh–Yan
mielke-geometrodynamics_F01134	228	■ 1.000	d C_{\mathrm{T}}	dC_{T}	term dC_{T} proportional to $d^* \mathscr{A}$ <i>vanishes identically</i> for this example of a spinning
mielke-geometrodynamics_F01135	228	■ 1.000	d^* \mathscr{A}	$d^* \mathscr{A}$	term dC_{T} proportional to $d^* \mathscr{A}$ <i>vanishes identically</i> for this example of a spinning
mielke-geometrodynamics_F01136	228	■ 1.000	\mathscr{N}=1	$\mathscr{N} = 1$	try generator, i.e., $\mathscr{N} = 1$, represents the simplest consistent coupling of a Rarita–
mielke-geometrodynamics_F01137	228	■ 0.961	\left.\Psi_{\alpha}:=e_{\alpha}\right\rfloor \Psi	$\Psi_\alpha := e_\alpha \rfloor \Psi$	as $\Psi_\alpha := e_\alpha \rfloor \Psi$ involves the tetrad that is inverse to the frame in the anholonomic
mielke-geometrodynamics_F01138	228	■ 1.000	i, j, k, \ldots=0,1,2	$i, j, k, \dots = 0, 1, 2$	run over $i, j, k, \dots = 0, 1, 2$, whereas $\alpha, \beta, \dots = \hat{0}, \hat{1}, \hat{2}$ for the anholonomic indices.
mielke-geometrodynamics_F01139	228	■ 1.000	\alpha, \beta, \ldots=\hat{0}, \hat{1}, \hat{2}	$\alpha, \beta, \dots = \hat{0}, \hat{1}, \hat{2}$	run over $i, j, k, \dots = 0, 1, 2$, whereas $\alpha, \beta, \dots = \hat{0}, \hat{1}, \hat{2}$ for the anholonomic indices.
mielke-geometrodynamics_F01140	228	■ 1.000	\bar{\Psi}:=\Psi^{\dagger} \gamma^0	$\bar{\Psi} := \Psi^\dagger \gamma^0$	$\bar{\Psi} := \Psi^\dagger \gamma^0$.
mielke-geometrodynamics_F01141	228	■ 1.000	\Psi:=\Psi_{\alpha} \vartheta^{\alpha}	$\Psi := \Psi_\alpha \vartheta^\alpha$	³ In four dimensions (4D), the RS field $\Psi := \Psi_\alpha \vartheta^\alpha$ entering (10.8.1) is a <i>Majorana-spinor</i> -valued
mielke-geometrodynamics_F01142	228	■ 0.700	\Psi=C \bar{\Psi}^{\mathrm{T}}	$\Psi = C \bar{\Psi}^{\text{T}}$	i.e., $\Psi = C \bar{\Psi}^{\text{T}}$, where C is the charge conjugation matrix given by $C = -i\gamma_0$ satisfying $C^\dagger =$

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mielke-geometrodynamics_F01143	228	0.700	$C=-i\ \gamma_{\{0\}}$	$C = -i\gamma_0$	i.e., $\Psi = C\bar{\Psi}^T$, where C is the charge conjugation matrix given by $C = -i\gamma_0$ satisfying $C^\dagger =$
mielke-geometrodynamics_F01144	228	0.700	$C^{\dagger}=$	$C^\dagger =$	i.e., $\Psi = C\bar{\Psi}^T$, where C is the charge conjugation matrix given by $C = -i\gamma_0$ satisfying $C^\dagger =$
mielke-geometrodynamics_F01145	228	1.000	$C^{-1},\ \quad C^T=-C$	$C^{-1},\ \quad C^T = -C$	$C^{-1},\ \quad C^T = -C$ and $C^{-1}\gamma^\alpha C = -(\gamma^\alpha)^T$. Consequently,
mielke-geometrodynamics_F01146	228	1.000	$C^{-1}\ \gamma^\alpha\ C=-\left(\gamma^\alpha\right)^T$	$C^{-1}\gamma^\alpha C = -(\gamma^\alpha)^T$	$C^{-1},\ \quad C^T = -C$ and $C^{-1}\gamma^\alpha C = -(\gamma^\alpha)^T$. Consequently,
mielke-geometrodynamics_F01147	228	1.000	γ^α	γ^α	For the real Majorana representation, all γ^α are purely imaginary, and the components of the
mielke-geometrodynamics_F01148	229	0.803	L_{RS}	L_{RS}	The generalization of L_{RS} analyzed by MIELKE & MAGGIOLO (2012) exists only
mielke-geometrodynamics_F01149	230	0.995	$\vartheta^\alpha,\Gamma_\alpha^\star,\Psi$	$\vartheta^\alpha,\Gamma_\alpha^\star,\Psi$	independent variables $(\vartheta^\alpha,\Gamma_\alpha^\star,\Psi)$ yields
mielke-geometrodynamics_F01150	230	1.000	c	c	where σ stands in for a spinor-valued zero-form and c is a real constant. Let us probe
mielke-geometrodynamics_F01151	230	1.000	$\overline{c\ \gamma\ \sigma}=c\ \overline{\sigma}\ \gamma$	$\overline{c\gamma\sigma} = c\overline{\sigma}\gamma$	where $\overline{c\gamma\sigma} = c\overline{\sigma}\gamma$ holds for the Dirac adjoint.
mielke-geometrodynamics_F01152	230	1.000	$\delta L\ /\ \delta \Gamma_\alpha^\star\ \cong\ 0$	$\delta L/\delta\Gamma_\alpha^\star \cong 0$	In the following, let us assume that the second field equation $\delta L/\delta\Gamma_\alpha^\star \cong 0$ satisfies,
mielke-geometrodynamics_F01153	231	1.000	$c=-m$	$c = -m$	$\tau_\alpha^\star$, vanish. Moreover, we require $c = -m$, in order to eliminate kinetic terms like
mielke-geometrodynamics_F01154	231	1.000	$\gamma\ \wedge\ D\ \Psi$	$\gamma \wedge D\Psi$	$\gamma \wedge D\Psi$. Then, using the 3D formula
mielke-geometrodynamics_F01155	232	0.999	L_{∞}	L_∞	for the coupling constants of the bosonic part of L_∞ . Consequently, massless RS
mielke-geometrodynamics_F01156	232	1.000	$\mathscr{N}=2$	$\mathcal{N} = 2$	More recently, ALVAREZ et al. (2015) considered an $\mathcal{N} = 2$ supersymmetry with-
mielke-geometrodynamics_F01157	233	0.988	$\mathrm{D}=3$	$\text{D} = 3$	Deser S, McCarthy J (1990) Self-dual formulations of $\text{D} = 3$ gravity theories. Phys Lett B
mielke-geometrodynamics_F01158	233	0.666	$(2+1)$	$(2 + 1)$	Garcia-Compean H, Obregon O, Ramirez C, Sabido M (2001) On S duality in (2+1) Chern–Simons
mielke-geometrodynamics_F01159	237	0.772	$1\ /\ 2$	$1/2$	$1/2$ are described by bispinor representations $D^{(-\frac{1}{2},\frac{3}{2})} \otimes D^{(\frac{1}{2},\frac{3}{2})}(\widetilde{SO}_\circ(1,3))$ of the
mielke-geometrodynamics_F01160	237	0.772	$D^{\left(-\frac{1}{2},\frac{3}{2}\right)}\otimes D^{\left(\frac{1}{2},\frac{3}{2}\right)}\left(\widetilde{SO}_\circ(1,3)\right)$	$D^{(-\frac{1}{2},\frac{3}{2})} \otimes D^{(\frac{1}{2},\frac{3}{2})} \left(\widetilde{SO}_\circ(1,3)\right)$	$1/2$ are described by bispinor representations $D^{(-\frac{1}{2},\frac{3}{2})} \otimes D^{(\frac{1}{2},\frac{3}{2})}(\widetilde{SO}_\circ(1,3))$ of the
mielke-geometrodynamics_F01161	237	1.000	$S\ O_{\circ}(1,3)$	$SO_\circ(1,3)$	proper Lorentz group $SO_\circ(1,3)$ (GEL’FAND et al. 1963). ² These representations
mielke-geometrodynamics_F01162	237	0.983	$P_{\circ}:=\mathbb{R}^4\otimes S\ O_{\circ}(1,3)$	$P_\circ := \mathbb{R}^4 \otimes SO_\circ(1,3)$	resentations of the (restricted) Poincaré group $P_\circ := \mathbb{R}^4 \otimes SO_\circ(1,3)$ of flat
mielke-geometrodynamics_F01163	237	1.000	$\psi_{\infty}:=\psi_{1\ /\ 2}\otimes\psi_{5\ /\ 2}\otimes\psi_{9\ /\ 2}\otimes\ldots$	$\psi_\infty := \psi_{1/2} \otimes \psi_{5/2} \otimes \psi_{9/2} \otimes \dots$	$\psi_\infty := \psi_{1/2} \otimes \psi_{5/2} \otimes \psi_{9/2} \otimes \dots$
mielke-geometrodynamics_F01164	237	0.706	$\mathrm{SO}(s,\ n-s)$	$\text{SO}(s, n - s)$	(pseudo-orthogonal) structure group $\text{SO}(s, n - s)$ of the tangent bundle of an
mielke-geometrodynamics_F01165	237	1.000	$s\ \neq\ 0$	$s \neq 0$	n-dimensional manifold ³ of signature s . Provided $s \neq 0$, this topological space
mielke-geometrodynamics_F01166	237	1.000	$S\ O_{\circ}(s,\ n-s)$	$SO_\circ(s, n - s)$	ing to homotopy theory. Concerning the connected component $SO_\circ(s, n - s)$ of the
mielke-geometrodynamics_F01167	237	0.929	$\widetilde{S\ O}_{\circ}(s,\ n-s)(=:\operatorname{Spin}(n))$	$\widetilde{SO}_\circ(s, n - s)(=: \text{Spin}(n))$	$\widetilde{SO}_\circ(s, n - s) (=:\text{Spin}(n)$ for $s = 0$), whose irreducible representations provide
mielke-geometrodynamics_F01168	237	0.929	$s=0)$	$s = 0)$	$\widetilde{SO}_\circ(s, n - s) (=:\text{Spin}(n)$ for $s = 0$), whose irreducible representations provide
mielke-geometrodynamics_F01169	238	1.000	$\Lambda:\ \widetilde{\mathrm{G}}\rightarrow\mathrm{G}$	$\Lambda : \tilde{G} \rightarrow G$	This transition is achieved by a group homomorphism $\Lambda : \tilde{G} \rightarrow G$, which is

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mielke-geometrodynamics_F01170	238	■ 1.000	$C(s, n-s)$	$C(s, n-s)$	based on the following explicit construction: Let $C(s, n-s)$ denote the so-called
mielke-geometrodynamics_F01171	238	■ 1.000	γ_{α}	γ_{α}	Clifford algebra, which is spanned by <i>generalized</i> complex Dirac matrices γ_{α} . These
mielke-geometrodynamics_F01172	238	■ 1.000	$N \times N$	$N \times N$	and admit realizations by $N \times N$ matrices for which $N = 2^{\lfloor n/2 \rfloor}$ corresponds to the
mielke-geometrodynamics_F01173	238	■ 1.000	$N=2^{\lfloor n/2 \rfloor}$	$N = 2^{\lfloor n/2 \rfloor}$	and admit realizations by $N \times N$ matrices for which $N = 2^{\lfloor n/2 \rfloor}$ corresponds to the
mielke-geometrodynamics_F01174	238	■ 1.000	$\mathbf{E}^n$	$\mathbf{E}^n$	Let the transformations of the Euclidean vector space $\mathbf{E}^n$ with coordinates x^{α} be
mielke-geometrodynamics_F01175	238	■ 0.963	$\mathbf{g} \in SO(s, n-s)$	$\mathbf{g} \in SO(s, n-s)$	determined by the (Lorentz) rotations $\mathbf{g} \in SO(s, n-s)$ on behalf of the group action
mielke-geometrodynamics_F01176	238	■ 0.983	2^N	2^N	to a 2^N -dimensional bundle, the so-called Clifford bundle CV. For details, see ATIYAH
mielke-geometrodynamics_F01177	239	■ 0.976	$\widetilde{SO}_{\circ}(1,3) \approx SL(2, \mathbb{C})$	$\widetilde{SO}_{\circ}(1,3) \approx SL(2, \mathbb{C})$	orthochronous Lorentz group $SO_{\circ}(1,3)$ is represented by $\widetilde{SO}_{\circ}(1,3) \approx SL(2, \mathbb{C})$
mielke-geometrodynamics_F01178	239	■ 1.000	$\widetilde{SO}_{\circ}(s, n-s)$	$\widetilde{SO}_{\circ}(s, n-s)$	infinitesimal generators of the group $\widetilde{SO}_{\circ}(s, n-s)$ are
mielke-geometrodynamics_F01179	239	■ 0.588	$\mathbf{P}_0 := \mathbb{R}^n \otimes \widetilde{SO}$	$\mathbf{P}_0 := \mathbb{R}^n \otimes \widetilde{SO}$	as induced representations of the “covering” Poincaré group $\mathbf{P}_0 := \mathbb{R}^n \otimes \widetilde{SO}_{\circ}$.
mielke-geometrodynamics_F01180	239	■ 1.000	$(s, n-s)$	$(s, n-s)$	$(s, n-s)$ in a curved spacetime. It is necessary to obtain from a more general
mielke-geometrodynamics_F01181	239	■ 1.000	$L^{\mathbf{g}}(M)$	$L^{\mathbf{g}}(M)$	“geometric arena” the bundle $L^{\mathbf{g}}(M)$ of orthogonal frames concerning the foundation
mielke-geometrodynamics_F01182	239	■ 0.974	$L_{\circ}^{\mathbf{g}}(M)$	$L_{\circ}^{\mathbf{g}}(M)$	a <i>spin structure</i> is imposed on a base space M as follows: Besides $L_{\circ}^{\mathbf{g}}(M)$, the
mielke-geometrodynamics_F01183	239	■ 0.998	$\mathbf{f}: \tilde{L}(M) \rightarrow L_{\circ}^{\mathbf{g}}(M)$	$\mathbf{f}: \tilde{L}(M) \rightarrow L_{\circ}^{\mathbf{g}}(M)$	is introduced (MILNOR 1963) together with the bundle map $\mathbf{f}: \tilde{L}(M) \rightarrow L_{\circ}^{\mathbf{g}}(M)$,
mielke-geometrodynamics_F01184	239	■ 0.950	$L_{\circ}^{\mathbf{g}}(M)$	$L_{\circ}^{\mathbf{g}}(M)$	A spin structure over M is equivalent to a double covering of $L_{\circ}^{\mathbf{g}}(M)$. This spin
mielke-geometrodynamics_F01185	239	■ 0.949	$\tilde{L}'(M), f'$	$\tilde{L}'(M), f'$	structure is identical to a second one that is represented by $(\tilde{L}'(M), f')$ if the iso-
mielke-geometrodynamics_F01186	239	■ 1.000	f'	f'	bundle mapping f' . Additionally, a <i>spin manifold</i> is understood as a Riemannian
mielke-geometrodynamics_F01187	239	■ 1.000	$\mathbf{w}_2 \in H^2(M, \mathbb{Z}_2)$	$\mathbf{w}_2 \in H^2(M, \mathbb{Z}_2)$	it is necessary and sufficient that the Stiefel–Whitney class $\mathbf{w}_2 \in H^2(M, \mathbb{Z}_2)$ of
mielke-geometrodynamics_F01188	239	■ 1.000	$\mathbf{g}^{\text{th}}$	$\mathbf{g}^{\text{th}}$	the base space M be zero. Here the $\mathbf{g}^{\text{th}}$ Stiefel–Whitney class is considered as the
mielke-geometrodynamics_F01189	240	■ 1.000	$H^{\mathbf{g}}(M, \mathbb{Z}_2)$	$H^{\mathbf{g}}(M, \mathbb{Z}_2)$	characteristic class of the $\mathbf{g}^{\text{th}}$ relative cohomology group $H^{\mathbf{g}}(M, \mathbb{Z}_2)$ of the manifold
mielke-geometrodynamics_F01190	240	■ 1.000	$\tilde{s} = \dim H^1(M, \mathbb{Z}_2)$	$\tilde{s} = \dim H^1(M, \mathbb{Z}_2)$	defined by the dimension $\tilde{s} = \dim H^1(M, \mathbb{Z}_2)$ of the first relative cohomology group
mielke-geometrodynamics_F01191	240	■ 0.813	$\tilde{s} = 2^v$	$\tilde{s} = 2^v$	(MILNOR 1963). Accordingly, there are exactly $\tilde{s} = 2^v$ spin structures possible
mielke-geometrodynamics_F01192	240	■ 0.997	ν	ν	hypersurface consists of the connected sum of ν handlebodies (“wormholes”) $W^3 :=$
mielke-geometrodynamics_F01193	240	■ 0.997	$W^3 :=$	$W^3 :=$	hypersurface consists of the connected sum of ν handlebodies (“wormholes”) $W^3 :=$
mielke-geometrodynamics_F01194	240	■ 0.999	$\mathbf{w}_1 = 0$	$\mathbf{w}_1 = 0$	s) bundles. The last bundle can be oriented if $\mathbf{w}_1 = 0$ is valid, while the number of
mielke-geometrodynamics_F01195	240	■ 0.993	$H^{\circ}(M, \mathbb{Z}_2)$	$H^{\circ}(M, \mathbb{Z}_2)$	distinguishable orientations has to be processed out of the dimension of $H^{\circ}(M, \mathbb{Z}_2)$
mielke-geometrodynamics_F01196	240	■ 1.000	S^7	S^7	dimensional sphere S^7 (MILNOR 1965). Attention can be drawn to other topologically

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mielke-geometrodynamics_F01197	240	■ 0.998	(1968,1970)	(1968, 1970)	However, it was GEROCH (1968, 1970) who pointed out an intuitive and in the
mielke-geometrodynamics_F01198	240	■ 0.969	$T(M)$	$T(M)$	structures on $T(M)$ is dependent only on the mapping f (ISHAM 1978).
mielke-geometrodynamics_F01199	240	■ 0.925	$S\ U(3)$	$SU(3)$	invariance or Gell-Mann’s $SU(3)$ -group as well into these global constructions, the
mielke-geometrodynamics_F01200	241	■ 1.000	$\tilde{\mathrm{L}}(\mathrm{M})$	$\tilde{\mathbb{L}}(M)$	<i>associated</i> to the spin bundle $\tilde{\mathbb{L}}(M)$. For this reason, it is also called an <i>associ-</i>
mielke-geometrodynamics_F01201	241	■ 0.614	D^{F}	D^{F}	plex) representation D^{F} of dimension $N = 2^{[n/2]}$ of the structure group $\widetilde{SO}_{\circ}$.
mielke-geometrodynamics_F01202	241	■ 0.614	$\widetilde{S\ 0}_{\circ}$	$\widetilde{SO}_{\circ}$	plex) representation D^{F} of dimension $N = 2^{[n/2]}$ of the structure group $\widetilde{SO}_{\circ}$.
mielke-geometrodynamics_F01203	241	■ 0.986	$\widetilde{S\ 0}_{\circ}(1,3)$	$\widetilde{SO}_{\circ}(1,3)$	$(s, n - s)$, which contains the physical Lorentz group $\widetilde{SO}_{\circ}(1, 3)$ as a subgroup.
mielke-geometrodynamics_F01204	241	■ 0.996	$S\ L(2, \mathbb{C}) \approx \widetilde{S\ 0}_{\circ}(1,3)$	$SL(2, \mathbb{C}) \approx \widetilde{SO}_{\circ}(1,3)$	denotes the bispinor representation of the covering group $SL(2, \mathbb{C}) \approx \widetilde{SO}_{\circ}(1, 3)$ of
mielke-geometrodynamics_F01205	241	■ 1.000	$\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$	$\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$	the proper Lorentz group, which itself is the direct sum $\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$ of spin-
mielke-geometrodynamics_F01206	241	■ 0.999	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$ representations labeled according to GEL’FAND et al. (1963). This formulation
mielke-geometrodynamics_F01207	241	■ 1.000	$\tilde{0}(1,3)$	$\tilde{O}(1,3)$	group $\tilde{O}(1, 3)$ if required. Concerning the spinor basis $\tilde{b}_{(1)}, i = 1, \dots, N$, their cross
mielke-geometrodynamics_F01208	241	■ 1.000	$\tilde{b}_{(1)}, i=1, \ldots, N$	$\tilde{b}_{(1)}, i = 1, \dots, N$	group $\tilde{O}(1, 3)$ if required. Concerning the spinor basis $\tilde{b}_{(1)}, i = 1, \dots, N$, their cross
mielke-geometrodynamics_F01209	242	■ 0.999	$\mathscr{G}_S$	$\mathscr{G}_S$	of the group $\mathscr{G}_S$ of (local) spin gauge transformations. This group results from $\mathscr{G}_p$, a
mielke-geometrodynamics_F01210	242	■ 0.999	$\mathscr{G}_p$	$\mathscr{G}_p$	of the group $\mathscr{G}_S$ of (local) spin gauge transformations. This group results from $\mathscr{G}_p$, a
mielke-geometrodynamics_F01211	242	■ 1.000	$\tilde{L}(M)$	$\tilde{L}(M)$	The spin bundle $\tilde{L}(M)$ can, analogously to the bundle $L^g(M)$ of the orthonormal
mielke-geometrodynamics_F01212	242	■ 1.000	$\gamma^{\mathrm{n}+1}$	$\gamma^{\mathrm{n}+1}$	The matrix $\gamma^{\mathrm{n}+1}$ that is thereby implicitly defined is an axial Lorentz scalar and can
mielke-geometrodynamics_F01213	243	■ 1.000	$\gamma^{\mathrm{n}+1}$	$\gamma^{\mathrm{n}+1}$	If one realizes the Dirac matrices in the “Cartan basis,” then $\gamma^{\mathrm{n}+1}$ can be regarded as
mielke-geometrodynamics_F01214	243	■ 0.999	$\tilde{\omega}$	$\tilde{\omega}$	The introduction of a $\widetilde{SO}_{\circ}(s, n - s)$ -valued 1-form $\tilde{\omega}$ of the connection in the asso-
mielke-geometrodynamics_F01215	243	■ 0.790	$\Gamma_i^{\alpha\beta}$	$\Gamma_i^{\alpha\beta}$	The local components of the spin connection, i.e., Ricci’s rotation coefficients $\Gamma_i^{\alpha\beta}$,
mielke-geometrodynamics_F01216	243	■ 0.958	$\Gamma_{ij}^{\cdot k}$	$\Gamma_{ij}^{\cdot k}$	can be generated from the holonomic connection coefficients $\Gamma_{ij}^{\cdot k}$ by means of
mielke-geometrodynamics_F01217	244	■ 0.995	$\mathbb{R}^n \otimes S\ 0$	$\mathbb{R}^n \otimes SO$	If we had started from a spinor representation of the Poincaré group $\mathbb{R}^n \ltimes SO_{\circ}$.
mielke-geometrodynamics_F01218	244	■ 0.996	$\mathrm{e}_{\alpha}^{-i}(m)$	$\mathbf{e}_{\alpha}^{-i}(m)$	strived for an extension by inserting the tetrad fields $\mathbf{e}_{\alpha}^{-i}(m)$, i.e., the local bundle coor-
mielke-geometrodynamics_F01219	245	■ 1.000	$\mathcal{D}(\mathrm{M})$	$\mathcal{D}(M)$	with respect to the group $\mathcal{D}(M)$ of differentiable coordinate transformations is again
mielke-geometrodynamics_F01220	245	■ 1.000	$\delta L_{\mathrm{D}} / \delta \bar{\psi}$	$\delta L_{\mathrm{D}} / \delta \bar{\psi}$	By variation of (11.2.1) for $\delta L_{\mathrm{D}} / \delta \bar{\psi}$ or for $\delta L_{\mathrm{D}} / \delta \psi$, respectively, the generally
mielke-geometrodynamics_F01221	245	■ 1.000	$\delta L_{\mathrm{D}} / \delta \psi$	$\delta L_{\mathrm{D}} / \delta \psi$	By variation of (11.2.1) for $\delta L_{\mathrm{D}} / \delta \bar{\psi}$ or for $\delta L_{\mathrm{D}} / \delta \psi$, respectively, the generally
mielke-geometrodynamics_F01222	246	■ 1.000	$\mathrm{D}\gamma = 0$	$\mathbf{D}\gamma = 0$	for a covariant constant γ , i.e., for $\mathbf{D}\gamma = 0$. However, the latter condition is globally

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mielke-geometrodynamics_F01223	246	■ 1.000	$\tau := \delta L / \delta \omega$	$\tau := \delta L / \delta \omega$	using the rules concerning exterior forms. According to the definitions $\tau := \delta L / \delta \omega$
mielke-geometrodynamics_F01224	246	■ 1.000	$\tau = i \bar{\psi} \gamma \psi$	$\tau = i \bar{\psi} \gamma \psi$	$\tau = i \bar{\psi} \gamma \psi$ of its charge current, but also on the (Lorentz) rotational gauge fields of
mielke-geometrodynamics_F01225	246	■ 0.992	τ_s	τ_s	respect to the current τ or τ_s , and this according to whether the “internal” Yang–
mielke-geometrodynamics_F01226	246	■ 0.999	$j_{n+1} := i \bar{\psi} \gamma \gamma^{n+1} \psi$	$j_{n+1} := i \bar{\psi} \gamma \gamma^{n+1} \psi$	⁶ A corresponding conservation law is also valid for the axial vector current $j_{n+1} := i \bar{\psi} \gamma \gamma^{n+1} \psi$
mielke-geometrodynamics_F01227	247	■ 1.000	H^{n-1}	H^{n-1}	hypersurfaces H^{n-1} , this is not only independent of time, but even <i>positive definite</i> ,
mielke-geometrodynamics_F01228	247	■ 1.000	τ_s^0	τ_s^0	Nevertheless, one could not interpret τ_s^0 as the probability density in a non quantized
mielke-geometrodynamics_F01229	247	■ 0.999	$\tilde{\omega}^{\{\}}$	$\tilde{\omega}^{\{\}}$	into the Christoffel-like connection $\tilde{\omega}^{\{\}}$ and the spin contortion that is projected by
mielke-geometrodynamics_F01230	247	■ 0.826	$D^{\{\}}$	$D^{\{\}}$	Here $D^{\{\}}$ denotes—similarly as in HEHL et al. (1976)—the covariant derivative with
mielke-geometrodynamics_F01231	247	■ 0.993	$\Theta = [K, \vartheta]$	$\Theta = [K, \vartheta]$	occurs in RC spacetime. Considering the resolution of the relation $\Theta = [K, \vartheta]$ for
mielke-geometrodynamics_F01232	248	■ 1.000	$[\Theta, \vartheta]$	$[\Theta, \vartheta]$	it follows that the modified torsion $[\Theta, \vartheta]$ couples <i>algebraically</i> to the canonical
mielke-geometrodynamics_F01233	248	■ 1.000	$\delta L_D / \delta \bar{\psi}$	$\delta L_D / \delta \bar{\psi}$	$\delta L_D / \delta \bar{\psi}$. This equation will be referred to as the <i>nonlinear Heisenberg–Pauli–Weyl</i>
mielke-geometrodynamics_F01234	249	■ 1.000	$D = \vartheta^\alpha \nabla_\alpha$	$D = \vartheta^\alpha \nabla_\alpha$	with $D = \vartheta^\alpha \nabla_\alpha$ taken into considerations, as follows:
mielke-geometrodynamics_F01235	249	■ 1.000	(1938, 1957)	(1938, 1957)	was IVANENKO (1938, 1957) who, for quantum-theoretic reasons, pointed out the
mielke-geometrodynamics_F01236	249	■ 1.000	PC	PC	reflection and charge conjugation PC , and the time reflection T . To be added is the
mielke-geometrodynamics_F01237	249	■ 0.520	$\psi := \binom{\psi_p}{\psi_N}$	$\psi := \binom{\psi_p}{\psi_N}$	fundamental equation for the isotopic doublet $\psi := \binom{\psi_p}{\psi_N}$, this equation had to
mielke-geometrodynamics_F01238	249	■ 1.000	$SL(2, \mathbb{C})$	$SL(2, \mathbb{C})$	group $SL(2, \mathbb{C})$ of Weyl to a gauge theory with $\overline{G} = SL(2f, \mathbb{C})$ as a structure group,
mielke-geometrodynamics_F01239	249	■ 1.000	$\overline{G} = SL(2f, \mathbb{C})$	$\overline{G} = SL(2f, \mathbb{C})$	group $SL(2, \mathbb{C})$ of Weyl to a gauge theory with $\overline{G} = SL(2f, \mathbb{C})$ as a structure group,
mielke-geometrodynamics_F01240	249	■ 0.994	$\overline{G}$	$\overline{G}$	results in a $\overline{G}$ -invariant nonlinear Heisenberg–Pauli–Weyl spinor equation (MIELKE
mielke-geometrodynamics_F01241	251	■ 1.000	$1 / r^2$	$1 / r^2$	and thus replaces Coulomb’s $1 / r^2$ -dependence on the distance $r := \vec{\mathbf{x}} $ from the cen-
mielke-geometrodynamics_F01242	251	■ 1.000	$r := \overrightarrow{\mathbf{x}} $	$r := \vec{\mathbf{x}} $	and thus replaces Coulomb’s $1 / r^2$ -dependence on the distance $r := \vec{\mathbf{x}} $ from the cen-
mielke-geometrodynamics_F01243	251	■ 1.000	(1934, 1935)	(1934, 1935)	context. Theoretically, it was coined by BORN & INFELD (1934, 1935) and, as far as a quantum-
mielke-geometrodynamics_F01244	252	■ 0.979	$n=4$	$n = 4$	that is given by (11.3.7) exclusively in the physically(!) relevant case of $n = 4$ dimen-
mielke-geometrodynamics_F01245	252	■ 1.000	$\varepsilon = \pm 1, \ell$	$\varepsilon = \pm 1, \ell$	the following analysis. On the other hand, the constants $\varepsilon = \pm 1, \ell$, and a are to be
mielke-geometrodynamics_F01246	253	■ 0.999	$v = v(\rho)$	$v = v(\rho)$	Concerning <i>static</i> spinor fields, the functions $v = v(\rho)$ and $\mu = \mu(\rho)$ depend only
mielke-geometrodynamics_F01247	253	■ 0.999	$\mu = \mu(\rho)$	$\mu = \mu(\rho)$	Concerning <i>static</i> spinor fields, the functions $v = v(\rho)$ and $\mu = \mu(\rho)$ depend only
mielke-geometrodynamics_F01248	253	■ 1.000	$\mathbf{r} := \overrightarrow{\mathbf{x}} , \vartheta$	$\mathbf{r} := \vec{\mathbf{x}} , \vartheta$	and the metric becomes static itself. Thereby the spherical coordinates $\mathbf{r} := \vec{\mathbf{x}} , \vartheta$,

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mielke-geometrodynamics_F01249	253	■ 1.000	$\overrightarrow{\mathrm{x}}$	$\overrightarrow{\mathbf{x}}$	and ϕ are related to the Cartesian coordinates $\overrightarrow{\mathbf{x}}$ in the familiar manner. The search for
mielke-geometrodynamics_F01250	253	■ 1.000	g_{kl}	g_{kl}	can be simplified by conformally relating the components g_{kl} of the metric with those
mielke-geometrodynamics_F01251	253	■ 0.919	$\Gamma_{\kappa}^{\{f \cdot \alpha \beta\}}$	$\Gamma_{\kappa}^{\{f \cdot \alpha \beta\}}$	between the Christoffel symbols and the torsion-free rotation coefficients $\Gamma_{\kappa}^{\{ \cdot \} \alpha \beta}$, the
mielke-geometrodynamics_F01252	254	■ 0.742	$\mathrm{R}_{\circ}$	$\mathrm{R}_{\circ}$	$\mathrm{R}_{\circ}$. For this purpose, the de Sitter space is to be covered with the so-called horospher-
mielke-geometrodynamics_F01253	254	■ 0.990	$\lambda, \mathbf{y}$	$\lambda, \mathbf{y}$	ical coordinate system $(\lambda, \mathbf{y})$. Concerning these coordinates, the line element of the
mielke-geometrodynamics_F01254	254	■ 0.984	$S_{0_{\circ}}(1, 4)$	$SO_{\circ}(1, 4)$	de Sitter group $SO_{\circ}(1, 4)$ (see MIELKE (1977c), Eqs. (1.23) and (4.5) in the case of
mielke-geometrodynamics_F01255	254	■ 1.000	$\bar{\Gamma}_k$	$\bar{\Gamma}_k$	For the sake of completeness, the conformally related component $\bar{\Gamma}_k$ of the spinor
mielke-geometrodynamics_F01256	254	■ 1.000	$i \vec{\gamma} \cdot \vec{\partial}$	$i \vec{\gamma} \cdot \vec{\partial}$	spacelike Dirac operator $i \vec{\gamma} \cdot \vec{\partial}$ as being written again in Cartesian coordinates.
mielke-geometrodynamics_F01257	255	■ 0.999	$\exp(-\mu/2 - \nu/4)$	$\exp(-\mu/2 - \nu/4)$	$\exp(-\mu/2 - \nu/4)$, as stated, into the separation ansatz
mielke-geometrodynamics_F01258	255	■ 0.643	χ_k^m	χ_k^m	Following the notation of ROSE (1961), the spin-weighted spherical harmonics χ_k^m
mielke-geometrodynamics_F01259	255	■ 1.000	$\mathrm{P} = (-1)^l$	$\mathrm{P} = (-1)^l$	of parity $\mathrm{P} = (-1)^l$ are given by
mielke-geometrodynamics_F01260	255	■ 1.000	l	l	Here the parameter κ is related to the quantum numbers j and l of the total angular
mielke-geometrodynamics_F01261	255	■ 1.000	$\mathrm{C}\left(l\frac{1}{2}j; m - \bar{m}, \bar{m}\right)$	$\mathrm{C}\left(l\frac{1}{2}j; m - \bar{m}, \bar{m}\right)$	Here $\mathrm{C}(l\frac{1}{2}j; m - \bar{m}, \bar{m})$ denote the Clebsch–Gordan coefficients of the rotation group
mielke-geometrodynamics_F01262	255	■ 0.996	$\mathrm{Y}_l^m(\vartheta, \phi)$	$\mathrm{Y}_l^m(\vartheta, \phi)$	SU(2), and $\mathrm{Y}_l^m(\vartheta, \phi)$ are the spherical harmonics. Concerning the operators $\vec{J}$ and
mielke-geometrodynamics_F01263	255	■ 0.996	$\vec{J}$	$\vec{J}$	SU(2), and $\mathrm{Y}_l^m(\vartheta, \phi)$ are the spherical harmonics. Concerning the operators $\vec{J}$ and
mielke-geometrodynamics_F01264	255	■ 1.000	$\vec{L}$	$\vec{L}$	$\vec{L}$ of the total spin and the orbital angular momentum, these spinors satisfy the
mielke-geometrodynamics_F01265	256	■ 1.000	$ \kappa =1$	$ \kappa =1$	follows from the properties of the spherical harmonics that this is the case for $ \kappa =1$
mielke-geometrodynamics_F01266	256	■ 1.000	$j=\frac{1}{2}, l=0, 1$	$j=\frac{1}{2}, l=0, 1$	to the values $j=\frac{1}{2}, l=0, 1$, and $m=\pm\frac{1}{2}$, respectively. States of higher quantum
mielke-geometrodynamics_F01267	256	■ 1.000	$m=\pm\frac{1}{2}$	$m=\pm\frac{1}{2}$	to the values $j=\frac{1}{2}, l=0, 1$, and $m=\pm\frac{1}{2}$, respectively. States of higher quantum
mielke-geometrodynamics_F01268	256	■ 1.000	$\int L_{D'}$	$\int L_{D'}$	functional $\int L_{D'}$ belonging to (11.4.3). After integrating over the angular variables
mielke-geometrodynamics_F01269	256	■ 1.000	ρ^*	ρ^*	schild solution, an utmost slow tending of ρ^* to the “coordinate singularity” is
mielke-geometrodynamics_F01270	257	■ 1.000	M^*	M^*	mass M^* of the (possibly “strong”) gravitational field is absorbed in the coefficients
mielke-geometrodynamics_F01271	257	■ 0.999	$j=\frac{1}{2}$	$j=\frac{1}{2}$	localized and regular solutions. For a soliton with total spin $j=\frac{1}{2}$, i.e., for $\kappa=-1$,
mielke-geometrodynamics_F01272	257	■ 0.999	$\kappa=-1$	$\kappa=-1$	localized and regular solutions. For a soliton with total spin $j=\frac{1}{2}$, i.e., for $\kappa=-1$,
mielke-geometrodynamics_F01273	257	■ 1.000	$H(\rho)$	$H(\rho)$	For a soliton with total spin $j=\frac{1}{2}$ and $\kappa=-1$, the radial functions $H(\rho)$ in the
mielke-geometrodynamics_F01274	257	■ 1.000	$l=0$	$l=0$	ansatz (11.4.15) account for a spherically symmetric distribution with $l=0$, while

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mielke-geometrodynamics_F01275	257	■ 1.000	$l=1$	$l = 1$	according to (11.4.17), the quantum number $l = 1$ of angular momentum can be
mielke-geometrodynamics_F01276	257	■ 1.000	$F(\rho)$	$F(\rho)$	ascribed to $F(\rho)$. Let us compare this with the situation in a nonlocal version of the
mielke-geometrodynamics_F01277	258	■ 0.943	$ \varphi ^6$	$ \varphi ^6$	$ \varphi ^6$ -model of ROSEN (1965). With reference to an expansion in terms of spherical
mielke-geometrodynamics_F01278	258	■ 1.000	$\mathrm{Y}_{\mathrm{l}}^{\mathrm{m}}(\vartheta, \phi)$	$\mathrm{Y}_l^{\mathrm{m}}(\vartheta, \phi)$	harmonics $\mathrm{Y}_l^{\mathrm{m}}(\vartheta, \phi)$, the occurring <i>exact</i> radial solutions
mielke-geometrodynamics_F01279	259	■ 0.726	$ \omega <1$	$ \omega < 1$	ansatz (11.4.15) satisfies $ \omega < 1$. On the other hand, it is known (VÁZQUEZ 1977)
mielke-geometrodynamics_F01280	259	■ 1.000	$ \omega >1$	$ \omega > 1$	that bounded and square-integrable radial solutions <i>do not</i> exist for $ \omega > 1$.
mielke-geometrodynamics_F01281	259	■ 1.000	$ \psi \sim e^{-m_{\pi} r}$	$ \psi \sim e^{-m_{\pi} r}$	$ \psi \sim e^{-m_{\pi} r}$ between the centers of charge of the solitons. This is the case if the
mielke-geometrodynamics_F01282	259	■ 1.000	m_{s}	$\mathbf{m}_{\mathrm{s}}$	The total mass m_{s} of a spinor soliton is obtained by the integration
mielke-geometrodynamics_F01283	259	■ 1.000	Σ_{D}	Σ_{D}	of the canonical energy momentum current Σ_{D} over a spacelike hypersurface H^3 .
mielke-geometrodynamics_F01284	259	■ 1.000	m_{s}	m_s	(11.4.40) is inversely proportional to the “fundamental” length ℓ^* and that m_s would
mielke-geometrodynamics_F01285	260	■ 0.979	(1951,1956)	(1951, 1956)	Following the preceding studies of FINKELSTEIN et al. (1951, 1956) and SOLER
mielke-geometrodynamics_F01286	260	■ 0.356	$a=0$	$a = 0$	(1970), which were mainly concerned with a model (11.4.3) with $a = 0$, RAÑADA
mielke-geometrodynamics_F01287	260	■ 0.988	$a=1$	$a = 1$	equation (11.3.7), i.e., on the Eq.(11.4.3) with $a = 1$ but with the opposite
mielke-geometrodynamics_F01288	260	■ 1.000	$\varepsilon=+1$	$\varepsilon = +1$	sign $\varepsilon = +1$. For solitons corresponding to an $S_{1/2}$ -bound state with normalized
mielke-geometrodynamics_F01289	260	■ 1.000	$S_{1/2}$	$S_{1/2}$	sign $\varepsilon = +1$. For solitons corresponding to an $S_{1/2}$ -bound state with normalized
mielke-geometrodynamics_F01290	260	■ 1.000	$Q=e$	$Q = e$	charge $Q = e$, only nodeless radial solutions whose total rest mass shows an
mielke-geometrodynamics_F01291	263	■ 1.000	$D_{\mathrm{F}}(x)$	$D_{\mathrm{F}}(x)$	tudes are not yet subjected to field (“second”) quantization. Here $D_{\mathrm{F}}(x)$ denotes
mielke-geometrodynamics_F01292	263	■ 1.000	(1957,1958)	(1957, 1958)	A further reason for IVANENKO (1957, 1958) to consider the necessity of nonlin-
mielke-geometrodynamics_F01293	263	■ 1.000	$\ell^*=8.1 \times 10^{-33} \mathrm{~cm}$	$\ell^* = 8.1 \times 10^{-33} \mathrm{~cm}$	$\ell^* = 8.1 \times 10^{-33} \mathrm{~cm}$ to the far larger amount of the Compton wavelength $\ell = 10^{-13}$
mielke-geometrodynamics_F01294	263	■ 1.000	$\ell=10^{-13}$	$\ell = 10^{-13}$	$\ell^* = 8.1 \times 10^{-33} \mathrm{~cm}$ to the far larger amount of the Compton wavelength $\ell = 10^{-13}$
mielke-geometrodynamics_F01295	263	■ 1.000	$\ell=10^{-13} \mathrm{~cm}$	$\ell = 10^{-13} \mathrm{~cm}$	field theory (1967), which starts from $\ell = 10^{-13} \mathrm{~cm}$, can refrain from using a “bare”
mielke-geometrodynamics_F01296	269	■ 1.000	$\langle dj \rangle = 0$	$\langle dj \rangle = 0$	charge current j can be conserved, i.e., $\langle dj \rangle = 0$, whereas the conservation of the
mielke-geometrodynamics_F01297	269	■ 1.000	$\langle dj_5 \rangle \neq 0$	$\langle dj_5 \rangle \neq 0$	axial current j_5 is broken, $\langle dj_5 \rangle \neq 0$, in the most common convention.
mielke-geometrodynamics_F01298	269	■ 0.999	$j_5(x; \varepsilon) := \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon)$	$j_5(x; \varepsilon) := \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon)$	sidering the point-split current $j_5(x; \varepsilon) := \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon)$, where ε is an infin-
mielke-geometrodynamics_F01299	269	■ 1.000	$\delta / \delta A$	$\delta / \delta A$	The variation $\delta / \delta A$ of the current $j_5(x; \varepsilon)$ is compensated by the variation of the
mielke-geometrodynamics_F01300	269	■ 1.000	$j_5(x; \varepsilon)$	$j_5(x; \varepsilon)$	The variation $\delta / \delta A$ of the current $j_5(x; \varepsilon)$ is compensated by the variation of the
mielke-geometrodynamics_F01301	269	■ 0.999	$x^i \rightarrow x^i + \varepsilon^i$	$x^i \rightarrow x^i + \varepsilon^i$	exponential. Since the parallel transport from $x^i \rightarrow x^i + \varepsilon^i$ along the infinitesimal

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mielke-geometrodynamics_F01302	270	■ 1.000	$P=\bar{\psi}\gamma_5\psi$	$P = \bar{\psi}\gamma_5\psi$	for massive fermions, where $P = \overline{\psi}\gamma_5\psi$ is the pseudoscalar current and $F := dA$
mielke-geometrodynamics_F01303	270	■ 1.000	$C:=A\wedge dA$	$C := A \wedge dA$	Simons (CS) term $C := A \wedge dA$ corresponds to the spin or helicity of the photon,
mielke-geometrodynamics_F01304	270	■ 0.760	$\mathbf{A}\cdot\mathbf{B}$	$\mathbf{A}\cdot\mathbf{B}$	with its spacelike part $\mathbf{A}\cdot\mathbf{B}$ known as magnetic helicity (JACKIW & Pi 2000). Since
mielke-geometrodynamics_F01305	270	■ 1.000	$\mathscr{D}:=D+iA\wedge$	$\mathscr{D} := D + iA\wedge$	potential $A = A_idx^i$ is accounted for via $\mathscr{D} := D + iA\wedge$, where $D\psi := d\psi + \Gamma\wedge$
mielke-geometrodynamics_F01306	270	■ 1.000	$D\psi:=d\psi+\Gamma\wedge$	$D\psi := d\psi + \Gamma\wedge$	potential $A = A_idx^i$ is accounted for via $\mathscr{D} := D + iA\wedge$, where $D\psi := d\psi + \Gamma\wedge$
mielke-geometrodynamics_F01307	270	■ 0.566	$\Gamma^{\alpha\beta}=\Gamma_i^{\alpha\beta}dx^i$	$\Gamma^{\alpha\beta} = \Gamma_i^{\alpha\beta}dx^i$	nection one-form $\Gamma^{\alpha\beta} = \Gamma_i^{\alpha\beta}dx^i$.
mielke-geometrodynamics_F01308	270	■ 1.000	$\Theta:=D\gamma$	$\Theta := D\gamma$	with respect to $\overline{\psi}$ and ψ . Making use of the torsion $\Theta := D\gamma$ and of the properties
mielke-geometrodynamics_F01309	271	■ 0.652	$\left.\left.T:=\frac{1}{4}\operatorname{Tr}(\check{\gamma}\rfloor\Theta)\right _{\alpha}=e_{\alpha}\right\lrcorner T^{\alpha}$	$T := \frac{1}{4}\operatorname{Tr}(\check{\gamma}\rfloor\Theta) = e_{\alpha}\lrcorner T^{\alpha}$	where $T := \frac{1}{4}Tr(\check{\gamma}\rfloor\Theta) = e_{\alpha}\lrcorner T^{\alpha}$ is the one-form of the trace (or vector) torsion.
mielke-geometrodynamics_F01310	271	■ 1.000	$K=\frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$	$K = \frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$	(or Christoffel) connection $\Gamma^{(\cdot)}$ and the <i>contortion</i> one-form $K = \frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$, sat-
mielke-geometrodynamics_F01311	271	■ 1.000	$D\gamma=[\gamma,K]=\gamma_{\alpha}T^{\alpha}$	$D\gamma = [\gamma, K] = \gamma_{\alpha}T^{\alpha}$	isfying $D\gamma = [\gamma, K] = \gamma_{\alpha}T^{\alpha}$. Accordingly, the Dirac Lagrangian (12.2.1) splits
mielke-geometrodynamics_F01312	271	■ 0.509	$D^{(t)}\gamma=0$	$D^{(t)}\gamma = 0$	$D^{(\cdot)}\gamma = 0$. Hence, in an RC spacetime, a Dirac spinor feels only the <i>axial torsion</i> ,
mielke-geometrodynamics_F01313	271	■ 1.000	$\gamma\rightarrow\gamma^{\beta}=$	$\gamma \rightarrow \gamma^{\beta} =$	which is invariant under Weyl rescalings and <i>chiral transformations</i> $\gamma \rightarrow \gamma^{\beta} =$
mielke-geometrodynamics_F01314	271	■ 1.000	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P : \vartheta^{\vec{B}} \rightarrow -\vartheta^{\vec{B}}$, where $B =$
mielke-geometrodynamics_F01315	271	■ 1.000	$P:\vartheta^B\rightarrow-\vartheta^B$	$P : \vartheta^B \rightarrow -\vartheta^B$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P : \vartheta^{\vec{B}} \rightarrow -\vartheta^{\vec{B}}$, where $B =$
mielke-geometrodynamics_F01316	271	■ 1.000	$B=$	$B =$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P : \vartheta^{\vec{B}} \rightarrow -\vartheta^{\vec{B}}$, where $B =$
mielke-geometrodynamics_F01317	271	■ 0.992	1, 2, 3	1, 2, 3	1, 2, 3.
mielke-geometrodynamics_F01318	271	■ 1.000	$dj\simeq 0$	$dj \simeq 0$	From the Dirac equation (12.2.2) and its adjoint one can readily deduce that $dj \simeq 0$
mielke-geometrodynamics_F01319	271	■ 1.000	$m\rightarrow 0$	$m \rightarrow 0$	spacetime. If we restore chiral symmetry in the limit $m \rightarrow 0$, this leads to the classical
mielke-geometrodynamics_F01320	272	■ 1.000	$dj_5=0$	$dj_5 = 0$	conservation law $dj_5 = 0$ of the axial current for massless Weyl spinors, or since
mielke-geometrodynamics_F01321	272	■ 1.000	$\mathscr{A}=2d\theta$	$\mathscr{A} = 2d\theta$	torsion via $\mathscr{A} = 2d\theta$. Then there arises in (12.2.3) a derivative coupling of the
mielke-geometrodynamics_F01322	272	■ 1.000	$a=\theta f_a$	$a = \theta f_a$	would-be axion $a = \theta f_a$ to two fermions via the CPT-invariant term
mielke-geometrodynamics_F01323	272	■ 0.972	$U(1)_{\mathrm{PQ}}$	$U(1)_{\mathrm{PQ}}$	associated with a spontaneously broken Peccei–Quinn symmetry $U(1)_{\mathrm{PQ}}$; cf. KIM &
mielke-geometrodynamics_F01324	273	■ 1.000	$\eta^{\alpha\beta}:=*(\vartheta^{\alpha}\wedge\vartheta^{\beta})$	$\eta^{\alpha\beta} := *(\vartheta^{\alpha} \wedge \vartheta^{\beta})$	where $\eta^{\alpha\beta} := *(\vartheta^{\alpha} \wedge \vartheta^{\beta})$ is dual to the unit two-form.
mielke-geometrodynamics_F01325	273	■ 1.000	$\langle dj_5\rangle$	$\langle dj_5 \rangle$	quantization, where the vacuum expectation value $\langle dj_5 \rangle$ of the axial current picks up
mielke-geometrodynamics_F01326	273	■ 0.999	$dC_{\mathrm{TT}}\cong(1/4)dj_5\rightarrow 0$	$dC_{\mathrm{TT}} \cong (1/4)dj_5 \rightarrow 0$	NY four-form tends to zero “on shell,” i.e., $dC_{\mathrm{TT}} \cong (1/4)dj_5 \rightarrow 0$. This is consistent
mielke-geometrodynamics_F01327	273	■ 0.997	$*j_5$	$*j_5$	Here we implicitly assume that the light-cone structure of the axial covector $*j_5$

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mielke-geometrodynamics_F01328	273	■ 1.000	n=4 k+2	$n = 4k + 2$	$n = 4k + 2$ dimensions (LEUTWYLER 1986).
mielke-geometrodynamics_F01329	274	■ 0.999	$d^{*}\mathscr{A} \sim dC_{\mathrm{TT}}$	$d^{*}\mathscr{A} \sim dC_{\mathrm{TT}}$	arises in the axial anomaly, but <i>not</i> the NY-type term $d^{*}\mathscr{A} \sim dC_{\mathrm{TT}}$ as was claimed
mielke-geometrodynamics_F01330	274	■ 1.000	$d\mathscr{A} \wedge d\mathscr{A} = -2\mathcal{E} \cdot \mathcal{B}\eta$	$d\mathscr{A} \wedge d\mathscr{A} = -2\mathcal{E} \cdot \mathcal{B}\eta$	arises in the anomaly. Thus only the Weyl invariant term $d\mathscr{A} \wedge d\mathscr{A} = -2\mathcal{E} \cdot \mathcal{B}\eta$
mielke-geometrodynamics_F01331	274	■ 1.000	$F \wedge F = dA \wedge dA$	$F \wedge F = dA \wedge dA$	$F \wedge F = dA \wedge dA$ of the Pontryagin term (12.4.8). Torsion terms like $d\mathscr{A} \wedge d\mathscr{A}$
mielke-geometrodynamics_F01332	274	■ 0.835	$d^{*}\mathscr{A} \wedge \{^{*}\left(d^{*}\mathscr{A}\right)=4\ell^4V_{\mathrm{NY}} \wedge ^{*}V_{\mathrm{NY}}\}$	$d^{*}\mathscr{A} \wedge (d^{*}\mathscr{A}) = 4\ell^4V_{\mathrm{NY}} \wedge ^{*}V_{\mathrm{NY}}$	and $d^{*}\mathscr{A} \wedge (d^{*}\mathscr{A}) = 4\ell^4V_{\mathrm{NY}} \wedge ^{*}V_{\mathrm{NY}}$ have been considered previously, as part of
mielke-geometrodynamics_F01333	274	■ 0.984	$d^{*}\mathscr{A}=2\ell^2dC_{\mathrm{TT}}=2\ell^2V_{\mathrm{NY}}$	$d^{*}\mathscr{A} = 2\ell^2dC_{\mathrm{TT}} = 2\ell^2V_{\mathrm{NY}}$	identity (12.4.10) for the NY term $d^{*}\mathscr{A} = 2\ell^2dC_{\mathrm{TT}} = 2\ell^2V_{\mathrm{NY}}$, the second term is
mielke-geometrodynamics_F01334	274	■ 1.000	Z	Z	Z-factor, which would creep into the definition of the NY term. This is in sharp con-
mielke-geometrodynamics_F01335	274	■ 0.588	$\Psi=\Psi_i dx^i$	$\Psi = \Psi_i dx^i$	Schwinger spinor-valued one-form $\Psi = \Psi_i dx^i$; cf. URRUTIA & VERGARA (1991),
mielke-geometrodynamics_F01336	275	■ 0.999	$J_5:=i\bar{\Psi}\wedge\gamma\wedge\Psi$	$J_5 := i\bar{\Psi} \wedge \gamma \wedge \Psi$	The anomaly for the corresponding <i>axial current</i> $J_5 := i\bar{\Psi} \wedge \gamma \wedge \Psi$ is −21 times
mielke-geometrodynamics_F01337	275	■ 1.000	$t \rightarrow +0$	$t \rightarrow +0$	In the heat kernel approach, there exists for small $t \rightarrow +0$ the asymptotic expansion
mielke-geometrodynamics_F01338	275	■ 0.988	$\mathbb{A}:=$	$\mathbb{A} :=$	of the kernel in n dimensions, where the usual Feynman “dagger” convention $\mathcal{A} :=$
mielke-geometrodynamics_F01339	275	■ 0.794	$\check{\gamma}\rfloor A=\gamma^\alpha e_\alpha\rfloor A=(-1)^{s+1}\left[*\gamma\wedge A\right]$	$\check{\gamma}\rfloor A = \gamma^\alpha e_\alpha \rfloor A = (-1)^{s+1} [* \gamma \wedge A]$	$\check{\gamma}\rfloor A = \gamma^\alpha e_\alpha \rfloor A = (-1)^{s+1} [* \gamma \wedge A]$ for one-forms is used.
mielke-geometrodynamics_F01340	275	■ 0.996	$\square:=$	$\square :=$	terms additional to the generally covariant Riemannian d’Alembertian operator $\square :=$
mielke-geometrodynamics_F01341	275	■ 0.980	$\partial_i\left(\sqrt{ g }g^{ij}\partial_j\right)/\sqrt{ g }$	$\partial_i\left(\sqrt{ g }g^{ij}\partial_j\right)/\sqrt{ g }$	$\partial_i\left(\sqrt{ g }g^{ij}\partial_j\right)/\sqrt{ g }$ can be explicitly identified (MIELKE & KREIMER 1998). Not
mielke-geometrodynamics_F01342	275	■ 0.998	$K_k(x,\not{D}^2)$	$K_k(x,\not{D}^2)$	The coefficients $K_k(x,\not{D}^2)$ are completely determined by the form of the second-
mielke-geometrodynamics_F01343	275	■ 0.952	$\not{D}^2$	$\not{D}^2$	order differential operator $\not{D}^2$, which is positive for Euclidean signature $\mathrm{diag}\,\mathrm{o}_{\alpha\beta} =$
mielke-geometrodynamics_F01344	275	■ 0.952	$\mathrm{o}_{\alpha\beta}=$	$\mathrm{o}_{\alpha\beta} =$	order differential operator $\not{D}^2$, which is positive for Euclidean signature $\mathrm{diag}\,\mathrm{o}_{\alpha\beta} =$
mielke-geometrodynamics_F01345	276	■ 0.950	$(-1,\ldots,-1)$	$(-1,\ldots,-1)$	$(-1,\ldots,-1)$. For odd $k = 1, 3, \ldots$, these coefficients are zero, while the first non-
mielke-geometrodynamics_F01346	276	■ 0.950	$k=1,3,\ldots$	$k = 1, 3, \ldots$	$(-1,\ldots,-1)$. For odd $k = 1, 3, \ldots$, these coefficients are zero, while the first non-
mielke-geometrodynamics_F01347	276	■ 0.483	$d\mathscr{K}=d^{*}\widetilde{\not{D}}\wedge^{*}\breve{\not{D}}\mathscr{A}$	$d\mathscr{K} = d^{*}\widetilde{\not{D}} \wedge ^{*}\breve{\not{D}}\mathscr{A}$	where the higher-order term $d\mathscr{K} = d^{*}\widetilde{\not{D}} \wedge ^{*}\breve{\not{D}}\mathscr{A}$ involves the covariant derivative
mielke-geometrodynamics_F01348	276	■ 0.507	$D=\not{D}^{\{\}}+i\mathscr{A}\gamma_5/4$	$D = D^{\{\}} + i\mathscr{A}\gamma_5/4$	$\widetilde{\not{D}}=D^{\{\}}+i\mathscr{A}\gamma_5/4$ modified by the axial torsion.
mielke-geometrodynamics_F01349	276	■ 1.000	K_2	K_2	terms K_2 and K_4 . Whereas in $n = 4$ dimensions, the Pontryagin-type term K_4 is
mielke-geometrodynamics_F01350	276	■ 1.000	K_4	K_4	terms K_2 and K_4 . Whereas in $n = 4$ dimensions, the Pontryagin-type term K_4 is
mielke-geometrodynamics_F01351	276	■ 1.000	$k=4$	$k = 4$	dimensionless and thus for $k = 4$ gets multiplied by $t^{(k-4)/2} = 1$, the term $K_2 \sim$
mielke-geometrodynamics_F01352	276	■ 1.000	$t^{\{(k-4)/2\}}=1$	$t^{(k-4)/2} = 1$	dimensionless and thus for $k = 4$ gets multiplied by $t^{(k-4)/2} = 1$, the term $K_2 \sim$
mielke-geometrodynamics_F01353	276	■ 1.000	$K_2\sim$	$K_2 \sim$	dimensionless and thus for $k = 4$ gets multiplied by $t^{(k-4)/2} = 1$, the term $K_2 \sim$

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mielke-geometrodynamics_F01354	276	■ 0.960	$d^{*}\!\mathscr{A}=2\,\ell^2\,dC_{\mathrm{TT}}$	$d^{*}\mathscr{A}=2\ell^2dC_{\mathrm{TT}}$	$d^{*}\mathscr{A}=2\ell^2dC_{\mathrm{TT}}$ carries dimensions. Since a <i>massive</i> Dirac spinor has canonical
mielke-geometrodynamics_F01355	276	■ 0.986	$[\mathrm{length}]^{-3/2}$	$[\mathrm{length}]^{-3/2}$	dimension $[\mathrm{length}]^{-3/2}$, it scales as $\psi\sim m^{3/2}$. Moreover, the term $t=1/M^2$ is
mielke-geometrodynamics_F01356	276	■ 0.986	$\psi\sim m^{3/2}$	$\psi\sim m^{3/2}$	dimension $[\mathrm{length}]^{-3/2}$, it scales as $\psi\sim m^{3/2}$. Moreover, the term $t=1/M^2$ is
mielke-geometrodynamics_F01357	276	■ 0.986	$t=1/M^2$	$t=1/M^2$	dimension $[\mathrm{length}]^{-3/2}$, it scales as $\psi\sim m^{3/2}$. Moreover, the term $t=1/M^2$ is
mielke-geometrodynamics_F01358	276	■ 1.000	$M\rightarrow\infty$	$M\rightarrow\infty$	related to the regulator mass $M\rightarrow\infty$ in the FUJIKAWA (1979) method. Then the
mielke-geometrodynamics_F01359	276	■ 1.000	$m\neq 0$	$m\neq 0$	heat kernel expansion will tend to zero in the chiral limit $m\rightarrow 0$. In the case $m\neq 0$,
mielke-geometrodynamics_F01360	276	■ 1.000	$\vartheta^{\alpha}\rightarrow\widetilde{\vartheta}^{\alpha}=M\vartheta^{\alpha}$	$\vartheta^{\alpha}\rightarrow\widetilde{\vartheta}^{\alpha}=M\vartheta^{\alpha}$	To rescale the coframe by $\vartheta^{\alpha}\rightarrow\widetilde{\vartheta}^{\alpha}=M\vartheta^{\alpha}$ does not help, since that would also
mielke-geometrodynamics_F01361	276	■ 1.000	$[\hbar]$	$[\hbar]$	change the dimension of the Dirac field in order to retain the physical dimension $[\hbar]$
mielke-geometrodynamics_F01362	276	■ 1.000	$(n=2)$	$(n=2)$	sions, neither classically nor in QFT. One would surmise that in $(n=2)$ -dimensional
mielke-geometrodynamics_F01363	276	■ 1.000	$\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}$	$\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}$	where $\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}$ is the energy–momentum current Belinfante symmetrized
mielke-geometrodynamics_F01364	277	■ 0.996	$\stackrel{(\pm)}{\Gamma}$	$\stackrel{(\pm)}{\Gamma}$	anti-self-dual variables $\stackrel{(\pm)}{\Gamma}$ in EC theory as well as in simple supergravity (MIELKE
mielke-geometrodynamics_F01365	277	■ 0.839	$dC_{\mathrm{RR}}^{\{\}}$	$dC_{\mathrm{RR}}^{\{\}}$	The appearance of the Riemannian Pontryagin term $dC_{\mathrm{RR}}^{\{\}}$ in the anomaly (12.3.1)
mielke-geometrodynamics_F01366	277	■ 1.000	$\Lambda\neq 0$	$\Lambda\neq 0$	essary condition $\Lambda\neq 0$ could even be the dominating configurations, one gets a net
mielke-geometrodynamics_F01367	277	■ 0.570	$\underline{C}_{\mathrm{RR}}^{\{\}}$	$\underline{C}_{\mathrm{RR}}^{\{\}}$	to gravity, the tangential complexified CS term $\underline{C}_{\mathrm{RR}}^{\{\}}$ is known (KODAMA 1990) to
mielke-geometrodynamics_F01368	277	■ 0.936	$\stackrel{(\pm)}{\underline{\Gamma}}$	$\stackrel{(\pm)}{\underline{\Gamma}}$	of gravity with cosmological constant, where the complex Ashtekar variable $\stackrel{(\pm)}{\underline{\Gamma}}$
mielke-geometrodynamics_F01369	277	■ 1.000	$\mathscr{C}:=\Gamma/\mathscr{G}$	$\mathscr{C}:=\Gamma/\mathscr{G}$	theory on a three-dimensional hypersurface, with $\mathscr{C}:=\Gamma/\mathscr{G}$ of inequivalent gauge
mielke-geometrodynamics_F01370	278	■ 0.423	$\mathbb{L}_n:=n\downarrow D+Dn$	$\mathbb{L}_n:=n\downarrow D+Dn$	where $\mathbb{L}_n:=n\downarrow D+Dn$ is the gauge-covariant Lie derivative along the normal
mielke-geometrodynamics_F01371	278	■ 0.523	τ^A	τ^A	for the matter current τ^A on the spacelike hypersurface (JIANG 1991). Only when
mielke-geometrodynamics_F01372	278	■ 1.000	$\tau^A:=(1/2)\eta^{0A\beta\gamma}\tau_{\beta\gamma}$	$\tau^A:=(1/2)\eta^{0A\beta\gamma}\tau_{\beta\gamma}$	field, it is the canonical spin $\tau^A:=(1/2)\eta^{0A\beta\gamma}\tau_{\beta\gamma}$ that appears on the right-hand
mielke-geometrodynamics_F01373	278	■ 1.000	$\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathbf{o}_{\alpha\beta}$	$\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathbf{o}_{\alpha\beta}$	When the constant Dirac matrices γ_{α} satisfying $\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathbf{o}_{\alpha\beta}$ saturate the
mielke-geometrodynamics_F01374	278	■ 1.000	$\eta^{\alpha}:=$	$\eta^{\alpha}:=$	index of the orthonormal coframe one-form $\vartheta^{\alpha}=E_j{}^{\alpha}dx^j$ and its Hodge dual $\eta^{\alpha}:=$
mielke-geometrodynamics_F01375	278	■ 0.996	$*\vartheta^{\alpha}$	$*\vartheta^{\alpha}$	$*\vartheta^{\alpha}$, we obtain a basis of Clifford-algebra-valued exterior forms via
mielke-geometrodynamics_F01376	278	■ 0.500	$\Gamma:=\frac{i}{4}\Gamma_i^{\alpha\beta}\sigma_{\alpha\beta}dx^i$	$\Gamma:=\frac{i}{4}\Gamma_i^{\alpha\beta}\sigma_{\alpha\beta}dx^i$	In terms of the Clifford-algebra-valued <i>connection</i> $\Gamma:=\frac{i}{4}\Gamma_i^{\alpha\beta}\sigma_{\alpha\beta}dx^i$, the $SL(2,\mathbb{C})$ -
mielke-geometrodynamics_F01377	278	■ 1.000	$D=d+\Gamma\wedge$	$D=d+\Gamma\wedge$	covariant exterior derivative is given by $D=d+\Gamma\wedge$, where $\sigma_{\alpha\beta}=\frac{i}{2}(\gamma_{\alpha}\gamma_{\beta}-\gamma_{\beta}\gamma_{\alpha})$
mielke-geometrodynamics_F01378	278	■ 1.000	$\sigma_{\alpha\beta}=\frac{i}{2}(\gamma_{\alpha}\gamma_{\beta}-\gamma_{\beta}\gamma_{\alpha})$	$\sigma_{\alpha\beta}=\frac{i}{2}(\gamma_{\alpha}\gamma_{\beta}-\gamma_{\beta}\gamma_{\alpha})$	covariant exterior derivative is given by $D=d+\Gamma\wedge$, where $\sigma_{\alpha\beta}=\frac{i}{2}(\gamma_{\alpha}\gamma_{\beta}-\gamma_{\beta}\gamma_{\alpha})$
mielke-geometrodynamics_F01379	278	■ 1.000	$\sigma:=\frac{i}{2}\gamma\wedge\gamma=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}$	$\sigma:=\frac{i}{2}\gamma\wedge\gamma=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}$	are the Lorentz generators entering in the two-form $\sigma:=\frac{i}{2}\gamma\wedge\gamma=\frac{1}{2}\sigma_{\alpha\beta}\vartheta^{\alpha}\wedge\vartheta^{\beta}$.

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mielke-geometrodynamics_F01380	279	■ 1.000	$R:=\{ \}^*\!\!\left(R^{\alpha}\!_{\beta}\wedge \eta_{\beta}\!_{\alpha}\right)$	$R:=\ast\left(R^{\alpha\beta}\wedge\eta_{\beta\alpha}\right)$	to the curvature scalar $R:=\ast(R^{\alpha\beta}\wedge\eta_{\beta\alpha})$ and the axial torsion piece $d\mathscr{A}\wedge d\mathscr{A}$ of
mielke-geometrodynamics_F01381	279	■ 1.000	$\mathfrak{sl}(5,\mathbb{R})$	$\mathfrak{sl}(5,\mathbb{R})$	the $\mathfrak{sl}(5,\mathbb{R})$ -valued connection $\hat{F}=\overline{\Gamma}+(\vartheta^{\alpha}L^4_{\alpha}+\vartheta_{\beta}\overline{L^{\beta}}_4)/\ell$ is expanded into the
mielke-geometrodynamics_F01382	279	■ 1.000	$\hat{\Gamma}=\Gamma+\left(\vartheta^{\alpha}L^4_{\alpha}+\vartheta_{\beta}\overline{L^{\beta}}_4\right)/\ell$	$\hat{\Gamma}=\Gamma+(\vartheta^{\alpha}L^4_{\alpha}+\vartheta_{\beta}\overline{L^{\beta}}_4)/\ell$	the $\mathfrak{sl}(5,\mathbb{R})$ -valued connection $\hat{F}=\overline{\Gamma}+(\vartheta^{\alpha}L^4_{\alpha}+\vartheta_{\beta}\overline{L^{\beta}}_4)/\ell$ is expanded into the
mielke-geometrodynamics_F01383	279	■ 0.763	$\hat{C}_{\mathrm{RR}}$	$\hat{C}_{\mathrm{RR}}$	canonical dimension $[length]$. The corresponding Pontryagin term $\hat{C}_{\mathrm{RR}}$ splits via
mielke-geometrodynamics_F01384	280	■ 1.000	$\pi^0\rightarrow 2\gamma$	$\pi^0\rightarrow 2\gamma$	Bell JS, Jackiw R (1969) A PCAC puzzle: $\pi^0\rightarrow 2\gamma$ in the σ model. Il Nuovo Cimento A 60(1):
mielke-geometrodynamics_F01385	282	■ 0.775	$\mathbb{R}^4\otimes\mathrm{SO}(1,3)$	$\mathbb{R}^4\otimes\mathrm{SO}(1,3)$	Lorentz or Poincaré group $\mathbb{R}^4\ltimes\mathrm{SO}(1,3)$ to a metalinear group $\mathrm{SL}(5,\mathbb{R})$ as a gauge
mielke-geometrodynamics_F01386	283	■ 1.000	$dB=0$	$dB=0$	Independent variations with respect to A and B lead to $dB=0$ and to the <i>constraint</i>
mielke-geometrodynamics_F01387	284	■ 0.999	$\sigma=\sigma_i dx^i$	$\sigma=\sigma_i dx^i$	where $\sigma=\sigma_i dx^i$ is a one-form or a covector; cf. LUCCHESI et al. (1993). Since Bianchi
mielke-geometrodynamics_F01388	284	■ 0.672	$\mathrm{SL}(5,\mathbb{R})$	$\mathrm{SL}(5,\mathbb{R})$	“semi-topological” BF scheme based on the metric-free structure group $\mathrm{SL}(5,\mathbb{R})$
mielke-geometrodynamics_F01389	284	■ 1.000	$\mathrm{SL}(n+1,\mathbb{R})$	$\mathrm{SL}(n+1,\mathbb{R})$	In general, the metalinear group $\mathrm{SL}(n+1,\mathbb{R})$ with $\det g=1$ is generated by
mielke-geometrodynamics_F01390	284	■ 1.000	$\det g=1$	$\det g=1$	In general, the metalinear group $\mathrm{SL}(n+1,\mathbb{R})$ with $\det g=1$ is generated by
mielke-geometrodynamics_F01391	284	■ 1.000	$n(n+2)$	$n(n+2)$	$n(n+2)$ trace-free generators
mielke-geometrodynamics_F01392	284	■ 1.000	$\mathfrak{gl}(n,\mathbb{R})$	$\mathfrak{gl}(n,\mathbb{R})$	comprise those of a $\mathfrak{gl}(n,\mathbb{R})$ subalgebra
mielke-geometrodynamics_F01393	284	■ 1.000	$\alpha,\ldots,\delta=0,\ldots,n-1$	$\alpha,\ldots,\delta=0,\ldots,n-1$	where $\alpha,\ldots,\delta=0,\ldots,n-1$.
mielke-geometrodynamics_F01394	284	■ 0.367	$\ell P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$	$\ell P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$	The generators $\ell P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$ and $\ell P^{\beta}_{\ast}:=\mathbb{L}^{\beta}_{n+1}=L^{\beta}_{n+1}$ transform
mielke-geometrodynamics_F01395	284	■ 0.367	$\ell P_{\ast}^{\beta}:=\mathbb{L}^{\beta}_{n+1}=L^{\beta}_{n+1}$	$\ell P_{\ast}^{\beta}:=\mathbb{L}^{\beta}_{n+1}=L^{\beta}_{n+1}$	The generators $\ell P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$ and $\ell P_{\ast}^{\beta}:=\mathbb{L}^{\beta}_{n+1}=L^{\beta}_{n+1}$ transform
mielke-geometrodynamics_F01396	285	■ 0.998	$P_{\ast}^{\alpha}$	$P_{\ast}^{\alpha}$	This reveals that $P_{\ast}^{\alpha}$ is a vector and P_{β} is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$
mielke-geometrodynamics_F01397	285	■ 0.998	P_{β}	P_{β}	This reveals that $P_{\ast}^{\alpha}$ is a vector and P_{β} is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$
mielke-geometrodynamics_F01398	285	■ 0.998	$\mathrm{GL}(n,\mathbb{R})$	$\mathrm{GL}(n,\mathbb{R})$	This reveals that $P_{\ast}^{\alpha}$ is a vector and P_{β} is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$
mielke-geometrodynamics_F01399	285	■ 0.996	$\mathbf{A}_{\ast}(n,\mathbb{R})$	$\mathbf{A}_{\ast}(n,\mathbb{R})$	to the semisimple <i>graded affine</i> group $\mathbf{A}_{\ast}(n,\mathbb{R})$ of the same rank:
mielke-geometrodynamics_F01400	285	■ 0.998	$\mathrm{SO}(n+1)$	$\mathrm{SO}(n+1)$	(Pseudo)orthogonal groups like $\mathrm{SO}(n+1)$ involve a Cartan–Killing metric with
mielke-geometrodynamics_F01401	285	■ 1.000	$\hat{g}_{AB}$	$\hat{g}_{AB}$	components $\hat{g}_{AB}$. Then a basis of the Lie generators
mielke-geometrodynamics_F01402	285	■ 0.998	$n(n+1)/2$	$n(n+1)/2$	is only $n(n+1)/2$ -dimensional. The same would apply to the respective connections
mielke-geometrodynamics_F01403	285	■ 0.999	$A(x)\in\mathbf{A}_{\ast}(n,R)$	$A(x)\in\mathbf{A}_{\ast}(n,R)$	where $A(x)\in\mathbf{A}_{\ast}(n,R)$ and the $\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$ are linear gauge transformations,
mielke-geometrodynamics_F01404	285	■ 0.999	$\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$	$\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$	where $A(x)\in\mathbf{A}_{\ast}(n,R)$ and the $\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$ are linear gauge transformations,
mielke-geometrodynamics_F01405	285	■ 1.000	$\tau(x):=\exp\left[\xi^{\alpha}\!_{\alpha}P_{\alpha}\right]\in\mathscr{T}(n,R)$	$\tau(x):=\exp\left[\xi^{\alpha}P_{\alpha}\right]\in\mathscr{T}(n,R)$	while $\tau(x):=\exp[\xi^{\alpha}P_{\alpha}]\in\mathscr{T}(n,R)$ and $\tau_{\ast}(x):=\exp[\xi^{\beta}_{\ast}P^{\beta}_{\ast}]\in\mathscr{T}_{\ast}(n,R)$ represent
mielke-geometrodynamics_F01406	285	■ 1.000	$\tau_{\ast}(x):=\exp\left[\xi^{\beta}_{\ast}P^{\beta}_{\ast}\right]\in\mathscr{T}_{\ast}(n,R)$	$\tau_{\ast}(x):=\exp\left[\xi^{\beta}_{\ast}P^{\beta}_{\ast}\right]\in\mathscr{T}_{\ast}(n,R)$	while $\tau(x):=\exp[\xi^{\alpha}P_{\alpha}]\in\mathscr{T}(n,R)$ and $\tau_{\ast}(x):=\exp[\xi^{\beta}_{\ast}P^{\beta}_{\ast}]\in\mathscr{T}_{\ast}(n,R)$ represent

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mielke-geometrodynamics_F01407	286	■ 1.000	<code>\ell \rightarrow \infty</code>	$\ell \rightarrow \infty$	For $\ell \rightarrow \infty$, this holds automatically due to the degeneracy of the commutation
mielke-geometrodynamics_F01408	286	■ 1.000	<code>\operatorname{Ad} A(B)=[A, B]</code>	$\operatorname{Ad} A(B) = [A, B]$	the adjoint representation $\operatorname{Ad} A(B) = [A, B]$. The curvature
mielke-geometrodynamics_F01409	286	■ 1.000	<code>\hat{\Gamma}</code>	$\hat{\Gamma}$	The inhomogeneous transformation law (13.4.3) for the graded affine connection $\hat{\Gamma}$
mielke-geometrodynamics_F01410	286	■ 1.000	<code>\Gamma^{(L)}</code>	$\Gamma^{(L)}$	can be split into that of the linear connection $\Gamma^{(L)}$, which acquires the conventional
mielke-geometrodynamics_F01411	286	■ 0.990	<code>\mathrm{GL}(n, \mathbb{R})</code>	$\mathrm{GL}(n, \mathbb{R})$	of a Yang–Mills-type connection for the gauge group $\mathrm{GL}(n, \mathbb{R})$. The remaining trans-
mielke-geometrodynamics_F01412	287	■ 0.998	<code>D \tau(x):=d \tau(x)+\Gamma^{(L)} \tau(x)</code>	$D\tau(x) := d\tau(x) + \Gamma^{(L)}\tau(x)$	derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01413	287	■ 0.998	<code>\Gamma^{(T)}</code>	$\Gamma^{(T)}$	derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01414	287	■ 0.998	<code>\Gamma_*^{(T)}</code>	$\Gamma_*^{(T)}$	derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01415	287	■ 1.000	<code>\xi=\xi^{\alpha} P_{\alpha}</code>	$\xi = \xi^\alpha P_\alpha$	Since $\xi = \xi^\alpha P_\alpha$ and $\xi_* = \xi_\beta^* P_\beta^*$ acquire their values in the coset space
mielke-geometrodynamics_F01416	287	■ 1.000	<code>\xi_{*}=\xi_{\beta}^{*} P_{*}^{\beta}</code>	$\xi_* = \xi_\beta^* P_\beta^*$	Since $\xi = \xi^\alpha P_\alpha$ and $\xi_* = \xi_\beta^* P_\beta^*$ acquire their values in the coset space
mielke-geometrodynamics_F01417	287	■ 1.000	<code>\mathscr{T}(n, R) \otimes \mathscr{T}_*(n, R)</code>	$\mathcal{T}(n, R) \otimes \mathcal{T}_*(n, R)$	“hide” the action of the local pseudotranslations $\mathcal{T}(n, R) \otimes \mathcal{T}_*(n, R)$ on ϑ and θ .
mielke-geometrodynamics_F01418	287	■ 1.000	<code>\xi=\xi^{*}=0</code>	$\xi = \xi^* = 0$	If we postulate even stronger constraints $\xi = \xi^* = 0$ of “zero section” vector
mielke-geometrodynamics_F01419	287	■ 1.000	<code>\theta_{\alpha}=g_{\alpha \beta} \vartheta^{\beta}</code>	$\theta_\alpha = g_{\alpha\beta}\vartheta^\beta$	mutually transposed one-forms, i.e., $\theta_\alpha = g_{\alpha\beta}\vartheta^\beta$.
mielke-geometrodynamics_F01420	287	■ 1.000	<code>D \xi^{\alpha} \stackrel{*}{=} \vartheta^{\alpha}</code>	$D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$	When $D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$, a Cartan transport of a radius vector ξ^α occurs via “rolling
mielke-geometrodynamics_F01421	287	■ 1.000	<code>\xi^{\alpha}</code>	ξ^α	When $D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$, a Cartan transport of a radius vector ξ^α occurs via “rolling
mielke-geometrodynamics_F01422	288	■ 0.593	<code>1 / [</code>	$1/[$	In view of the canonical dimension 1/[length] of the generators P_α and P_β^* of pseudo-
mielke-geometrodynamics_F01423	288	■ 0.593	<code>]</code>	$]$	In view of the canonical dimension 1/[length] of the generators P_α and P_β^* of pseudo-
mielke-geometrodynamics_F01424	288	■ 0.593	<code>P_{*}^{\beta}</code>	P_β^*	In view of the canonical dimension 1/[length] of the generators P_α and P_β^* of pseudo-
mielke-geometrodynamics_F01425	288	■ 0.647	<code>\vartheta^{\alpha}=\ell \Gamma_{\hat{4}}^{\alpha}</code>	$\vartheta^\alpha = \ell \Gamma_4^\alpha$	i.e., it becomes a Cartan connection, where $\vartheta^\alpha = \ell \Gamma_4^\alpha$ and $\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$ are two inde-
mielke-geometrodynamics_F01426	288	■ 0.647	<code>\theta_{\beta}=\ell \Gamma_{\beta}^{\hat{4}}</code>	$\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$	i.e., it becomes a Cartan connection, where $\vartheta^\alpha = \ell \Gamma_4^\alpha$ and $\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$ are two inde-
mielke-geometrodynamics_F01427	288	■ 1.000	<code>\Lambda=3 /\ell^2</code>	$\Lambda = 3/\ell^2$	where $\Lambda = 3/\ell^2$ will later be identified with the cosmological constant.
mielke-geometrodynamics_F01428	289	■ 0.998	<code>d \hat{C}</code>	$d\hat{C}$	$d\hat{C}$ provides us, after variation with respect to Γ_A^B , with the Bianchi identity
mielke-geometrodynamics_F01429	289	■ 0.998	<code>\Gamma_A{}^B</code>	Γ_A^B	$d\hat{C}$ provides us, after variation with respect to Γ_A^B , with the Bianchi identity
mielke-geometrodynamics_F01430	289	■ 1.000	<code>\hat{B}</code>	$\hat{B}$	The variation with respect to $\hat{B}$ would lead to $\hat{R} = 0$, a physically too strong constraint.
mielke-geometrodynamics_F01431	289	■ 1.000	<code>\hat{R}=0</code>	$\hat{R} = 0$	The variation with respect to $\hat{B}$ would lead to $\hat{R} = 0$, a physically too strong constraint.
mielke-geometrodynamics_F01432	289	■ 0.994	<code>\eta_{A B C D E}:=\sqrt{\hat{g}} \varepsilon_{A B C D E}</code>	$\eta_{ABCDE} := \sqrt{\hat{g}}\varepsilon_{ABCDE}$	Here, the antisymmetric unit tensor $\eta_{ABCDE} := \sqrt{\hat{g}}\varepsilon_{ABCDE}$ is constructed from the

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mielke-geometrodynamics_F01433	289	■ 1.000	<code>\varepsilon_{A B C D E}</code>	ε_{ABCDE}	<i>metric-free</i> Levi-Civita symbol ε_{ABCDE} , being invariant under the five-dimensional
mielke-geometrodynamics_F01434	289	■ 0.913	<code>\hat{g}</code>	$\hat{g}$	structure group $\mathrm{SL}(5, \mathbb{R})$, together with the determinant $\hat{g}$ of the invertible group
mielke-geometrodynamics_F01435	289	■ 1.000	<code>B^{A B}:=\hat{g}^{A C} B_{C}\{ \}^{B}</code>	$B^{AB} := \hat{g}^{AC} B_C{}^B$	$B^{AB} := \hat{g}^{AC} B_C{}^B$. Thus, the additional quadratic term provides a <i>dynamical symmetry</i> -
mielke-geometrodynamics_F01436	289	■ 0.853	<code>\operatorname{SO}(2,3)</code>	$\mathrm{SO}(2, 3)$	or $\mathrm{SO}(2, 3)$, respectively. Moreover, the five-index structure of η_{ABCDE} has forced us
mielke-geometrodynamics_F01437	289	■ 0.853	<code>\eta_{A B C D E}</code>	η_{ABCDE}	or $\mathrm{SO}(2, 3)$, respectively. Moreover, the five-index structure of η_{ABCDE} has forced us
mielke-geometrodynamics_F01438	289	■ 0.997	<code>\Phi^A=\left\{\phi^{\alpha}, \phi^{\hat{4}}\right\}</code>	$\Phi^A = \left\{\phi^\alpha, \phi^{\hat{4}}\right\}$	to complement the construction by a quintet of pseudoscalars $\Phi^A = \{\phi^\alpha, \phi^{\hat{4}}\}$, our
mielke-geometrodynamics_F01439	289	■ 0.988	<code>U\left(\Phi^B \Phi_B\right)</code>	$U\left(\Phi^B \Phi_B\right)$	arise from a minimum of a Higgs-type potential $U(\Phi^B \Phi_B)$; cf. PAGELS (1984) and
mielke-geometrodynamics_F01440	290	■ 1.000	<code>\Phi=\Re \Phi+i \Im \Phi</code>	$\Phi = \Re \Phi + i \Im \Phi$	$\Phi = \Re \Phi + i \Im \Phi$ is invariant
mielke-geometrodynamics_F01441	290	■ 0.995	<code>d C_{\mathrm{RR}}</code>	dC_{RR}	where the resulting Pontryagin term dC_{RR} serves merely to reproduce the Bianchi
mielke-geometrodynamics_F01442	290	■ 0.989	<code>\stackrel{\circ}{\Gamma}^{\alpha \beta} =</code>	$\overset{\circ}{\Gamma}^{\alpha \beta} =$	identity (13.5.5) for the curvature, now in terms of the Lorentz connection $\overset{\circ}{\Gamma}^{\alpha \beta} =$
mielke-geometrodynamics_F01443	290	■ 1.000	<code>-\stackrel{\circ}{\Gamma}^{\beta \alpha}</code>	$-\overset{\circ}{\Gamma}^{\beta \alpha}$	$-\overset{\circ}{\Gamma}^{\beta \alpha}$. The variation with respect to the Lorentz-valued two-form $B^{\alpha \beta}$ leads to
mielke-geometrodynamics_F01444	290	■ 1.000	<code>B^{\alpha \beta}</code>	$B^{\alpha \beta}$	$-\overset{\circ}{\Gamma}^{\beta \alpha}$. The variation with respect to the Lorentz-valued two-form $B^{\alpha \beta}$ leads to
mielke-geometrodynamics_F01445	290	■ 1.000	<code>\mu \neq 0</code>	$\mu \neq 0$	as an equation of motion. Thus, for $\mu \neq 0$, our Lagrangian is “on shell” equivalent
mielke-geometrodynamics_F01446	291	■ 1.000	<code>\vartheta^{\alpha}=g^{\alpha \beta} \theta_{\beta}</code>	$\vartheta^\alpha = g^{\alpha \beta} \theta_\beta$	For a nondegenerate coframe $\vartheta^\alpha = g^{\alpha \beta} \theta_\beta$ and vanishing torsion, the covariant
mielke-geometrodynamics_F01447	291	■ 1.000	<code>\kappa =</code>	$\kappa =$	Pontryagin four-forms (here suppressed). Einstein’s gravitational constant $\kappa =$
mielke-geometrodynamics_F01448	291	■ 1.000	<code>8 \pi G_{\mathrm{N}}=8 \pi \ell_{\text {Planck }}^2</code>	$8 \pi G_{\mathrm{N}} = 8 \pi \ell_{\mathrm{Planck}}^2$	$8 \pi G_{\mathrm{N}} = 8 \pi \ell_{\mathrm{Planck}}^2$ and the density $\rho_\Lambda = \Lambda / \kappa = \Omega_\Lambda \rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$ corresponding to
mielke-geometrodynamics_F01449	291	■ 1.000	<code>\rho_{\Lambda}=\Lambda / \kappa=\Omega_{\Lambda} \rho_{\text {crit }}<\rho_{\text {crit }}</code>	$\rho_\Lambda = \Lambda / \kappa = \Omega_\Lambda \rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$	$8 \pi G_{\mathrm{N}} = 8 \pi \ell_{\mathrm{Planck}}^2$ and the density $\rho_\Lambda = \Lambda / \kappa = \Omega_\Lambda \rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$ corresponding to
mielke-geometrodynamics_F01450	291	■ 1.000	<code>\kappa=\mu \ell^2 / 4</code>	$\kappa = \mu \ell^2 / 4$	implies that these constants are related via $\kappa = \mu \ell^2 / 4$ and $\Lambda = 12 \kappa / \mu \ell^4 = 3 / \ell^2$ to
mielke-geometrodynamics_F01451	291	■ 1.000	<code>\Lambda=12 \kappa / \mu \ell^4=3 / \ell^2</code>	$\Lambda = 12 \kappa / \mu \ell^4 = 3 / \ell^2$	implies that these constants are related via $\kappa = \mu \ell^2 / 4$ and $\Lambda = 12 \kappa / \mu \ell^4 = 3 / \ell^2$ to
mielke-geometrodynamics_F01452	291	■ 1.000	<code>\mu \simeq 10^{-123}</code>	$\mu \simeq 10^{-123}$	The symmetry-breaking parameter $\mu \simeq 10^{-123}$ is observationally an extremely small
mielke-geometrodynamics_F01453	291	■ 1.000	<code>\hat{B} \wedge \hat{B}</code>	$\hat{B} \wedge \hat{B}$	breaking $\hat{B} \wedge \hat{B}$ term, as anticipated by SMOLIN (2000) in the restricted case of
mielke-geometrodynamics_F01454	291	■ 1.000	<code>G_{\mathrm{N}}</code>	G_{N}	gravitational constant G_{N} has its origin in SSB, as suggested already by ENGLERT et
mielke-geometrodynamics_F01455	292	■ 0.997	<code>O(5)</code>	$O(5)$	Since our $\mathrm{SL}(5, \mathbb{R})$ gauge model is broken into two steps from $O(5)$ down to the
mielke-geometrodynamics_F01456	292	■ 0.466	<code>\operatorname{SO}(1,3)</code>	$\mathrm{SO}(1, 3)$	Lorentz group $\mathrm{SO}(1, 3)$ as an exact subgroup, a local holonomic spacetime metric
mielke-geometrodynamics_F01457	292	■ 1.000	<code>g_{i j}</code>	g_{ij}	g_{ij} is then induced. In the last step, we assumed that the translational connections
mielke-geometrodynamics_F01458	292	■ 0.999	<code>\Gamma_4^{\alpha} \otimes_{\mathrm{S}} \Gamma_{\alpha}^4</code>	$\Gamma_4{}^\alpha \otimes_{\mathrm{S}} \Gamma_\alpha{}^4$	$\Gamma_4{}^\alpha \otimes_{\mathrm{S}} \Gamma_\alpha{}^4$ is dimensionless, we had to multiply the vacuum expectation (13.6.1) of
mielke-geometrodynamics_F01459	292	■ 1.000	<code>\ell^2 / \mu^2=\left(\kappa / 4 \Lambda_{\mathrm{H}}\right)</code>	$\ell^2 / \mu^2 = (\kappa / 4 \Lambda_{\mathrm{H}})$	the derivative of the Higgs field by a <i>huge</i> dimensional factor $\ell^2 / \mu^2 = (\kappa / 4 \Lambda_{\mathrm{H}})$. This

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mielke-geometrodynamics_F01460	294	■ 1.000	$W^{\{+\}}$	W^+	W^+ and W^- gauge bosons or even “heavy photons” Z , again via pair creation. These
mielke-geometrodynamics_F01461	294	■ 1.000	$W^{\{-\}}$	W^-	W^+ and W^- gauge bosons or even “heavy photons” Z , again via pair creation. These
mielke-geometrodynamics_F01462	294	■ 0.633	$\mu \rightarrow 0$	$\mu \rightarrow 0$	In the limit $\mu \rightarrow 0$ of $\mathrm{SL}(5, \mathbb{R})$ symmetry restoration, concepts like macroscopic
mielke-geometrodynamics_F01463	294	■ 1.000	$d\,s$	ds	logical field theory does a line element ds emerge as a necessary means for measuring
mielke-geometrodynamics_F01464	294	■ 1.000	$1/\alpha :=$	$1/\alpha :=$	slowly decay (ZEE 2004). The inverse <i>dimensionless</i> coupling constants $1/\alpha :=$
mielke-geometrodynamics_F01465	295	■ 1.000	$\hbar c/e^2 \simeq 137$	$\hbar c/e^2 \simeq 137$	$\hbar c/e^2 \simeq 137$ of Sommerfeld and $1/\mu \simeq 10^{123}$ of gravity seem to be on a par except
mielke-geometrodynamics_F01466	295	■ 1.000	$1/\mu \simeq 10^{123}$	$1/\mu \simeq 10^{123}$	$\hbar c/e^2 \simeq 137$ of Sommerfeld and $1/\mu \simeq 10^{123}$ of gravity seem to be on a par except
mielke-geometrodynamics_F01467	295	■ 0.991	C_{TT}	C_{TT}	contains a translational CS three-form C_{TT} that gives rise to the <i>parity-violating</i>
mielke-geometrodynamics_F01468	295	■ 1.000	$\mathrm{gl}(4, \mathbb{R})$	$\mathrm{gl}(4, \mathbb{R})$	of Nieh and Yan (NY) (NIEH 2007). On the other hand, the $\mathrm{gl}(4, \mathbb{R})$ -valued Pontryagin
mielke-geometrodynamics_F01469	296	■ 0.865	$\mathrm{Ric}_{\alpha\beta} := (-1)^{\mathrm{sig}} *$	$\mathrm{Ric}_{\alpha\beta} := (-1)^{\mathrm{sig}} *$	sion involving the symmetric Ricci tensor, i.e., the zero-form $\mathrm{Ric}_{\alpha\beta} := (-1)^{\mathrm{sig}} *$
mielke-geometrodynamics_F01470	296	■ 0.998	$\left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)}\right)$	$(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$	$(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$, is known as the <i>Gauss-Bonnet identity</i> .
mielke-geometrodynamics_F01471	300	■ 1.000	$\mathscr{S} := \mathscr{M} / \mathscr{D}$	$\mathscr{S} := \mathscr{M} / \mathscr{D}$	² The quotient space $\mathscr{S} := \mathscr{M} / \mathscr{D}$ of all Riemannian metrics on a 3-dimensional manifold modulo the
mielke-geometrodynamics_F01472	300	■ 1.000	$\mathscr{D}$	$\mathscr{D}$	group $\mathscr{D}$ of the diffeomorphisms of M^3 is called the “superspace”. Accordingly, it has a completely
mielke-geometrodynamics_F01473	300	■ 1.000	M^3	M^3	group $\mathscr{D}$ of the diffeomorphisms of M^3 is called the “superspace”. Accordingly, it has a completely
mielke-geometrodynamics_F01474	301	■ 1.000	F_{ij}	F_{ij}	ponents F_{ij} of the electromagnetic field strengths. Such a theory was developed not
mielke-geometrodynamics_F01475	301	■ 1.000	$g_{[ij]} g^{[ij]}$	$g_{[ij]} g^{[ij]}$	term proportional to $g_{[ij]} g^{[ij]}$ occurs side by side with Hilbert’s Lagrangian 4-form,
mielke-geometrodynamics_F01476	301	■ 1.000	$\mathrm{g}_{[ij]} := \ell^{*2} F_{ij}$	$\mathfrak{g}_{[ij]} := \ell^{*2} F_{ij}$	FAT (1979a), in a more recent study, has given up the identification $\mathfrak{g}_{[ij]} := \ell^{*2} F_{ij}$.
mielke-geometrodynamics_F01477	301	■ 1.000	$m \in M$	$m \in M$	be dependent not only on the point $m \in M$ but also on locally distinguished direc-
mielke-geometrodynamics_F01478	303	■ 1.000	Σ_{ω}	Σ_{ω}	mathematical structure of the canonical energy–momentum current Σ_{ω} of the Yang–
mielke-geometrodynamics_F01479	304	■ 0.995	$\Sigma_{\mathrm{mat}} = 0$	$\Sigma_{\mathrm{mat}} = 0$	$\Sigma_{\mathrm{mat}} = 0$.
mielke-geometrodynamics_F01480	304	■ 1.000	$\delta = \pi/2$	$\delta = \pi/2$	$\delta = \pi/2$ is exactly equal to the operation of taking the Hodge dual ³
mielke-geometrodynamics_F01481	305	■ 1.000	Σ_{YM}	Σ_{YM}	transformation). The <i>invariance</i> of Σ_{YM} with respect to duality rotations can easily
mielke-geometrodynamics_F01482	305	■ 1.000	$\cos^2 \delta +$	$\cos^2 \delta +$	be proved by writing out and applying the elementary functional relation $\cos^2 \delta +$
mielke-geometrodynamics_F01483	305	■ 1.000	$\sin^2 \delta = 1$	$\sin^2 \delta = 1$	$\sin^2 \delta = 1$ for trigonometric functions.
mielke-geometrodynamics_F01484	307	■ 0.999	$R_{ij}^{\{ \}}$	$R_{ij}^{\{ \}}$	for the holonomically written components $R_{ij}^{\{ \}}$ or the Ricci tensor. Combined with
mielke-geometrodynamics_F01485	307	■ 0.993	$\bar{\tau}_e$	$\bar{\tau}_e$	out of the matter currents $\bar{\tau}_e$ and $\bar{\tau}_p$ of a theory (SCHWINGER 1968; JULIA &
mielke-geometrodynamics_F01486	307	■ 0.993	$\bar{\tau}_p$	$\bar{\tau}_p$	out of the matter currents $\bar{\tau}_e$ and $\bar{\tau}_p$ of a theory (SCHWINGER 1968; JULIA &

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mielke-geometrodynamics_F01487	308	■ 1.000	$\delta=0$	$\delta = 0$	traditional electrodynamics, which is defined by the choice $\delta = 0$ of a “dual gauge”
mielke-geometrodynamics_F01488	308	■ 0.562	$\xi \wedge \vartheta$	$\xi \wedge \vartheta$	After the exterior multiplication of (14.1.32) by $\xi \wedge \vartheta$ and (14.1.33) by $\ast \xi \wedge \vartheta$ and
mielke-geometrodynamics_F01489	308	■ 0.562	$\{ \}^{\ast} \xi \wedge \vartheta$	$\ast \xi \wedge \vartheta$	After the exterior multiplication of (14.1.32) by $\xi \wedge \vartheta$ and (14.1.33) by $\ast \xi \wedge \vartheta$ and
mielke-geometrodynamics_F01490	308	■ 0.966	$d \delta$	$d\delta$	$d\delta$, Eq. (14.1.37) has to be contracted by $\ast \Sigma_{\mathrm{YM}}(\xi)$. On account of the invariance
mielke-geometrodynamics_F01491	308	■ 0.966	$\{ \}^{\ast} \Sigma_{\mathrm{YM}}(\xi)$	$\ast \Sigma_{\mathrm{YM}}(\xi)$	$d\delta$, Eq. (14.1.37) has to be contracted by $\ast \Sigma_{\mathrm{YM}}(\xi)$. On account of the invariance
mielke-geometrodynamics_F01492	309	■ 1.000	$d \delta \equiv 0$	$dd\delta \equiv 0$	Due to the identity $dd\delta \equiv 0$, this follows in an unequivocal way from the exactness
mielke-geometrodynamics_F01493	309	■ 0.928	$b_1=\operatorname{dim} H^1(M)=\nu$	$b_1 = \dim H^1(M) = \nu$	of the form $d\delta$. For manifolds with Betti number $b_1 = \dim H^1(M) = \nu$, e.g., for
mielke-geometrodynamics_F01494	309	■ 1.000	ν	ν	those that exhibit ν -fold global connectivity, however,
mielke-geometrodynamics_F01495	310	■ 1.000	(1962,1969)	(1962, 1969)	Encouraged by this remarkable result, WHEELER (1962, 1969) was inclined not
mielke-geometrodynamics_F01496	311	■ 1.000	$\mathrm{n}=4+\mathrm{K}$	$\mathbf{n} = 4 + \mathbf{K}$	(or supergravity) in $\mathbf{n} = 4 + \mathbf{K}$ dimensions, which is then shown or assumed to
mielke-geometrodynamics_F01497	311	■ 0.844	(1926,1928)	(1926, 1928)	(1926, 1928) proved that this is possible after proceeding to 5-dimensional ⁵ man-
mielke-geometrodynamics_F01498	312	■ 1.000	M^{4+K}	M^{4+K}	M^{4+K} that is endowed with a bundle $\hat{L}(M^{4+K})$ of linear frames and is furthermore
mielke-geometrodynamics_F01499	312	■ 1.000	$\hat{L}(M^{4+K})$	$\hat{L}(M^{4+K})$	M^{4+K} that is endowed with a bundle $\hat{L}(M^{4+K})$ of linear frames and is furthermore
mielke-geometrodynamics_F01500	312	■ 0.998	$\hat{\vartheta}$	$\hat{\vartheta}$	equipped with a canonical 1-form $\hat{\vartheta}$ associated with $\hat{L}(M^{4+K})$. Only later does one
mielke-geometrodynamics_F01501	312	■ 1.000	$\hat{E} \approx M^4 \times B$	$\hat{E} \approx M^4 \times B$	such that M^{4+K} takes on the form $\hat{E} \approx M^4 \times B$, where B is a (non)compact K-
mielke-geometrodynamics_F01502	312	■ 1.000	G/H	G/H	dimensional homogeneous space, G/H . Here G denotes a semisimple Lie group of
mielke-geometrodynamics_F01503	312	■ 0.951	$\mathrm{K}=\operatorname{dim} \mathrm{G}$	$\mathbf{K} = \dim \mathbf{G}$	dimension $\mathbf{K} = \dim \mathbf{G}$, and H an isotropic subgroup. As shown by COQUEREAUX &
mielke-geometrodynamics_F01504	312	■ 0.600	$P(M^4, G, \pi, \delta)$	$P(M^4, G, \pi, \delta)$	JADCZYK (1983), the “geometric arena” is not the fiber bundle $P(M^4, G, \pi, \delta)$ but
mielke-geometrodynamics_F01505	312	■ 0.983	$P(M^4, N/H, \pi, \delta)$	$P(M^4, N/H, \pi, \delta)$	$P(M^4, N/H, \pi, \delta)$, where N denotes the normalizer of H in G. The extended space
mielke-geometrodynamics_F01506	312	■ 1.000	$\hat{E}$	$\hat{E}$	$\hat{E}$, which is locally isomorphic to $M^4 \times G/H$, is globally the associated bundle
mielke-geometrodynamics_F01507	312	■ 1.000	$M^4 \times G/H$	$M^4 \times G/H$	$\hat{E}$, which is locally isomorphic to $M^4 \times G/H$, is globally the associated bundle
mielke-geometrodynamics_F01508	312	■ 0.996	$\hat{E}(M^4, F=G/H, N/H, P)$	$\hat{E}(M^4, F = G/H, N/H, P)$	$\hat{E}(M^4, F = G/H, N/H, P)$ with N/H as structure group.
mielke-geometrodynamics_F01509	312	■ 0.996	N/H	N/H	$\hat{E}(M^4, F = G/H, N/H, P)$ with N/H as structure group.
mielke-geometrodynamics_F01510	312	■ 0.858	$\omega=\omega^j_{\alpha} L_j \otimes \vartheta^{\alpha}$	$\omega = \omega^j_{\alpha} L_j \otimes \vartheta^{\alpha}$	and $\omega = \omega^j_{\alpha} L_j \otimes \vartheta^{\alpha}$ is a 1-form with values in the Lie algebra of G or N/H ,
mielke-geometrodynamics_F01511	312	■ 0.760	$\overset{\circ}{\vartheta}$	$\overset{\circ}{\vartheta}$	Mills-type gauge fields. The 1-form $\overset{\circ}{\vartheta}$ is defined on the cotangent bundle $T_e^*(G)$ of
mielke-geometrodynamics_F01512	312	■ 0.760	$T_e^*(G)$	$T_e^*(G)$	Mills-type gauge fields. The 1-form $\overset{\circ}{\vartheta}$ is defined on the cotangent bundle $T_e^*(G)$ of

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mielke-geometrodynamics_F01513	312	■ 1.000	ξ^j	ξ^j	the group G , being parametrized by the coordinates ξ^j . In its expansion
mielke-geometrodynamics_F01514	312	■ 1.000	φ_j	φ_j	the coefficients φ_j may be regarded as a multiplet of (Higgs-type) scalar fields trans-
mielke-geometrodynamics_F01515	312	■ 1.000	$d \xi^j$	$d\xi^j$	(holonomic) basis $d\xi^j$ is “dual” to the infinitesimal generators I_j of G . Moreover,
mielke-geometrodynamics_F01516	312	■ 1.000	$I_{\{j\}}$	I_j	(holonomic) basis $d\xi^j$ is “dual” to the infinitesimal generators I_j of G . Moreover,
mielke-geometrodynamics_F01517	312	■ 0.881	$V=V^{\{j\}} I_{\{j\}}, \stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{\vartheta}$	$V = V^j I_j, \overset{\circ}{\vartheta}$	as a frame “dual” to the Lie vector field $V = V^j I_j$, $\overset{\circ}{\vartheta}$ satisfies the Maurer–Cartan
mielke-geometrodynamics_F01518	312	■ 1.000	$\stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{\omega}$	$\overset{\circ}{\omega}$	A connection $\overset{\circ}{\omega}$ on G arises naturally via the Ansatz
mielke-geometrodynamics_F01519	313	■ 1.000	$\stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{\Theta}$	$\overset{\circ}{\Theta}$	The <i>internal</i> torsion $\overset{\circ}{\Theta}$ and curvature $\overset{\circ}{\Omega}$, respectively, are obtained from Cartan’s
mielke-geometrodynamics_F01520	313	■ 1.000	$\stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{\Omega}$	$\overset{\circ}{\Omega}$	The <i>internal</i> torsion $\overset{\circ}{\Theta}$ and curvature $\overset{\circ}{\Omega}$, respectively, are obtained from Cartan’s
mielke-geometrodynamics_F01521	313	■ 0.811	$b=1 / 2$	$b = 1/2$	$b = 1/2$: torsionless connection in a symmetric coset space B of constant
mielke-geometrodynamics_F01522	313	■ 1.000	$\stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{R}=1 / 4$	$\overset{\circ}{R}= 1/4$	curvature $\overset{\circ}{R} = 1/4$.
mielke-geometrodynamics_F01523	313	■ 1.000	$b=0, 1$	$b = 0, 1$	$b = 0, 1$: Flat connection <i>with</i> parallelizing <i>torsion</i> .
mielke-geometrodynamics_F01524	313	■ 1.000	$\operatorname{Tr}(\stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{\vartheta} \otimes \stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{\vartheta})$	$\operatorname{Tr}(\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta})$	G . The term $\operatorname{Tr}(\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta})$ just gives the internal line element
mielke-geometrodynamics_F01525	313	■ 1.000	$\stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{g}_{\{i j\}}$	$\overset{\circ}{g}_{ij}$	where $\overset{\circ}{g}_{ij}$ denotes the Cartan–Killing metric of G . As far as <i>simple</i> Lie groups are
mielke-geometrodynamics_F01526	313	■ 0.997	$\operatorname{Tr}(\omega \otimes \stackrel{\circ}{\stackrel{\circ}{\text{rel}}}\{\text{circ}\}{\vartheta})$	$\operatorname{Tr}(\omega \otimes \overset{\circ}{\vartheta})$	the terms $\operatorname{Tr}(\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta})$ and $\operatorname{Tr}(\omega \otimes \overset{\circ}{\vartheta})$ but also in $\operatorname{Tr}(\omega \otimes \omega)$, since ω takes values in
mielke-geometrodynamics_F01527	313	■ 0.997	$\operatorname{Tr}(\omega \otimes \omega)$	$\operatorname{Tr}(\omega \otimes \omega)$	the terms $\operatorname{Tr}(\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta})$ and $\operatorname{Tr}(\omega \otimes \overset{\circ}{\vartheta})$ but also in $\operatorname{Tr}(\omega \otimes \omega)$, since ω takes values in
mielke-geometrodynamics_F01528	313	■ 1.000	$d \hat{s}^2$	$d\hat{s}^2$	classical, i.e., nonquantized models, the total signature of the line element $d\hat{s}^2$ has
mielke-geometrodynamics_F01529	313	■ 0.983	$\mathrm{G} \times \mathrm{G} / \mathrm{G}_{\mathrm{D}}$	$G \times G/G_D$	⁶ More precisely, the group manifold of G is the symmetric space $G \times G/G_D$, where G_D is the
mielke-geometrodynamics_F01530	313	■ 0.983	G_{D}	G_D	⁶ More precisely, the group manifold of G is the symmetric space $G \times G/G_D$, where G_D is the
mielke-geometrodynamics_F01531	314	■ 1.000	$\hat{L}(\hat{E})$	$\hat{L}(\hat{E})$	may be imprinted on the bundle $\hat{L}(\hat{E})$ of the linear frames. Here ω^g denotes the
mielke-geometrodynamics_F01532	314	■ 0.998	$L\left(M^4\right), \stackrel{\circ}{\omega}$	$L\left(M^4\right), \overset{\circ}{\omega}$	gravitational connection in $L(M^4)$, $\overset{\circ}{\omega}$ a connection in the group manifold, and $\hat{\alpha}$
mielke-geometrodynamics_F01533	314	■ 0.998	$\hat{\alpha}$	$\hat{\alpha}$	gravitational connection in $L(M^4)$, $\overset{\circ}{\omega}$ a connection in the group manifold, and $\hat{\alpha}$
mielke-geometrodynamics_F01534	314	■ 0.514	$4+\mathrm{K}$	$4 + K$	is valid. Connections possessing a nonvanishing $(4 + K)$ -dimensional torsion and
mielke-geometrodynamics_F01535	314	■ 1.000	$\hat{\Omega}$	$\hat{\Omega}$	Accordingly, the higher-dimensional Riemann–Cartan curvature 2-form $\hat{\Omega}$, given
mielke-geometrodynamics_F01536	314	■ 0.917	$(4+\mathrm{K})$	$(4 + K)$	In our matrix notation, the $(4 + K)$ -dimensional exterior derivative $\hat{d}$ is given by
mielke-geometrodynamics_F01537	314	■ 0.917	$\hat{\mathrm{d}}$	$\hat{d}$	In our matrix notation, the $(4 + K)$ -dimensional exterior derivative $\hat{d}$ is given by
mielke-geometrodynamics_F01538	315	■ 0.980	M^4	M^4	internal space B relative to the four-dimensional spacetime M^4 in the bundle $\hat{E}$.

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mielke-geometrodynamics_F01539	315	■ 0.999	$\Omega^{\prime}$	Ω'	$\hat{\alpha}$ and the 2-form Ω' of the Yang–Mills field strengths, we have obtained the decisive
mielke-geometrodynamics_F01540	316	■ 0.988	$\alpha_g:=g^2/\hbar c$	$\alpha_g := g^2/\hbar c$	modification by the gauge-coupling constant $\alpha_g := g^2/\hbar c$. The term $\tilde{L}_W^G$ depends
mielke-geometrodynamics_F01541	316	■ 0.988	$\hat{L}_{\mathrm{W}}^G$	$\hat{L}_W^G$	modification by the gauge-coupling constant $\alpha_g := g^2/\hbar c$. The term $\tilde{L}_W^G$ depends
mielke-geometrodynamics_F01542	316	■ 0.928	$\stackrel{\circ}{R}$	$\overset{\circ}{R}$	whose value is determined by the scalar curvature $\overset{\circ}{R}$ and by the density $\sqrt{\det \overset{\circ}{g}}$ of
mielke-geometrodynamics_F01543	316	■ 0.928	$\sqrt{\operatorname{det} \stackrel{\circ}{g}}$	$\sqrt{\det \overset{\circ}{g}}$	whose value is determined by the scalar curvature $\overset{\circ}{R}$ and by the density $\sqrt{\det \overset{\circ}{g}}$ of
mielke-geometrodynamics_F01544	316	■ 1.000	$M^{4+K} \approx M^4 \times B$	$M^{4+K} \approx M^4 \times B$	(i) $M^{4+K} \approx M^4 \times B$ locally and
mielke-geometrodynamics_F01545	316	■ 1.000	$R_4 \gg R_K$	$R_4 \gg R_K$	(ii) $R_4 \gg R_K$ for the corresponding characteristic radii.
mielke-geometrodynamics_F01546	316	■ 0.611	$\mathscr{D}\bigl(M^{4+K}\bigr)$	$\mathscr{D}\left(M^{4+K}\right)$	“spontaneous” symmetry breaking of the general $(4 + K)$ -dimensional covariance group $\mathscr{D}(M^{4+K})$
mielke-geometrodynamics_F01547	316	■ 0.686	$R_{\circ}$	$R_{\circ}$	which tends to diminish the characteristic radius $R_{\circ}$ of B to the order of the Planck length ℓ^*
mielke-geometrodynamics_F01548	317	■ 0.998	$Y_n(B)$	$Y_n(B)$	into a complete set of harmonic functions $Y_n(B)$ on $B = G/H$; cf. SALAM &
mielke-geometrodynamics_F01549	317	■ 0.998	$B=G/H$	$B = G/H$	into a complete set of harmonic functions $Y_n(B)$ on $B = G/H$; cf. SALAM &
mielke-geometrodynamics_F01550	317	■ 1.000	$M^4 \times S^1$	$M^4 \times S^1$	closing up of the five-dimensional world to a cylinder $M^4 \times S^1$ of an exceedingly
mielke-geometrodynamics_F01551	317	■ 0.725	R_{o}	$R_{\circ}$	small radius $R_{\circ}$ is taken for granted, Eq. (14.2.21) degenerates to a Fourier expansion
mielke-geometrodynamics_F01552	317	■ 1.000	S^1	S^1	on the circle S^1 . The coefficients in the expansion (14.2.21) are fields of short range
mielke-geometrodynamics_F01553	318	■ 1.000	$S\,U(2) \times U(1)$	$SU(2) \times U(1)$	of strong interactions and the $SU(2) \times U(1)$ gauge group of the Weinberg–Salam
mielke-geometrodynamics_F01554	318	■ 0.948	$\mathbb{C}P^2=S\,U(3)/U(2)$	$\mathbb{C}P^2 = SU(3)/U(2)$	where $\mathbb{C}P^2 = SU(3)/U(2)$ denotes the complex projective two-space and S^n the
mielke-geometrodynamics_F01555	318	■ 0.948	S^n	S^n	where $\mathbb{C}P^2 = SU(3)/U(2)$ denotes the complex projective two-space and S^n the
mielke-geometrodynamics_F01556	318	■ 0.994	$M^4 \times B$	$M^4 \times B$	difficulty arises due to the fact that the Cartesian product $M^4 \times B$ of the Minkowski
mielke-geometrodynamics_F01557	318	■ 1.000	$\mathrm{n}=11$	$\mathfrak{n} = 11$	Einstein space. In order to remedy this deficiency in, e.g., $\mathfrak{n} = 11$ dimensions, it is
mielke-geometrodynamics_F01558	318	■ 0.874	$S^7=S\,O(8)/S\,O(7)$	$S^7 = SO(8)/SO(7)$	necessary to “squash” the (round) sphere $S^7 = SO(8)/SO(7)$ by partial rescalings
mielke-geometrodynamics_F01559	318	■ 0.999	$\mathrm{K}=4+2+1=7$	$\mathsf{K} = 4 + 2 + 1 = 7$	As observed by WITTEN (1981), $\mathsf{K} = 4 + 2 + 1 = 7$ of (14.2.25) represents the min-
mielke-geometrodynamics_F01560	319	■ 1.000	$\mathrm{n}=4+7=11$	$\mathfrak{n} = 4 + 7 = 11$	metric, belong to a single representation. Most remarkably, $\mathfrak{n} = 4 + 7 = 11$ is the
mielke-geometrodynamics_F01561	319	■ 1.000	$(\mathscr{N}=1)$	$(\mathscr{N} = 1)$	<i>maximal</i> dimension of a theory of simple ($\mathscr{N} = 1$) supergravity, cf. DEWITT &
mielke-geometrodynamics_F01562	319	■ 1.000	$\overline{\mathrm{s}} \leq 3/2$	$\bar{\mathfrak{s}} \leq 3/2$	STORA (1984), that contains the necessary fermion fields of spin $\bar{\mathfrak{s}} \leq 3/2$ without
mielke-geometrodynamics_F01563	319	■ 1.000	$\overline{\mathrm{s}}=2$	$\bar{\mathfrak{s}} = 2$	the bosonic fields exceeding the spin or helicity $\bar{\mathfrak{s}} = 2$ of the graviton. The bosonic
mielke-geometrodynamics_F01564	319	■ 0.928	A_{IJK}	A_{IJK}	metric tensor potential A_{IJK} , a remnant of the internal torsion, which may induce a
mielke-geometrodynamics_F01565	319	■ 0.835	$M^{11}=(\mathrm{AdS})^4 \times S^7$	$M^{11} = (\mathrm{AdS})^4 \times S^7$	tions, the underlying vacuum geometry factors into the product $M^{11} = (\mathrm{AdS})^4 \times S^7$

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mielke-geometrodynamics_F01566	319	■ 1.000	S^7	$\tilde{\mathbf{S}}^7$	ical model of the real world. For the “squashed” seven-sphere $\tilde{S}^7$, there exists a
mielke-geometrodynamics_F01567	319	■ 1.000	$M^5=\mathbb{R}^4\times S^1$	$M^5=\mathbb{R}^4\times S^1$	BERGMANN (1938), who utilized the topology $M^5=\mathbb{R}^4\times S^1$ of a cylinder for the
mielke-geometrodynamics_F01568	320	■ 0.994	$\mathrm{E}_{2,4}$	$\mathbf{E}_{2,4}$	pseudo-Euclidean space $\mathbf{E}_{2,4}$ of conformal symmetries was pointed out by BUDINI
mielke-geometrodynamics_F01569	320	■ 0.998	$\hat{\omega}$	$\hat{\omega}$	point. Employing the dissociation (14.2.9) of the connection 1-form $\hat{\omega}$, one obtains
mielke-geometrodynamics_F01570	320	■ 1.000	$\hat{\psi}$	$\hat{\psi}$	field $\hat{\psi}$ with $2^{[2+K/2]^+}$ components is adopted:
mielke-geometrodynamics_F01571	320	■ 1.000	$2^{\{[2+\mathrm{K}]/2\}}$	$2^{[2+K/2]}$	field $\hat{\psi}$ with $2^{[2+K/2]^+}$ components is adopted:
mielke-geometrodynamics_F01572	320	■ 1.000	$\chi_{\rho F}^{\pm}(\xi)$	$\chi_{\rho F}^{\pm}(\xi)$	Here the spinorial harmonics $\chi_{\rho F}^{\pm}(\xi)$ furnish a representation ρ of the symmetry
mielke-geometrodynamics_F01573	320	■ 0.989	$\stackrel{\circ}{\gamma}^{K+1}$	$\stackrel{\circ}{\gamma}^{K+1}$	of the internal helicity operator $\stackrel{\circ}{\gamma}^{K+1}$. After projection π to four-dimensional compo-
mielke-geometrodynamics_F01574	321	■ 1.000	$\tilde{\omega}^g$	$\tilde{\omega}^g$	the (spinor) connections $\tilde{\omega}^g$ and ω , respectively, but that in addition, inherits a gauge
mielke-geometrodynamics_F01575	321	■ 1.000	$\gamma\wedge\gamma\wedge^*\Omega\psi$	$\gamma\wedge\gamma\wedge^*\Omega\psi$	covariant term $\gamma\wedge\gamma\wedge^*\Omega\psi$ of electric dipole type. This term in the Lagrangian
mielke-geometrodynamics_F01576	321	■ 0.527	$\hat{\mathbf{m}}=0$	$\hat{\mathbf{m}}=0$	which follows from the Lagrangian (4 + K)-form in the case $\hat{\mathbf{m}}=0$. This equation
mielke-geometrodynamics_F01577	322	■ 1.000	$\stackrel{\circ}{D}$	$\stackrel{\circ}{D}$	depends via the covariant derivative $\stackrel{\circ}{D}$ on the spinor connection $\stackrel{\circ}{\omega}$ and therefore acts
mielke-geometrodynamics_F01578	322	■ 1.000	$\stackrel{\circ}{\omega}$	$\stackrel{\circ}{\omega}$	depends via the covariant derivative $\stackrel{\circ}{D}$ on the spinor connection $\stackrel{\circ}{\omega}$ and therefore acts
mielke-geometrodynamics_F01579	322	■ 0.978	$i\not{D}^{\{\text{(int)}\}}$	$i\mathcal{D}^{(\text{int})}$	only on the internal space B. Provided B is compact, $i\mathcal{D}^{(\text{int})}$ has a discrete spectrum
mielke-geometrodynamics_F01580	322	■ 1.000	$\gamma^5\psi=(\mp)\psi$	$\gamma^5\psi=(\mp)\psi$	obeying $\gamma^5\psi=(\mp)\psi$, transform according to inequivalent representations of the
mielke-geometrodynamics_F01581	322	■ 1.000	$n=4+K$	$n=4+K$	dimension of the total space. For $n=4+K$, the spinor fields transform accord-
mielke-geometrodynamics_F01582	322	■ 0.999	$\widetilde{SO}(1,3+K)$	$\widetilde{SO}(1,3+K)$	ing to the fundamental representation of the covering tangent group $\widetilde{SO}(1,3+K)$.
mielke-geometrodynamics_F01583	322	■ 0.995	$\hat{\gamma}^{n+1}$	$\hat{\gamma}^{n+1}$	eigenvalues of the n-dimensional chirality operator $\hat{\gamma}^{n+1}$. Under the subgroup
mielke-geometrodynamics_F01584	322	■ 0.615	$\overparen{SO}(1,3)\otimes\widetilde{SO}(K)$	$\widehat{SO}(1,3)\otimes\widetilde{SO}(K)$	$\widetilde{SO}(1,3)\otimes\widetilde{SO}(K)$, such a representation branches according to
mielke-geometrodynamics_F01585	322	■ 1.000	$n=2p+1$	$n=2p+1$	(i) $n=2p+1$: The transformation $\hat{\gamma}^A\rightarrow-\hat{\gamma}^A$ does not affect the infinitesimal
mielke-geometrodynamics_F01586	322	■ 1.000	$\hat{\gamma}^A\rightarrow-\hat{\gamma}^A$	$\hat{\gamma}^A\rightarrow-\hat{\gamma}^A$	(i) $n=2p+1$: The transformation $\hat{\gamma}^A\rightarrow-\hat{\gamma}^A$ does not affect the infinitesimal
mielke-geometrodynamics_F01587	322	■ 1.000	$\hat{\sigma}^{AB}$	$\hat{\sigma}^{AB}$	generators $\hat{\sigma}^{AB}$ but $\hat{\gamma}^{n+1}\rightarrow-\hat{\gamma}^{n+1}$. Since G-transformations are
mielke-geometrodynamics_F01588	322	■ 1.000	$\hat{\gamma}^{n+1}\rightarrow-\hat{\gamma}^{n+1}$	$\hat{\gamma}^{n+1}\rightarrow-\hat{\gamma}^{n+1}$	generators $\hat{\sigma}^{AB}$ but $\hat{\gamma}^{n+1}\rightarrow-\hat{\gamma}^{n+1}$. Since G-transformations are
mielke-geometrodynamics_F01589	322	■ 1.000	$n=4p$	$n=4p$	(ii) $n=4p$: This implies that
mielke-geometrodynamics_F01590	322	■ 1.000	$n=4p+2$	$n=4p+2$	(iii) $n=4p+2$: Then it follows that
mielke-geometrodynamics_F01591	323	■ 0.543	$i\not{D}^{\{\text{(int)}\}}$	$i\mathcal{D}^{(\text{int})}$	Then it remains to reveal the conditions under which $i\mathcal{D}^{(\text{int})}$ has zero eigenmodes.

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mielke-geometrodynamics_F01592	323	■ 1.000	$\mathrm{D}^{\wedge\{*\}}=\{ \ }^{\wedge\{*\}} \mathrm{D}$	$\mathrm{D}^* = *\mathrm{D}$	Since $\mathrm{D}^* = *\mathrm{D}$ holds for a metric-compatible (spinor) connection,
mielke-geometrodynamics_F01593	323	■ 0.922	$\stackrel{\circ}{\Theta}=0$	$\overset{\circ}{\Theta}=0$	for vanishing torsion, i.e., $\overset{\circ}{\Theta}=0$, Eq.(14.3.18) is the sum of a nonnegative and a
mielke-geometrodynamics_F01594	324	■ 1.000	$n>2$	$n>2$	izable in $n>2$ dimensions according to perturbation theory. For supergravity with
mielke-geometrodynamics_F01595	324	■ 0.999	$n=11$	$n=11$	turbative expansion. Moreover, simple ($\mathcal{N}=1$) supergravity in $n=11$ dimensions
mielke-geometrodynamics_F01596	324	■ 0.999	$G=\mathrm{SO}(32)$	$G=\mathrm{SO}(32)$	ical strings. These <i>superstring theories</i> with $G=\mathrm{SO}(32)$ or the exceptional group
mielke-geometrodynamics_F01597	324	■ 1.000	$\mathrm{E}_{_8} \times \mathrm{E}_{_8}$	$\mathrm{E}_8 \times \mathrm{E}_8$	$\mathrm{E}_8 \times \mathrm{E}_8$ as structure group seem to provide a unification of all physical interactions
mielke-geometrodynamics_F01598	324	■ 1.000	$n=10+\operatorname{dim} G=506$	$n=10+\operatorname{dim} G=506$	in $n=10+\operatorname{dim} G=506$ dimensions (DUFF et al. 1985) and a consistent and <i>finite</i>
mielke-geometrodynamics_F01599	324	■ 0.993	$\{ \ }^{\wedge\{10\}} J_{_3} \hbar$	$^{10}J_3\hbar$	by the mass M, the z-component ¹⁰ $J_3\hbar$ of the angular momentum, and the total angu-
mielke-geometrodynamics_F01600	324	■ 1.000	$\sqrt{J(J+1)} \hbar$	$\sqrt{J(J+1)}\hbar$	lar momentum $\sqrt{J(J+1)}\hbar$. Furthermore, it may carry the generalized “internal”
mielke-geometrodynamics_F01601	324	■ 1.000	$Q^{\{i\}}$	$Q^{(i)}$	Here $Q^{(i)}$ and $P^{(i)}$ denote the “electric”- and “magnetic”-type charges of a “dually”
mielke-geometrodynamics_F01602	324	■ 1.000	$P^{\{i\}}$	$P^{(i)}$	Here $Q^{(i)}$ and $P^{(i)}$ denote the “electric”- and “magnetic”-type charges of a “dually”
mielke-geometrodynamics_F01603	324	■ 1.000	$J_{_3} \hat{=} J$	$J_3\hat{=}J$	rotation group. As for our still purely <i>classical</i> considerations, $J_3\hat{=}J$ remains valid.
mielke-geometrodynamics_F01604	325	■ 0.798	$r^{\wedge\{2\}} e^{\wedge\{v\}}$	r^2e^v	substitutes the function r^2e^v in the Schwarzschild metric. The generalization of
mielke-geometrodynamics_F01605	326	■ 0.976	$\bar{Q}^{\wedge\{(\mathrm{o})\}}$	$\bar{Q}^{(\mathrm{o})}$	mass M, generalized charge $\bar{Q}^{(\mathrm{o})}$, and (classical) angular momentum component $J_3\hbar$
mielke-geometrodynamics_F01606	326	■ 0.976	$J_{_3} \hbar$	$J_3\hbar$	mass M, generalized charge $\bar{Q}^{(\mathrm{o})}$, and (classical) angular momentum component $J_3\hbar$
mielke-geometrodynamics_F01607	326	■ 1.000	$\mu_{_m}=\frac{Q_{_e}}{M} J$	$\mu_m=\frac{Q_e}{M}J$	with respect to the axis of rotation, but it also has the magnetic moment $\mu_m=\frac{Q_e}{M}J$,
mielke-geometrodynamics_F01608	326	■ 0.823	$\mathrm{M}_{_M\mathrm{ir}}$	M_{ir}	The total mass M of a black hole depends on its geometric remnant mass M_{ir} ,
mielke-geometrodynamics_F01609	326	■ 0.938	$\mathrm{M}_{_M\mathrm{ir}}, \bar{Q}^{\wedge\{(\mathrm{c})\}}$	$\mathrm{M}_{\mathrm{ir}}, \bar{Q}^{(\mathrm{o})}$	dependence of M on the characteristic values M_{ir} , $\bar{Q}^{(\mathrm{o})}$, and J is nonlinear. More
mielke-geometrodynamics_F01610	326	■ 0.811	$J \hat{=} J_{_3}=0$	$J\hat{=}J_3=0$	formulated by REISSNER (1916) and NORDSTRÖM (1918) if $J\hat{=}J_3=0$ holds.
mielke-geometrodynamics_F01611	327	■ 1.000	$\lambda_{_3}$	λ_3	The generalized Gell-Mann matrix λ_3 , which is identical to the Pauli matrix σ_3 for
mielke-geometrodynamics_F01612	327	■ 1.000	$\sigma_{_3}$	σ_3	The generalized Gell-Mann matrix λ_3 , which is identical to the Pauli matrix σ_3 for
mielke-geometrodynamics_F01613	327	■ 1.000	Δ	Δ	Here Δ is to be substituted by the term (14.4.5) with $J=0$. Apart from the horizon
mielke-geometrodynamics_F01614	327	■ 0.993	$r=r_{\mathrm{o}}$	$r=r_{\mathrm{o}}$	given by (14.4.6), this spacetime geometry has obviously still a further one at $r=r_{\mathrm{o}}$,
mielke-geometrodynamics_F01615	327	■ 1.000	$r=0$	$r=0$	away from the essential singularity at $r=0$.
mielke-geometrodynamics_F01616	328	■ 0.672	$Q=-\bar{Q} \cos \delta$	$Q=-\bar{Q} \cos \delta$	a particle of mass M with an electric charge $Q=-\bar{Q} \cos \delta$ and a magnetic monopole
mielke-geometrodynamics_F01617	328	■ 1.000	$P=-\bar{Q} \sin \delta$	$P=-\bar{Q} \sin \delta$	strength $P=-\bar{Q} \sin \delta$. If the surface area S_{A} of the event horizon, which is defined
mielke-geometrodynamics_F01618	328	■ 1.000	$\mathrm{S}_{_M\mathrm{A}}$	S_{A}	strength $P=-\bar{Q} \sin \delta$. If the surface area S_{A} of the event horizon, which is defined

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mielke-geometrodynamics_F01619	328	■ 0.546	$\backslash\mathrm{bar}\{Q\}$	$\overline{Q}$	and a large dual charge $\overline{Q}$, the charge–mass relation and the total mass M of the
mielke-geometrodynamics_F01620	328	■ 0.990	$1 / \backslash\kappa$	$1/\kappa$	necessarily fixed by the coupling constant $1/\kappa$ of the Lorentz “rotons”.
mielke-geometrodynamics_F01621	330	■ 1.000	$\mathrm{d}=10$	$\mathbf{d} = 10$	Green MB, Schwarz JH (1984) Anomaly cancellations in supersymmetric $\mathbf{d} = 10$ gauge theory and
mielke-geometrodynamics_F01622	335	■ 1.000	$1 / \backslash\alpha_{\mathrm{e}}$	$1/\alpha_e$	than electromagnetic forces by the factor $1/\alpha_e$. On the other hand, all experiments
mielke-geometrodynamics_F01623	335	■ 0.967	$m_{\pi}=2\ m_{\mathrm{e}} / \backslash\alpha_{\mathrm{e}}$	$m_{\pi} = 2m_e/\alpha_e$	(1964), is explained by a virtual exchange of scalar π -mesons of mass $m_{\pi} = 2\ m_e/\alpha_e$.
mielke-geometrodynamics_F01624	335	■ 0.829	$\mathrm{SU}(3)^{\mathrm{c}}$	$SU(3)^c$	ory with an exact $SU(3)^c$ structure group. The introduction of these additional color
mielke-geometrodynamics_F01625	336	■ 0.924	$\mathrm{f}_{\mathrm{i}\ \mathrm{j}}$	f_{ij}	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01626	336	■ 0.924	$\mathrm{\vartheta}^{\mathrm{f}}$	ϑ^f	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01627	336	■ 0.924	$\mathrm{\vartheta}^{\mathrm{g}}$	ϑ^g	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01628	336	■ 1.000	$\mathrm{L}_{\mathrm{f}\ \mathrm{g}}$	L_{fg}	necessitates the introduction of an additional mixing term L_{fg} . Were it not for this
mielke-geometrodynamics_F01629	336	■ 1.000	$\mathrm{\vartheta}^{\mathrm{f}}$	ϑ^{f}	mixing term is understood to have only a polynomial dependence on ϑ^{f} and ϑ^{g} of
mielke-geometrodynamics_F01630	336	■ 1.000	$\mathbf{b}-4$	$b - 4$	degree a or $b - 4$,
mielke-geometrodynamics_F01631	337	■ 1.000	$\sum^{\mathrm{f}}$	$\sum^f$	Let $\sum^f$ and $\sum^g$ be the canonical energy–momentum currents that are generated
mielke-geometrodynamics_F01632	337	■ 1.000	$\sum^{\mathrm{g}}$	$\sum^g$	Let $\sum^f$ and $\sum^g$ be the canonical energy–momentum currents that are generated
mielke-geometrodynamics_F01633	338	■ 0.921	G_{N}	G_{N}	of the (weak) Newtonian constant G_{N} of gravity to the coupling constant G_{S} of strong
mielke-geometrodynamics_F01634	338	■ 0.921	G_{S}	G_{S}	of the (weak) Newtonian constant G_{N} of gravity to the coupling constant G_{S} of strong
mielke-geometrodynamics_F01635	338	■ 1.000	$=10^{\{-13\}}\ \mathrm{cm}$	$= 10^{-13}\ \mathrm{cm}$	spacetime structure beyond hadronic distances of the size of one fermi = $10^{-13}\ \mathrm{cm}$.
mielke-geometrodynamics_F01636	338	■ 1.000	f^0	$\mathbf{f}^0$	$\mathbf{f}^0$ -meson and the massless graviton in a first approximation.
mielke-geometrodynamics_F01637	339	■ 1.000	$\mu^{\mathrm{g}},\ \mu^{\mathrm{f}}$	μ^g, μ^f	is the result for the f-metric. The integration constants are μ^g, μ^f , and γ . For $\mathbf{u} = 1/2$,
mielke-geometrodynamics_F01638	339	■ 1.000	$\mathrm{u}=1 / 2$	$\mathbf{u} = 1/2$	is the result for the f-metric. The integration constants are μ^g, μ^f , and γ . For $\mathbf{u} = 1/2$,
mielke-geometrodynamics_F01639	339	■ 0.992	$\mu^{\mathrm{g}}=\Lambda=0$	$\mu^g = \Lambda = 0$	$\mu^g = \Lambda = 0$, this solution reduces to the configuration that was given by SALAM &
mielke-geometrodynamics_F01640	339	■ 1.000	$m_{\mathrm{f}}=1-2\ \mathrm{GeV} / c^2$	$m_f = 1 - 2\mathrm{GeV}/c^2$	f-mesons of mass $m_f = 1 - 2\ \mathrm{GeV}/c^2$. This additional property of the tensor dom-
mielke-geometrodynamics_F01641	340	■ 1.000	$\mathrm{E}_{\mathrm{j}}^{\beta}$	E_j^{β}	the Poincaré gauge theory. These are given by the tetrads E_j^{β} and Ricci’s rotation
mielke-geometrodynamics_F01642	340	■ 1.000	$\Gamma_i^{\alpha\beta}$	$\Gamma_i^{\alpha\beta}$	coefficients $\Gamma_i^{\alpha\beta}$, respectively. With respect to anholonomic frame, these potentials
mielke-geometrodynamics_F01643	340	■ 1.000	$\left(\frac{1}{2},\ \frac{1}{2}\right)$	$\left(\frac{1}{2},\ \frac{1}{2}\right)$	transform under the Lorentz group according to the representation $\left(\frac{1}{2},\ \frac{1}{2}\right)$ or $(1, 0) \oplus$
mielke-geometrodynamics_F01644	340	■ 1.000	$(1,0)\ \oplus$	$(1,0)\oplus$	transform under the Lorentz group according to the representation $\left(\frac{1}{2},\ \frac{1}{2}\right)$ or $(1, 0) \oplus$

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mielke-geometrodynamics_F01645	340	■ 0.609	$(0,1)$	$(0,1)$	$(0,1)$, respectively, and are therefore have an holonomic spin $\bar{s}_a = 1$. After the tran-
mielke-geometrodynamics_F01646	340	■ 0.609	$\operatorname{spin} \bar{s}_a=1$	$\operatorname{spin} \bar{s}_a = 1$	$(0,1)$, respectively, and are therefore have an holonomic spin $\bar{s}_a = 1$. After the tran-
mielke-geometrodynamics_F01647	340	■ 1.000	$\bar{s}=2$	$\bar{s} = 2$	induces gauge potentials of maximal (reducible) spin or helicity $\bar{s} = 2$ (HEHL et al.
mielke-geometrodynamics_F01648	340	■ 1.000	$\Lambda_{\text{eff}} \neq 0$	$\Lambda_{\text{eff}} \neq 0$	nonlinear theory with $\Lambda_{\text{eff}} \neq 0$. Other quantization procedures are needed in order to isolate the
mielke-geometrodynamics_F01649	341	■ 1.000	$\ell=\ell^{*} / \sqrt{\kappa}$	$\ell = \ell^{*} / \sqrt{\kappa}$	proportional to $\ell = \ell^{*} / \sqrt{\kappa}$ is superposed in the metric background of the solutions.
mielke-geometrodynamics_F01650	341	■ 1.000	$1 / \sqrt{\kappa}$	$1 / \sqrt{\kappa}$	ℓ^{*} scaled by $1 / \sqrt{\kappa}$ is identical to the Compton wavelength of a proton according to
mielke-geometrodynamics_F01651	341	■ 1.000	$10^{13} \mathrm{~cm}$	10^{13} cm	minimum radius would be increased to about 10^{13} cm as a result of the hypothesis of
mielke-geometrodynamics_F01652	341	■ 0.999	ℓ^2	ℓ^2	“strong” gravity. The relative strength of the coupling constants ℓ^2 and ℓ^{*2} , which
mielke-geometrodynamics_F01653	341	■ 0.999	ℓ^{*2}	ℓ^{*2}	“strong” gravity. The relative strength of the coupling constants ℓ^2 and ℓ^{*2} , which
mielke-geometrodynamics_F01654	341	■ 1.000	$\kappa=0,38 \times 10^{-38}$	$\kappa = 0,38 \times 10^{-38}$	small number of order of magnitude $\kappa = 0,38 \times 10^{-38}$. It was again EINSTEIN in
mielke-geometrodynamics_F01655	342	■ 1.000	10^{20}	10^{20}	there would be a fundamental length ℓ being enlarged by about 10^{20} . This hypothe-
mielke-geometrodynamics_F01656	342	■ 1.000	$\operatorname{M}_{\operatorname{U}}$	$\operatorname{M}_{\operatorname{U}}$	between the total mass $\operatorname{M}_{\operatorname{U}}$ of the universe and the mass $\operatorname{M}_{\operatorname{P}}$ of the proton while
mielke-geometrodynamics_F01657	342	■ 1.000	$\operatorname{M}_{\operatorname{P}}$	$\operatorname{M}_{\operatorname{P}}$	between the total mass $\operatorname{M}_{\operatorname{U}}$ of the universe and the mass $\operatorname{M}_{\operatorname{P}}$ of the proton while
mielke-geometrodynamics_F01658	342	■ 1.000	ℓ_{H}	ℓ_{H}	relating it to the product of the relative Hubble radius ℓ_{H} and the relative Compton
mielke-geometrodynamics_F01659	342	■ 1.000	$S \operatorname{U}(f)$	$SU(f)$	resentations of the unitary groups $SU(f)$. Concerning this aspect, particular success
mielke-geometrodynamics_F01660	342	■ 0.995	$S \operatorname{U}(4)$	$SU(4)$	to a far larger extent for the enlarged group $SU(4)$ that is postulated in the context of the charm
mielke-geometrodynamics_F01661	342	■ 0.997	$S \operatorname{U}(5)$	$SU(5)$	hypothesis, and also for the grand unified theories (GUTs) that embarks upon $SU(5)$ as internal
mielke-geometrodynamics_F01662	343	■ 0.999	$S \operatorname{O}(3) \approx S \operatorname{U}(2)$	$SO(3) \approx SU(2)$	transform according to spinor representations of the group $SO(3) \approx SU(2)$ of the
mielke-geometrodynamics_F01663	343	■ 1.000	$S \operatorname{U}(4) \approx S \operatorname{O}(6) \supset$	$SU(4) \approx SO(6) \supset$	nal spin within a common group-theoretic scheme, the group $SU(4) \approx SO(6) \supset$
mielke-geometrodynamics_F01664	343	■ 1.000	$S \operatorname{O}(4) \approx S \operatorname{U}(2) \times S \operatorname{U}(2)$	$SO(4) \approx SU(2) \times SU(2)$	$SO(4) \approx SU(2) \times SU(2)$ was proposed by Wigner. Later on, the group $SU(6)$ was
mielke-geometrodynamics_F01665	343	■ 1.000	$S \operatorname{U}(6)$	$SU(6)$	$SO(4) \approx SU(2) \times SU(2)$ was proposed by Wigner. Later on, the group $SU(6)$ was
mielke-geometrodynamics_F01666	343	■ 1.000	$J=1 / 2$	$J = 1/2$	$J = 1/2$ and $J = 3/2$ of total angular momentum are thought to fill out a 56-fold
mielke-geometrodynamics_F01667	343	■ 1.000	$J=3 / 2$	$J = 3/2$	$J = 1/2$ and $J = 3/2$ of total angular momentum are thought to fill out a 56-fold
mielke-geometrodynamics_F01668	343	■ 1.000	$S \operatorname{L}(6, \mathbb{C}) \supset S \operatorname{L}(2, \mathbb{C}) \times S \operatorname{U}(3)$	$SL(6, \mathbb{C}) \supset SL(2, \mathbb{C}) \times SU(3)$	this problem is to extend $SU(6)$ to the linear group $SL(6, \mathbb{C}) \supset SL(2, \mathbb{C}) \times SU(3)$.
mielke-geometrodynamics_F01669	343	■ 0.962	$\operatorname{SU}(3)$	$\operatorname{SU}(3)$	Thereby a relativistic union of the internal symmetry group $\operatorname{SU}(3)$ with the (simply
mielke-geometrodynamics_F01670	343	■ 1.000	$S \operatorname{L}(6, \mathbb{C})$	$SL(6, \mathbb{C})$	spacetime, is achieved. If the symmetry is regarded only as a local one, an $SL(6, \mathbb{C})$
mielke-geometrodynamics_F01671	343	■ 0.933	$\operatorname{P}(\operatorname{M}, \overline{\operatorname{G}}, \pi, \delta)$	$\operatorname{P}(\operatorname{M}, \overline{\operatorname{G}}, \pi, \delta)$	fiber bundle $\operatorname{P}(\operatorname{M}, \overline{\operatorname{G}}, \pi, \delta)$ as a geometric arena. But this time, it is
mielke-geometrodynamics_F01672	343	■ 1.000	$[S \operatorname{U}(4) \otimes]^4$	$[SU(4) \otimes]^4$	chiral $[SU(4) \otimes]^4$ model of ELIAS et al. (1978) ready to hand for the case $f = c = 4$.

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mielke-geometrodynamics_F01673	343	■ 1.000	f=c=4	$f = c = 4$	chiral $[SU(4)\otimes]^4$ model of ELIAS et al. (1978) ready to hand for the case $f = c = 4$.
mielke-geometrodynamics_F01674	344	■ 0.993	$\overline{\mathrm{L}}(\mathrm{M})$	$\bar{\mathrm{L}}(\mathrm{M})$	with $\bar{\mathrm{L}}(\mathrm{M})$. This 1-form is supposed to have holonomic components in the local
mielke-geometrodynamics_F01675	344	■ 0.993	A_2	A_2	fields—as is, for instance, realized in nature via the $SU(3)$ -nonet f, f', A_2 , and K^*
mielke-geometrodynamics_F01676	344	■ 0.995	$S\mathrm{L}(2\mathrm{N},\mathbb{C})\supset S\mathrm{L}(2,\mathbb{C})\times U(\mathrm{N})$	$SL(2N,\mathbb{C})\supset SL(2,\mathbb{C})\times U(N)$	erators of $SL(2N,\mathbb{C})\supset SL(2,\mathbb{C})\times U(N)$ can be realized by the matrices
mielke-geometrodynamics_F01677	344	■ 1.000	λ_{i}	λ_i	where the generalized Gell-Mann matrices λ_i are the N^2-1 vector operators of the
mielke-geometrodynamics_F01678	344	■ 1.000	N^2-1	$N^2 - 1$	where the generalized Gell-Mann matrices λ_i are the N^2-1 vector operators of the
mielke-geometrodynamics_F01679	344	■ 0.983	$\mathrm{SU}(\mathrm{N})$	$SU(N)$	$SU(N)$ subgroup. If the gauge potentials of the unitary groups—which are contained
mielke-geometrodynamics_F01680	345	■ 1.000	A_{i}	A_i	The Yang–Mills-like gauge potentials A_i aside, this total connection 1-form com-
mielke-geometrodynamics_F01681	345	■ 0.998	$\bar{\gamma}$	$\bar{\gamma}$	field strengths that belong to the 1-forms $\bar{\gamma}$ and $\bar{\omega}$ are the 2-form
mielke-geometrodynamics_F01682	345	■ 1.000	$L_{\mathrm{W}}(\mathrm{f})$	$L_W(f)$	freedom, this is achieved by replacing the Weyl term $L_W(f)$ in the total Lagrangian
mielke-geometrodynamics_F01683	345	■ 1.000	f=c=0	$f = c = 0$	If the internal degrees of freedom are suppressed, i.e., for the case $f = c = 0$, the
mielke-geometrodynamics_F01684	345	■ 1.000	f=3	$f = 3$	as a structure group. In the framework of an $SL(6,\mathbb{C})$ gauge theory, i.e., for $f = 3$
mielke-geometrodynamics_F01685	345	■ 0.987	c=0	$c = 0$	and $c = 0$, they also proposed a mixing term $\bar{L}_{\mathrm{fg}}$ that is responsible for the medi-
mielke-geometrodynamics_F01686	345	■ 0.987	$\bar{L}_{\mathrm{fg}}$	$\bar{L}_{\mathrm{fg}}$	and $c = 0$, they also proposed a mixing term $\bar{L}_{\mathrm{fg}}$ that is responsible for the medi-
mielke-geometrodynamics_F01687	346	■ 1.000	G=S U(2)	$G = SU(2)$	made allowance only for the isospin group, i.e., $G = SU(2)$. If we followed that
mielke-geometrodynamics_F01688	346	■ 0.989	$S\mathrm{U}(4)^{\mathrm{f}}\otimes S\mathrm{U}(4)^{\mathrm{c}}$	$SU(4)^f\otimes SU(4)^c$	theories. Within the scheme A of this $SU(4)^f\otimes SU(4)^c$ -gauge theory, the quarklike
mielke-geometrodynamics_F01689	346	■ 1.000	$S\mathrm{L}(8,\mathbb{C})^{\mathrm{f}}\otimes S\mathrm{U}(8,\mathbb{C})^{\mathrm{c}}$	$SL(8,\mathbb{C})^f\otimes SU(8,\mathbb{C})^c$	$SL(8,\mathbb{C})^f\otimes SU(8,\mathbb{C})^c$ -gauge theory, however, is rather more related—following
mielke-geometrodynamics_F01690	347	■ 1.000	$M=\mathbb{R}\times S^1\times S^2$	$M = \mathbb{R} \times S^1 \times S^2$	fields within the nontrivial topology of the space $M = \mathbb{R} \times S^1 \times S^2$. Its apparent
mielke-geometrodynamics_F01691	348	■ 0.994	$GL(8,\mathbb{C})^{\mathrm{f}}\otimes$	$GL(8,\mathbb{C})^f\otimes$	1980) as indicated previously; see also CASTRO (2012). Thereby, a $GL(8,\mathbb{C})^f\otimes$
mielke-geometrodynamics_F01692	348	■ 1.000	$GL(8,\mathbb{C})^{\mathrm{c}}$	$GL(8,\mathbb{C})^c$	$GL(8,\mathbb{C})^c$ gauge unification of all basic particle forces emerged into a rather spec-
mielke-geometrodynamics_F01693	348	■ 1.000	U(5)	$U(5)$	Since $SU(4)$ and $U(5)$ together with the local Lorentz group can also be embedded
mielke-geometrodynamics_F01694	348	■ 0.987	$SO(1,9)$	$SO(1,9)$	(FADDEEV 2011) such as $SO(1,9)$ and the embedding of the $SO(10)$ GUT into the
mielke-geometrodynamics_F01695	348	■ 0.987	$SO(10)$	$SO(10)$	(FADDEEV 2011) such as $SO(1,9)$ and the embedding of the $SO(10)$ GUT into the
mielke-geometrodynamics_F01696	348	■ 0.994	$SO(1,13)$	$SO(1,13)$	group $SO(1,13)$ have been proposed (PERCACCI 1991). Even “graviweak” groups
mielke-geometrodynamics_F01697	348	■ 0.998	$SO(4,\mathbb{C})$	$SO(4,\mathbb{C})$	such as $SO(4,\mathbb{C})$ have been considered by NESTI & PERCACCI (2010). Again, in
mielke-geometrodynamics_F01698	348	■ 0.995	E(8)	$E(8)$	such as $E(8)$ and hyperbolic Kac–Moody algebras (KLEINSCHMIDT & NICOLAI
mielke-geometrodynamics_F01699	348	■ 0.995	E(10)	$E(10)$	2010) like $E(10)$ have been suggested to achieve this. However, these approaches
mielke-geometrodynamics_F01700	349	■ 0.984	$(5,\mathbb{C})$	$(5,\mathbb{C})$	Castro C (2012) A Clifford $Cl(5,\mathbb{C})$ unified gauge field theory of conformal gravity, Maxwell and

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mielke-geometrodynamics_F01701	349	■ 0.992	$\operatorname{SL}(2\ \mathrm{n},\ \mathrm{mathbb{C}})\ \otimes\ \operatorname{SL}(2,\ \mathrm{mathbb{C}})$	$\mathrm{SL}(2\mathrm{n},\mathbb{C})\otimes\mathrm{SL}(2,\mathbb{C})$	Dennis PW, Huang JC (1977) Spontaneous symmetry breakdown in an $\mathrm{SL}(2\mathrm{n},\mathbb{C})\otimes\mathrm{SL}(2,\mathbb{C})$ -
mielke-geometrodynamics_F01702	349	■ 0.998	$1000\ \mathrm{m}_{\mathrm{W}}$	1000 m_{W}	1000 m_{W} . Phys Rev Lett 40(14):920
mielke-geometrodynamics_F01703	350	■ 0.473	$2\ \mathrm{mathrm{n}},\ \mathrm{mathbb{C}}$	$2\mathrm{n},\mathbb{C}$	Huang JC, Dennis PW (1981a) Spin-1 and spin-2 gauge fields in unified $\mathrm{SL}(2\mathrm{n},\mathbb{C})$ model. Phys
mielke-geometrodynamics_F01704	350	■ 0.805	2^{+}	2^{+}	Isham C, Salam A, Strathdee J (1973a) 2^{+} nonet as gauge particles for $\mathrm{SL}(6,\mathbb{C})$ symmetry. Phys
mielke-geometrodynamics_F01705	354	■ 0.952	$\mathrm{mathrm{R}}-\mathrm{mathrm{R}}_{\mathrm{o}}$	$\mathrm{R}-\mathrm{R}_{\mathrm{o}}$	$\mathrm{R}-\mathrm{R}_{\mathrm{o}}$) whose gradient balances the electrodynamic force. ¹
mielke-geometrodynamics_F01706	354	■ 1.000	(1966,1967)	(1966,1967)	for instance, PRASAD (1966, 1967) and ROMAN & KOH (1966). These notions have
mielke-geometrodynamics_F01707	354	■ 1.000	$\kappa>0$	$\kappa>0$	responsible for the negative (for $\kappa>0$) pressure term
mielke-geometrodynamics_F01708	354	■ 0.983	$2\ \mu\ /\ \mathrm{mathrm{r}}$	$2\mu/\mathrm{r}$	sented by a Schwarzschild–de Sitter microuniverse, whereby the $2\mu/\mathrm{r}$ -singularity
mielke-geometrodynamics_F01709	355	■ 1.000	ℓ_{m}	ℓ_m	coordinate. The (dimensionless) ratio of the Compton wavelength ℓ_m of the test
mielke-geometrodynamics_F01710	355	■ 0.998	$\mathrm{mathrm{F}}=\mathrm{mathrm{F}}_{\mathrm{nm}}^{\mathrm{mathrm{mathbb{I}}}}\left(\rho^*\right)$	$\mathrm{F}=\mathrm{F}_{\mathrm{nm}}^{\iota}(\rho^*)$	with reference to the still to be determined functions $\mathrm{F}=\mathrm{F}_{\mathrm{nm}}^{\iota}(\rho^*)$.
mielke-geometrodynamics_F01711	356	■ 0.877	$\mathrm{iota}=0$	$\iota=0$	For the sake of illustration (Fig. 16.1), the simplest case $\iota=0$ for $\kappa>0$, i.e., the
mielke-geometrodynamics_F01712	357	■ 1.000	$\rho^*=\frac{\pi}{2}$	$\rho^*=\frac{\pi}{2}$	At $\rho^*=\frac{\pi}{2}$ there arises for $\beta^2<2$ a steeply rising potential wall that can never be
mielke-geometrodynamics_F01713	357	■ 1.000	$\beta^2<2$	$\beta^2<2$	At $\rho^*=\frac{\pi}{2}$ there arises for $\beta^2<2$ a steeply rising potential wall that can never be
mielke-geometrodynamics_F01714	357	■ 1.000	$\mathrm{mathrm{V}}(\mathrm{mathrm{r}})=2\ \mu\ /\ \mathrm{mathrm{r}}+\mathrm{mathrm{c}}\mathrm{r}$	$\mathrm{V}(\mathrm{r})=2\mu/\mathrm{r}+\mathrm{c}\mathrm{r}$	In the latter, it is usually the potential $\mathrm{V}(\mathrm{r})=2\mu/\mathrm{r}+\mathrm{c}\mathrm{r}$ that is postulated for the
mielke-geometrodynamics_F01715	358	■ 0.997	$\overline{\mathrm{mathrm{Q}}}$	$\bar{\mathrm{Q}}$	quantum number J of total angular momentum, and generalized charge $\bar{\mathrm{Q}}$ is (almost)
mielke-geometrodynamics_F01716	358	■ 1.000	$\mathrm{mathrm{Q}}_{\mathrm{e}}$	Q_{e}	while the electric charge Q_{e} is determined via the generalized Gell-Mann–Nishijima
mielke-geometrodynamics_F01717	358	■ 0.522	$\mathrm{mathrm{M}}_{\mathrm{ir}}$	M_{ir}	irreducible mass M_{ir} ; concerning the tensor dominance model, it is thus suggestive
mielke-geometrodynamics_F01718	358	■ 0.630	$\mathrm{mathrm{M}}^*:=\hbar\sqrt{8\ \pi}\ /\ \ell^*\mathrm{c}$	$\mathrm{M}^*:=\hbar\sqrt{8\pi}/\ell^*\mathrm{c}$	to propose that M_{ir} as well as the modified Planck mass $\mathrm{M}^*:=\hbar\sqrt{8\pi}/\ell^*\mathrm{c}$ are
mielke-geometrodynamics_F01719	358	■ 0.966	$\mathrm{mathrm{M}}_{\mathrm{ir}}=\mathrm{mathrm{M}}^*=1\ \mathrm{GeV}\ /\ \mathrm{mathrm{c}}^2$	$\mathrm{M}_{\mathrm{ir}}=\mathrm{M}^*=1\mathrm{GeV}/\mathrm{c}^2$	approximately identical to the mass of a proton, i.e., $\mathrm{M}_{\mathrm{ir}}=\mathrm{M}^*=1\mathrm{GeV}/\mathrm{c}^2$. For
mielke-geometrodynamics_F01720	359	■ 0.999	$\mathrm{M},\ \mathrm{J}$	$\mathrm{M},\ \mathrm{J}$	M , J , and I are the characteristics of particles that are here defined <i>canonically</i> in
mielke-geometrodynamics_F01721	359	■ 0.999	I	I	M , J , and I are the characteristics of particles that are here defined <i>canonically</i> in
mielke-geometrodynamics_F01722	359	■ 1.000	$\mathrm{mathrm{J}}=0$	$\mathrm{J}=0$	(However, if $\mathrm{J}=0$, this leads precisely back to the linear Gell-Mann–Okubo formula!)
mielke-geometrodynamics_F01723	359	■ 0.999	$\mathrm{b}=\frac{1}{5}$	$\mathrm{b}=\frac{1}{5}$	surprisingly well represented by (16.3.3). (In Fig. 16.2, $\mathrm{b}=\frac{1}{5}$ has been chosen, while
mielke-geometrodynamics_F01724	359	■ 0.998	$\Lambda(1116)$	$\Lambda(1116)$	M^* has been accommodated to the state $\Lambda(1116)$ of excited baryons.) A detailed
mielke-geometrodynamics_F01725	360	■ 1.000	$k_{\mathrm{B}}=8.6\ \times\ 10^{-5}\ \mathrm{eV}\ /\ \mathrm{mathrm{K}}$	$k_B=8.6\times10^{-5}\mathrm{eV}/\mathrm{K}$	Here $k_B=8.6\times10^{-5}\mathrm{eV}/\mathrm{K}$ denotes the Boltzmann constant. Under the assump-
mielke-geometrodynamics_F01726	362	■ 0.877	$\mathrm{S}\ \mathrm{U}(8)$	$\mathrm{SU}(8)$	Okubo S (1975) $\mathrm{SU}(4)$ and $\mathrm{SU}(8)$ mass formulas and weak interactions. Phys Rev D 11(11):3261
mielke-geometrodynamics_F01727	362	■ 0.824	$(6,\ \mathrm{mathbb{C}})$	$(6,\mathbb{C})$	Salam A (1973). On $\mathrm{SL}(6,\mathbb{C})$ gauge invariance, In: Fundamental Interactions in Physics, 1973 Coral

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mielke-geometrodynamics_F01728	364	■ 0.591	<code>\equiv</code>	$\equiv$	:= : Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $G_{\circ}$: Connected component of the group $\tilde{G}$: Simply connected covering group $[A, B] := AB - BA$: Commutator $\{A, B\} := AB + BA$: Anticommutator $A^* = C A$: Complex-conjugate matrix $A^+ = A^{*T}$: Hermitian adjoint matrix $\pi^{-1}(N)$: Original domain of N with respect to the mapping $\pi : M \rightarrow N$ $T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$: Symmetrized second-rank tensor $T_{[\mu\nu]} := \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})$: Antisymmetrized second-rank tensor

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mielke-geometrodynamics_F01731	364	■ 0.591	$[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$	$[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$	:= ≡ ≈ ≅ ∼ × ⊗ ⊕ ⊗ G_o G̃ : Equal by definition : Identically equal : Isomorphic : Approximately equal : Asymptotic expansion : Topological product : Direct product : Direct sum : Semidirect product : Connected component of the group : Simply connected covering group [A, B] := AB − BA {A, B} := AB + BA A* = C A A⁺ = A*^T π^{−1}(N) T_(μν) := $\frac{1}{2}$(T_{μν} + T_{νμ}) T_[μν] := $\frac{1}{2}$(T_{μν} − T_{νμ}) : Commutator : Anticommutator : Complex-conjugate matrix : Hermitian adjoint matrix : Original domain of N with respect to the mapping π : M → N : Symmetrized second-rank tensor : Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01732	364	■ 0.591	$\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$	$\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$	:= ≡ ≈ ≅ ∼ × ⊗ ⊕ ⊗ G_o G̃ : Equal by definition : Identically equal : Isomorphic : Approximately equal : Asymptotic expansion : Topological product : Direct product : Direct sum : Semidirect product : Connected component of the group : Simply connected covering group [A, B] := AB − BA {A, B} := AB + BA A* = C A A⁺ = A*^T π^{−1}(N) T_(μν) := $\frac{1}{2}$(T_{μν} + T_{νμ}) T_[μν] := $\frac{1}{2}$(T_{μν} − T_{νμ}) : Commutator : Anticommutator : Complex-conjugate matrix : Hermitian adjoint matrix : Original domain of N with respect to the mapping π : M → N : Symmetrized second-rank tensor : Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01733	364	■ 0.591	$\mathrm{A}^{+} = \mathrm{A}^{* \mathrm{T}}$	$\mathrm{A}^{+} = \mathrm{A}^{* \mathrm{T}}$	:= ≡ ≈ ≅ ∼ × ⊗ ⊕ ⊗ G_o G̃ : Equal by definition : Identically equal : Isomorphic : Approximately equal : Asymptotic expansion : Topological product : Direct product : Direct sum : Semidirect product : Connected component of the group : Simply connected covering group [A, B] := AB − BA {A, B} := AB + BA A* = C A A⁺ = A*^T π^{−1}(N) T_(μν) := $\frac{1}{2}$(T_{μν} + T_{νμ}) T_[μν] := $\frac{1}{2}$(T_{μν} − T_{νμ}) : Commutator : Anticommutator : Complex-conjugate matrix : Hermitian adjoint matrix : Original domain of N with respect to the mapping π : M → N : Symmetrized second-rank tensor : Antisymmetrized second-rank tensor

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mielke-geometrodynamics_F01734	364	■ 0.591	$\pi^{-1}(\mathrm{~N})$	$\pi^{-1}(\mathrm{~N})$	:= : Equal by definition ≡ : Identically equal ≈ : Isomorphic ≅ : Approximately equal ~ : Asymptotic expansion × : Topological product ⊗ : Direct product ⊕ : Direct sum ⋈ : Semidirect product G_o : Connected component of the group $\tilde{G}$: Simply connected covering group [A, B] := AB − BA : Commutator {A, B} := AB + BA : Anticommutator A* = C A : Complex-conjugate matrix A⁺ = A*^T : Hermitian adjoint matrix $\pi^{-1}(\mathrm{~N})$: Original domain of N with respect to the mapping $\pi : \mathrm{~M} \rightarrow \mathrm{~N}$ $T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$: Symmetrized second-rank tensor $T_{[\mu\nu]} := \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01735	364	■ 0.591	$\pi: \mathrm{~M} \rightarrow \mathrm{~N}$	$\pi : \mathrm{~M} \rightarrow \mathrm{~N}$	:= : Equal by definition ≡ : Identically equal ≈ : Isomorphic ≅ : Approximately equal ~ : Asymptotic expansion × : Topological product ⊗ : Direct product ⊕ : Direct sum ⋈ : Semidirect product G_o : Connected component of the group $\tilde{G}$: Simply connected covering group [A, B] := AB − BA : Commutator {A, B} := AB + BA : Anticommutator A* = C A : Complex-conjugate matrix A⁺ = A*^T : Hermitian adjoint matrix $\pi^{-1}(\mathrm{~N})$: Original domain of N with respect to the mapping $\pi : \mathrm{~M} \rightarrow \mathrm{~N}$ $T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$: Symmetrized second-rank tensor $T_{[\mu\nu]} := \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01736	364	■ 0.591	$\mathrm{T}_{(\mu\nu)}:=\frac{1}{2}\left(\mathrm{T}_{\mu\nu}+\mathrm{T}_{\nu\mu}\right)$	$T_{(\mu\nu)} := \frac{1}{2}(\mathrm{~T}_{\mu\nu} + \mathrm{T}_{\nu\mu})$	:= : Equal by definition ≡ : Identically equal ≈ : Isomorphic ≅ : Approximately equal ~ : Asymptotic expansion × : Topological product ⊗ : Direct product ⊕ : Direct sum ⋈ : Semidirect product G_o : Connected component of the group $\tilde{G}$: Simply connected covering group [A, B] := AB − BA : Commutator {A, B} := AB + BA : Anticommutator A* = C A : Complex-conjugate matrix A⁺ = A*^T : Hermitian adjoint matrix $\pi^{-1}(\mathrm{~N})$: Original domain of N with respect to the mapping $\pi : \mathrm{~M} \rightarrow \mathrm{~N}$ $T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$: Symmetrized second-rank tensor $T_{[\mu\nu]} := \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})$: Antisymmetrized second-rank tensor

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mielke-geometrodynamics_F01737	364	0.591	$\mathrm{T}_{[\mu \nu]}:=\frac{1}{2}\left(\mathrm{T}_{\sim \mu \nu}-\mathrm{T}_{\nu \mu}\right)$	$T_{[\mu \nu]}:=\frac{1}{2}\left(T_{\mu \nu}-T_{\nu \mu}\right)$	:= ≡ ≈ ≈ ~ × ⊗ ⊕ ⊗ G。 G̃ [A, B] := AB − BA {A, B} := AB + BA A* = C A A⁺ = A[*]T π^{−1}(N) T_(μν) := ½(T_{μν} + T_{νμ}) T_[μν] := ½(T_{μν} − T_{νμ}) : Equal by definition : Identically equal : Isomorphic : Approximately equal : Asymptotic expansion : Topological product : Direct product : Direct sum : Semidirect product : Connected component of the group : Simply connected covering group : Commutator : Anticommutator : Complex-conjugate matrix : Hermitian adjoint matrix : Original domain of N with respect to the mapping π : M → N : Symmetrized second-rank tensor : Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01738	365	0.998	2 \times 2	2 × 2	2 × 2 Pauli matrices
mielke-geometrodynamics_F01739	365	0.984	$\mathrm{C}(0,2)=\mathbb{H}$	C(0, 2) = ℍ	of the Clifford algebra C(0, 2) = ℍ (field of quaternions).
mielke-geometrodynamics_F01740	365	0.989	4 \times 4	4 × 4	4 × 4 Dirac matrices in the so-called Pauli realization:
mielke-geometrodynamics_F01741	366	1.000	4<2 \mathrm{p}+1<2 \mathrm{p}+2 \leq n=2 l+2	4 < 2p + 1 < 2p + 2 ≤ n = 2l + 2	where 4 < 2p + 1 < 2p + 2 ≤ n = 2l + 2; cf. KALINOWSKI (1984).
mielke-geometrodynamics_F01742	366	0.997	\emptyset	∅	(iii) Both X itself and the empty set ∅ are open.
mielke-geometrodynamics_F01743	366	1.000	T_1	T ₁	T ₁ : For each two distinct points m ₁ , m ₂ ∈ X, there exist neighborhoods U _i ∋ m _i of
mielke-geometrodynamics_F01744	366	1.000	$\mathrm{m}_1, \mathrm{m}_2 \in X$	m ₁ , m ₂ ∈ X	T ₁ : For each two distinct points m ₁ , m ₂ ∈ X, there exist neighborhoods U _i ∋ m _i of
mielke-geometrodynamics_F01745	366	1.000	$\mathrm{U}_i \ni \mathrm{m}_i$	U _i ∋ m _i	T ₁ : For each two distinct points m ₁ , m ₂ ∈ X, there exist neighborhoods U _i ∋ m _i of
mielke-geometrodynamics_F01746	366	1.000	T_2	T ₂	T ₂ : For any two distinct points m ₁ , m ₂ ∈ X there exist neighborhoods U _i ∋ m _i that
mielke-geometrodynamics_F01747	366	0.566	$\mathrm{U}_1 \bigcap \mathrm{U}_2=\varnothing$	U ₁ ∩ U ₂ = ∅	are disjoint, namely U ₁ ∩ U ₂ = ∅.
mielke-geometrodynamics_F01748	367	1.000	$\sigma: \mathrm{M} \rightarrow \mathrm{N}$	σ : M → N	A diffeomorphism σ : M → N between manifolds of the same dimension (e.g., M ≈
mielke-geometrodynamics_F01749	367	1.000	$\mathrm{M} \approx$	M ≈	A diffeomorphism σ : M → N between manifolds of the same dimension (e.g., M ≈
mielke-geometrodynamics_F01750	367	1.000	$\mathrm{N} \approx \mathbb{R}^n$	N ≈ ℝ ⁿ	N ≈ ℝ ⁿ) is an injective mapping for which σ and the corresponding inverse mapping
mielke-geometrodynamics_F01751	367	0.980	$\{ \}^{-1}$	−1	σ ^{−1} are C [∞] functions, i.e., every partial derivative of σ exists and is continuous in
mielke-geometrodynamics_F01752	367	0.990	$\sigma_*: \mathrm{T}_m(\mathrm{M}) \rightarrow \mathrm{T}_{\sigma(m)}(\mathrm{N})$	σ _* : T _m (M) → T _{σ(m)} (N)	m gives rise to a mapping σ _* : T _m (M) → T _{σ(m)} (N) within the space of tangent
mielke-geometrodynamics_F01753	367	0.999	$\sigma(\mathrm{s}(\mathrm{t}))$	σ(s(t))	vectors along the curve s(t) or σ(s(t)) if constructed according to the directional
mielke-geometrodynamics_F01754	367	1.000	e^f	e ^f	derivative e ^f . On the other hand, the transposed mapping of σ _* defines a linear
mielke-geometrodynamics_F01755	367	1.000	σ_*	σ _*	derivative e ^f . On the other hand, the transposed mapping of σ _* defines a linear
mielke-geometrodynamics_F01756	367	1.000	$\sigma_*^T: T_{\sigma(m)}^*(\mathrm{N}) \rightarrow \mathrm{T}_M^*(\mathrm{M})$	σ _* ^T : T _{σ(m)} [*] (N) → T _M [*] (M)	mapping σ _* ^T : T _{σ(m)} [*] (N) → T _M [*] (M) between the “dual” cotangent spaces. As such,

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mielke-geometrodynamics_F01757	367	■ 1.000	$\{ \}^{*} \backslash \alpha$	${}^*\alpha$	Consequently, ${}^*\alpha$ denotes the inverse image of the differential form α . For cross
mielke-geometrodynamics_F01758	368	■ 0.858	$(p, \ 0)$	$(p, 0)$	Completely <i>antisymmetric</i> covariant tensor fields of degree $(p, 0)$ are of prime impor-
mielke-geometrodynamics_F01759	369	■ 0.982	$\{ \}^{\{1\}} \backslash \mathrm{d} \mathrm{f} = \vartheta^{\alpha} \partial_{\alpha} \backslash \mathrm{d} \mathrm{f}$	${}^1 \mathrm{d} \mathrm{f} = \vartheta^{\alpha} \partial_{\alpha} \mathrm{f}$	f, it is nothing but the total differential ¹ $\mathrm{d} \mathrm{f} = \vartheta^{\alpha} \partial_{\alpha} \mathrm{f}$. This action is extended to
mielke-geometrodynamics_F01760	369	■ 0.937	$\backslash \sigma_{*} \alpha^{\{0\}} := \alpha^{\{0\}} \backslash \circ \backslash \sigma$	$\sigma_{*} \alpha^{(0)} := \alpha^{(0)} \circ \sigma$	that for a 0-form is obtained by composition the $\sigma_{*} \alpha^{(0)} := \alpha^{(0)} \circ \sigma$.
mielke-geometrodynamics_F01761	369	■ 1.000	$\backslash \mathrm{varepsilon}_{\{\alpha_1 \dots \alpha_n\}} \backslash \mathrm{d} \mathrm{ots} \backslash \alpha_{\{n\}}$	$\varepsilon_{\alpha_1 \dots \alpha_n}$	Let $\varepsilon_{\alpha_1 \dots \alpha_n}$ be the totally antisymmetric unit tensor (Levi-Civita tensor; see MTW,
mielke-geometrodynamics_F01762	369	■ 0.999	$\backslash \mathrm{varepsilon}_{\{\hat{1} \dots \tilde{n}\}} = +1$	$\varepsilon_{\hat{1} \dots \tilde{n}} = +1$	p. 87) with the conventional choice of signs $\varepsilon_{\hat{1} \dots \tilde{n}} = +1$. In a pseudo-Riemannian
mielke-geometrodynamics_F01763	369	■ 1.000	$g_{\{\mathrm{i} \mathrm{j}\}}$	g_{ij}	manifold endowed with the metric tensor g_{ij} , this unit tensor allows one to define the
mielke-geometrodynamics_F01764	369	■ 0.997	$\backslash \mathrm{d} \mathrm{f} = \left(\partial \backslash \mathrm{f} / \partial x^{\{\mathrm{i}\}} \right) \mathrm{d} x^{\{\mathrm{i}\}}$	$\mathrm{d} \mathrm{f} = \left(\partial \mathrm{f} / \partial x^i \right) dx^i$	to a coordinate basis, it would read $\mathrm{d} \mathrm{f} = \left(\partial \mathrm{f} / \partial x^i \right) dx^i$.
mielke-geometrodynamics_F01765	371	■ 1.000	$\backslash \delta \backslash \alpha = 0$	$\delta \alpha = 0$	if and only if it is closed as well as <i>coclosed</i> , i.e., $\delta \alpha = 0$.
mielke-geometrodynamics_F01766	371	■ 0.901	$\backslash \mathrm{B}(\backslash \mathrm{M}) := \backslash \mathrm{d} \backslash \mathrm{D}(\backslash \mathrm{M})$	$\mathrm{B}(\mathrm{M}) := \mathrm{d} \mathrm{D}(\mathrm{M})$	$\mathrm{B}(\mathrm{M}) := \mathrm{d} \mathrm{D}(\mathrm{M})$ of <i>exact</i> forms
mielke-geometrodynamics_F01767	371	■ 1.000	$\backslash \mathrm{B}^{\{0\}}(\backslash \mathrm{M}) = 0$	$\mathrm{B}^0(\mathrm{M}) = 0$	On the other hand, we have $\mathrm{B}^0(\mathrm{M}) = 0$. Consequently,
mielke-geometrodynamics_F01768	372	■ 0.995	$\backslash \mathrm{H}^{\{\mathrm{p}\}}(\backslash \mathrm{M})$	$\mathrm{H}^{\mathrm{p}}(\mathrm{M})$	manifolds), the dimensions of the spaces $\mathrm{H}^{\mathrm{p}}(\mathrm{M})$ are called the <i>Betti numbers</i>
mielke-geometrodynamics_F01769	372	■ 1.000	$\backslash \mathrm{t} = -1$	$\mathrm{t} = -1$	with the Betti numbers as coefficients, gives for $\mathrm{t} = -1$ the alternating sum
mielke-geometrodynamics_F01770	372	■ 0.998	$\backslash \eta_{\{\mathrm{i}\}}$	η_i	with a countable basis η_i , a p-dimensional standard simplex is defined by the set
mielke-geometrodynamics_F01771	373	■ 0.843	$\backslash \mathrm{c} : \backslash \Delta_{\mathrm{p}} \rightarrow \backslash \mathrm{M}$	$\mathrm{c} : \Delta_{\mathrm{p}} \rightarrow \mathrm{M}$	the C^{∞} -mapping $\mathrm{c} : \Delta_{\mathrm{p}} \rightarrow \mathrm{M}$. These simplices form a free $\mathbb{R}$ -module $\mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$,
mielke-geometrodynamics_F01772	373	■ 0.843	$\backslash \mathrm{C}_{\{\mathrm{p}\}}(\backslash \mathrm{M}, \backslash \mathrm{R})$	$\mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$	the C^{∞} -mapping $\mathrm{c} : \Delta_{\mathrm{p}} \rightarrow \mathrm{M}$. These simplices form a free $\mathbb{R}$ -module $\mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$,
mielke-geometrodynamics_F01773	373	■ 1.000	$\backslash \partial \backslash \mathrm{c}$	$\partial \mathrm{c}$	whose elements are called p-chains. The boundary $\partial \mathrm{c}$ of a chain will be generated
mielke-geometrodynamics_F01774	373	■ 0.997	$\backslash \mathrm{Z}_{\{\mathrm{p}\}}(\backslash \mathrm{M}, \backslash \mathrm{R}) = \left(\backslash \mathrm{operatorname{Ker}} \backslash \partial \right)_{\{\mathrm{p}\}}, \backslash \mathrm{B}_{\{\mathrm{p}\}}(\backslash \mathrm{M}, \backslash \mathrm{R}) := \left(\backslash \mathrm{operatorname{Im}} \backslash \partial \right)_{\{\mathrm{p}\}}$	$\mathrm{Z}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) = (\mathrm{Ker} \partial)_{\mathrm{p}}, \mathrm{B}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) := (\mathrm{Im} \partial)_{\mathrm{p}}$	tically.” The spaces $\mathrm{Z}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) = (\mathrm{Ker} \partial)_{\mathrm{p}}, \mathrm{B}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) := (\mathrm{Im} \partial)_{\mathrm{p}}$ as well as
mielke-geometrodynamics_F01775	373	■ 1.000	$\backslash \mathrm{N}_{\{\mathrm{p}\}} \backslash \subset \backslash \mathrm{M}$	$\mathrm{N}_{(\mathrm{p})} \subset \mathrm{M}$	dimensional (compact) submanifold $\mathrm{N}_{(\mathrm{p})} \subset \mathrm{M}$ of M . Consequently, they can be
mielke-geometrodynamics_F01776	373	■ 1.000	$\backslash \mathrm{N}_{\{\mathrm{p}\}}$	$\mathrm{N}_{(\mathrm{p})}$	integrated, provided $\mathrm{N}_{(\mathrm{p})}$ admits a simplicial covering by the chain $\mathrm{c}_{\mathrm{p}} \in \mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$.
mielke-geometrodynamics_F01777	373	■ 1.000	$\backslash \mathrm{c}_{\{\mathrm{p}\}} \in \backslash \mathrm{C}_{\{\mathrm{p}\}}(\backslash \mathrm{M}, \backslash \mathrm{R})$	$\mathrm{c}_{\mathrm{p}} \in \mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$	integrated, provided $\mathrm{N}_{(\mathrm{p})}$ admits a simplicial covering by the chain $\mathrm{c}_{\mathrm{p}} \in \mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$.
mielke-geometrodynamics_F01778	374	■ 1.000	$\backslash \mathrm{H}^{\{\mathrm{P}\}}(\backslash \mathrm{M}) \backslash \times \backslash \mathrm{H}_{\{\mathrm{p}\}}(\backslash \mathrm{M}, \backslash \mathrm{R}) \rightarrow \backslash \mathrm{R}$	$\mathrm{H}^{\mathrm{P}}(\mathrm{M}) \times \mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) \rightarrow \mathbb{R}$	ping $\mathrm{H}^{\mathrm{P}}(\mathrm{M}) \times \mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) \rightarrow \mathbb{R}$ from the Cartesian product of the cohomology and
mielke-geometrodynamics_F01779	375	■ 1.000	$\backslash \mathrm{g} \in \backslash \mathrm{G}$	$\mathrm{g} \in \mathrm{G}$	(i) the elements $\mathrm{g} \in \mathrm{G}$ form an (analytic) manifold,
mielke-geometrodynamics_F01780	375	■ 1.000	$\backslash \mathrm{G} \backslash \times \backslash \mathrm{G} \rightarrow \backslash \mathrm{G}$	$\mathrm{G} \times \mathrm{G} \rightarrow \mathrm{G}$	(ii) the mapping $\mathrm{G} \times \mathrm{G} \rightarrow \mathrm{G}$ originating from the pointwise multiplication
mielke-geometrodynamics_F01781	375	■ 0.999	$\backslash \left(\backslash \mathrm{g}_{\{1\}}, \backslash \mathrm{g}_{\{2\}} \right) \rightarrow \backslash \mathrm{g}_{\{1\}} \backslash \mathrm{g}_{\{2\}}^{-1}$	$(\mathrm{g}_1, \mathrm{g}_2) \rightarrow \mathrm{g}_1 \mathrm{g}_2^{-1}$	$(\mathrm{g}_1, \mathrm{g}_2) \rightarrow \mathrm{g}_1 \mathrm{g}_2^{-1}$ of (group) elements is analytic as well.
mielke-geometrodynamics_F01782	375	■ 1.000	$\backslash \mathrm{G} = \backslash \mathrm{GL}(\backslash \mathrm{N}, \backslash \mathrm{C})$	$\mathrm{G} = \mathrm{GL}(\mathrm{N}, \mathbb{C})$	The most general example is given by $\mathrm{G} = \mathrm{GL}(\mathrm{N}, \mathbb{C})$, i.e., by the general linear

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mielke-geometrodynamics_F01783	375	■ 0.999	$\mathrm{T}_{\mathrm{e}}(\mathrm{e})(\mathrm{G})$	$T_e(G)$	The tangent space $T_e(G)$ attached to the unit element $e \in G$ of the group possesses
mielke-geometrodynamics_F01784	375	■ 0.999	$\mathrm{e} \in \mathrm{G}$	$e \in G$	The tangent space $T_e(G)$ attached to the unit element $e \in G$ of the group possesses
mielke-geometrodynamics_F01785	375	■ 0.952	$\xi=\left\{\xi^{\mathrm{j}},\mathrm{j}=1,\ldots,\mathrm{K}\right\}$	$\xi = \left\{ \xi^j, j = 1, \dots, K \right\}$	the <i>Lie algebra</i> of $\mathfrak{g}$ of G (KNI., p. 38). Let $\xi = \{ \xi^j, j = 1, \dots, K \}$ be an appropriate
mielke-geometrodynamics_F01786	375	■ 0.943	$\mathrm{c}_{\mathrm{ij}}^{\mathrm{k}}$	c_{ij}^k	Here the real <i>structure constants</i> c_{ij}^k are characteristic of the group G . They are always
mielke-geometrodynamics_F01787	376	■ 1.000	$\mathrm{A}=\mathrm{A}^{\mathrm{i}}\mathrm{I}_{\mathrm{i}}\mathrm{I}_{\mathrm{i}}$	$A = A^i I_i$	defined for generic Lie-algebra-valued vectors $A = A^i I_i$ and $B = B^j I_j$ is linear and
mielke-geometrodynamics_F01788	376	■ 1.000	$\mathrm{B}=\mathrm{B}^{\mathrm{j}}\mathrm{I}_{\mathrm{j}}\mathrm{I}_{\mathrm{j}}$	$B = B^j I_j$	defined for generic Lie-algebra-valued vectors $A = A^i I_i$ and $B = B^j I_j$ is linear and
mielke-geometrodynamics_F01789	376	■ 1.000	$\stackrel{\circ}{g}_{ls}$	$\overset{\circ}{g}_{ls}$	may be introduced that is determined by the <i>Cartan–Killing metric</i> $\overset{\circ}{g}_{ls}$ of the Lie
mielke-geometrodynamics_F01790	376	■ 0.863	$\mathfrak{gl}(\mathrm{N},\mathbb{C})$	$\mathfrak{gl}(N, \mathbb{C})$	of $\mathfrak{g}$. In the case of the Lie algebra $\mathfrak{gl}(N, \mathbb{C})$ of the general linear group, this product
mielke-geometrodynamics_F01791	376	■ 0.737	$\mathrm{G}=\exp^{\mathrm{Trg}}$	$G = \exp^{\mathrm{Trg}}$	$\det G = \exp^{\mathrm{Trg}}$. The scalar product (C.12) is nondegenerate if and only if G is
mielke-geometrodynamics_F01792	377	■ 0.995	ξ^{j}	ξ^j	algebra. In terms of a real parametrization ξ^j , an arbitrary group element is generated
mielke-geometrodynamics_F01793	377	■ 1.000	λ_{j}	λ_j	Here the λ_j denote the generalized Gell-Mann matrices, which are Hermitian and